```
1   PROGRAM PLC_PRG
2   VAR
3       start_request : BOOL;
4       stop : BOOL;
5       start : BOOL;
6       TON_start : TON;
7       TON_start_value : TIME;
8       operate : BOOL;
9       TOF_stop : TOF;
10      TOF_stop_value : TIME;
11      operate_status : BOOL;
12      emergency : BOOL;
13      operate_status_led : BOOL;
14      alarm : BOOL;
15
16      conveyor_request_RS : RS;
17      item_entry_sensor : BOOL;
18      item_id_sensor : BOOL;
19      conveyor_request : BOOL;
20      conveyor_motor : BOOL;
21      conveyor_timeout : BOOL;
22      TON_conveyor_timeout : TON;
23      TON_conveyor_timeout_value : TIME;
24
25      reset : BOOL;
26      conveyor_fault_RS : RS;
27      conveyor_fault : BOOL;
28
29      (* Batch Tracking *)
30      expected_boxes : WORD;
31      boxes_entered : WORD;
32      all_boxes_entered : BOOL;
33      CTU_boxes_entered : CTU;
34      batch_reset : BOOL;
35
36      boxes_sorted : WORD;
37      all_boxes_sorted : BOOL;
38      CTU_boxes_sorted : CTU;
39      box_sorted_pulse  : BOOL;
40      box_sorted_R_TRIG  : R_TRIG;
41
42      (* Box Tracking *)
43      entry_box_id : WORD;
44      id_station_box_id : WORD;
45      id_station_occupied : BOOL;
46      id_station_occupied_RS : RS;
47
48
49      (* Barcode / SKU Identification *)
50      barcode_input  : INT;
51      barcode_scan  : BOOL;
52      barcode_scan_R_TRIG : R_TRIG;
```

```
53          barcode_scan_pulse  : BOOL ;
54
55          barcode_valid  : BOOL ;
56
57          CTU_barcode_attempts  : CTU ;
58          barcode_attempts  : WORD ;
59          barcode_attempt_limit  : BOOL ;
60
61          barcode_fault  : BOOL ;
62          barcode_fault_RS  : RS ;
63
64          id_scan_complete_RS  : RS ;
65
66          id_station_sku  : INT ;
67
68          (* Simulated Item Database - 6 Example SKUs *)
69          db_expected_weight  : ARRAY [ 1 .. 6 ] OF REAL := [ 5.0 , 1.0 , 3.0 , 2.0 ,
        5.0 , 2.0 ] ;
70
71          db_expected_length  : ARRAY [ 1 .. 6 ] OF REAL := [ 400.0 , 100.0 , 400.0 ,
        300.0 , 400.0 , 300.0 ] ;
72          db_expected_width  : ARRAY [ 1 .. 6 ] OF REAL := [ 250.0 , 100.0 , 300.0 ,
        250.0 , 300.0 , 300.0 ] ;
73          db_expected_height  : ARRAY [ 1 .. 6 ] OF REAL := [ 100.0 , 300.0 , 200.0 ,
        150.0 , 150.0 , 150.0 ] ;
74
75          db_temperature_class  : ARRAY [ 1 .. 6 ] OF WORD := [ 0 , 0 , 1 , 1 , 2 , 2 ]
        ;
76          db_fragility_class  : ARRAY [ 1 .. 6 ] OF WORD := [ 0 , 1 , 0 , 1 , 0 , 1 ] ;
77
78          (* ID Station Database Data *)
79          id_expected_weight  : REAL ;
80          id_expected_length  : REAL ;
81          id_expected_width  : REAL ;
82          id_expected_height  : REAL ;
83
84          id_temperature_class  : WORD ;
85          id_fragility_class  : WORD ;
86
87
88          (* ID to Weight Transfer *)
89          id_scan_complete : BOOL ;
90
91          id_to_weight_request_RS : RS ;
92          id_to_weight_request : BOOL ;
93          id_to_weight_motor : BOOL ;
94
95
96          (* ID to Weight Transfer Fault Handling *)
97          id_to_weight_timeout  : BOOL ;
98          TON_id_to_weight_timeout  : TON ;
```

```
 99          TON_id_to_weight_timeout_value  : TIME ;
100          id_to_weight_fault_RS  : RS ;
101          id_to_weight_fault  : BOOL ;
102
103          (* Wrong Destination Detection *)
104          wrong_destination_fault_RS  : RS ;
105          wrong_destination_fault  : BOOL ;
106          wrong_destination_storage  : WORD ;
107
108          (* Error / Holding Area *)
109          holding_request_RS  : RS ;
110          holding_request  : BOOL ;
111
112          holding_motor  : BOOL ;
113
114          holding_entry_sensor  : BOOL ;
115          holding_entry_confirm  : BOOL ;
116
117          holding_timeout  : BOOL ;
118          TON_holding_timeout  : TON ;
119          TON_holding_timeout_value  : TIME ;
120
121          holding_fault_RS  : RS ;
122          holding_fault  : BOOL ;
123
124          holding_box_id  : WORD ;
125
126          (* Holding Reason:
127             1 = Barcode Failure
128             2 = Measurement Failure
129             3 = No Suitable Storage
130             *)
131          holding_reason_code  : WORD ;
132
133
134          weight_station_sensor : BOOL ;
135          weight_station_box_id : WORD ;
136          weight_station_occupied : BOOL ;
137          weight_station_occupied_RS : RS ;
138
139
140          (* Weight Station Item Data *)
141          weight_station_sku  : INT ;
142
143          weight_station_expected_weight  : REAL ;
144          weight_station_expected_length  : REAL ;
145          weight_station_expected_width  : REAL ;
146          weight_station_expected_height  : REAL ;
147
148          weight_station_temperature_class  : WORD ;
149          weight_station_fragility_class  : WORD ;
```

```
150
151
152         (* Measurement Inputs *)
153         measured_weight  :  REAL ;
154         measured_length  :  REAL ;
155         measured_width  :  REAL ;
156         measured_height  :  REAL ;
157
158         measurement_check  :  BOOL ;
159         measurement_check_R_TRIG  :  R_TRIG ;
160         measurement_check_pulse  :  BOOL ;
161
162         (* Calculated Measurement Limits *)
163         weight_lower_limit  :  REAL ;
164         weight_upper_limit  :  REAL ;
165
166         length_lower_limit  :  REAL ;
167         length_upper_limit  :  REAL ;
168
169         width_lower_limit  :  REAL ;
170         width_upper_limit  :  REAL ;
171
172         height_lower_limit  :  REAL ;
173         height_upper_limit  :  REAL ;
174
175         (* Measurement Checks *)
176         weight_match  :  BOOL ;
177         length_match  :  BOOL ;
178         width_match  :  BOOL ;
179         height_match  :  BOOL ;
180
181         physical_weight_ok  :  BOOL ;
182         physical_dimensions_ok  :  BOOL ;
183
184         measurement_valid  :  BOOL ;
185
186         (* Measurement Retry / Fault *)
187         CTU_measurement_attempts  :  CTU ;
188         measurement_attempts  :  WORD ;
189         measurement_attempt_limit  :  BOOL ;
190
191         measurement_complete  :  BOOL ;
192         measurement_complete_RS  :  RS ;
193
194         measurement_fault  :  BOOL ;
195         measurement_fault_RS  :  RS ;
196
197         classification_complete  :  BOOL ;
198         classification_complete_RS  :  RS ;
199
200
```

```
201         (* Storage Type Configuration *)
202         storage_1_temperature_class  : WORD := 0 ;    (* Room *)
203         storage_1_fragility_class  : WORD := 0 ;      (* Non-fragile *)
204
205         storage_2_temperature_class  : WORD := 0 ;    (* Room *)
206         storage_2_fragility_class  : WORD := 1 ;      (* Fragile *)
207
208         storage_3_temperature_class  : WORD := 1 ;    (* Refrigerated *)
209         storage_3_fragility_class  : WORD := 0 ;      (* Non-fragile *)
210
211         storage_4_temperature_class  : WORD := 1 ;    (* Refrigerated *)
212         storage_4_fragility_class  : WORD := 1 ;      (* Fragile *)
213
214         storage_5_temperature_class  : WORD := 2 ;    (* Frozen *)
215         storage_5_fragility_class  : WORD := 0 ;      (* Non-fragile *)
216
217         storage_6_temperature_class  : WORD := 2 ;    (* Frozen *)
218         storage_6_fragility_class  : WORD := 1 ;      (* Fragile *)
219
220
221         (* Global Storage VARs *)
222         storage_assignment_request  : BOOL ;
223         assigned_storage  : WORD ;
224         storage_assigned  : BOOL ;
225         storage_assigned_RS  : RS ;
226         assignment_reset  : BOOL ;
227
228         storage_assigned_R_TRIG  : R_TRIG ;
229         weight_station_release  : BOOL ;
230
231         no_storage_available  : BOOL ;
232         no_storage_available_RS  : RS ;
233
234         (* Storage 1 - Room / Non-fragile *)
235         storage_1_match  : BOOL ;
236         storage_1_available  : BOOL ;
237         storage_1_max_weight  : REAL := 100.0 ;
238         storage_1_current_weight  : REAL ;
239         storage_1_weight_after_item  : REAL ;
240
241         (* Storage 2 - Room / Fragile *)
242         storage_2_match  : BOOL ;
243         storage_2_available  : BOOL ;
244         storage_2_max_weight  : REAL := 100.0 ;
245         storage_2_current_weight  : REAL ;
246         storage_2_weight_after_item  : REAL ;
247
248         (* Storage 3 - Refrigerated / Non-fragile *)
249         storage_3_match  : BOOL ;
250         storage_3_available  : BOOL ;
251         storage_3_max_weight  : REAL := 100.0 ;
```

```
252         storage_3_current_weight  : REAL ;
253         storage_3_weight_after_item  : REAL ;
254
255         (* Storage 4 - Refrigerated / Fragile *)
256         storage_4_match  : BOOL ;
257         storage_4_available  : BOOL ;
258         storage_4_max_weight  : REAL  := 100.0 ;
259         storage_4_current_weight  : REAL ;
260         storage_4_weight_after_item  : REAL ;
261
262         (* Storage 5 - Frozen / Non-fragile *)
263         storage_5_match  : BOOL ;
264         storage_5_available  : BOOL ;
265         storage_5_max_weight  : REAL  := 100.0 ;
266         storage_5_current_weight  : REAL ;
267         storage_5_weight_after_item  : REAL ;
268
269         (* Storage 6 - Frozen / Fragile *)
270         storage_6_match  : BOOL ;
271         storage_6_available  : BOOL ;
272         storage_6_max_weight  : REAL  := 100.0 ;
273         storage_6_current_weight  : REAL ;
274         storage_6_weight_after_item  : REAL ;
275
276         (* Item Classification / Sorting *)
277         item_temperature_class  : WORD ;    (* 0=Room, 1=Refrigerated, 2=Frozen *)
278         item_fragility_class  : WORD ;      (* 0=Non-fragile, 1=Fragile *)
279
280
281         item_weight  : REAL ;
282         item_sku  : INT ;
283         item_length  : REAL ;
284         item_width  : REAL ;
285         item_height  : REAL ;
286
287         (* Latched Sorting Item Data *)
288         sorting_box_id  : WORD ;
289         sorting_item_weight  : REAL ;
290         sorting_temperature_class  : WORD ;
291         sorting_fragility_class  : WORD ;
292         sorting_assigned_storage  : WORD ;
293
294         sorting_sku  : INT ;
295         sorting_item_length  : REAL ;
296         sorting_item_width  : REAL ;
297         sorting_item_height  : REAL ;
298
299         sorting_item_quantity  : WORD ;
300         sorting_expiry_day  : WORD ;
301
302
```

```
303
304
305         (* Sorting Control *)
306         (* Sorting Conveyor Control *)
307         sorting_conveyor_request_RS  : RS ;
308         sorting_conveyor_request  : BOOL ;
309         sorting_conveyor_motor  : BOOL ;
310
311         sorting_conveyor_timeout  : BOOL ;
312         TON_sorting_conveyor_timeout  : TON ;
313         TON_sorting_conveyor_timeout_value  : TIME ;
314
315         sorting_conveyor_fault_RS  : RS ;
316         sorting_conveyor_fault  : BOOL ;
317             (* Routing Request *)
318             sorting_request  : BOOL ;
319             sorting_complete  : BOOL ;
320             sorting_complete_RS  : RS ;
321             sorting_reset  : BOOL ;
322
323             (* Sorting Cycle Auto Reset *)
324             sorting_cycle_reset  : BOOL ;
325             TON_sorting_reset  : TON ;
326             TON_sorting_reset_value  : TIME ;
327
328             (* Storage Routing *)
329             route_to_storage_1  : BOOL ;
330             route_to_storage_2  : BOOL ;
331             route_to_storage_3  : BOOL ;
332             route_to_storage_4  : BOOL ;
333             route_to_storage_5  : BOOL ;
334             route_to_storage_6  : BOOL ;
335
336             (* Storage Entry Sensors *)
337             storage_1_entry_sensor  : BOOL ;
338             storage_2_entry_sensor  : BOOL ;
339             storage_3_entry_sensor  : BOOL ;
340             storage_4_entry_sensor  : BOOL ;
341             storage_5_entry_sensor  : BOOL ;
342             storage_6_entry_sensor  : BOOL ;
343
344             (* Storage Entry Confirmation *)
345             storage_1_entry_confirm  : BOOL ;
346             storage_2_entry_confirm  : BOOL ;
347             storage_3_entry_confirm  : BOOL ;
348             storage_4_entry_confirm  : BOOL ;
349             storage_5_entry_confirm  : BOOL ;
350             storage_6_entry_confirm  : BOOL ;
351
352             (* Storage Entry Trigger *)
353             storage_1_entry_R_TRIG  : R_TRIG ;
```

```
354             storage_2_entry_R_TRIG  : R_TRIG;
355             storage_3_entry_R_TRIG  : R_TRIG;
356             storage_4_entry_R_TRIG  : R_TRIG;
357             storage_5_entry_R_TRIG  : R_TRIG;
358             storage_6_entry_R_TRIG  : R_TRIG;
359
360
361
362
363             (* ========================================================  *)
364         (* INVENTORY DATABASE *)
365         (* 12 records = 2 batch records for each of 6 SKUs *)
366         (* ========================================================  *)
367
368         inventory_occupied  : ARRAY[1..12] OF BOOL;
369
370         inventory_box_id  : ARRAY[1..12] OF WORD;
371         inventory_sku  : ARRAY[1..12] OF INT;
372         inventory_quantity  : ARRAY[1..12] OF WORD;
373
374         inventory_storage_location  : ARRAY[1..12] OF WORD;
375
376         inventory_weight  : ARRAY[1..12] OF REAL;
377         inventory_length  : ARRAY[1..12] OF REAL;
378         inventory_width  : ARRAY[1..12] OF REAL;
379         inventory_height  : ARRAY[1..12] OF REAL;
380
381         inventory_temperature_class  : ARRAY[1..12] OF WORD;
382         inventory_fragility_class  : ARRAY[1..12] OF WORD;
383
384         inventory_expiry_day  : ARRAY[1..12] OF WORD;
385
386         inventory_entry_sequence  : ARRAY[1..12] OF WORD;
387
388         (* Status:
389            0 = Empty
390            1 = Stored
391            2 = Retrieval in progress
392            3 = Removed / Used
393            *)
394         inventory_status  : ARRAY[1..12] OF WORD;
395         (* Inventory Record Allocation *)
396         inventory_write_slot  : INT;
397         inventory_write_valid  : BOOL;
398
399         inventory_record_added  : BOOL;
400
401         inventory_sequence_counter  : WORD;
402         inventory_record_count  : WORD;
403         total_inventory_quantity  : WORD;
404
```

```
405         (* Inventory Database Capacity Fault *)
406         inventory_record_full_fault_RS  : RS;
407         inventory_record_full_fault  : BOOL;
408
409
410
411
412
413         (* Incoming Batch Information *)
414         item_quantity_input  : WORD  := 1;
415         item_expiry_day_input  : WORD;
416
417         item_quantity  : WORD;
418         item_expiry_day  : WORD;
419
420
421
422
423         sku_1_quantity  : WORD;
424         sku_2_quantity  : WORD;
425         sku_3_quantity  : WORD;
426         sku_4_quantity  : WORD;
427         sku_5_quantity  : WORD;
428         sku_6_quantity  : WORD;
429
430         sku_1_min_stock  : WORD  := 5;
431         sku_2_min_stock  : WORD  := 5;
432         sku_3_min_stock  : WORD  := 5;
433         sku_4_min_stock  : WORD  := 5;
434         sku_5_min_stock  : WORD  := 5;
435         sku_6_min_stock  : WORD  := 5;
436
437         sku_1_low_stock  : BOOL;
438         sku_2_low_stock  : BOOL;
439         sku_3_low_stock  : BOOL;
440         sku_4_low_stock  : BOOL;
441         sku_5_low_stock  : BOOL;
442         sku_6_low_stock  : BOOL;
443
444         sku_1_out_of_stock  : BOOL;
445         sku_2_out_of_stock  : BOOL;
446         sku_3_out_of_stock  : BOOL;
447         sku_4_out_of_stock  : BOOL;
448         sku_5_out_of_stock  : BOOL;
449         sku_6_out_of_stock  : BOOL;
450
451
452
453
454
455         (* ========================================================  *)
```

```
456         (* ENVIRONMENTAL MONITORING *)
457         (* ========================================================  *)
458
459         (* Storage Temperature Inputs *)
460         storage_3_temperature  : REAL := 4.0 ;
461         storage_4_temperature  : REAL := 4.0 ;
462         storage_5_temperature  : REAL := - 20.0 ;
463         storage_6_temperature  : REAL := - 20.0 ;
464
465         (* Maximum Temperature Limits *)
466         storage_3_temp_max  : REAL := 5.0 ;
467         storage_4_temp_max  : REAL := 5.0 ;
468
469         (* -18C used as configurable simulation assumption for frozen storage *)
470         storage_5_temp_max  : REAL := - 18.0 ;
471         storage_6_temp_max  : REAL := - 18.0 ;
472
473         (* Individual Temperature Faults *)
474         storage_3_temp_fault  : BOOL ;
475         storage_4_temp_fault  : BOOL ;
476         storage_5_temp_fault  : BOOL ;
477         storage_6_temp_fault  : BOOL ;
478
479         (* Global Temperature State *)
480         temperature_fault  : BOOL ;
481         temperature_fault_zone  : WORD ;
482
483         (* Temperature Fault Logging *)
484         CTU_storage_3_temp_faults  : CTU ;
485         CTU_storage_4_temp_faults  : CTU ;
486         CTU_storage_5_temp_faults  : CTU ;
487         CTU_storage_6_temp_faults  : CTU ;
488
489         storage_3_temp_fault_count  : WORD ;
490         storage_4_temp_fault_count  : WORD ;
491         storage_5_temp_fault_count  : WORD ;
492         storage_6_temp_fault_count  : WORD ;
493
494         (* Humidity Monitoring *)
495         humidity_monitor_enable  : BOOL ;
496         humidity_monitored_zone  : WORD := 3 ;
497
498         measured_humidity  : REAL := 50.0 ;
499         humidity_min_limit  : REAL := 40.0 ;
500         humidity_max_limit  : REAL := 60.0 ;
501
502         humidity_zone_valid  : BOOL ;
503         humidity_low_fault  : BOOL ;
504         humidity_high_fault  : BOOL ;
505         humidity_fault  : BOOL ;
506
```

```
507         CTU_humidity_faults  : CTU ;
508         humidity_fault_count  : WORD ;
509
510         (* Environmental Alarm / Inspection *)
511         environment_alarm_RS  : RS ;
512         environment_alarm  : BOOL ;
513
514         environment_inspection_required_RS  : RS ;
515         environment_inspection_required  : BOOL ;
516
517         inspection_storage_zone  : WORD ;
518
519         (* Environmental Log Reset *)
520         environment_log_reset  : BOOL ;
521
522
523         //Restaurant Implementation Code
524
525         //Ingredient request
526         ingredient_request_sku  : INT ;
527         ingredient_request_quantity  : WORD ;
528         ingredient_request_submit  : BOOL ;
529         ingredient_request_reset  : BOOL ;
530
531         ingredient_request_R_TRIG  : R_TRIG ;
532         ingredient_request_pulse  : BOOL ;
533
534         ingredient_available_quantity  : WORD ;
535         ingredient_request_sku_valid  : BOOL ;
536         ingredient_request_quantity_valid  : BOOL ;
537         ingredient_request_stock_sufficient  : BOOL ;
538
539         ingredient_request_status  : WORD ;
540
541         ingredient_sku_ge_1  : BOOL ;
542         ingredient_sku_le_6  : BOOL ;
543
544
545         ingredient_sku_1_selected  : BOOL ;
546         ingredient_sku_2_selected  : BOOL ;
547         ingredient_sku_3_selected  : BOOL ;
548         ingredient_sku_4_selected  : BOOL ;
549         ingredient_sku_5_selected  : BOOL ;
550         ingredient_sku_6_selected  : BOOL ;
551
552         ingredient_request_valid  : BOOL ;
553
554
555         //FIFO system
556         ingredient_selection_start  : BOOL ;
557
```

```
558         ingredient_slot_a  :  INT ;
559         ingredient_slot_b  :  INT ;
560
561         ingredient_slot_a_eligible  :  BOOL ;
562         ingredient_slot_b_eligible  :  BOOL ;
563
564         ingredient_selected_slot  :  INT ;
565         ingredient_selected_storage  :  WORD ;
566         ingredient_selected_quantity  :  WORD ;
567         ingredient_selected_expiry_day  :  WORD ;
568
569         ingredient_selection_found  :  BOOL ;
570         ingredient_selection_complete  :  BOOL ;
571
572             (* FEFO Configuration *)
573         restaurant_current_day  :  WORD  :=  1 ;
574
575         (* Selection Reset *)
576         ingredient_selection_clear  :  BOOL ;
577
578         (* Batch A Checks *)
579         ingredient_slot_a_sku_match  :  BOOL ;
580         ingredient_slot_a_status_ok  :  BOOL ;
581         ingredient_slot_a_quantity_ok  :  BOOL ;
582         ingredient_slot_a_expiry_ok  :  BOOL ;
583
584         (* Batch B Checks *)
585         ingredient_slot_b_sku_match  :  BOOL ;
586         ingredient_slot_b_status_ok  :  BOOL ;
587         ingredient_slot_b_quantity_ok  :  BOOL ;
588         ingredient_slot_b_expiry_ok  :  BOOL ;
589
590         (* FEFO Comparison *)
591         ingredient_expiry_a_before_b  :  BOOL ;
592         ingredient_expiry_equal  :  BOOL ;
593         ingredient_sequence_a_first  :  BOOL ;
594
595         (* Final Batch Decision *)
596         ingredient_choose_a  :  BOOL ;
597         ingredient_choose_b  :  BOOL ;
598         ingredient_selection_valid_now  :  BOOL ;
599         ingredient_status_accepted :  BOOL ;
600
601         ingredient_slot_a_raw  :  INT ;
602         ingredient_slot_b_raw  :  INT ;
603
604
605
606         (* Ingredient Retrieval *)
607         ingredient_retrieval_pending  :  BOOL ;
608         ingredient_retrieval_confirm  :  BOOL ;
```

```
609         ingredient_retrieval_cancel  : BOOL ;
610         ingredient_retrieval_complete  : BOOL ;
611
612         ingredient_retrieval_slot  : INT ;
613         ingredient_retrieval_storage  : WORD ;
614         ingredient_retrieval_quantity  : WORD ;
615         ingredient_retrieval_sku  : INT ;
616         ingredient_retrieval_available  : WORD ;
617
618         ingredient_retrieval_quantity_ok  : BOOL ;
619         ingredient_retrieval_start  : BOOL ;
620
621         ingredient_request_accepted  : BOOL ;
622         ingredient_request_accepted_RS  : RS ;
623         ingredient_retrieval_pending_RS  : RS ;
624         ingredient_retrieval_ready  : BOOL ;
625
626         (* Ingredient Retrieval Confirmation *)
627         ingredient_retrieval_safe_slot  : INT ;
628         ingredient_retrieval_slot_valid  : BOOL ;
629         ingredient_retrieval_occupied  : BOOL ;
630         ingredient_retrieval_sku_match  : BOOL ;
631         ingredient_retrieval_stock_ok  : BOOL ;
632         ingredient_retrieval_locked_quantity_ok  : BOOL ;
633
634         ingredient_sku_total_ok  : BOOL ;
635         ingredient_overall_total_ok  : BOOL ;
636
637         ingredient_retrieval_confirm_R_TRIG  : R_TRIG ;
638         ingredient_retrieval_confirm_pulse  : BOOL ;
639         ingredient_retrieval_checks_ok  : BOOL ;
640         ingredient_retrieval_commit_pulse  : BOOL ;
641
642         (* Ingredient Inventory Deduction *)
643         ingredient_deduction_done  : BOOL ;
644         ingredient_deduction_done_RS  : RS ;
645         ingredient_deduction_allowed  : BOOL ;
646
647         ingredient_batch_quantity_after  : WORD ;
648         ingredient_sku_quantity_after  : WORD ;
649         ingredient_total_quantity_after  : WORD ;
650
651         ingredient_batch_empty  : BOOL ;
652         ingredient_retrieval_success  : BOOL ;
653         ingredient_retrieval_success_R_TRIG  : R_TRIG ;
654         ingredient_retrieval_sku_total  : WORD ;
655         ingredient_retrieval_sku_valid  : BOOL ;
656         ingredient_retrieval_success_pulse  : BOOL ;
657
658         (* ========================================  *)
659         (* STORAGE TO KITCHEN TRANSPORT SYSTEM *)
```

```
660         (* ========================================  *)
661
662         ingredient_transport_pending_RS  :  RS ;
663         ingredient_transport_pending  :  BOOL ;
664
665         kitchen_transport_motor  :  BOOL ;
666
667         (* Transport Item Information *)
668         kitchen_transport_sku  :  INT ;
669         kitchen_transport_quantity  :  WORD ;
670         kitchen_transport_source_storage  :  WORD ;
671         kitchen_transport_source_slot  :  INT ;
672
673         (* Storage Routing *)
674         kitchen_route_from_storage_1  :  BOOL ;
675         kitchen_route_from_storage_2  :  BOOL ;
676         kitchen_route_from_storage_3  :  BOOL ;
677         kitchen_route_from_storage_4  :  BOOL ;
678         kitchen_route_from_storage_5  :  BOOL ;
679         kitchen_route_from_storage_6  :  BOOL ;
680
681         (* Kitchen Arrival Detection *)
682         kitchen_arrival_sensor  :  BOOL ;
683         kitchen_arrival_R_TRIG  :  R_TRIG ;
684         kitchen_arrival_pulse  :  BOOL ;
685         kitchen_transport_arrival_valid  :  BOOL ;
686
687         (* Transport Completion *)
688         kitchen_transport_complete  :  BOOL ;
689         kitchen_transport_complete_RS  :  RS ;
690
691         (* Transport Timeout *)
692         kitchen_transport_timeout  :  BOOL ;
693         TON_kitchen_transport_timeout  :  TON ;
694         TON_kitchen_transport_timeout_value  :  TIME ;
695
696         (* Transport Fault *)
697         kitchen_transport_fault  :  BOOL ;
698         kitchen_transport_fault_RS  :  RS ;
699
700         (* Transport Reset and Retry *)
701         kitchen_transport_reset  :  BOOL ;
702         kitchen_transport_retry  :  BOOL ;
703
704
705         kitchen_transport_request_RS :  RS ;
706         kitchen_transport_request :  BOOL ;
707
708
709         (* KITCHEN ORDER MANAGEMENT *)
710
```

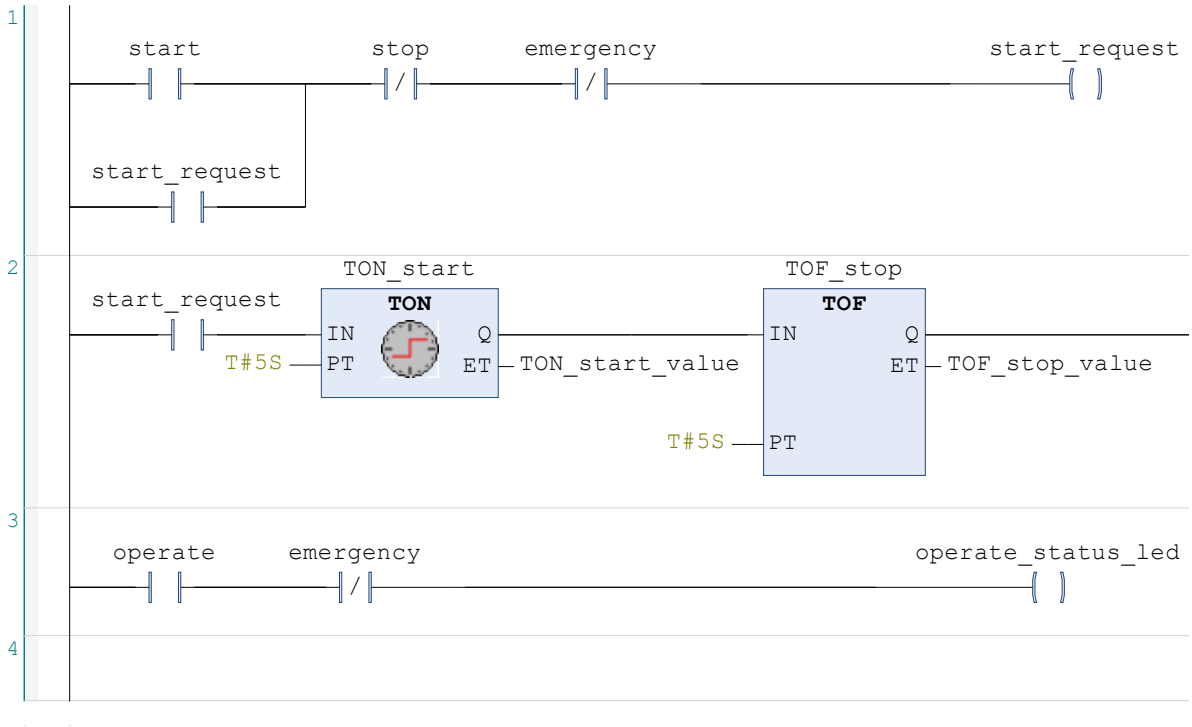
```
711         (* Customer Order Inputs *)
712         order_input_meal_id : INT;
713         order_input_quantity : WORD;
714         order_input_table_id : WORD;
715         order_submit : BOOL;
716         order_submit_R_TRIG : R_TRIG;
717         order_submit_pulse : BOOL;
718
719         (* Order Validation *)
720         order_meal_valid : BOOL;
721         order_quantity_valid : BOOL;
722         order_table_valid : BOOL;
723         order_valid : BOOL;
724         order_queue_full : BOOL;
725         order_accepted : BOOL;
726         order_rejected : BOOL;
727
728         (* FIFO Order Queue - 5 Orders *)
729         order_queue_count : INT;
730         order_queue_write_index : INT := 1;
731         order_queue_read_index : INT := 1;
732         order_id_counter : WORD;
733
734         order_queue_id : ARRAY[1..5] OF WORD;
735         order_queue_meal_id : ARRAY[1..5] OF INT;
736         order_queue_quantity : ARRAY[1..5] OF WORD;
737         order_queue_table_id : ARRAY[1..5] OF WORD;
738
739         (* Queue Operations *)
740         order_enqueue : BOOL;
741         order_dequeue : BOOL;
742
743         (* Current Kitchen Order *)
744         kitchen_order_active : BOOL;
745         kitchen_order_current_id : WORD;
746         kitchen_order_current_meal_id : INT;
747         kitchen_order_current_quantity : WORD;
748         kitchen_order_current_table_id : WORD;
749
750         (* Meal Preparation *)
751         kitchen_order_meal_ready : BOOL;
752         kitchen_order_meal_ready_R_TRIG : R_TRIG;
753         kitchen_order_meal_ready_pulse : BOOL;
754
755         (* Delivery Handoff *)
756         kitchen_order_ready_for_delivery : BOOL;
757         kitchen_order_delivery_ack : BOOL;
758
759         (* Reset *)
760         kitchen_order_reset : BOOL;
761
```

```
762         order_queue_has_items  : BOOL ;
763
764         kitchen_order_meal_ready_valid  : BOOL ;
765
766         kitchen_order_delivery_ack_R_TRIG  : R_TRIG ;
767         kitchen_order_delivery_ack_pulse  : BOOL ;
768         kitchen_order_handoff_complete  : BOOL ;
769
770         kitchen_order_ready_RS  : RS ;
771         kitchen_order_active_RS  : RS ;
772
773         order_reset_allowed  : BOOL ;
774
775
776
777         (* ======================================== *)
778         (* KITCHEN TO CUSTOMER DELIVERY SYSTEM *)
779         (* ======================================== *)
780
781         (* Handoff Validation *)
782         customer_delivery_table_ge_1  : BOOL ;
783         customer_delivery_table_le_6  : BOOL ;
784         customer_delivery_quantity_valid  : BOOL ;
785         customer_delivery_id_valid  : BOOL ;
786         customer_delivery_handoff_valid  : BOOL ;
787         customer_delivery_accept  : BOOL ;
788
789         (* Delivery Job Tracking *)
790         customer_delivery_job_pending_RS  : RS ;
791         customer_delivery_job_pending  : BOOL ;
792
793         customer_delivery_busy_RS  : RS ;
794         customer_delivery_busy  : BOOL ;
795
796         (* Delivery Information *)
797         customer_delivery_order_id  : WORD ;
798         customer_delivery_meal_id  : INT ;
799         customer_delivery_quantity  : WORD ;
800         customer_delivery_table_id  : WORD ;
801
802         (* Table Routing *)
803         customer_delivery_route_1  : BOOL ;
804         customer_delivery_route_2  : BOOL ;
805         customer_delivery_route_3  : BOOL ;
806         customer_delivery_route_4  : BOOL ;
807         customer_delivery_route_5  : BOOL ;
808         customer_delivery_route_6  : BOOL ;
809         customer_delivery_route_valid  : BOOL ;
810
811         (* Conveyor *)
812         customer_delivery_motor  : BOOL ;
```

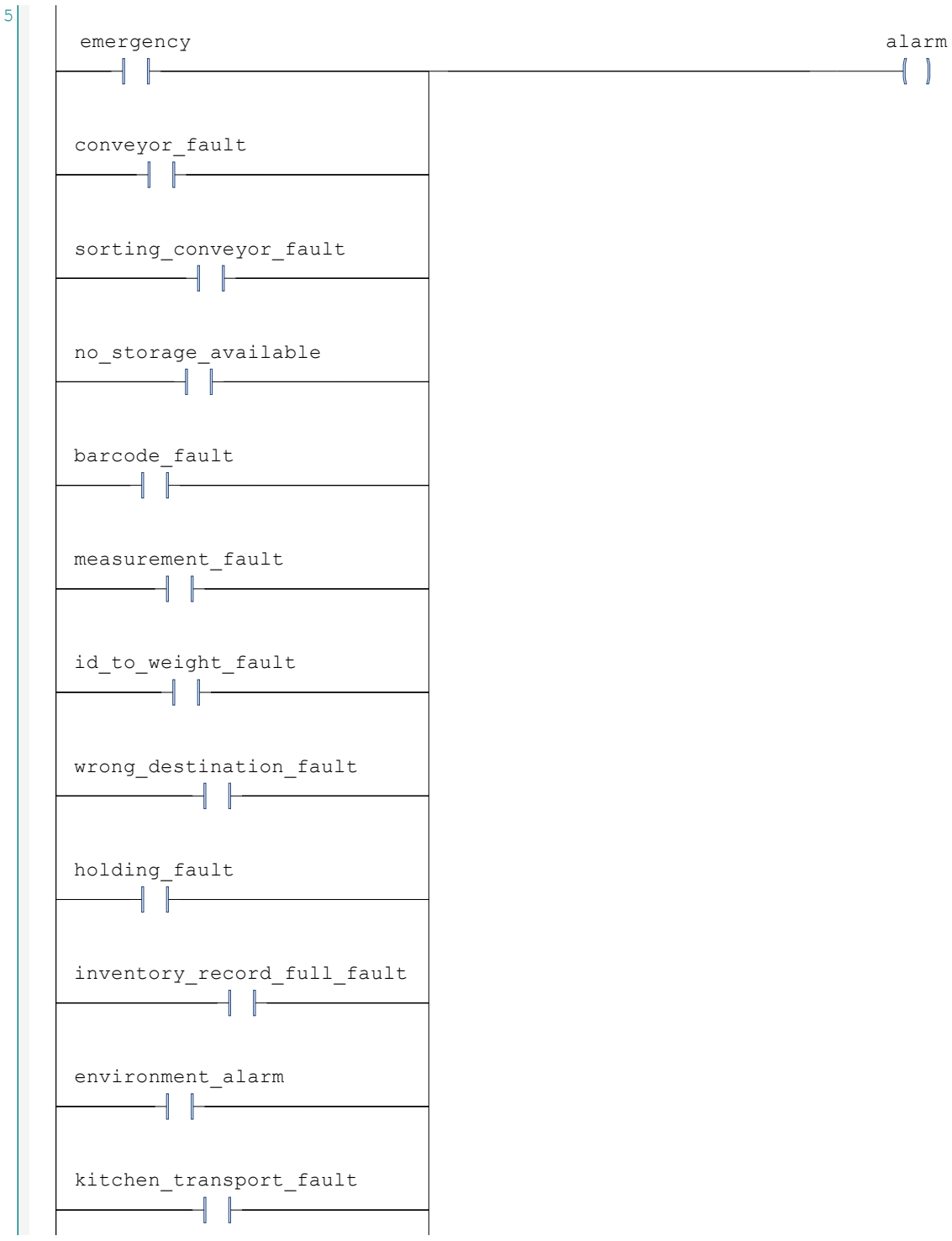
```
813
814         (* Arrival Sensors *)
815         customer_table_1_sensor  : BOOL ;
816         customer_table_2_sensor  : BOOL ;
817         customer_table_3_sensor  : BOOL ;
818         customer_table_4_sensor  : BOOL ;
819         customer_table_5_sensor  : BOOL ;
820         customer_table_6_sensor  : BOOL ;
821
822         customer_delivery_sensor_any  : BOOL ;
823         customer_delivery_sensors_clear  : BOOL ;
824         customer_delivery_arrival_R_TRIG  : R_TRIG ;
825         customer_delivery_arrival_pulse  : BOOL ;
826
827         (* Arrival Validation *)
828         customer_delivery_expected_sensor  : BOOL ;
829         customer_delivery_wrong_sensor  : BOOL ;
830         customer_delivery_arrival_valid  : BOOL ;
831         customer_delivery_no_route  : BOOL ;
832
833         (* Timeout *)
834         TON_customer_delivery_timeout  : TON ;
835         customer_delivery_timeout  : BOOL ;
836         TON_customer_delivery_timeout_value  : TIME ;
837
838         (* Fault Handling *)
839         customer_delivery_fault_RS  : RS ;
840         customer_delivery_fault  : BOOL ;
841
842         (* Completion *)
843         customer_delivery_complete_RS  : RS ;
844         customer_delivery_complete  : BOOL ;
845
846         (* Reset and Retry *)
847         customer_delivery_reset  : BOOL ;
848         customer_delivery_retry  : BOOL ;
849         customer_delivery_retry_R_TRIG  : R_TRIG ;
850         customer_delivery_retry_pulse  : BOOL ;
851         customer_delivery_retry_start  : BOOL ;
852         customer_delivery_stop_condition  : BOOL ;
853
854         (* Kitchen Order Validation Update *)
855         order_table_le_6  : BOOL ;
856
857
858         (* DIRTY DISH RETURN & CLEANING *)
859
860         (* Dish Return Request *)
861         dish_return_table_input  : WORD ;
862         dish_return_quantity_input  : WORD ;
863         dish_return_submit  : BOOL ;
```

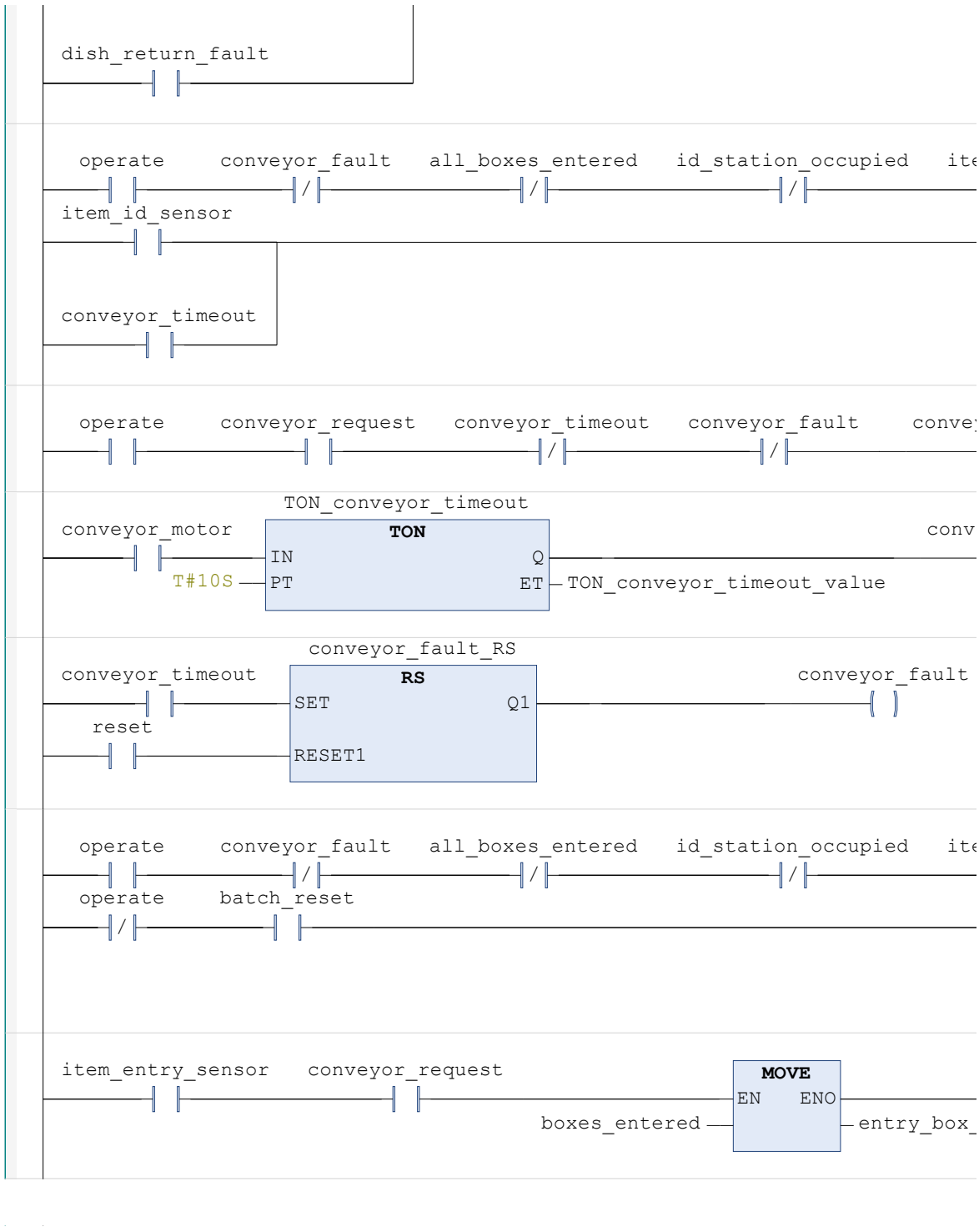
```
864         dish_return_submit_R_TRIG  :  R_TRIG ;
865         dish_return_submit_pulse  :  BOOL ;
866
867         (* Request Validation *)
868         dish_return_table_min_ok  :  BOOL ;
869         dish_return_table_max_ok  :  BOOL ;
870         dish_return_quantity_valid  :  BOOL ;
871         dish_return_input_valid  :  BOOL ;
872         dish_return_accept  :  BOOL ;
873         dish_return_rejected  :  BOOL ;
874
875         (* Active Job *)
876         dish_process_busy_RS  :  RS ;
877         dish_process_busy  :  BOOL ;
878         dish_transport_busy_RS  :  RS ;
879         dish_transport_busy  :  BOOL ;
880
881         dish_return_table_id  :  WORD ;
882         dish_return_quantity  :  WORD ;
883
884         (* Table Routing *)
885         dish_route_from_table_1  :  BOOL ;
886         dish_route_from_table_2  :  BOOL ;
887         dish_route_from_table_3  :  BOOL ;
888         dish_route_from_table_4  :  BOOL ;
889         dish_route_from_table_5  :  BOOL ;
890         dish_route_from_table_6  :  BOOL ;
891         dish_route_valid  :  BOOL ;
892         dish_return_no_route  :  BOOL ;
893
894         (* Return Conveyor *)
895         dish_return_motor  :  BOOL ;
896         dish_wash_station_sensor  :  BOOL ;
897         dish_wash_station_R_TRIG  :  R_TRIG ;
898         dish_wash_station_pulse  :  BOOL ;
899         dish_return_arrival_valid  :  BOOL ;
900
901         (* Transport Timeout *)
902         TON_dish_return_timeout  :  TON ;
903         dish_return_timeout  :  BOOL ;
904
905         (* Fault and Reset *)
906         dish_operation_interrupted  :  BOOL ;
907         dish_return_fault_RS  :  RS ;
908         dish_return_fault  :  BOOL ;
909         dish_return_reset  :  BOOL ;
910         dish_return_reset_allowed  :  BOOL ;
911
912         (* Cleaning Stage *)
913         dish_cleaning_stage  :  WORD  :=  0 ;
914         dish_wash_active  :  BOOL ;
```

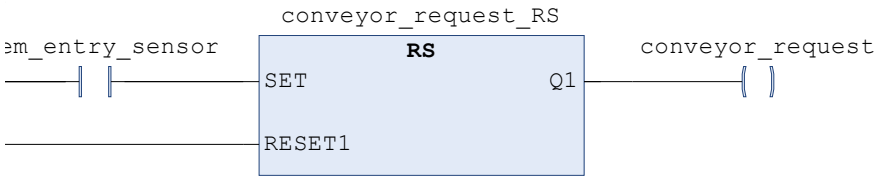
```
915         dish_rinse_active  :  BOOL ;
916         dish_sanitise_active  :  BOOL ;
917         dish_dry_active  :  BOOL ;
918         dish_stage_complete  :  BOOL ;
919
920         (* Cleaning Timers *)
921         TON_dish_wash  :  TON ;
922         TON_dish_rinse  :  TON ;
923         TON_dish_sanitise  :  TON ;
924         TON_dish_dry  :  TON ;
925
926         dish_wash_timer_done  :  BOOL ;
927         dish_rinse_timer_done  :  BOOL ;
928         dish_sanitise_timer_done  :  BOOL ;
929         dish_dry_timer_done  :  BOOL ;
930
931         (* Cleaning Completion *)
932         dish_cleaning_done_R_TRIG  :  R_TRIG ;
933         dish_cleaning_done_pulse  :  BOOL ;
934         dish_cleaning_complete_RS  :  RS ;
935         dish_cleaning_complete  :  BOOL ;
936
937     END_VAR
938
```




emergency
operate

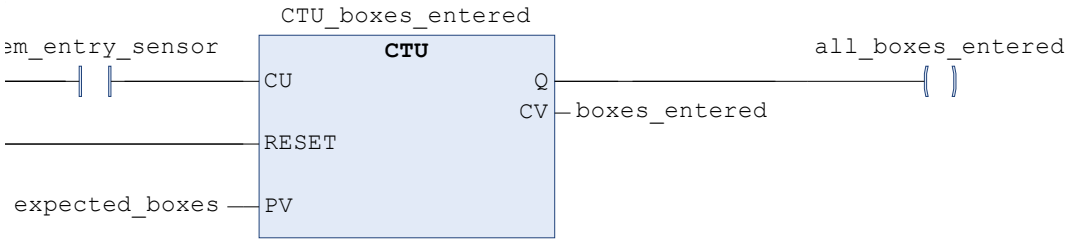

5
emergency
alarm
conveyor_fault
sorting_conveyor_fault
no_storage_available
barcode_fault
measurement_fault
id_to_weight_fault
wrong_destination_fault
holding_fault
inventory_record_full_fault
environment_alarm
kitchen_transport_fault


dish_return_fault
6
operate
conveyor_fault
all_boxes_entered
id_station_occupied
item_id_sensor
conveyor_timeout
7
operate
conveyor_request
conveyor_timeout
conveyor_fault
8
TON_conveyor_timeout
conveyor_motor
TON
IN
Q
T#10S
PT
ET
TON_conveyor_timeout_value
9
conveyor_fault_RS
conveyor_timeout
RS
SET
Q1
conveyor_fault
reset
RESET1
10
operate
conveyor_fault
all_boxes_entered
id_station_occupied
operate
batch_reset
11
item_entry_sensor
conveyor_request
MOVE
EN
ENO
boxes_entered
entry_box_

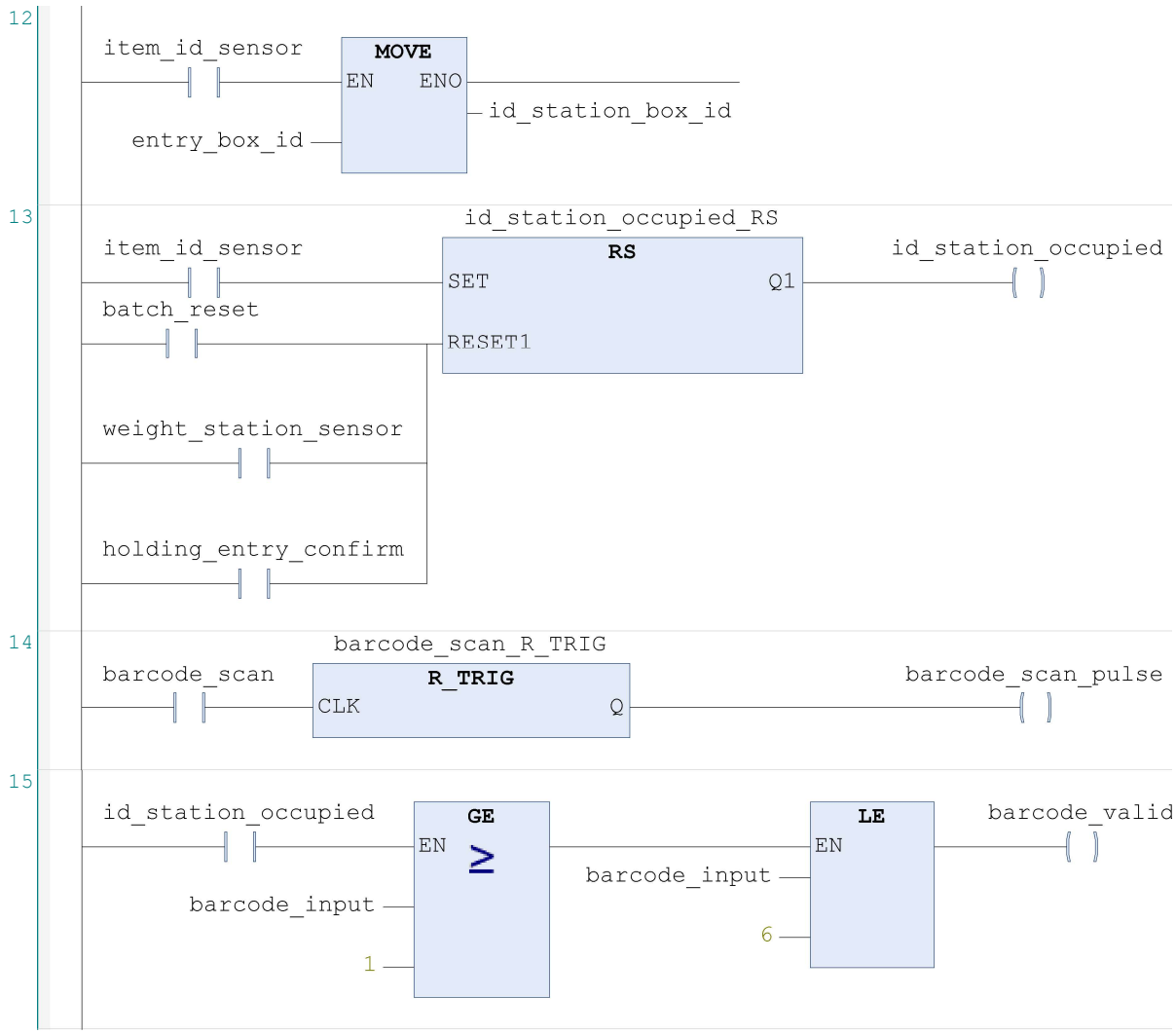


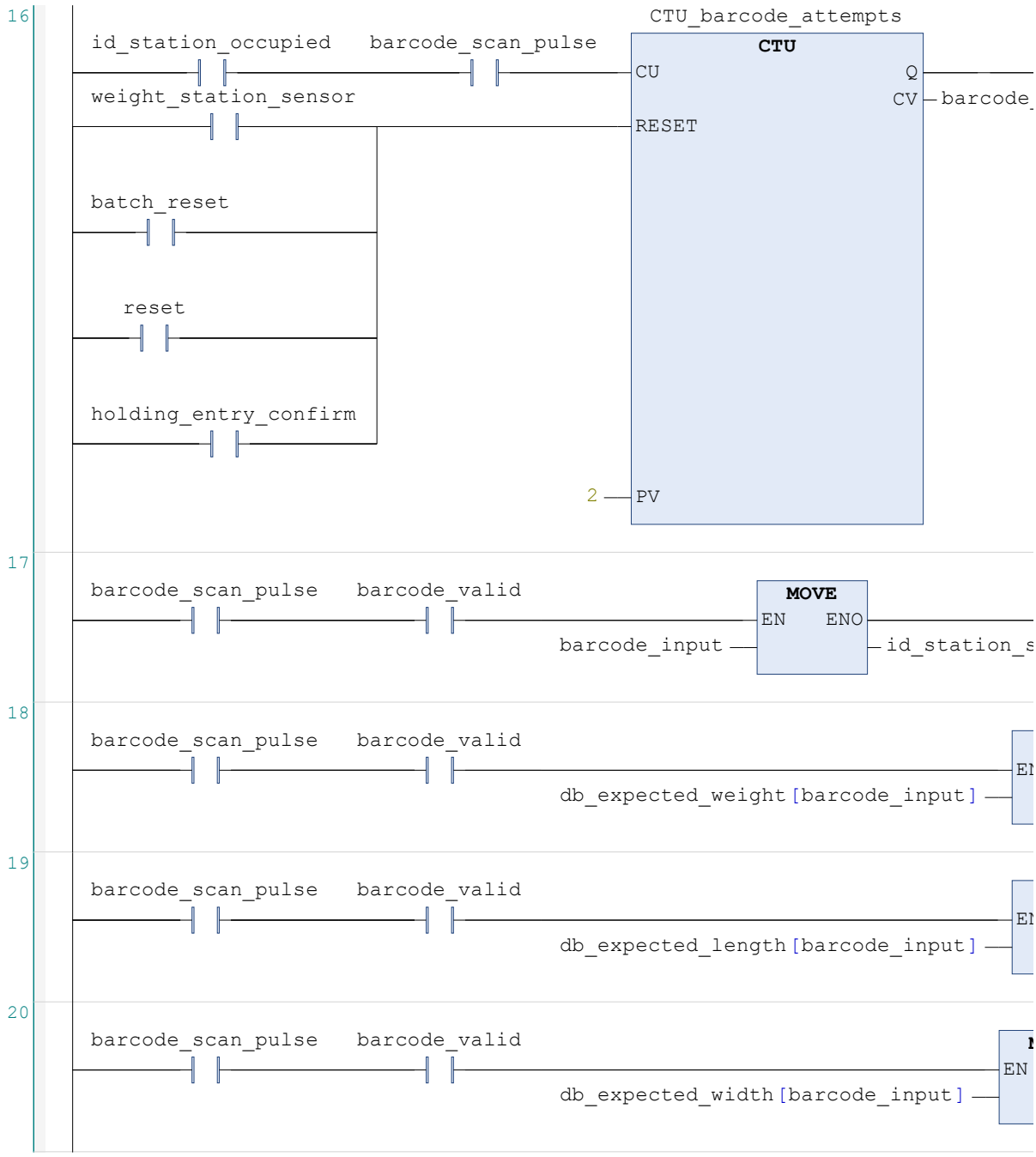
yor_motor

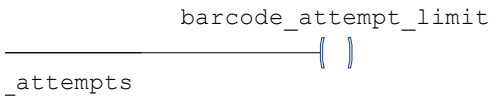
eyor_timeout

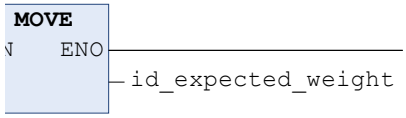


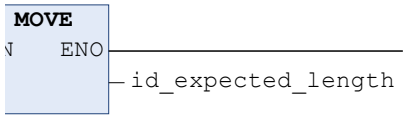
_id

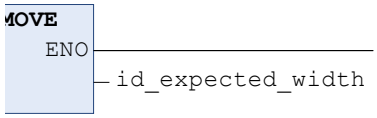

12
item_id_sensor
MOVE
EN
ENO
id_station_box_id
entry_box_id
13
id_station_occupied_RS
item_id_sensor
RS
SET
Q1
id_station_occupied
batch_reset
RESET1
weight_station_sensor
holding_entry_confirm
14
barcode_scan_R_TRIG
barcode_scan
R_TRIG
CLK
Q
barcode_scan_pulse
15
id_station_occupied
GE
EN
≥
LE
EN
barcode_valid
barcode_input
barcode_input
6
1

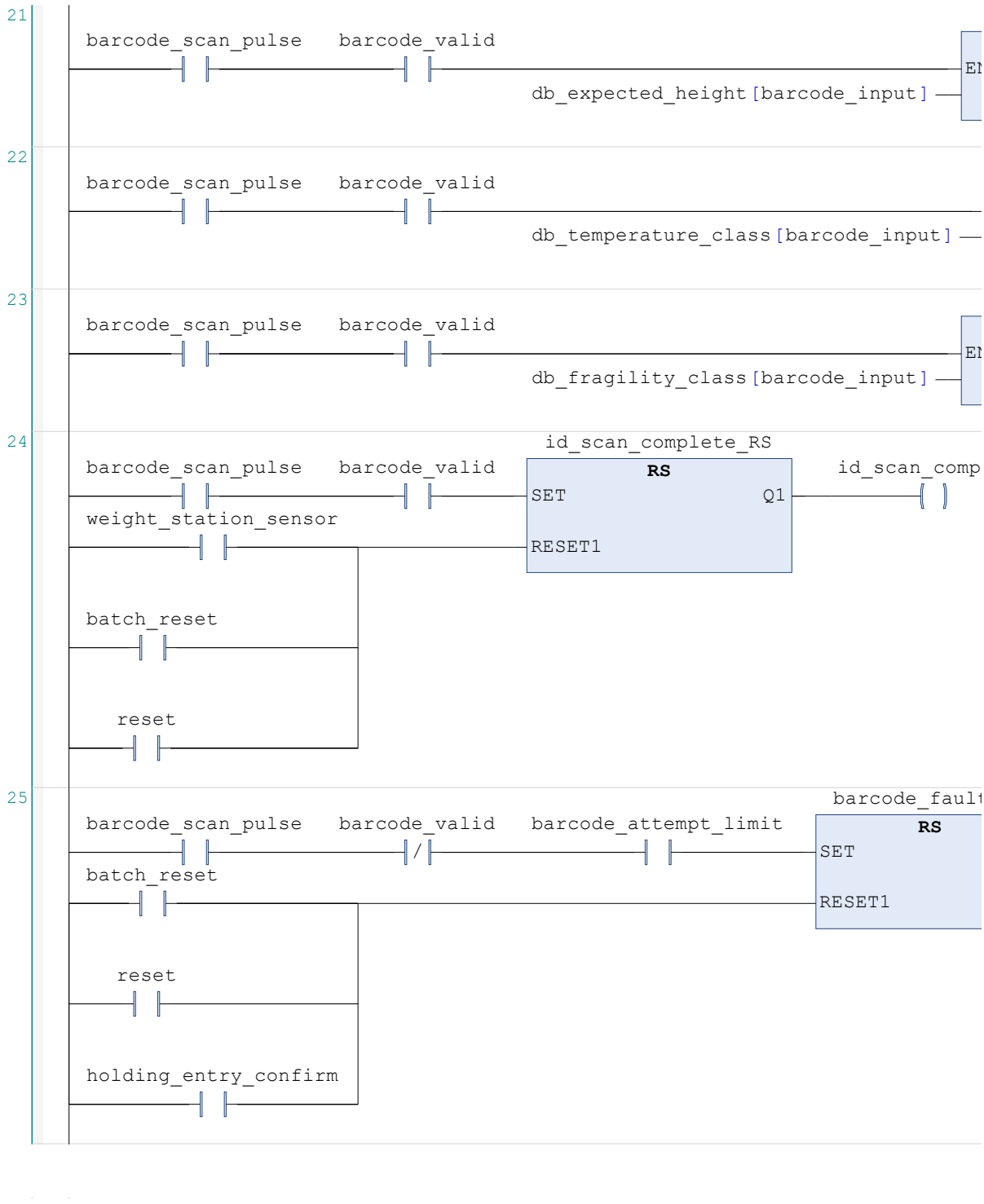

16
CTU_barcode_attempts
CTU
id_station_occupied
barcode_scan_pulse
CU
Q
weight_station_sensor
CV
barcode_
RESET
batch_reset
reset
holding_entry_confirm
2
PV
17
barcode_scan_pulse
barcode_valid
MOVE
EN
ENO
barcode_input
id_station_s
18
barcode_scan_pulse
barcode_valid
db_expected_weight[barcode_input]
19
barcode_scan_pulse
barcode_valid
db_expected_length[barcode_input]
20
barcode_scan_pulse
barcode_valid
EN
db_expected_width[barcode_input]

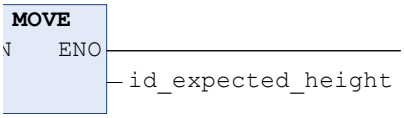

barcode_attempt_limit
_attempts

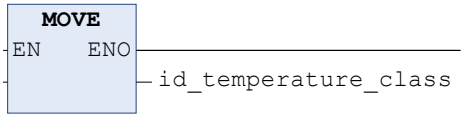

MOVE
ENO
id_expected_weight

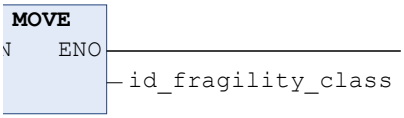

MOVE
ENO
id_expected_length


MOVE
ENO
id_expected_width

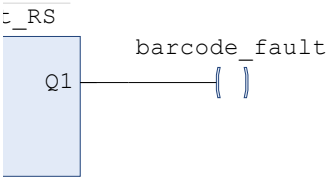

21
barcode_scan_pulse
barcode_valid
db_expected_height[barcode_input]
22
barcode_scan_pulse
barcode_valid
db_temperature_class[barcode_input]
23
barcode_scan_pulse
barcode_valid
db_fragility_class[barcode_input]
24
id_scan_complete_RS
barcode_scan_pulse
barcode_valid
RS
SET
Q1
RESET1
id_scan_comp
weight_station_sensor
batch_reset
reset
25
barcode_faul
barcode_scan_pulse
barcode_valid
barcode_attempt_limit
RS
SET
RESET1
batch_reset
reset
holding_entry_confirm

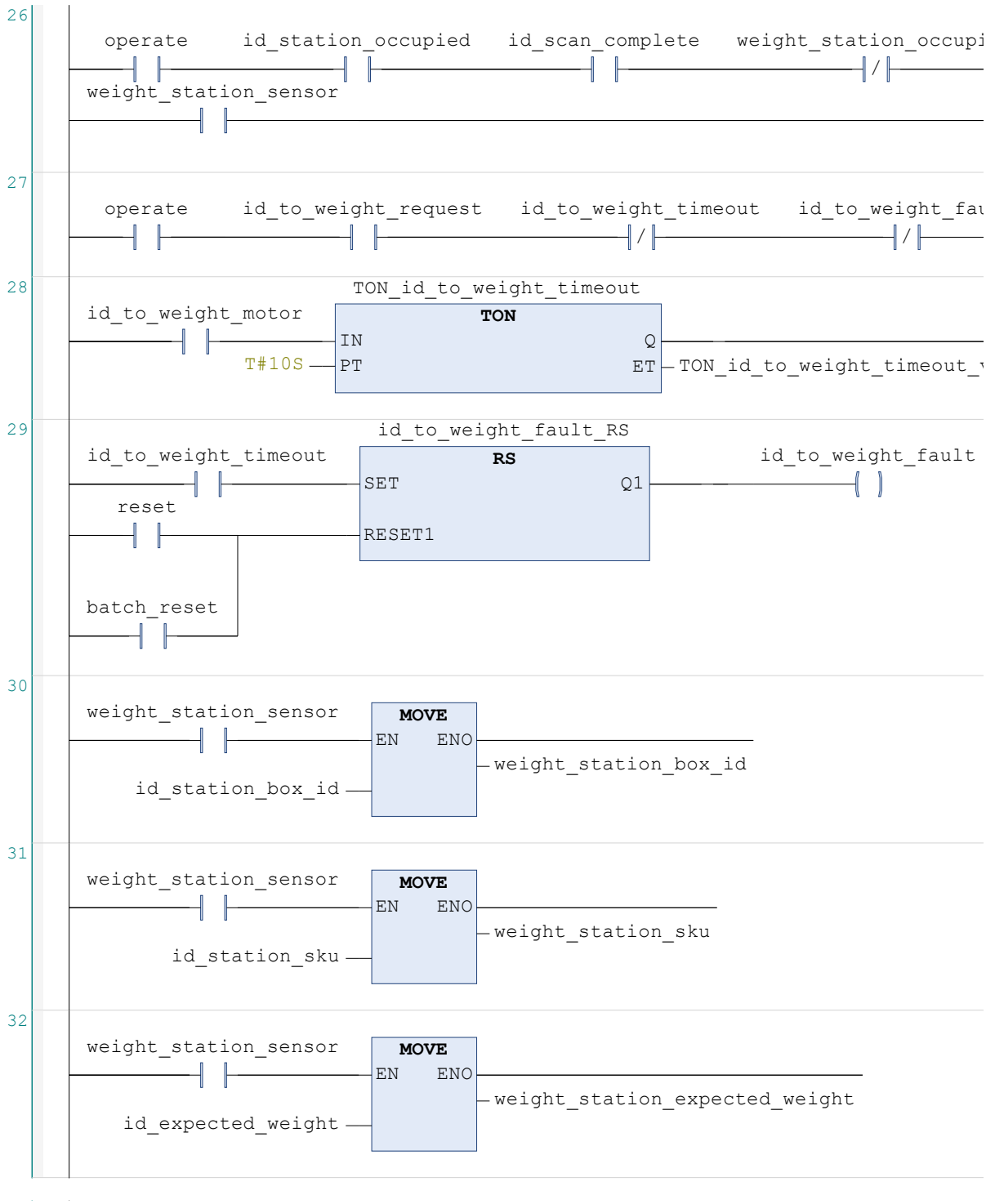

MOVE
ENO
id_expected_height

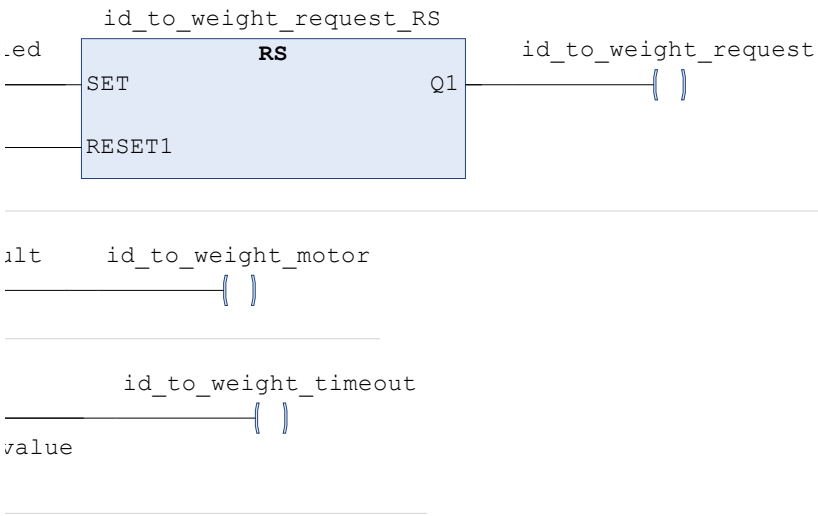

MOVE
EN
ENO
id_temperature_class

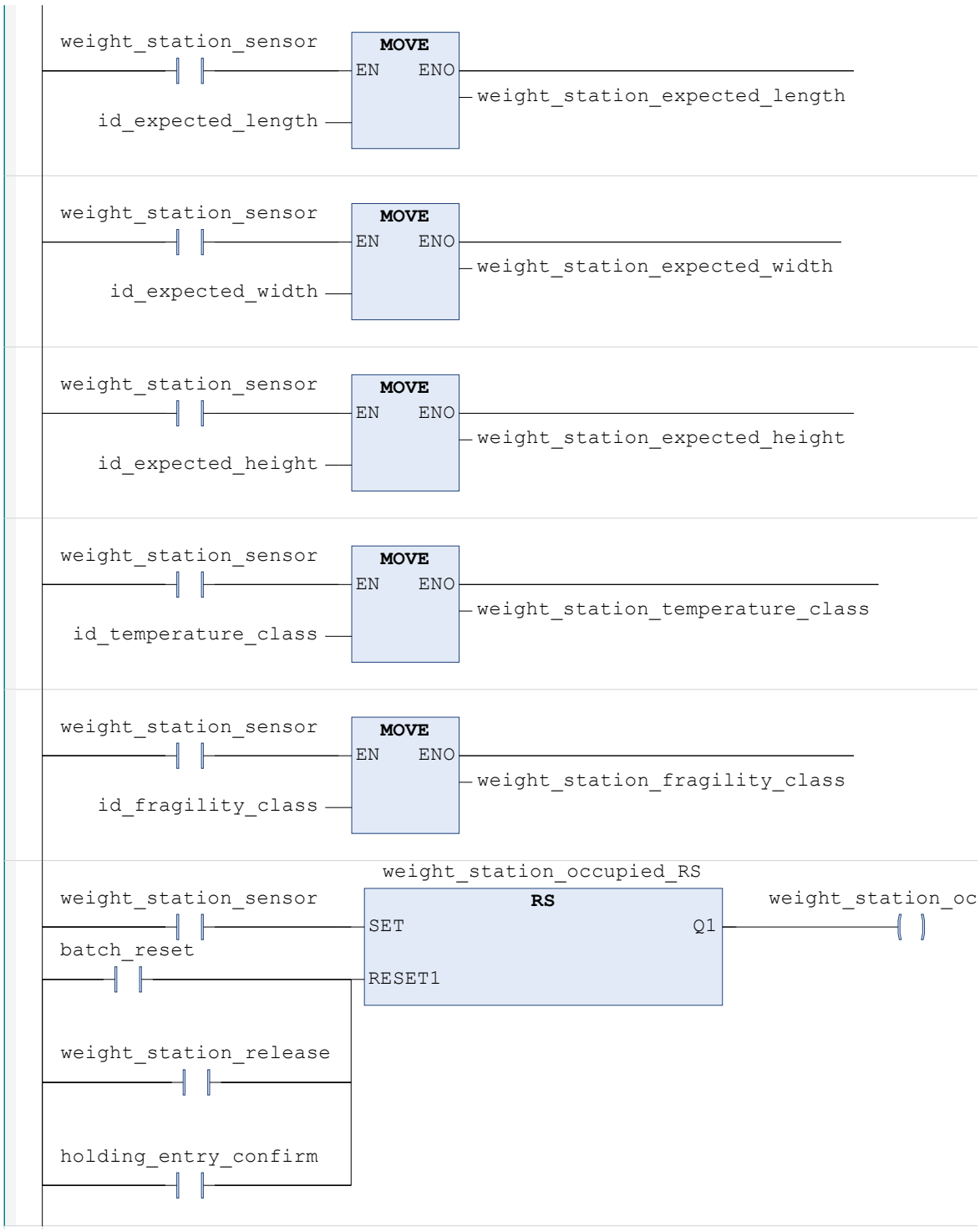

MOVE
ENO
id_fragility_class

lete

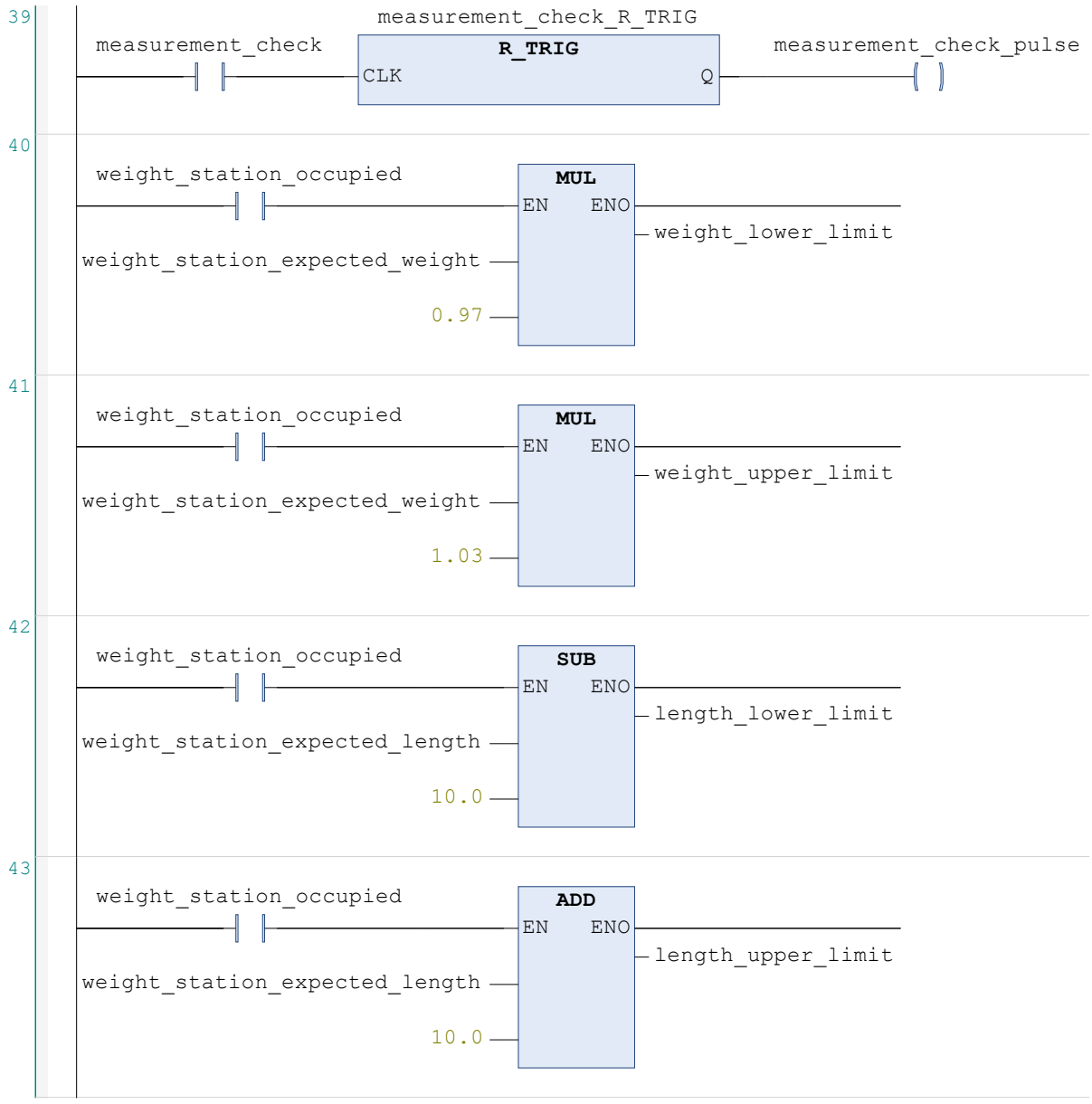

_RS
barcode_fault
Q1

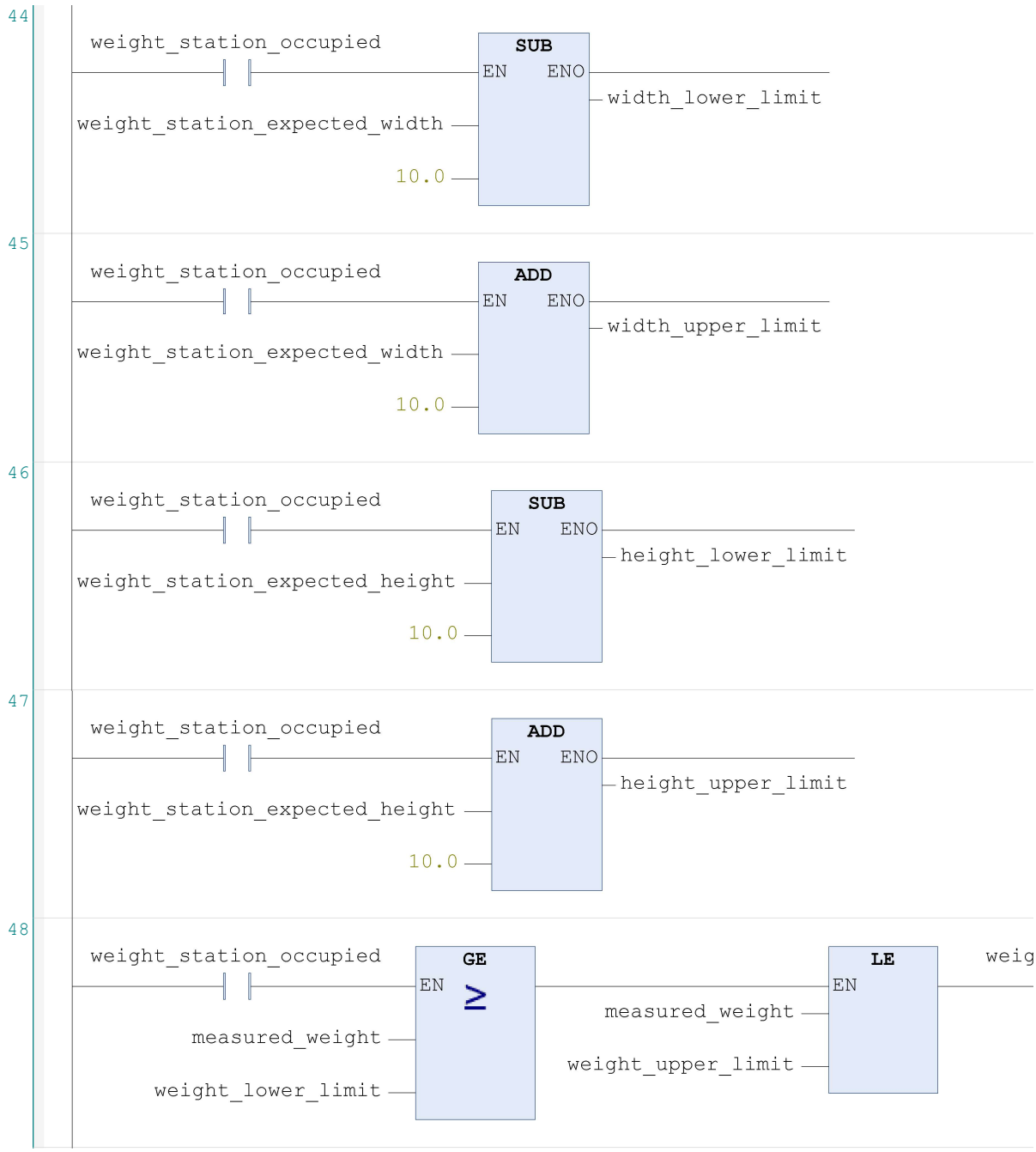

26
operate
id_station_occupied
id_scan_complete
weight_station_occupi
weight_station_sensor
27
operate
id_to_weight_request
id_to_weight_timeout
id_to_weight_fau
28
TON_id_to_weight_timeout
id_to_weight_motor
TON
IN
Q
T#10S
PT
ET
TON_id_to_weight_timeout_
29
id_to_weight_fault_RS
id_to_weight_timeout
RS
id_to_weight_fault
SET
Q1
reset
RESET1
batch_reset
30
weight_station_sensor
MOVE
EN
ENO
weight_station_box_id
id_station_box_id
31
weight_station_sensor
MOVE
EN
ENO
weight_station_sku
id_station_sku
32
weight_station_sensor
MOVE
EN
ENO
weight_station_expected_weight
id_expected_weight


id_to_weight_request_RS
RS
SET
Q1
RESET1
id_to_weight_request
id_to_weight_motor
id_to_weight_timeout

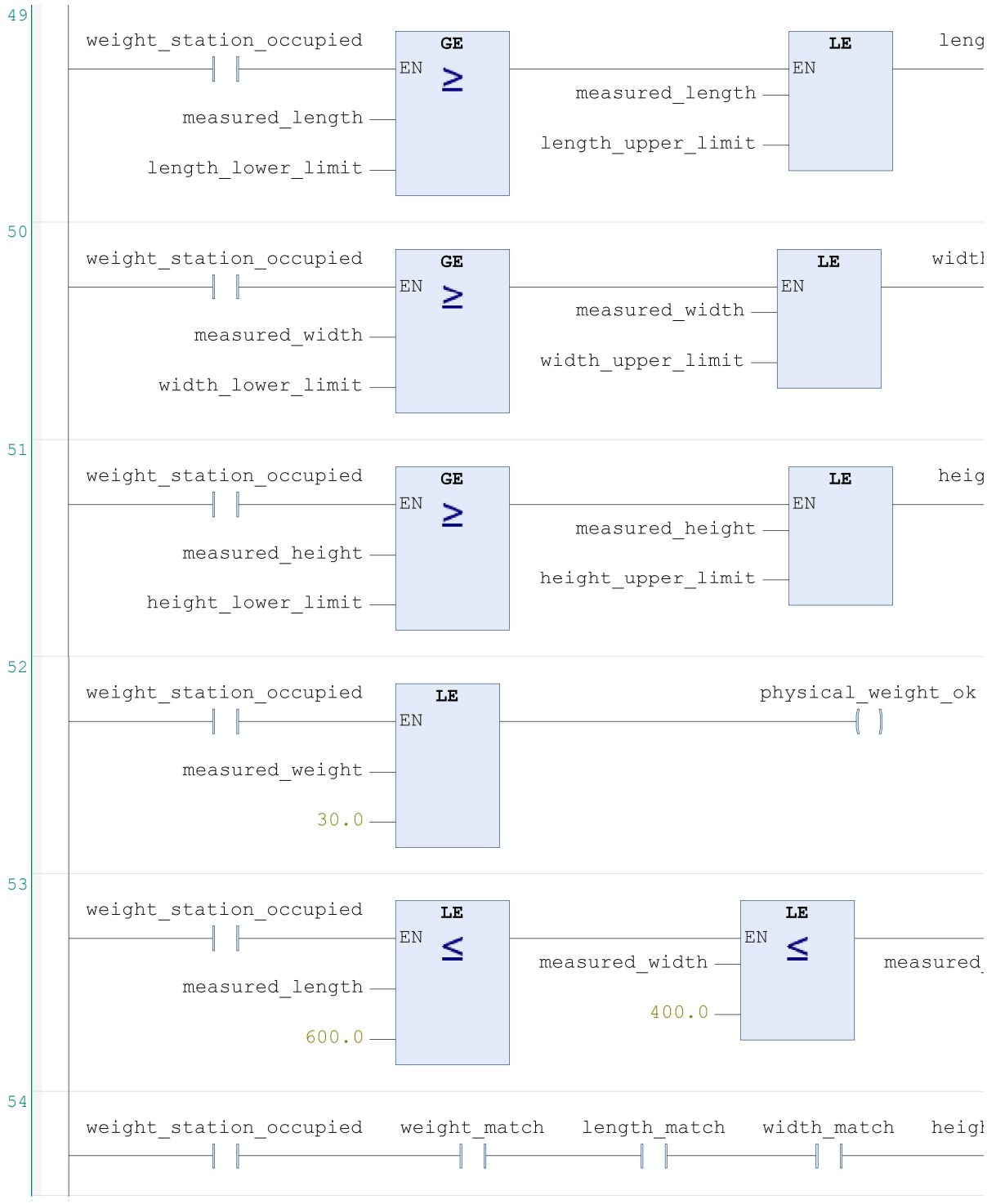

33
weight_station_sensor
MOVE
EN
ENO
weight_station_expected_length
id_expected_length
34
weight_station_sensor
MOVE
EN
ENO
weight_station_expected_width
id_expected_width
35
weight_station_sensor
MOVE
EN
ENO
weight_station_expected_height
id_expected_height
36
weight_station_sensor
MOVE
EN
ENO
weight_station_temperature_class
id_temperature_class
37
weight_station_sensor
MOVE
EN
ENO
weight_station_fragility_class
id_fragility_class
38
weight_station_occupied_RS
weight_station_sensor
RS
SET
Q1
weight_station_oc
batch_reset
RESET1
weight_station_release
holding_entry_confirm

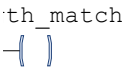
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cupied
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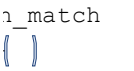

39
measurement_check_R_TRIG
measurement_check
R_TRIG
CLK
Q
measurement_check_pulse
40
weight_station_occupied
MUL
EN
ENO
weight_lower_limit
weight_station_expected_weight
0.97
41
weight_station_occupied
MUL
EN
ENO
weight_upper_limit
weight_station_expected_weight
1.03
42
weight_station_occupied
SUB
EN
ENO
length_lower_limit
weight_station_expected_length
10.0
43
weight_station_occupied
ADD
EN
ENO
length_upper_limit
weight_station_expected_length
10.0

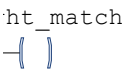

44
weight_station_occupied
SUB
EN
ENO
width_lower_limit
weight_station_expected_width
10.0
45
weight_station_occupied
ADD
EN
ENO
width_upper_limit
weight_station_expected_width
10.0
46
weight_station_occupied
SUB
EN
ENO
height_lower_limit
weight_station_expected_height
10.0
47
weight_station_occupied
ADD
EN
ENO
height_upper_limit
weight_station_expected_height
10.0
48
weight_station_occupied
GE
EN
≥
LE
EN
weig
measured_weight
weight_upper_limit
measured_weight
weight_lower_limit

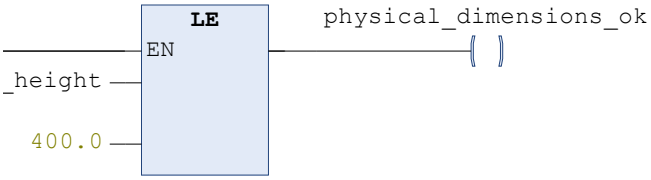
ht_match
)

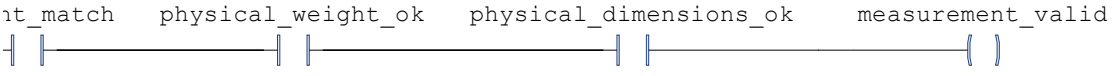

49
weight_station_occupied
GE
EN
≥
measured_length
length_lower_limit
LE
EN
measured_length
length_upper_limit
leng
50
weight_station_occupied
GE
EN
≥
measured_width
width_lower_limit
LE
EN
measured_width
width_upper_limit
widtl
51
weight_station_occupied
GE
EN
≥
measured_height
height_lower_limit
LE
EN
measured_height
height_upper_limit
heig
52
weight_station_occupied
LE
EN
measured_weight
30.0
physical_weight_ok
53
weight_station_occupied
LE
EN
≤
measured_length
600.0
LE
EN
≤
measured_width
400.0
measured_
54
weight_station_occupied
weight_match
length_match
width_match
heigh

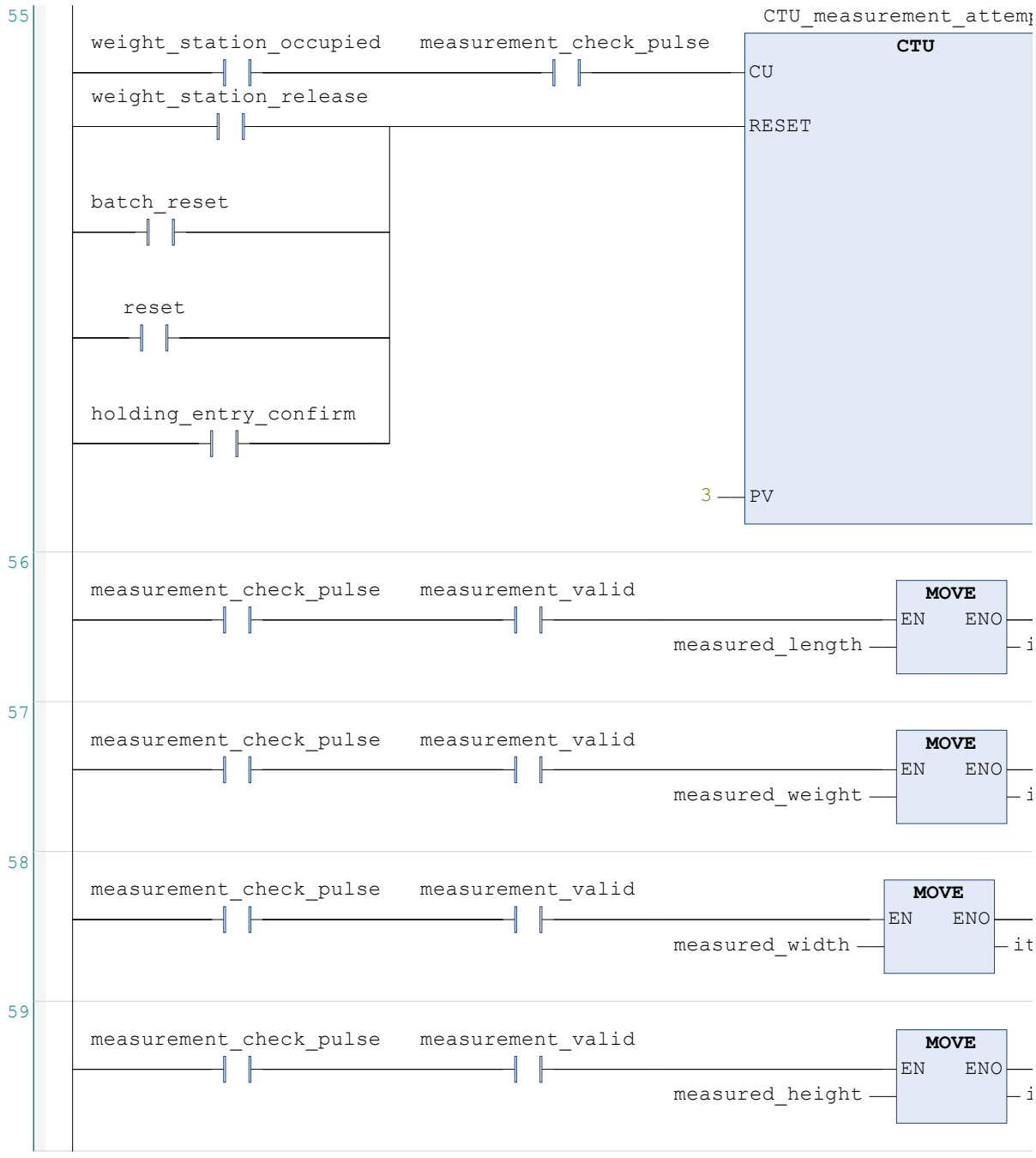

th_match

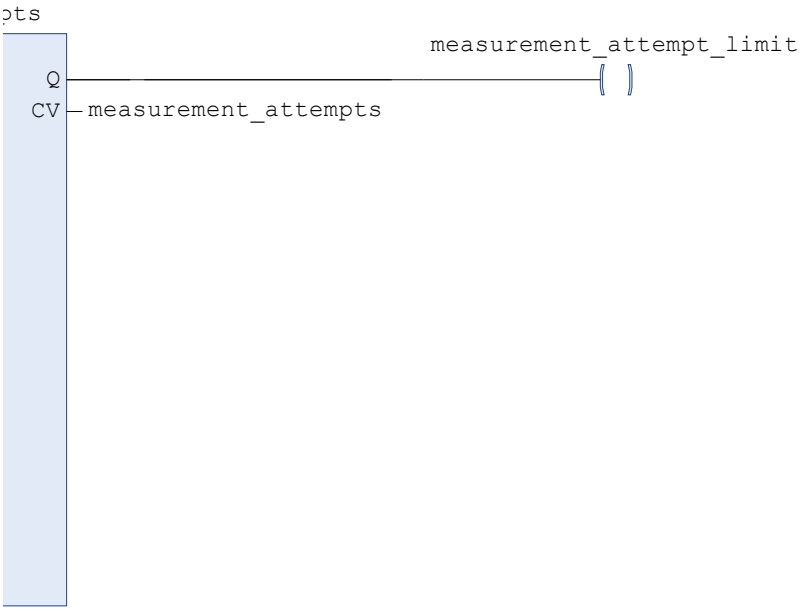

n_match


ht_match


LE
EN
physical_dimensions_ok
_height
400.0


nt_match
physical_weight_ok
physical_dimensions_ok
measurement_valid


55
CTU_measurement_attem
CTU
weight_station_occupied
measurement_check_pulse
CU
weight_station_release
RESET
batch_reset
reset
holding_entry_confirm
3
PV
56
measurement_check_pulse
measurement_valid
MOVE
EN
ENO
measured_length
57
measurement_check_pulse
measurement_valid
MOVE
EN
ENO
measured_weight
58
measurement_check_pulse
measurement_valid
MOVE
EN
ENO
measured_width
59
measurement_check_pulse
measurement_valid
MOVE
EN
ENO
measured_height

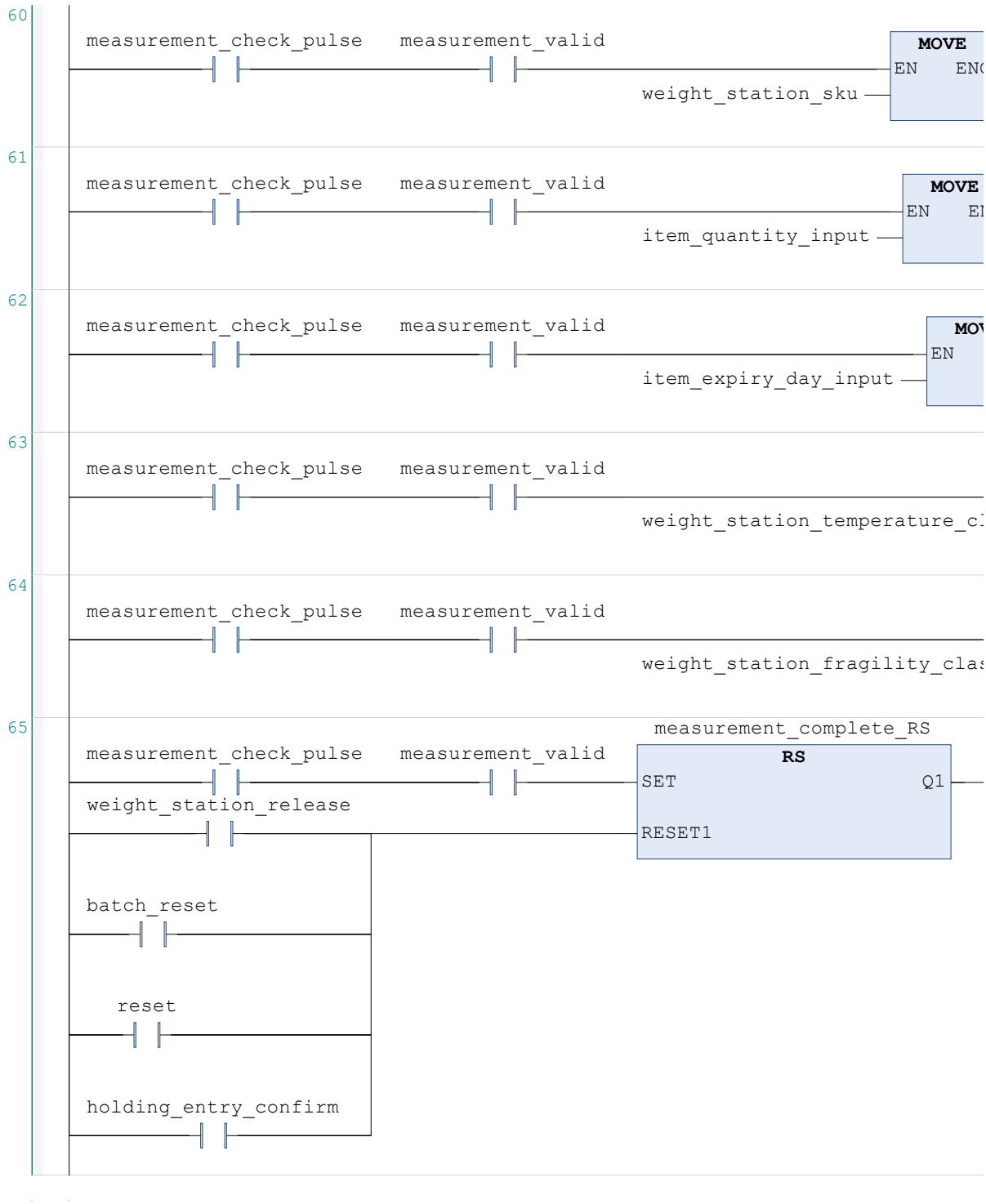


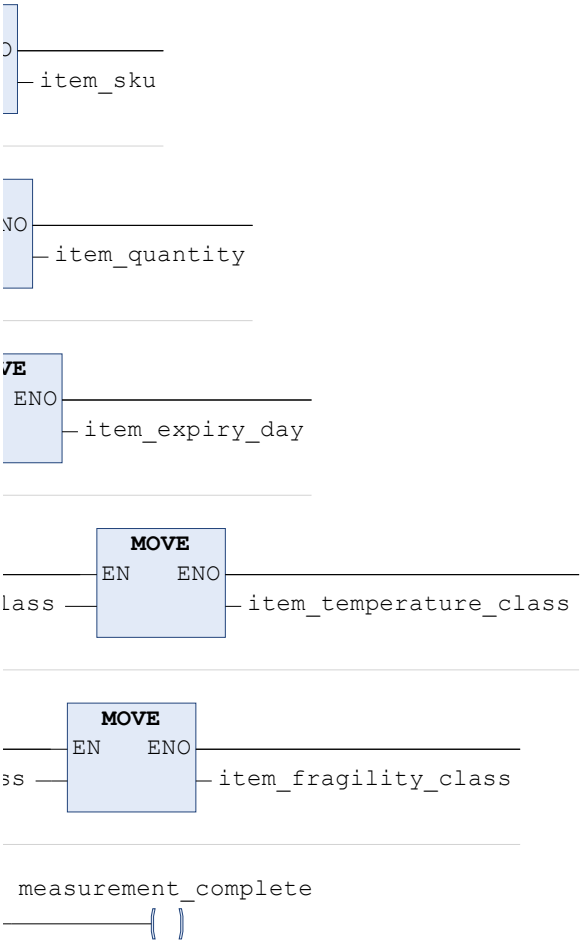
.tem_length

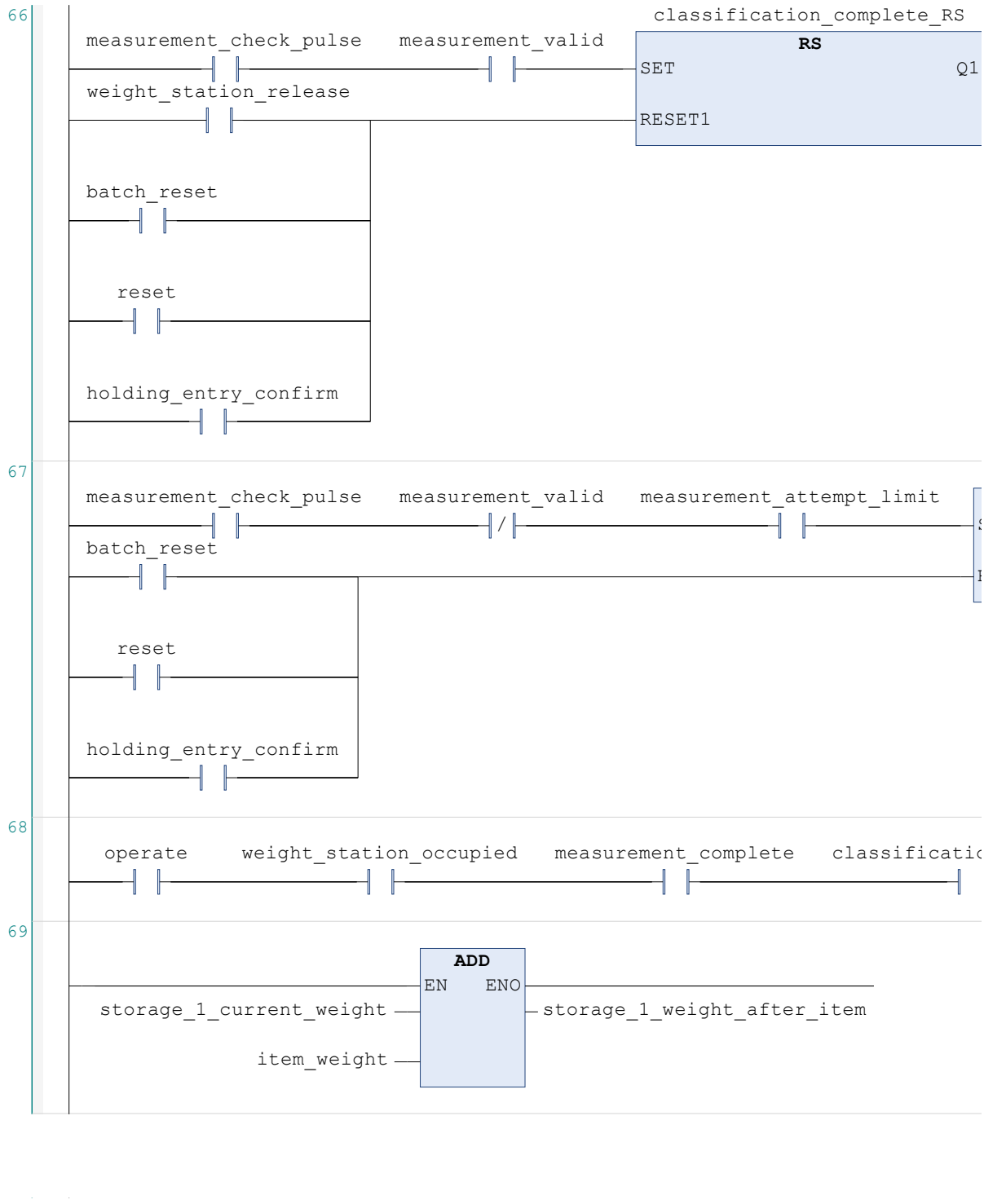
.tem_weight

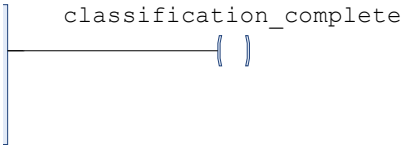
:em_width

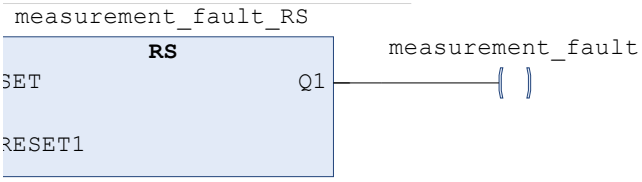
.tem_height

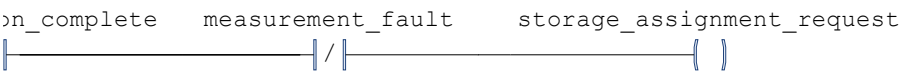

60
measurement_check_pulse
measurement_valid
MOVE
EN
weight_station_sku
61
measurement_check_pulse
measurement_valid
MOVE
EN
item_quantity_input
62
measurement_check_pulse
measurement_valid
EN
item_expiry_day_input
63
measurement_check_pulse
measurement_valid
64
measurement_check_pulse
measurement_valid
65
measurement_complete_RS
measurement_check_pulse
measurement_valid
RS
SET
Q1
RESET1
weight_station_release
batch_reset
reset
holding_entry_confirm


item_sku
item_quantity
ENO
item_expiry_day
MOVE
EN
ENO
item_temperature_class
MOVE
EN
ENO
item_fragility_class
measurement_complete

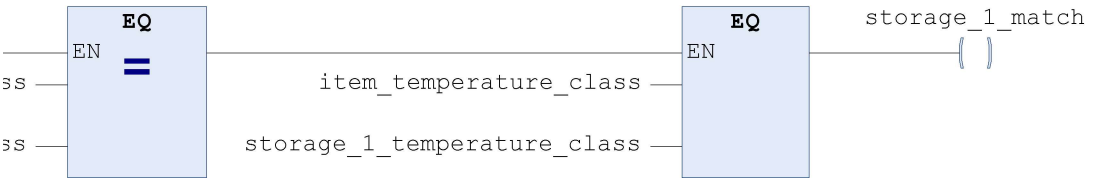

66
classification_complete_RS
RS
measurement_check_pulse
measurement_valid
SET
Q1
weight_station_release
RESET1
batch_reset
reset
holding_entry_confirm
67
measurement_check_pulse
measurement_valid
measurement_attempt_limit
batch_reset
reset
holding_entry_confirm
68
operate
weight_station_occupied
measurement_complete
69
ADD
EN
ENO
storage_1_current_weight
storage_1_weight_after_item
item_weight

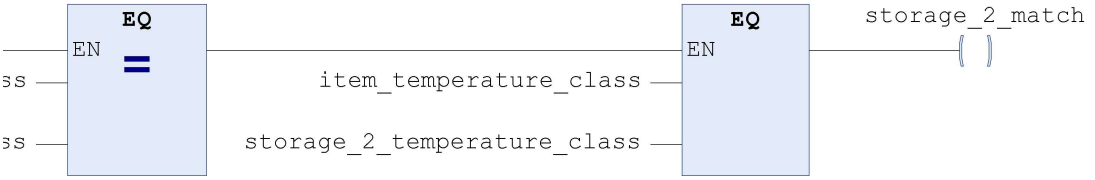

classification_complete

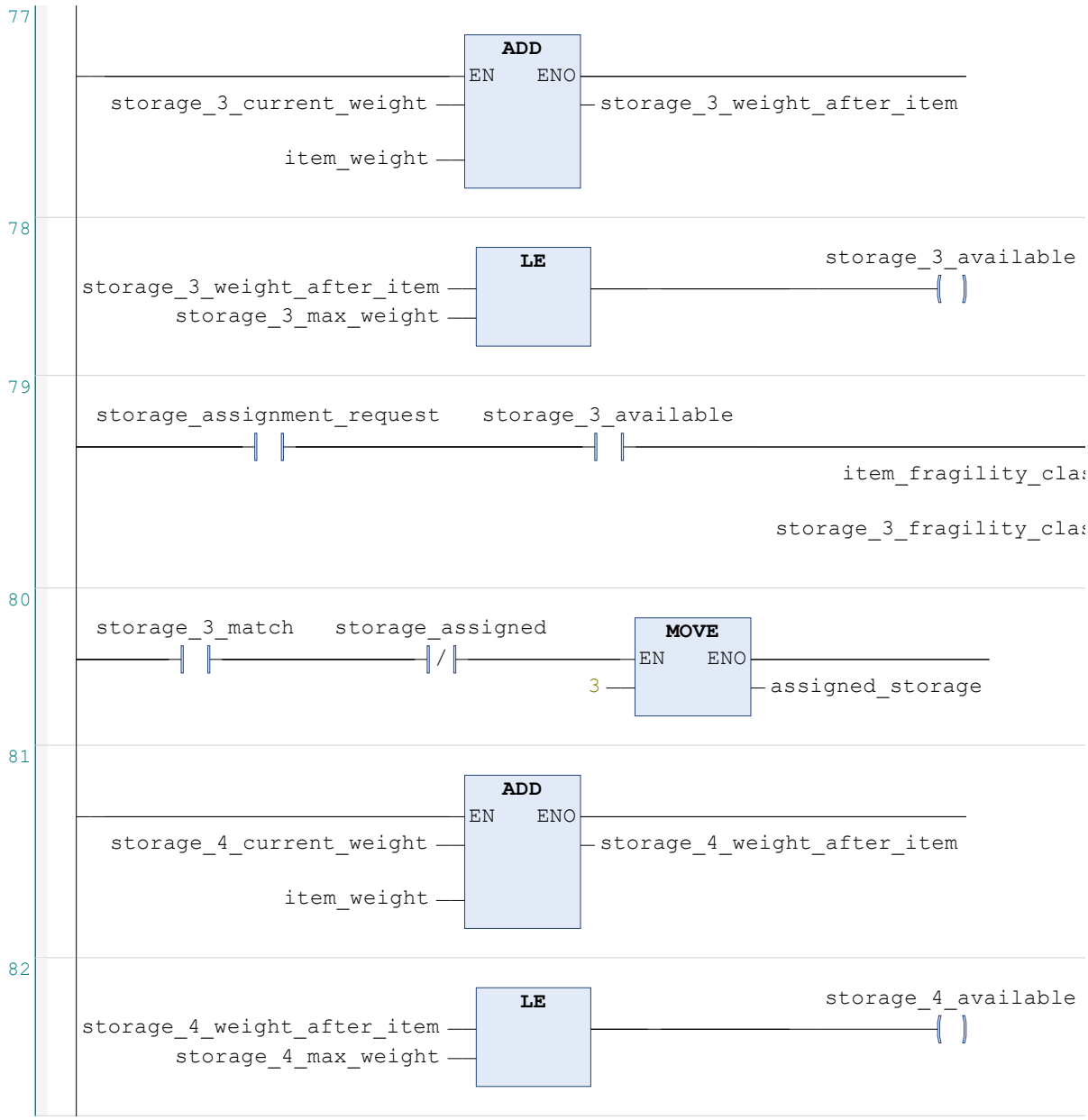

measurement_fault_RS
RS
SET
Q1
RESET1
measurement_fault

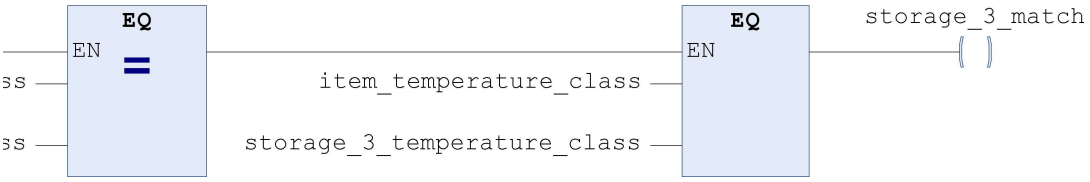

on_complete
measurement_fault
storage_assignment_request

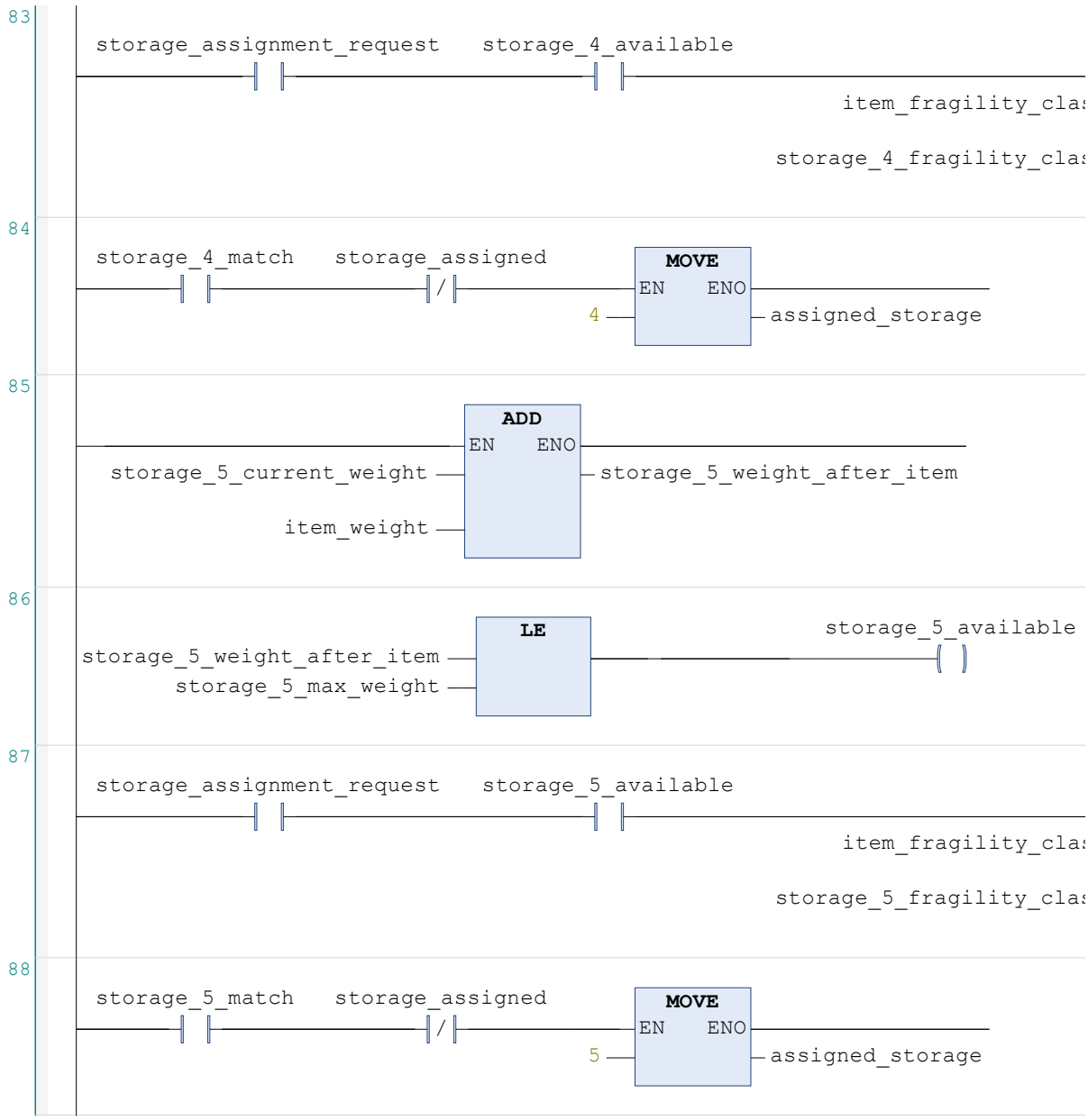

70
LE
storage_1_weight_after_item
storage_1_max_weight
storage_1_available
71
storage_assignment_request
storage_1_available
item_fragility_cla
storage_1_fragility_cla
72
storage_1_match
storage_assigned
MOVE
EN
ENO
1
assigned_storage
73
ADD
EN
ENO
storage_2_current_weight
item_weight
storage_2_weight_after_item
74
LE
storage_2_weight_after_item
storage_2_max_weight
storage_2_available
75
storage_assignment_request
storage_2_available
item_fragility_cla
storage_2_fragility_cla
76
storage_2_match
storage_assigned
MOVE
EN
ENO
2
assigned_storage

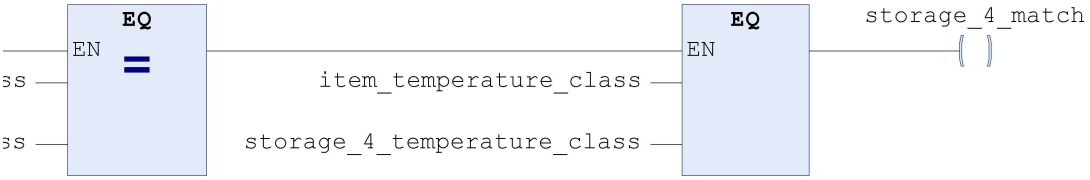

EQ
EN
=
ss
ss
item_temperature_class
storage_1_temperature_class
EQ
EN
storage_1_match

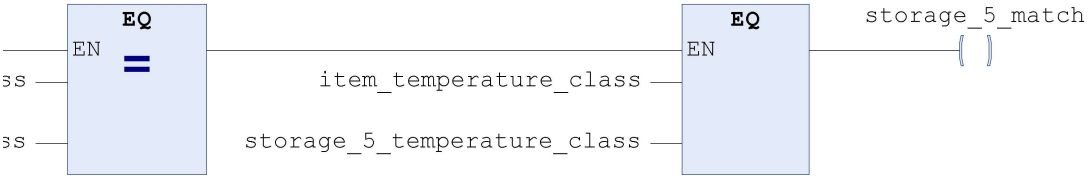

EQ
EN
=
ss
ss
item_temperature_class
storage_2_temperature_class
EQ
EN
storage_2_match

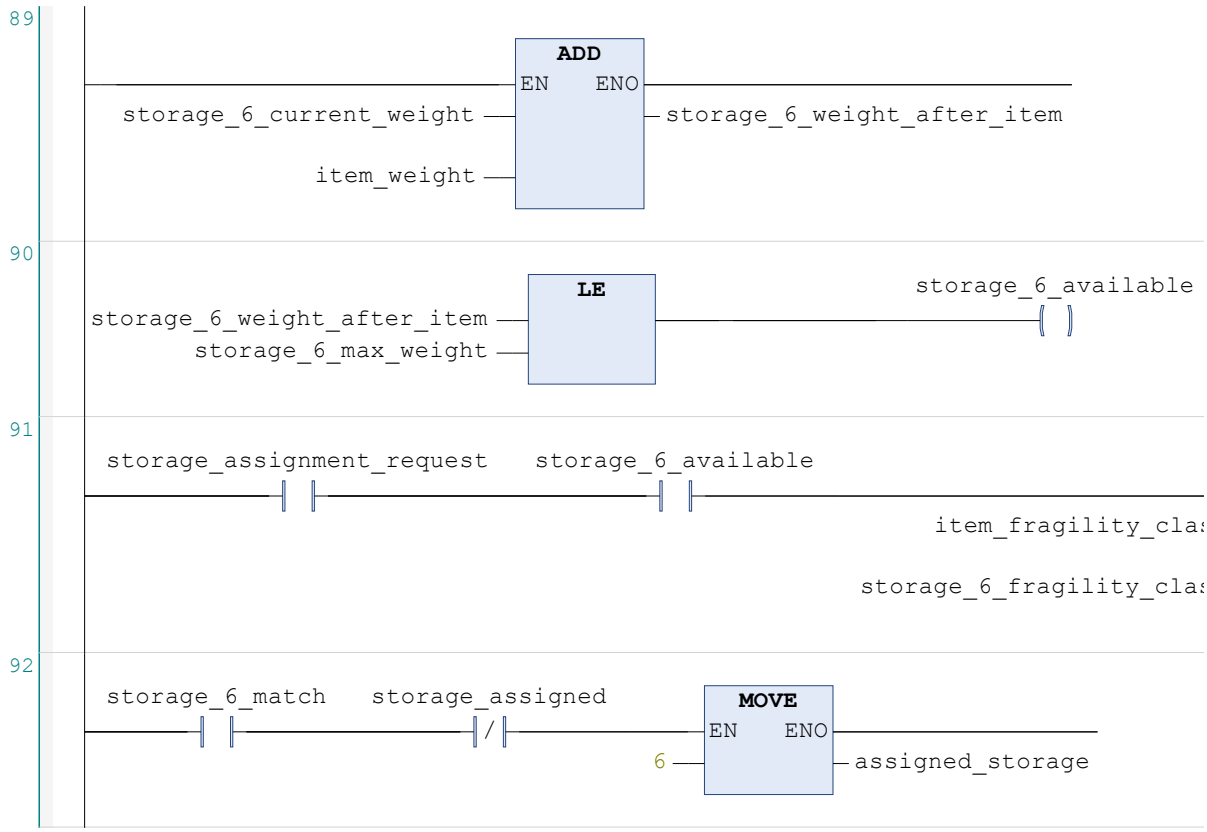

77
ADD
EN
ENO
storage_3_current_weight
item_weight
storage_3_weight_after_item
78
LE
storage_3_weight_after_item
storage_3_max_weight
storage_3_available
79
storage_assignment_request
storage_3_available
item_fragility_cla
storage_3_fragility_cla
80
storage_3_match
storage_assigned
/
MOVE
EN
ENO
3
assigned_storage
81
ADD
EN
ENO
storage_4_current_weight
item_weight
storage_4_weight_after_item
82
LE
storage_4_weight_after_item
storage_4_max_weight
storage_4_available

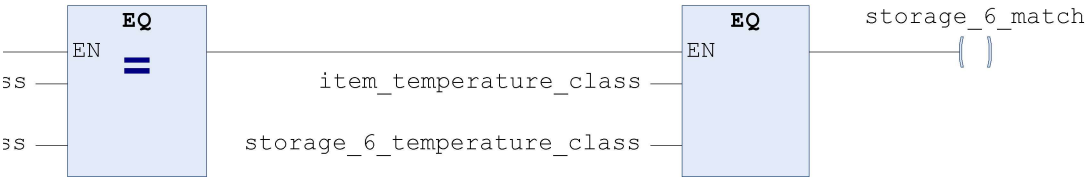

EQ
EN
=
ss
ss
item_temperature_class
storage_3_temperature_class
EQ
EN
storage_3_match

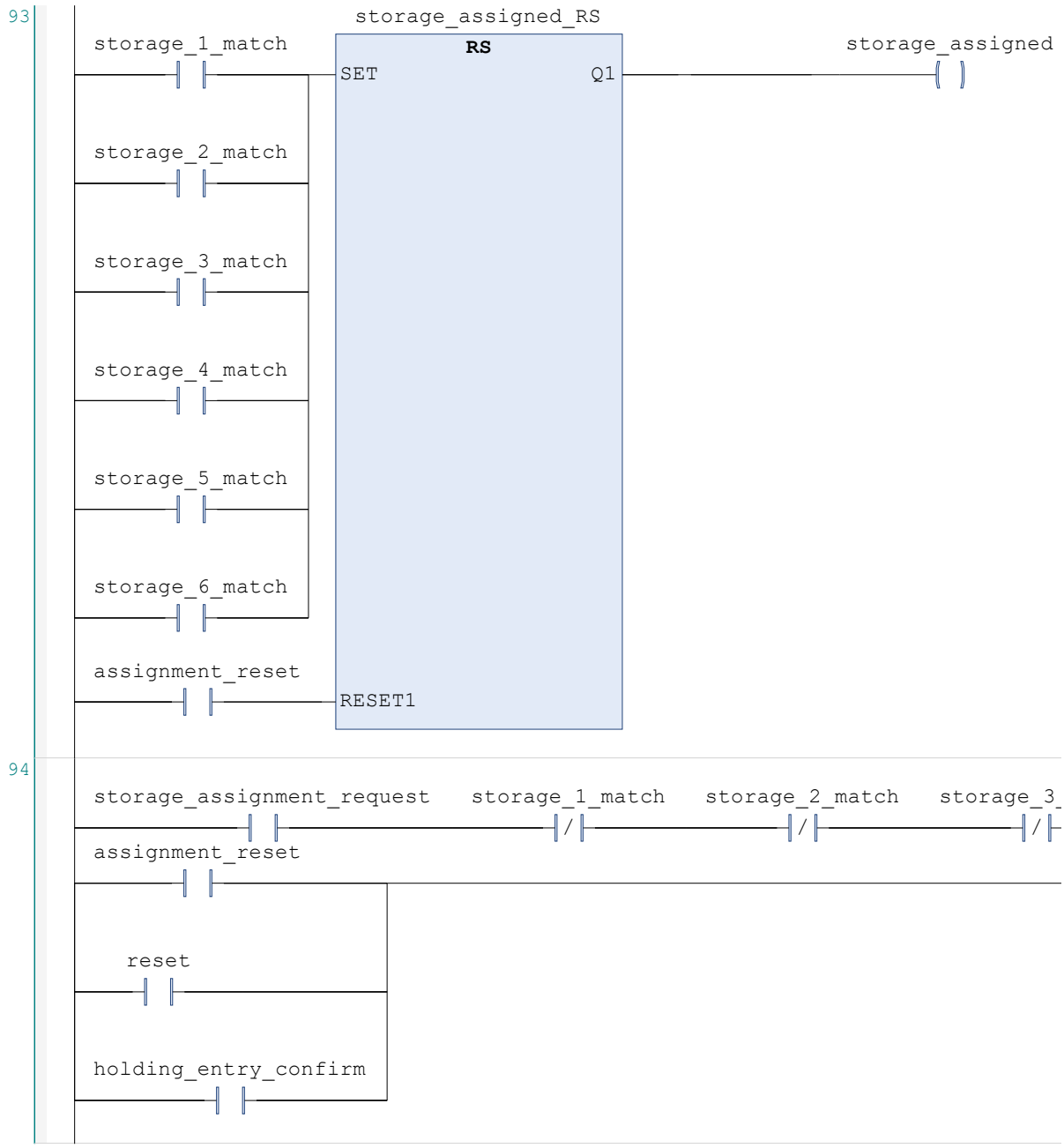

83
storage_assignment_request
storage_4_available
item_fragility_clas
storage_4_fragility_clas
84
storage_4_match
storage_assigned
MOVE
EN
ENO
4
assigned_storage
85
ADD
EN
ENO
storage_5_current_weight
item_weight
storage_5_weight_after_item
86
LE
storage_5_weight_after_item
storage_5_max_weight
storage_5_available
87
storage_assignment_request
storage_5_available
item_fragility_clas
storage_5_fragility_clas
88
storage_5_match
storage_assigned
MOVE
EN
ENO
5
assigned_storage

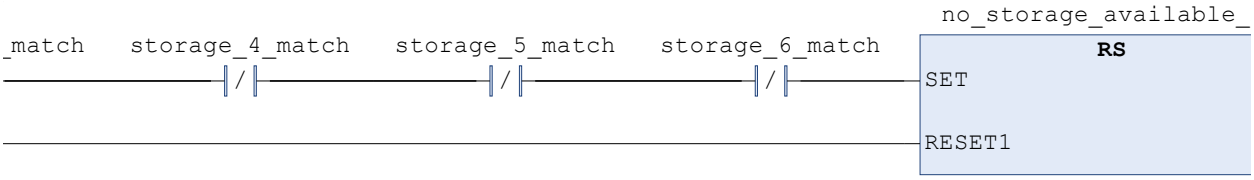

EQ
EN
=
ss
ss
item_temperature_class
storage_4_temperature_class
EQ
EN
storage_4_match

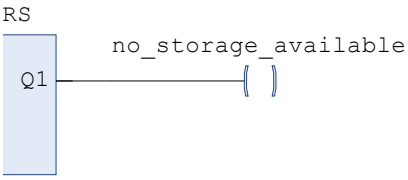

EQ
EN
=
ss
ss
item_temperature_class
storage_5_temperature_class
EQ
EN
storage_5_match

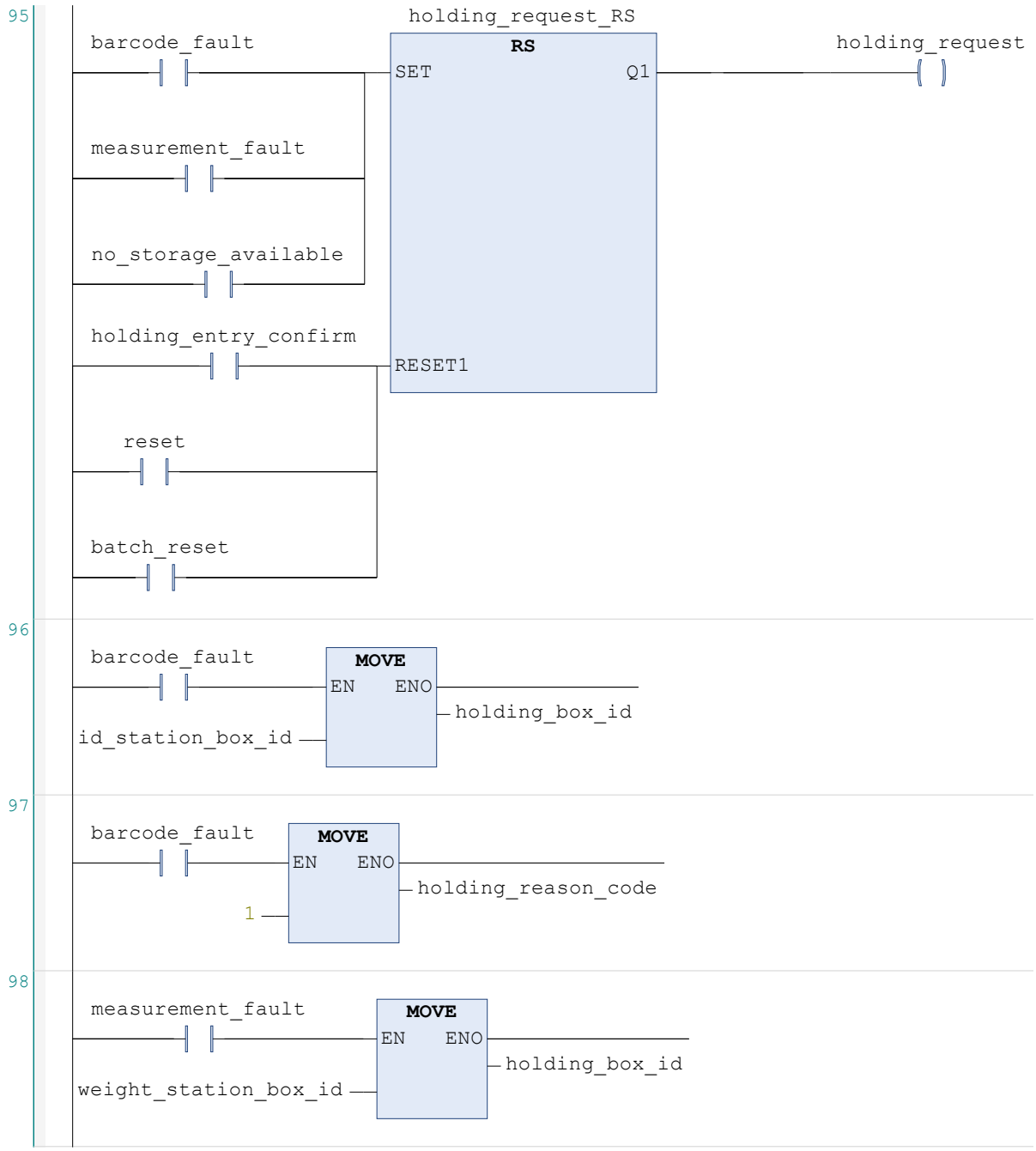

89
ADD
EN
ENO
storage_6_current_weight
item_weight
storage_6_weight_after_item
90
LE
storage_6_weight_after_item
storage_6_max_weight
storage_6_available
91
storage_assignment_request
storage_6_available
item_fragility_cla
storage_6_fragility_cla
92
storage_6_match
storage_assigned
MOVE
EN
ENO
6
assigned_storage

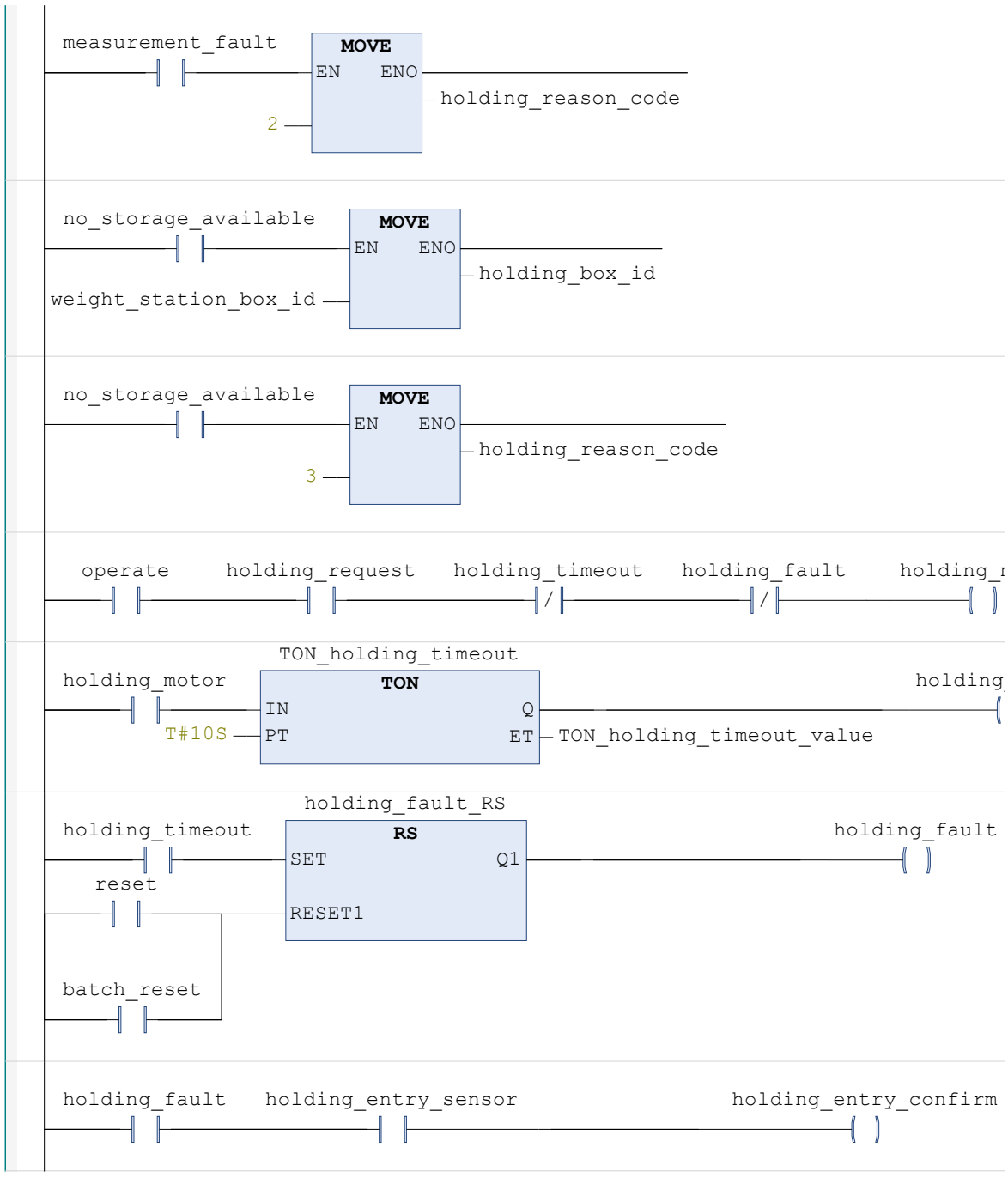

EQ
EN
=
ss
ss
item_temperature_class
storage_6_temperature_class
EQ
EN
storage_6_match


93
storage_assigned_RS
RS
storage_1_match
SET
Q1
storage_assigned
storage_2_match
storage_3_match
storage_4_match
storage_5_match
storage_6_match
assignment_reset
RESET1
94
storage_assignment_request
storage_1_match
storage_2_match
storage_3_
assignment_reset
reset
holding_entry_confirm


no_storage_available_
_match
storage_4_match
storage_5_match
storage_6_match
RS
SET
RESET1

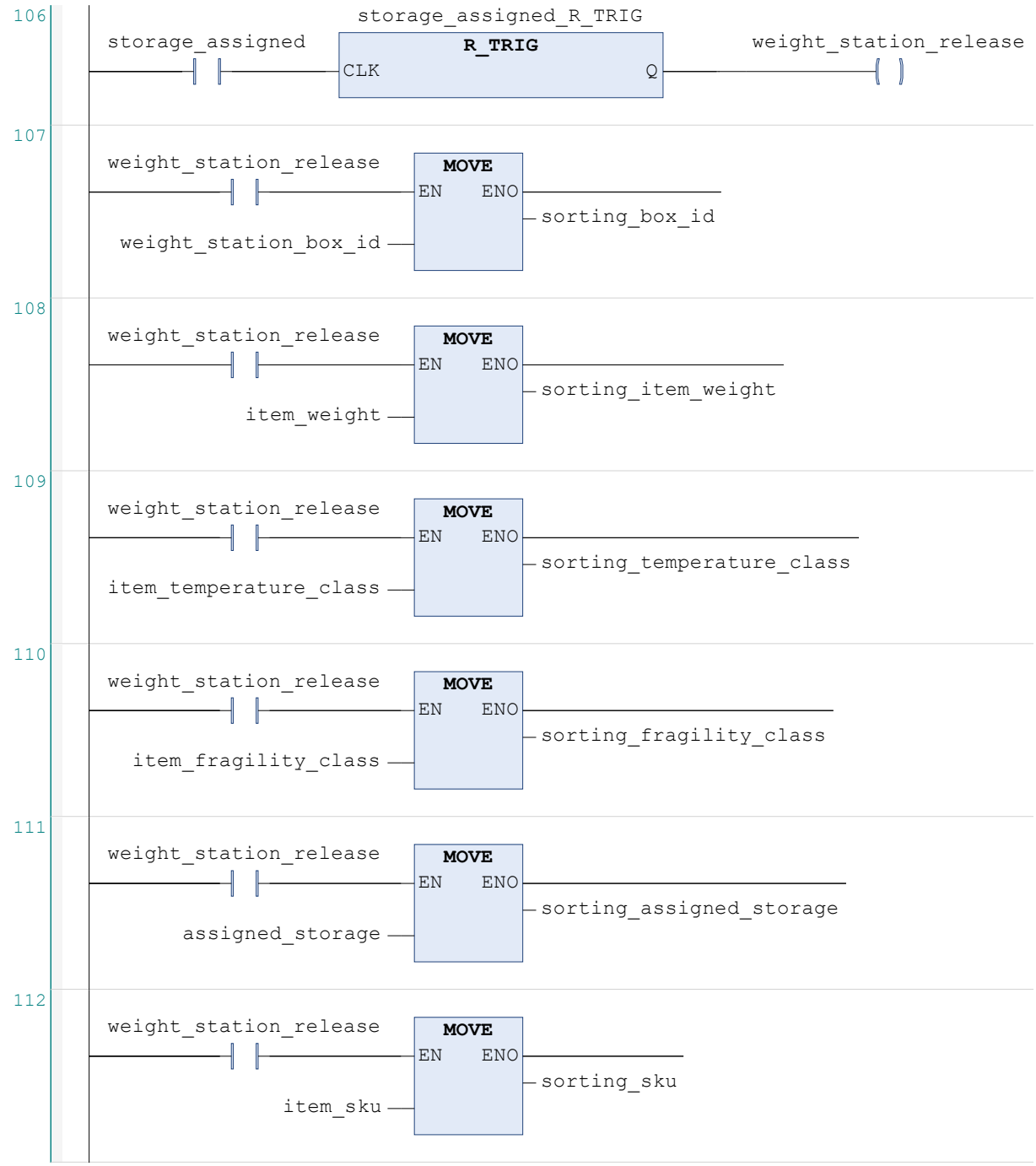

RS
Q1
no_storage_available

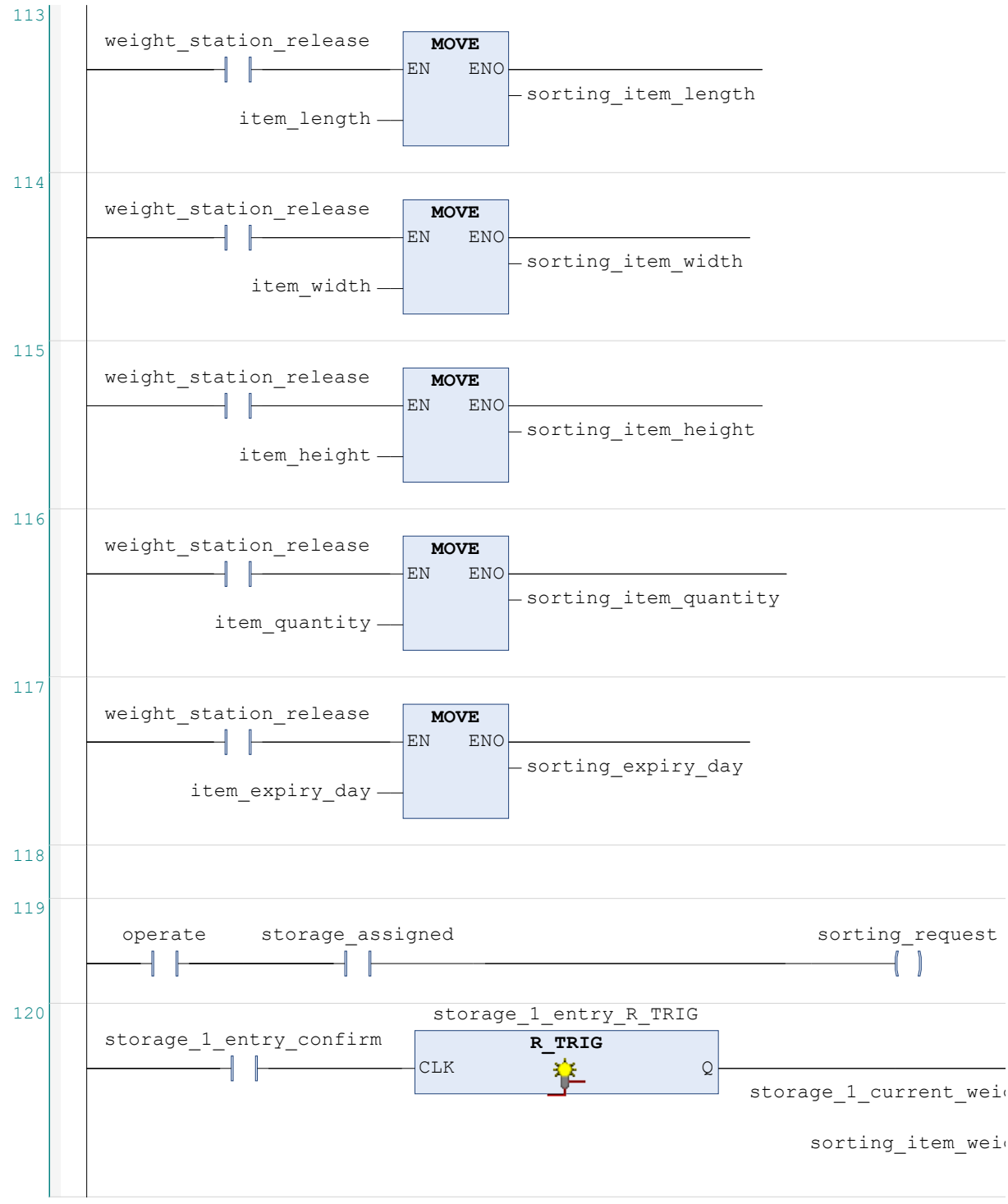

95
holding_request_RS
RS
barcode_fault
SET
Q1
holding_request
measurement_fault
no_storage_available
holding_entry_confirm
RESET1
reset
batch_reset
96
barcode_fault
MOVE
EN
ENO
holding_box_id
id_station_box_id
97
barcode_fault
MOVE
EN
ENO
holding_reason_code
1
98
measurement_fault
MOVE
EN
ENO
holding_box_id
weight_station_box_id

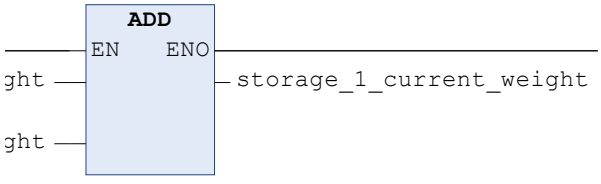

99
measurement_fault
MOVE
EN
ENO
2
holding_reason_code
100
no_storage_available
MOVE
EN
ENO
weight_station_box_id
holding_box_id
101
no_storage_available
MOVE
EN
ENO
3
holding_reason_code
102
operate
holding_request
holding_timeout
holding_fault
103
TON_holding_timeout
holding_motor
TON
IN
PT
T#10S
Q
ET
TON_holding_timeout_value
104
holding_fault_RS
holding_timeout
RS
SET
RESET1
Q1
holding_fault
reset
batch_reset
105
holding_fault
holding_entry_sensor
holding_entry_confirm

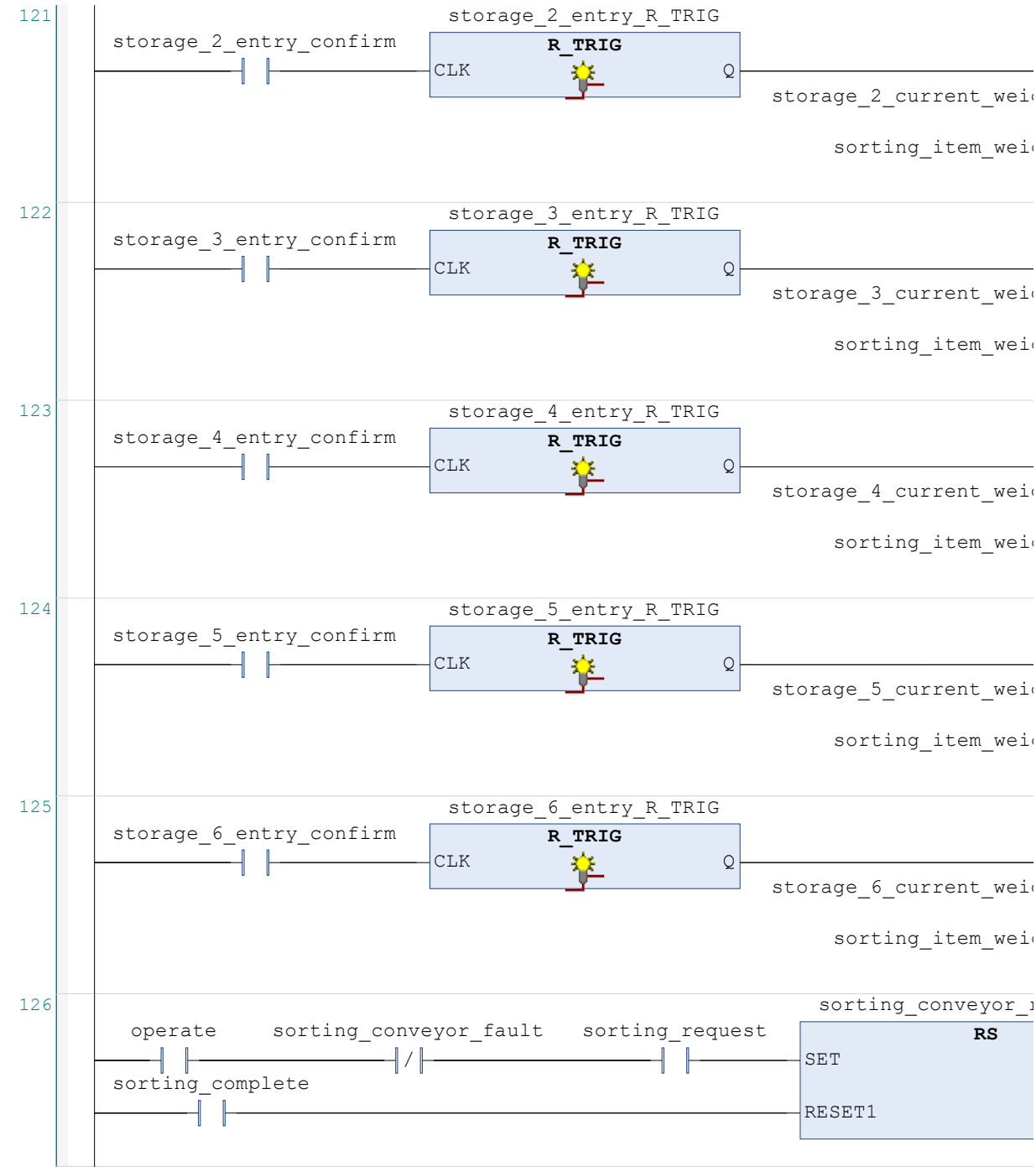
```
motor

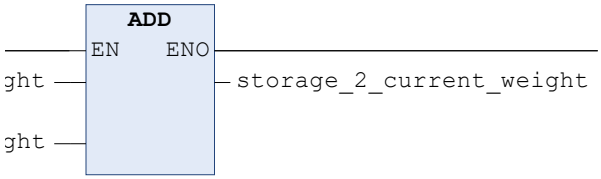

_timeout
 )
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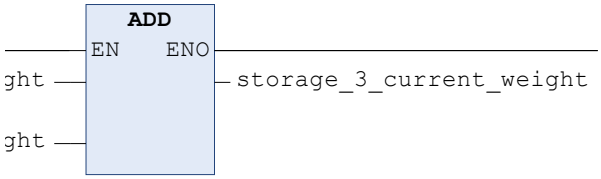

106
storage_assigned
storage_assigned_R_TRIG
R_TRIG
CLK
Q
weight_station_release
107
weight_station_release
MOVE
EN
ENO
sorting_box_id
weight_station_box_id
108
weight_station_release
MOVE
EN
ENO
sorting_item_weight
item_weight
109
weight_station_release
MOVE
EN
ENO
sorting_temperature_class
item_temperature_class
110
weight_station_release
MOVE
EN
ENO
sorting_fragility_class
item_fragility_class
111
weight_station_release
MOVE
EN
ENO
sorting_assigned_storage
assigned_storage
112
weight_station_release
MOVE
EN
ENO
sorting_sku
item_sku

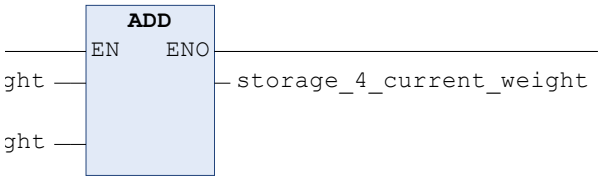

113
weight_station_release
MOVE
EN
ENO
sorting_item_length
item_length
114
weight_station_release
MOVE
EN
ENO
sorting_item_width
item_width
115
weight_station_release
MOVE
EN
ENO
sorting_item_height
item_height
116
weight_station_release
MOVE
EN
ENO
sorting_item_quantity
item_quantity
117
weight_station_release
MOVE
EN
ENO
sorting_expiry_day
item_expiry_day
118
119
operate
storage_assigned
sorting_request
120
storage_1_entry_R_TRIG
storage_1_entry_confirm
R_TRIG
CLK
Q
storage_1_current_wei
sorting_item_wei

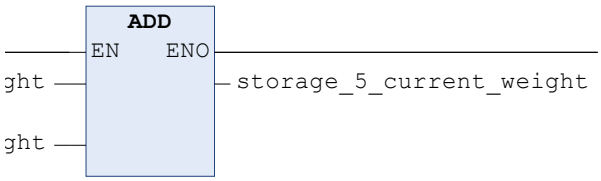

ADD
EN
ENO
ght
storage_1_current_weight
ght

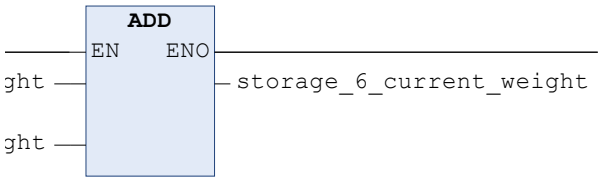

121
storage_2_entry_confirm
storage_2_entry_R_TRIG
R_TRIG
CLK
Q
storage_2_current_wei
sorting_item_wei
122
storage_3_entry_confirm
storage_3_entry_R_TRIG
R_TRIG
CLK
Q
storage_3_current_wei
sorting_item_wei
123
storage_4_entry_confirm
storage_4_entry_R_TRIG
R_TRIG
CLK
Q
storage_4_current_wei
sorting_item_wei
124
storage_5_entry_confirm
storage_5_entry_R_TRIG
R_TRIG
CLK
Q
storage_5_current_wei
sorting_item_wei
125
storage_6_entry_confirm
storage_6_entry_R_TRIG
R_TRIG
CLK
Q
storage_6_current_wei
sorting_item_wei
126
operate
sorting_conveyor_fault
sorting_request
sorting_complete
RS
SET
RESET1

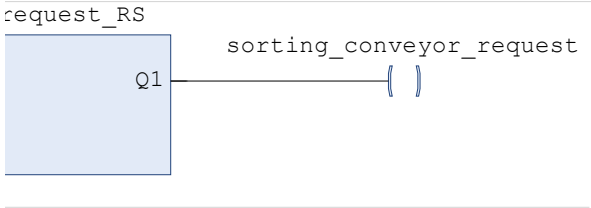

ADD
EN
ENO
ght
ght
storage_2_current_weight


ADD
EN
ENO
ght
ght
storage_3_current_weight

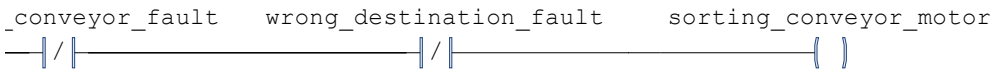

ADD
EN
ENO
ght
ght
storage_4_current_weight

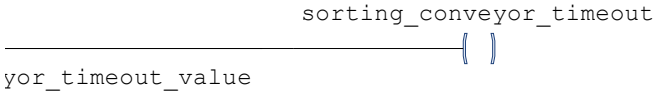

ADD
EN
ENO
ght
ght
storage_5_current_weight


ADD
EN
ENO
ght
ght
storage_6_current_weight

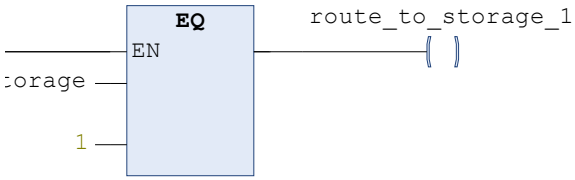

equest_RS
Q1
sorting_conveyor_request

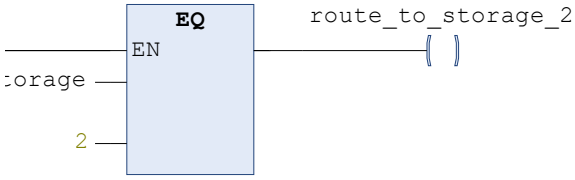

127
operate
sorting_conveyor_request
sorting_conveyor_timeout
sorting_
128
TON_sorting_conveyor_timeout
sorting_conveyor_motor
TON
IN
Q
T#10S
PT
ET
TON_sorting_conve
129
sorting_conveyor_fault_RS
sorting_conveyor_timeout
RS
SET
Q1
sorting_conveyor
reset
RESET1
130
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st
131
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st
132
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st
133
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st

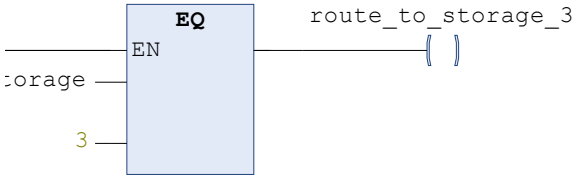

_conveyor_fault
wrong_destination_fault
sorting_conveyor_motor

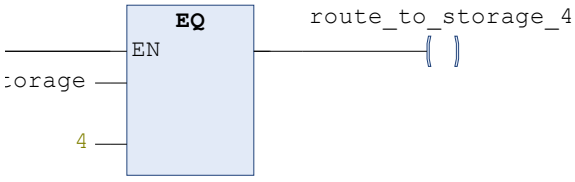

sorting_conveyor_timeout
yor_timeout_value

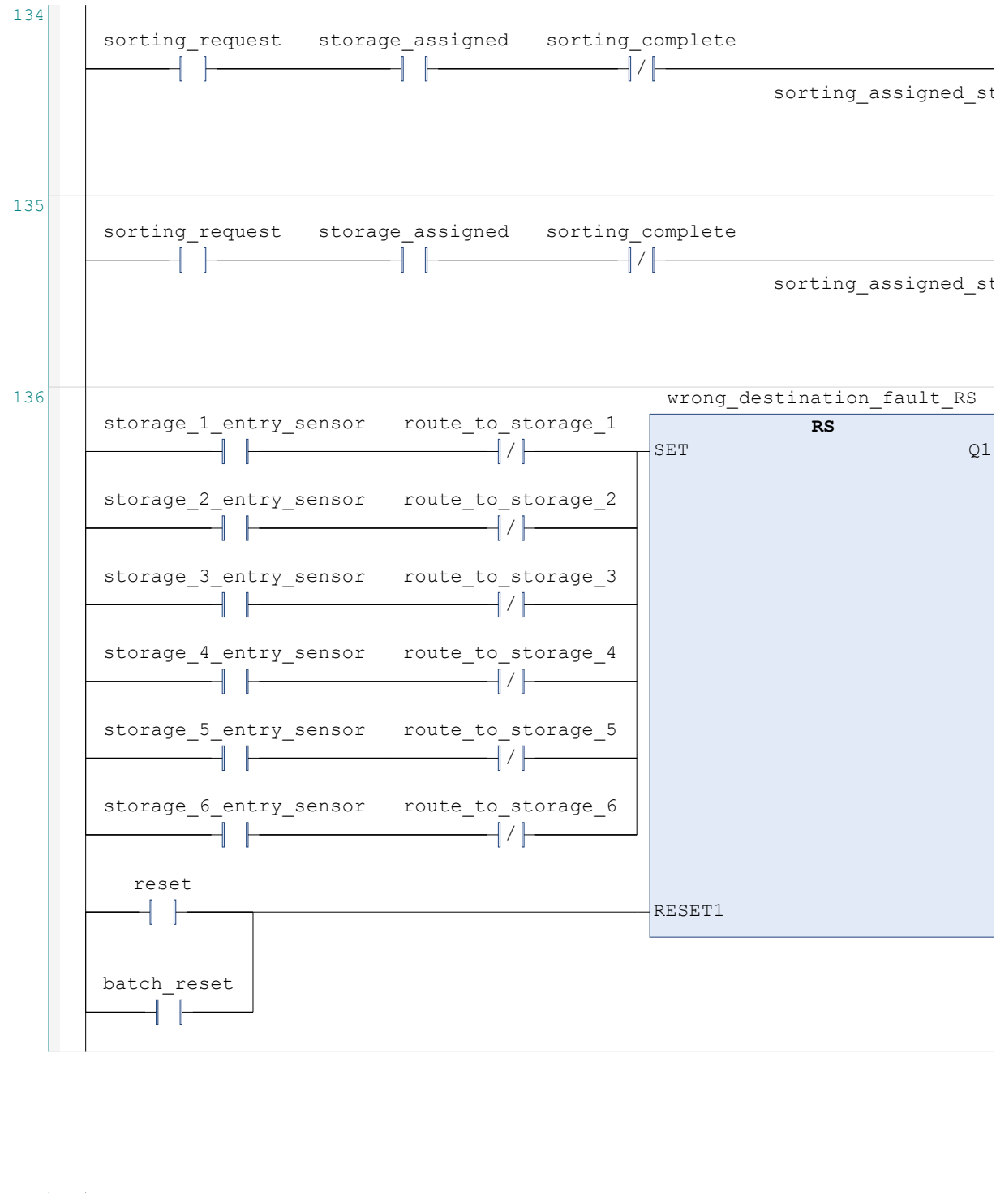

_fault

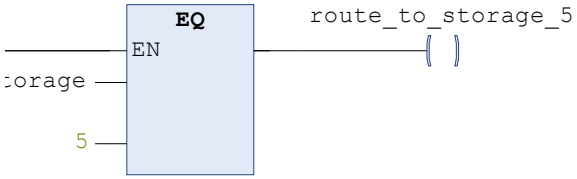

EQ
EN
orage
1
route_to_storage_1

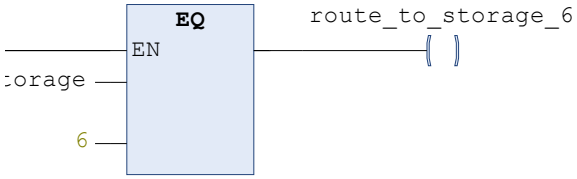

EQ
EN
orage
2
route_to_storage_2

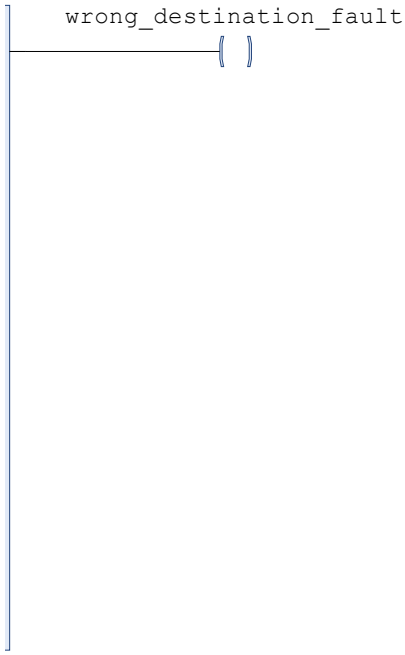

EQ
EN
orage
3
route_to_storage_3

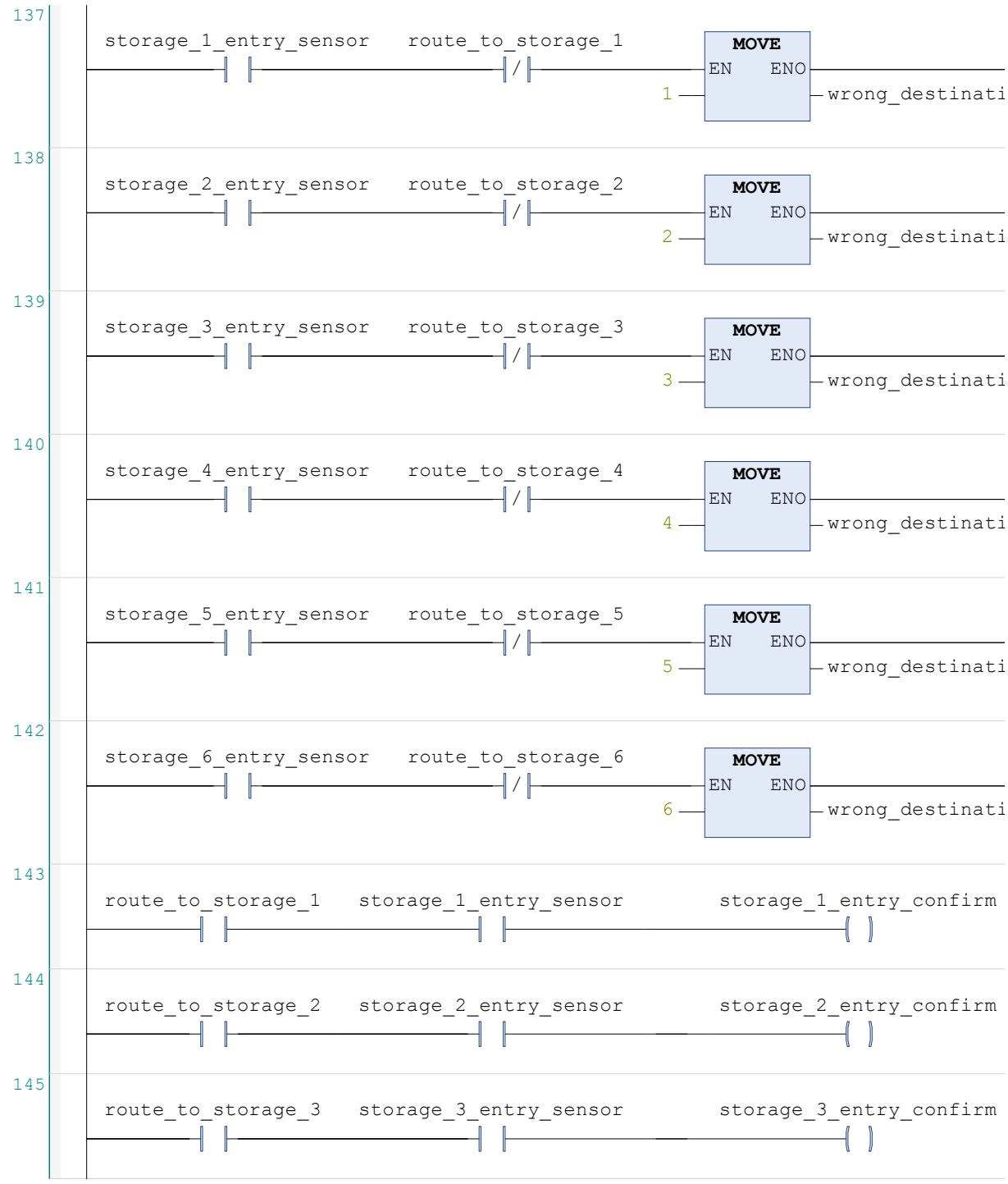

EQ
EN
orage
4
route_to_storage_4


134
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st
135
sorting_request
storage_assigned
sorting_complete
sorting_assigned_st
136
wrong_destination_fault_RS
RS
SET
Q1
storage_1_entry_sensor
route_to_storage_1
storage_2_entry_sensor
route_to_storage_2
storage_3_entry_sensor
route_to_storage_3
storage_4_entry_sensor
route_to_storage_4
storage_5_entry_sensor
route_to_storage_5
storage_6_entry_sensor
route_to_storage_6
reset
RESET1
batch_reset


EQ
EN
route_to_storage_5
5


EQ
EN
route_to_storage_6
6


wrong_destination_fault


137
storage_1_entry_sensor
route_to_storage_1
MOVE
EN
ENO
1
wrong_destinati
138
storage_2_entry_sensor
route_to_storage_2
MOVE
EN
ENO
2
wrong_destinati
139
storage_3_entry_sensor
route_to_storage_3
MOVE
EN
ENO
3
wrong_destinati
140
storage_4_entry_sensor
route_to_storage_4
MOVE
EN
ENO
4
wrong_destinati
141
storage_5_entry_sensor
route_to_storage_5
MOVE
EN
ENO
5
wrong_destinati
142
storage_6_entry_sensor
route_to_storage_6
MOVE
EN
ENO
6
wrong_destinati
143
route_to_storage_1
storage_1_entry_sensor
storage_1_entry_confirm
144
route_to_storage_2
storage_2_entry_sensor
storage_2_entry_confirm
145
route_to_storage_3
storage_3_entry_sensor
storage_3_entry_confirm

```
.on_storage
```

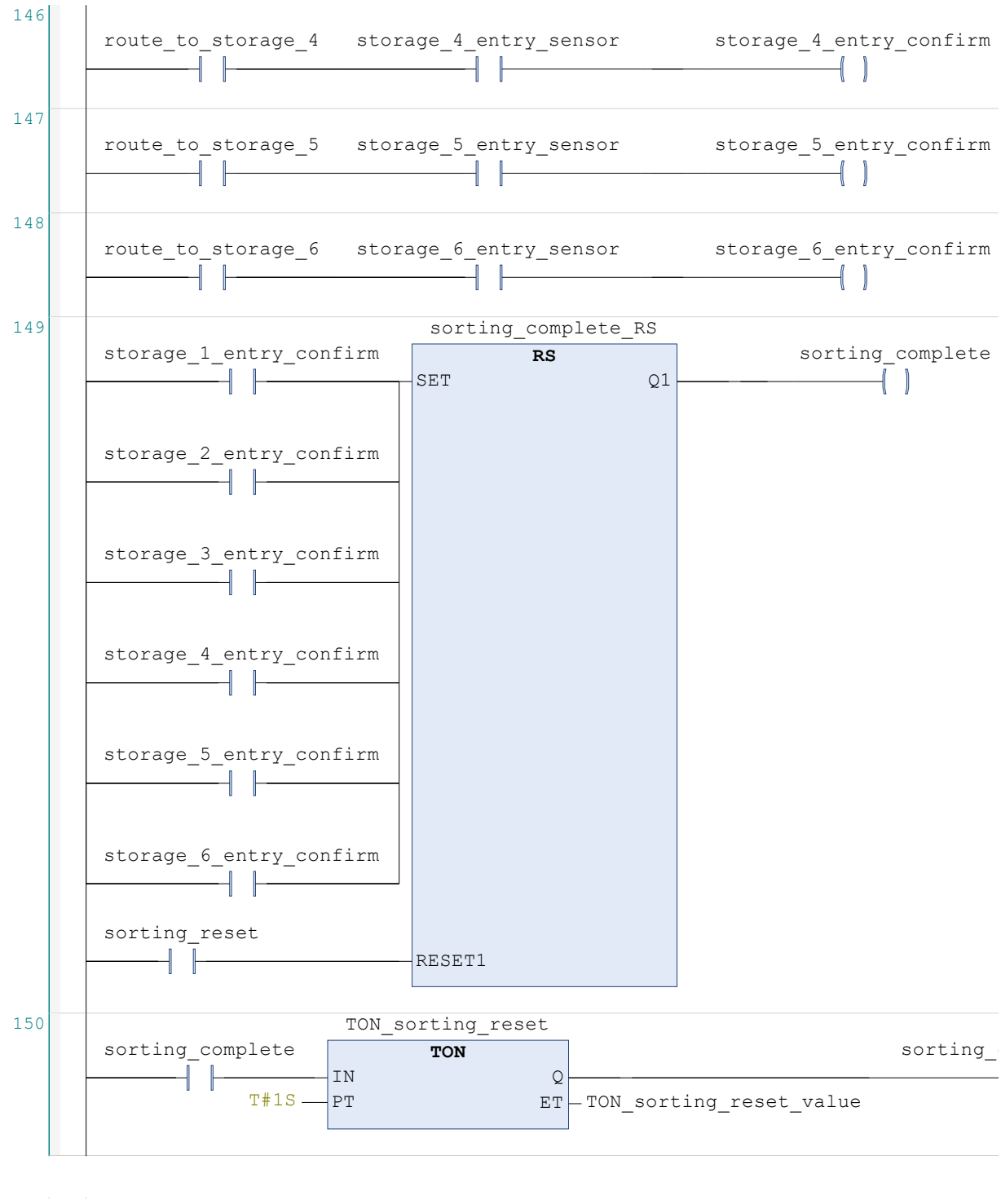
```
.on_storage
```

```
.on_storage
```

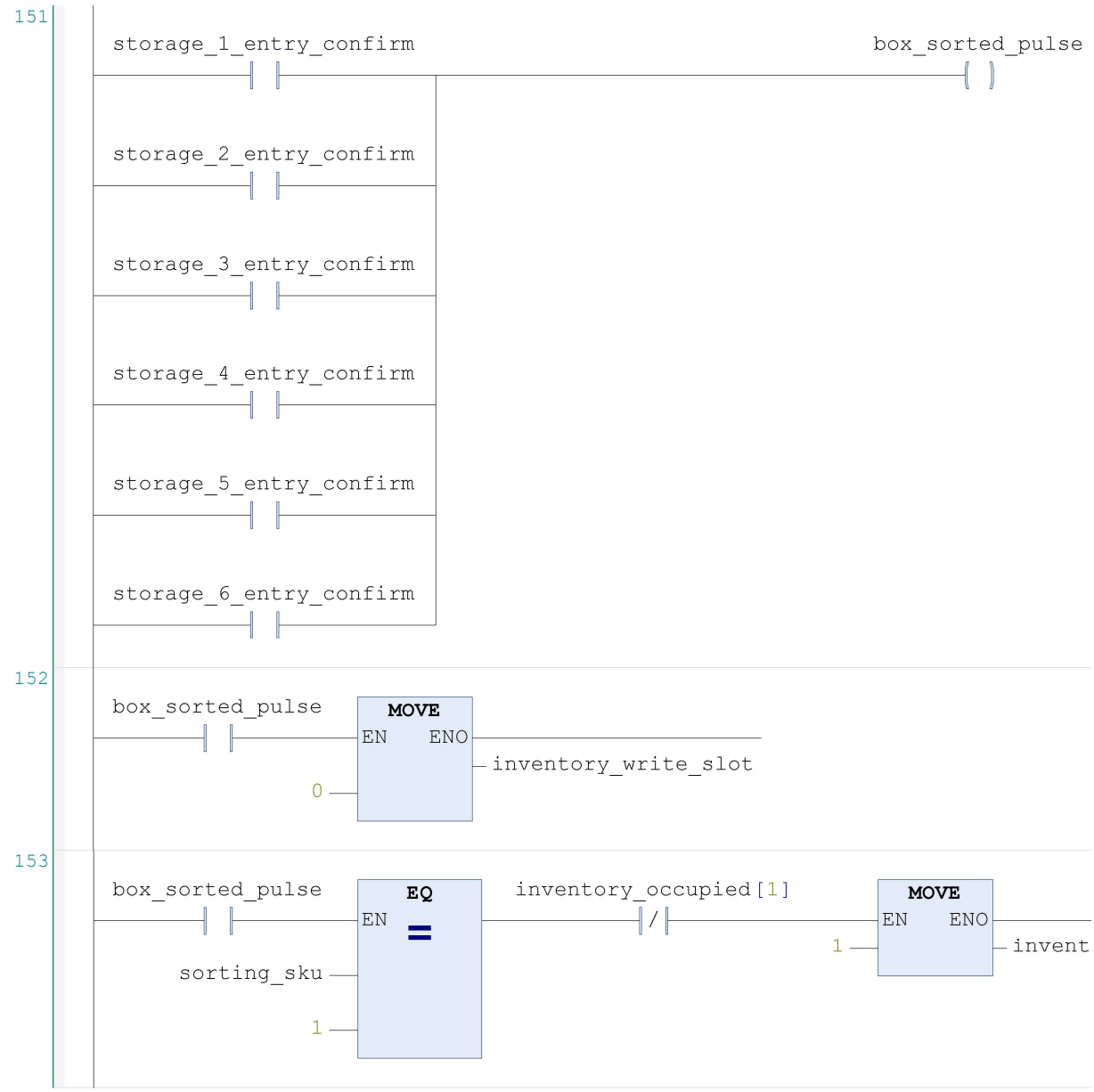
```
.on_storage
```

```
.on_storage
```

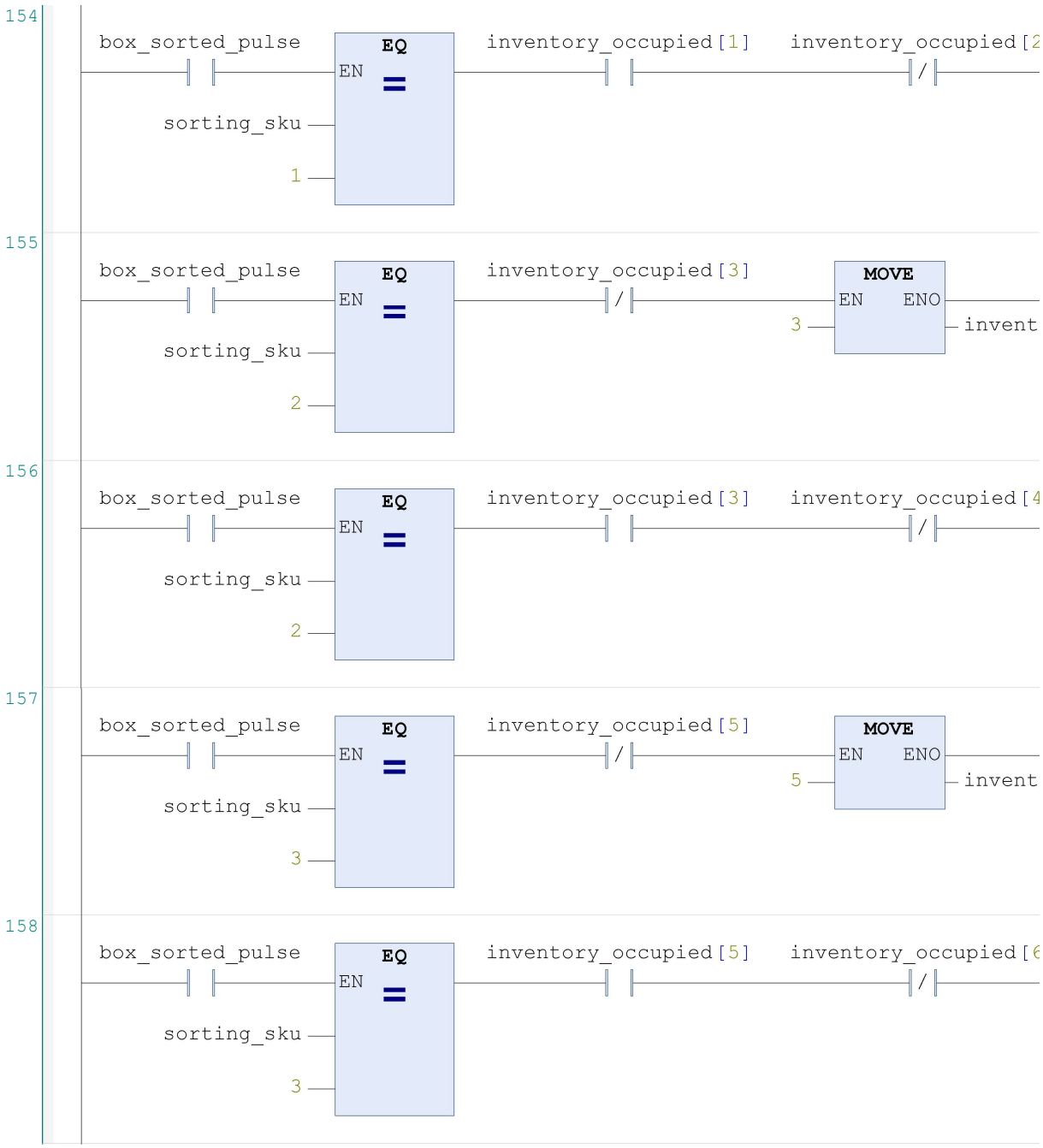
```
.on_storage
```

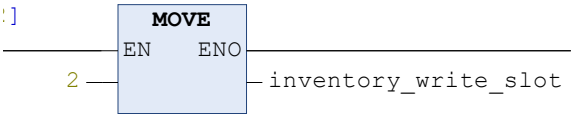

146
route_to_storage_4
storage_4_entry_sensor
storage_4_entry_confirm
147
route_to_storage_5
storage_5_entry_sensor
storage_5_entry_confirm
148
route_to_storage_6
storage_6_entry_sensor
storage_6_entry_confirm
149
sorting_complete_RS
RS
storage_1_entry_confirm
SET
Q1
sorting_complete
storage_2_entry_confirm
storage_3_entry_confirm
storage_4_entry_confirm
storage_5_entry_confirm
storage_6_entry_confirm
sorting_reset
RESET1
150
TON_sorting_reset
TON
sorting_complete
IN
Q
sorting_
T#1S
PT
ET
TON_sorting_reset_value

cycle_reset
( )

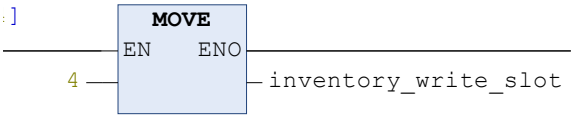

151
storage_1_entry_confirm
box_sorted_pulse
storage_2_entry_confirm
storage_3_entry_confirm
storage_4_entry_confirm
storage_5_entry_confirm
storage_6_entry_confirm
152
box_sorted_pulse
MOVE
EN
ENO
inventory_write_slot
0
153
box_sorted_pulse
EQ
EN
=
inventory_occupied[1]
/
MOVE
EN
ENO
1
invent
sorting_sku
1

```
ory_write_slot
```

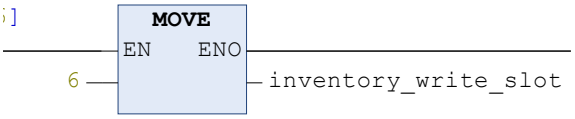

154
box_sorted_pulse
EQ
EN
=
sorting_sku
1
inventory_occupied[1]
inventory_occupied[2
/
155
box_sorted_pulse
EQ
EN
=
sorting_sku
2
inventory_occupied[3]
/
MOVE
EN
ENO
3
invent
156
box_sorted_pulse
EQ
EN
=
sorting_sku
2
inventory_occupied[3]
inventory_occupied[4
/
157
box_sorted_pulse
EQ
EN
=
sorting_sku
3
inventory_occupied[5]
/
MOVE
EN
ENO
5
invent
158
box_sorted_pulse
EQ
EN
=
sorting_sku
3
inventory_occupied[5]
inventory_occupied[6
/

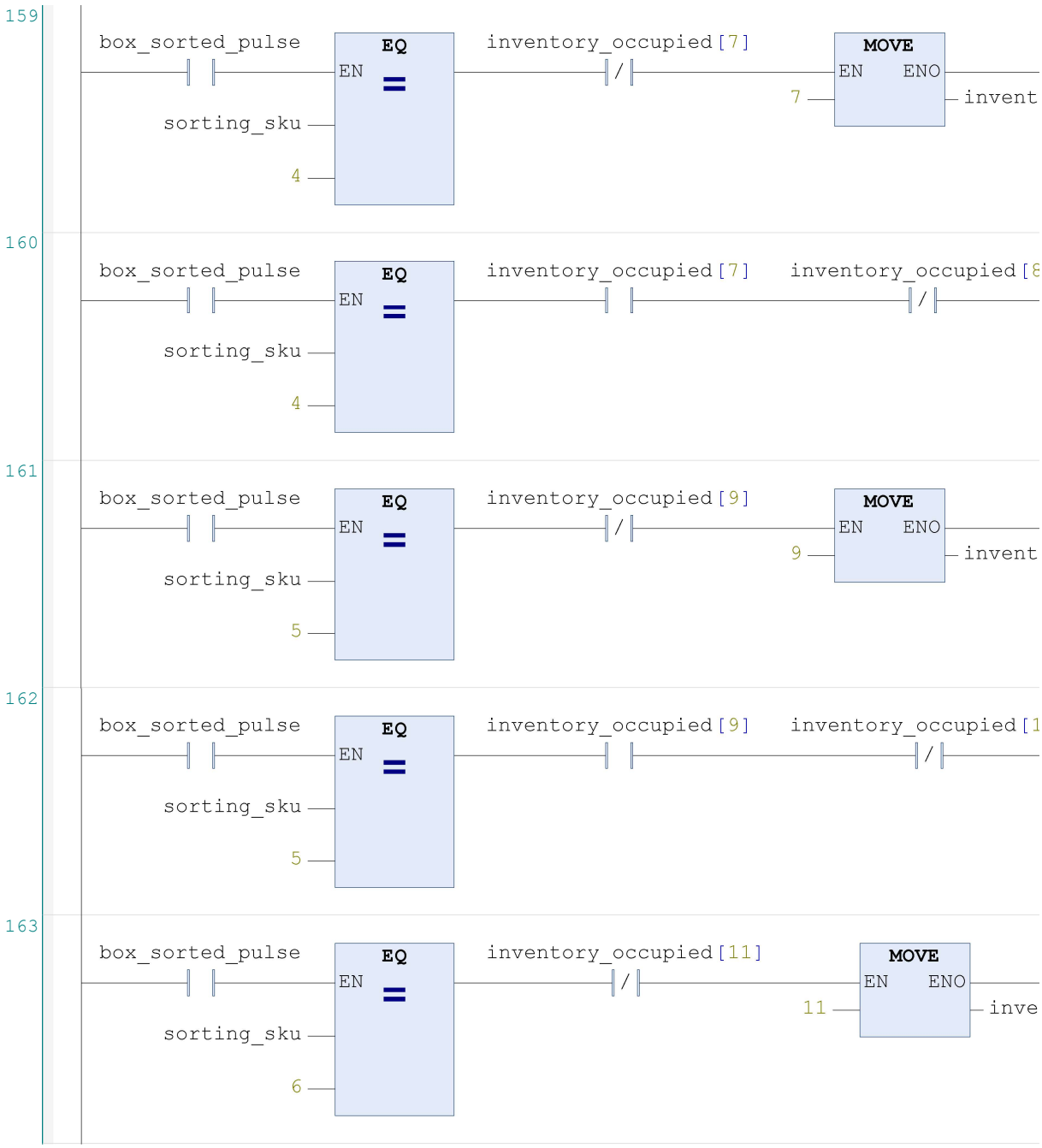


ory_write_slot

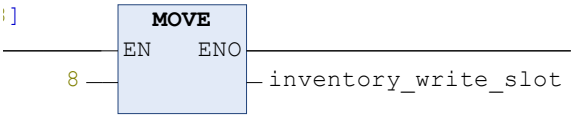


ory_write_slot

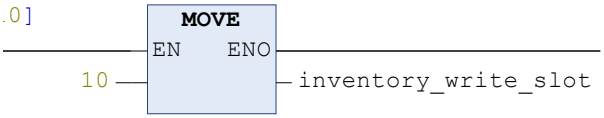



159
box_sorted_pulse
EQ
EN
=
sorting_sku
4
inventory_occupied[7]
/
MOVE
EN
ENO
7
invent
160
box_sorted_pulse
EQ
EN
=
sorting_sku
4
inventory_occupied[7]
inventory_occupied[8
/
161
box_sorted_pulse
EQ
EN
=
sorting_sku
5
inventory_occupied[9]
/
MOVE
EN
ENO
9
invent
162
box_sorted_pulse
EQ
EN
=
sorting_sku
5
inventory_occupied[9]
inventory_occupied[1
/
163
box_sorted_pulse
EQ
EN
=
sorting_sku
6
inventory_occupied[11]
/
MOVE
EN
ENO
11
inve

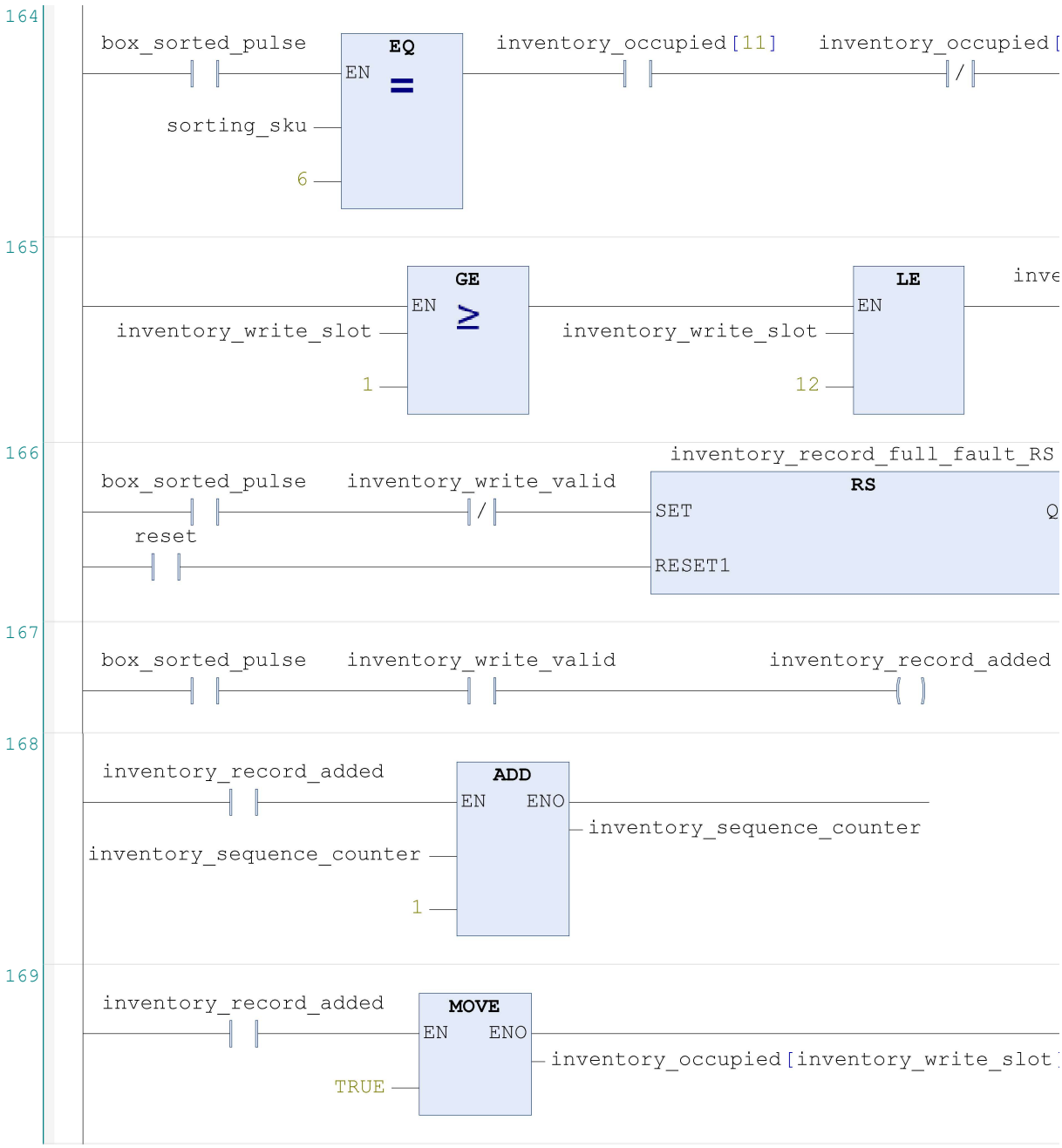
ory_write_slot

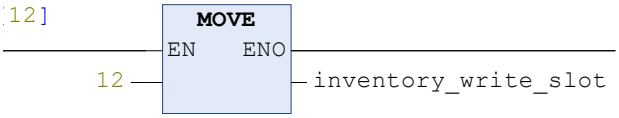


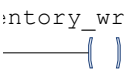
ory_write_slot

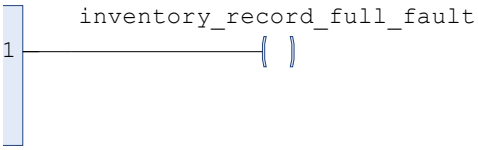


ntory_write_slot

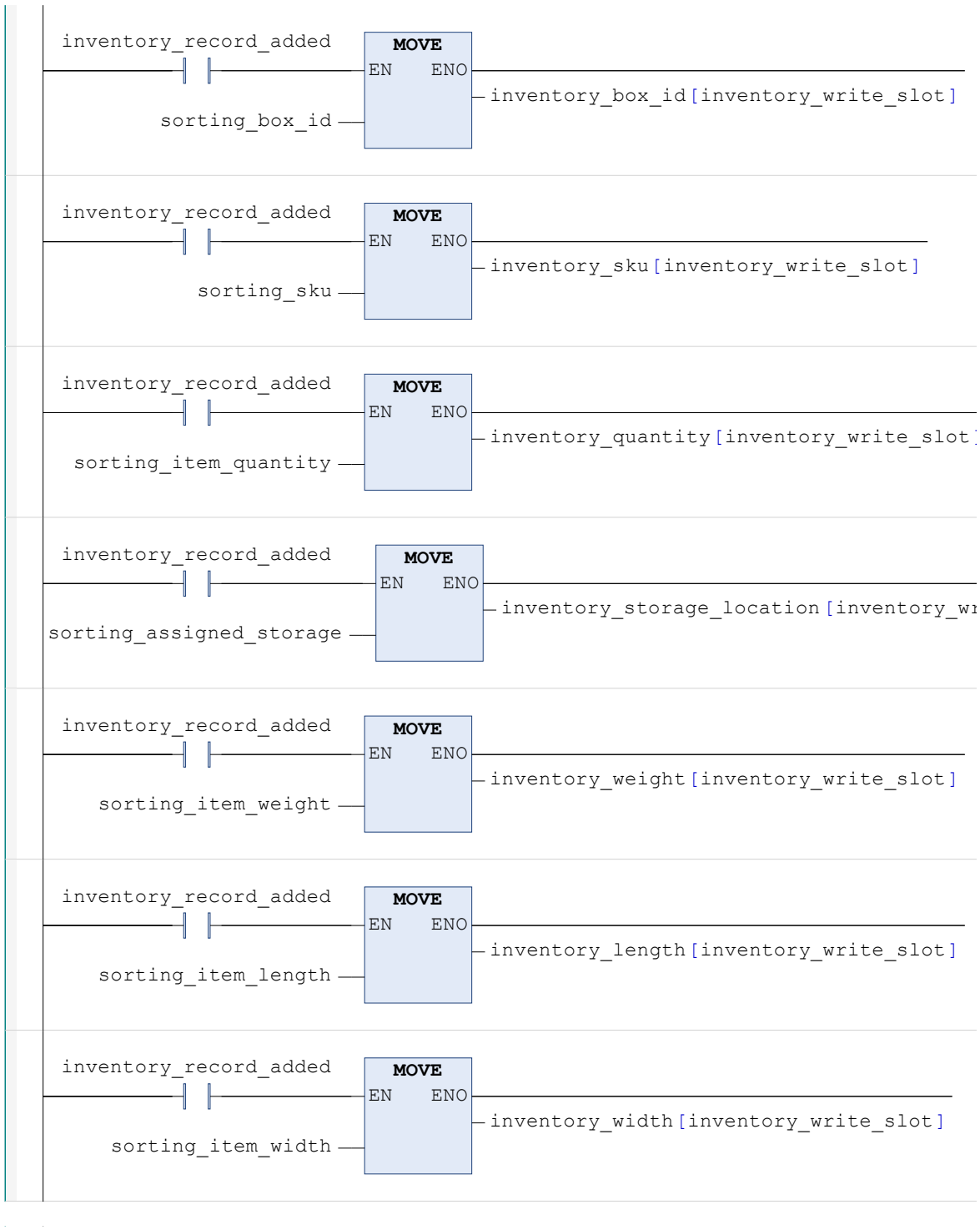

164
box_sorted_pulse
EQ
EN
=
sorting_sku
6
inventory_occupied[11]
inventory_occupied[
/
165
GE
EN
≥
inventory_write_slot
1
LE
EN
inventory_write_slot
12
inve
166
inventory_record_full_fault_RS
box_sorted_pulse
inventory_write_valid
/
RS
SET
Q
reset
RESET1
167
box_sorted_pulse
inventory_write_valid
inventory_record_added
168
inventory_record_added
ADD
EN
ENO
inventory_sequence_counter
inventory_sequence_counter
1
169
inventory_record_added
MOVE
EN
ENO
inventory_occupied[inventory_write_slot
TRUE


12]
MOVE
EN
ENO
12
inventory_write_slot


ntory_write_valid

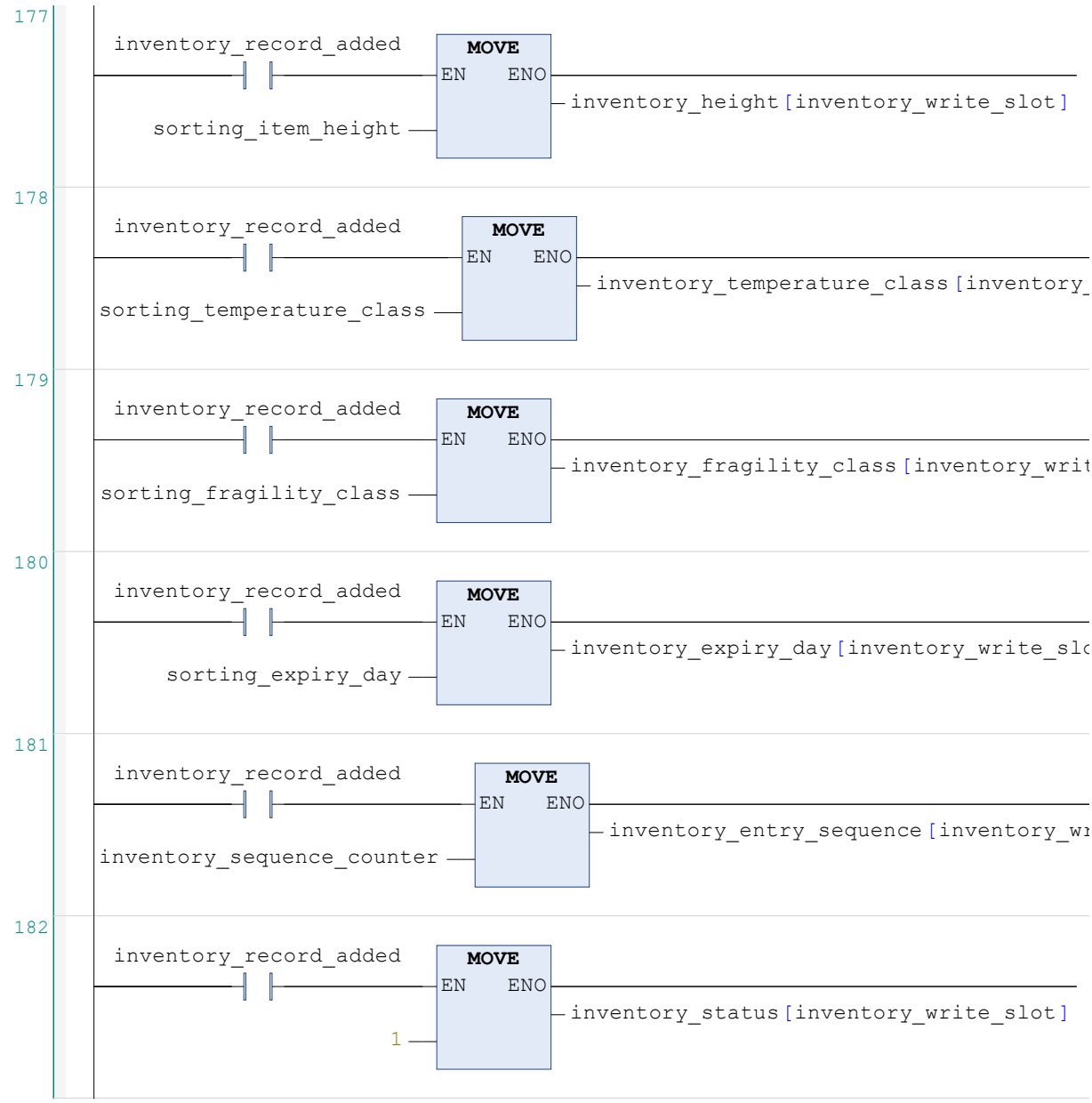

inventory_record_full_fault
1

]


170
inventory_record_added
sorting_box_id
MOVE
EN
ENO
inventory_box_id[inventory_write_slot]
171
inventory_record_added
sorting_sku
MOVE
EN
ENO
inventory_sku[inventory_write_slot]
172
inventory_record_added
sorting_item_quantity
MOVE
EN
ENO
inventory_quantity[inventory_write_slot
173
inventory_record_added
sorting_assigned_storage
MOVE
EN
ENO
inventory_storage_location[inventory_wr
174
inventory_record_added
sorting_item_weight
MOVE
EN
ENO
inventory_weight[inventory_write_slot]
175
inventory_record_added
sorting_item_length
MOVE
EN
ENO
inventory_length[inventory_write_slot]
176
inventory_record_added
sorting_item_width
MOVE
EN
ENO
inventory_width[inventory_write_slot]

_
]

:ite_slot]

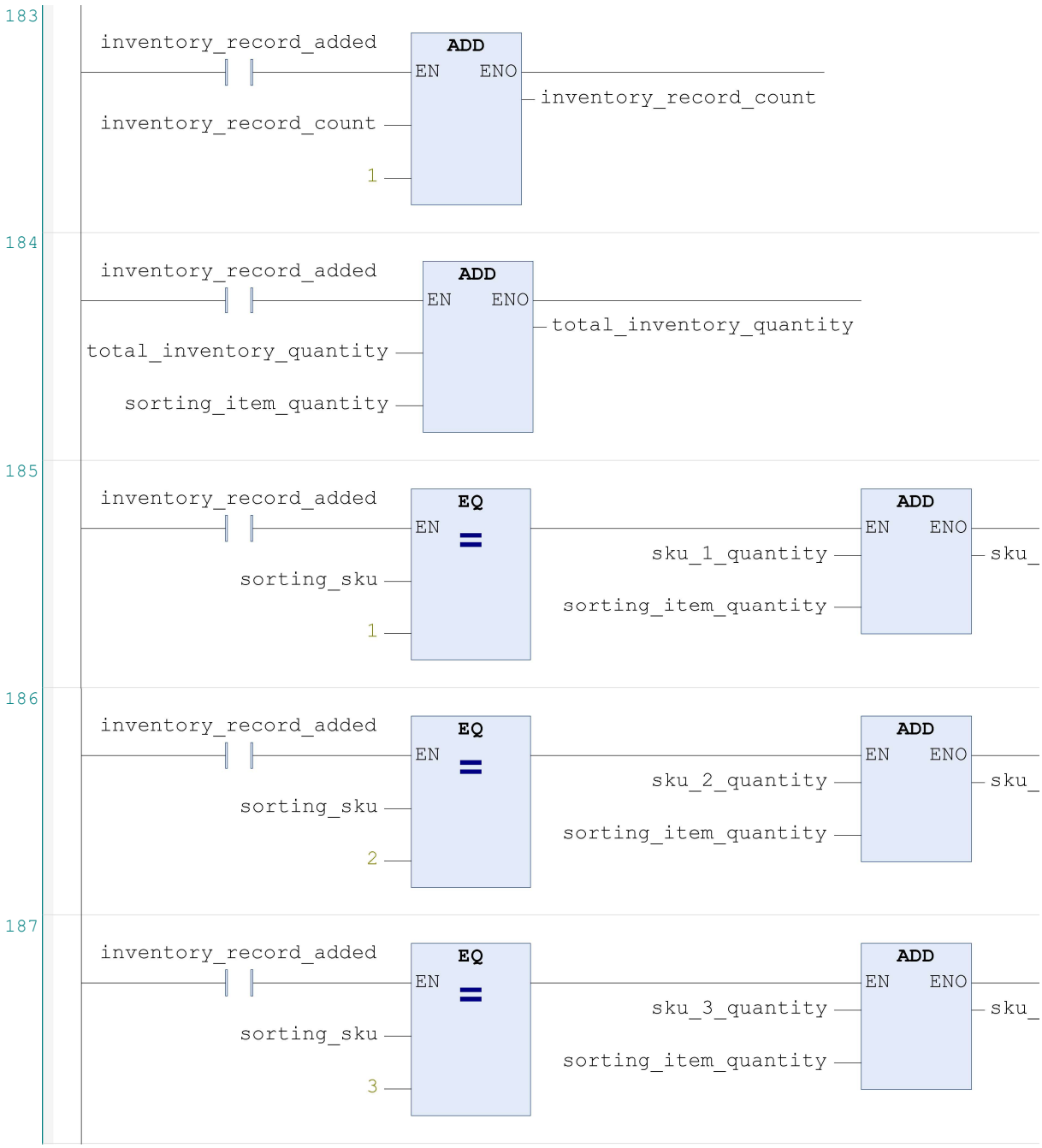

177
inventory_record_added
MOVE
EN
ENO
inventory_height[inventory_write_slot]
sorting_item_height
178
inventory_record_added
MOVE
EN
ENO
inventory_temperature_class[inventory_
sorting_temperature_class
179
inventory_record_added
MOVE
EN
ENO
inventory_fragility_class[inventory_writ
sorting_fragility_class
180
inventory_record_added
MOVE
EN
ENO
inventory_expiry_day[inventory_write_slo
sorting_expiry_day
181
inventory_record_added
MOVE
EN
ENO
inventory_entry_sequence[inventory_wr
inventory_sequence_counter
182
inventory_record_added
MOVE
EN
ENO
inventory_status[inventory_write_slot]
1

_write_slot]

:e_slot]

ɔt]

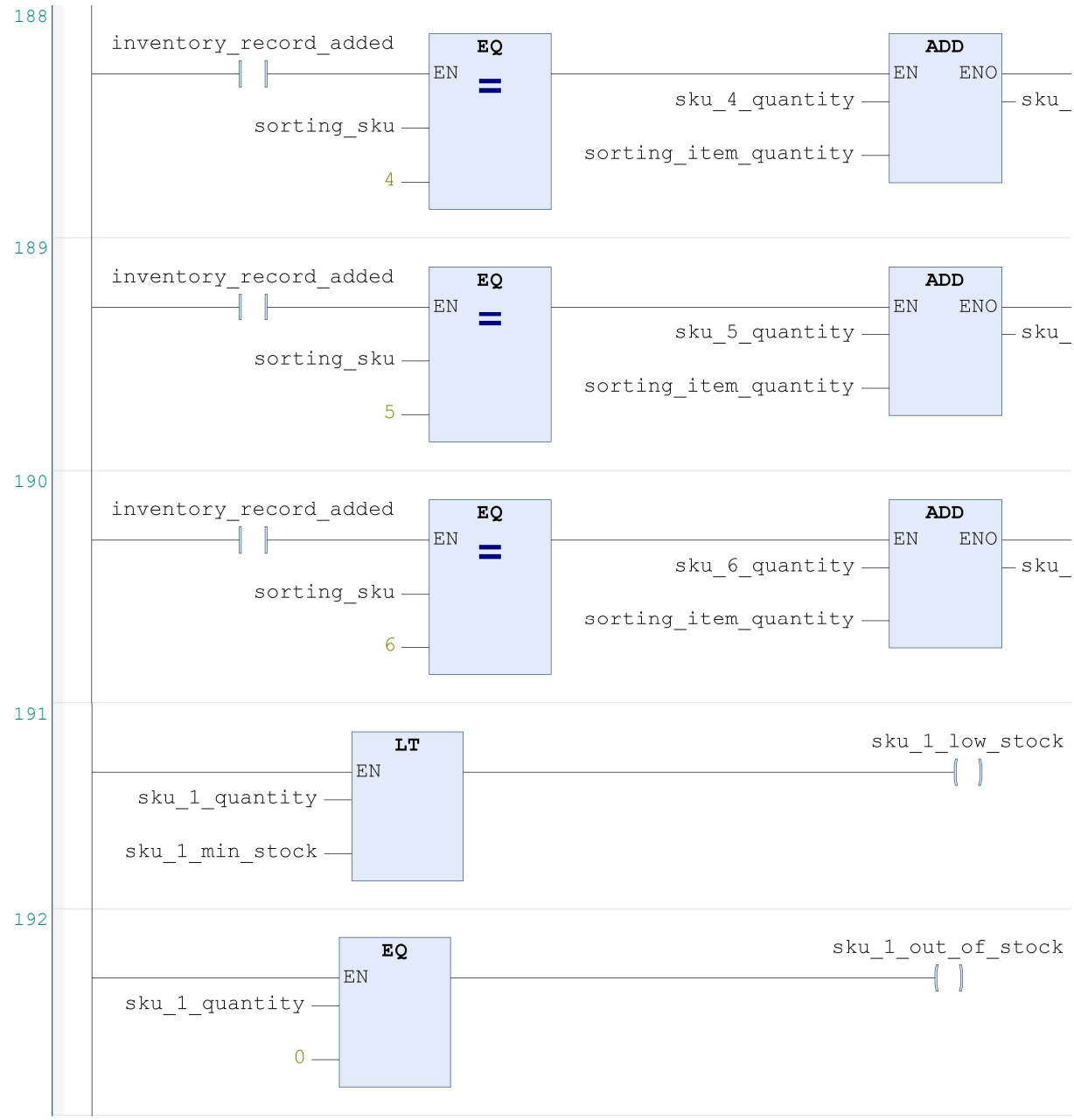
:ite_slot]


183
inventory_record_added
ADD
EN
ENO
inventory_record_count
inventory_record_count
1
184
inventory_record_added
ADD
EN
ENO
total_inventory_quantity
total_inventory_quantity
sorting_item_quantity
185
inventory_record_added
EQ
EN
=
ADD
EN
ENO
sku_1_quantity
sku_
sorting_sku
sorting_item_quantity
1
186
inventory_record_added
EQ
EN
=
ADD
EN
ENO
sku_2_quantity
sku_
sorting_sku
sorting_item_quantity
2
187
inventory_record_added
EQ
EN
=
ADD
EN
ENO
sku_3_quantity
sku_
sorting_sku
sorting_item_quantity
3

1_quantity

2_quantity

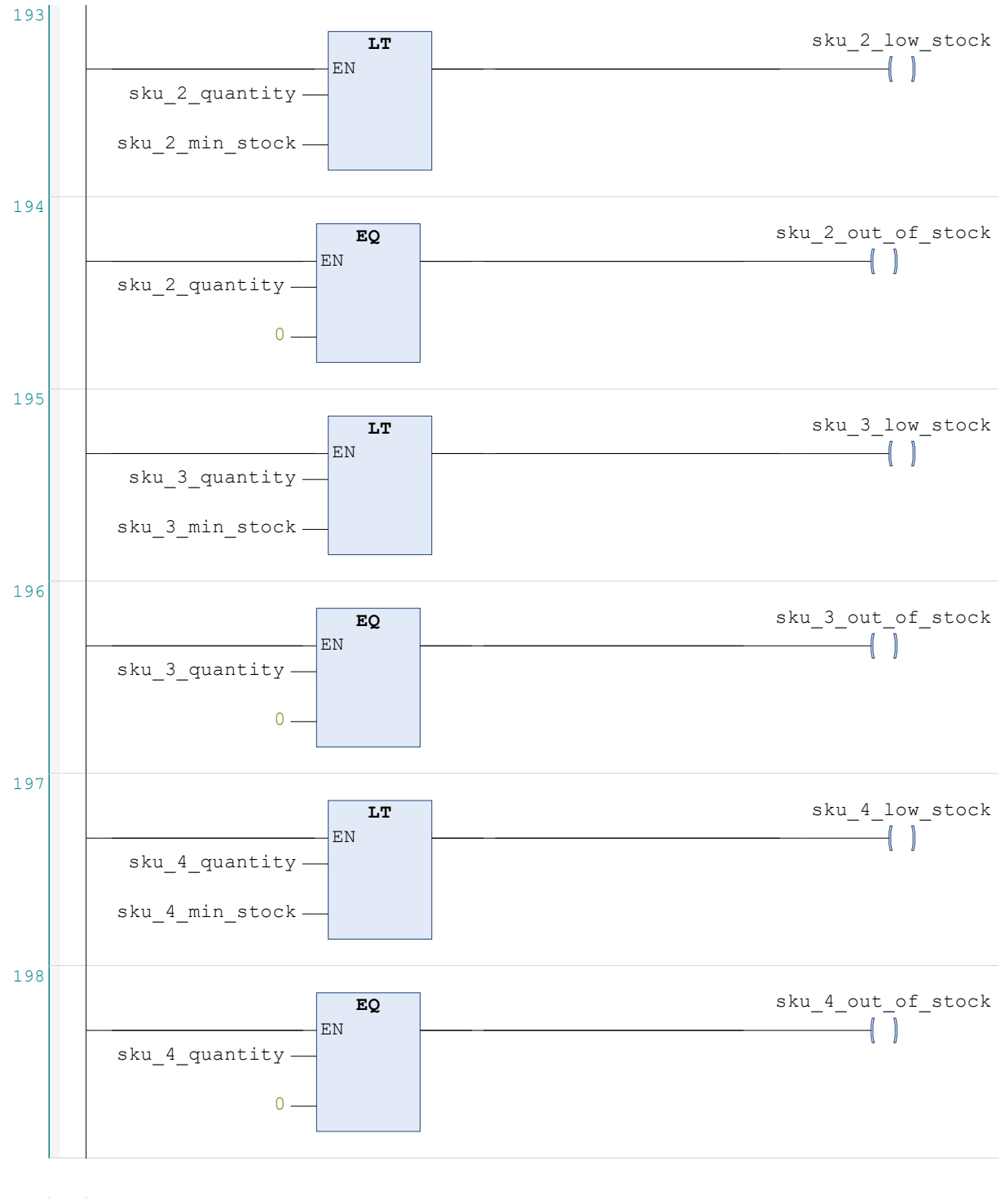
3_quantity

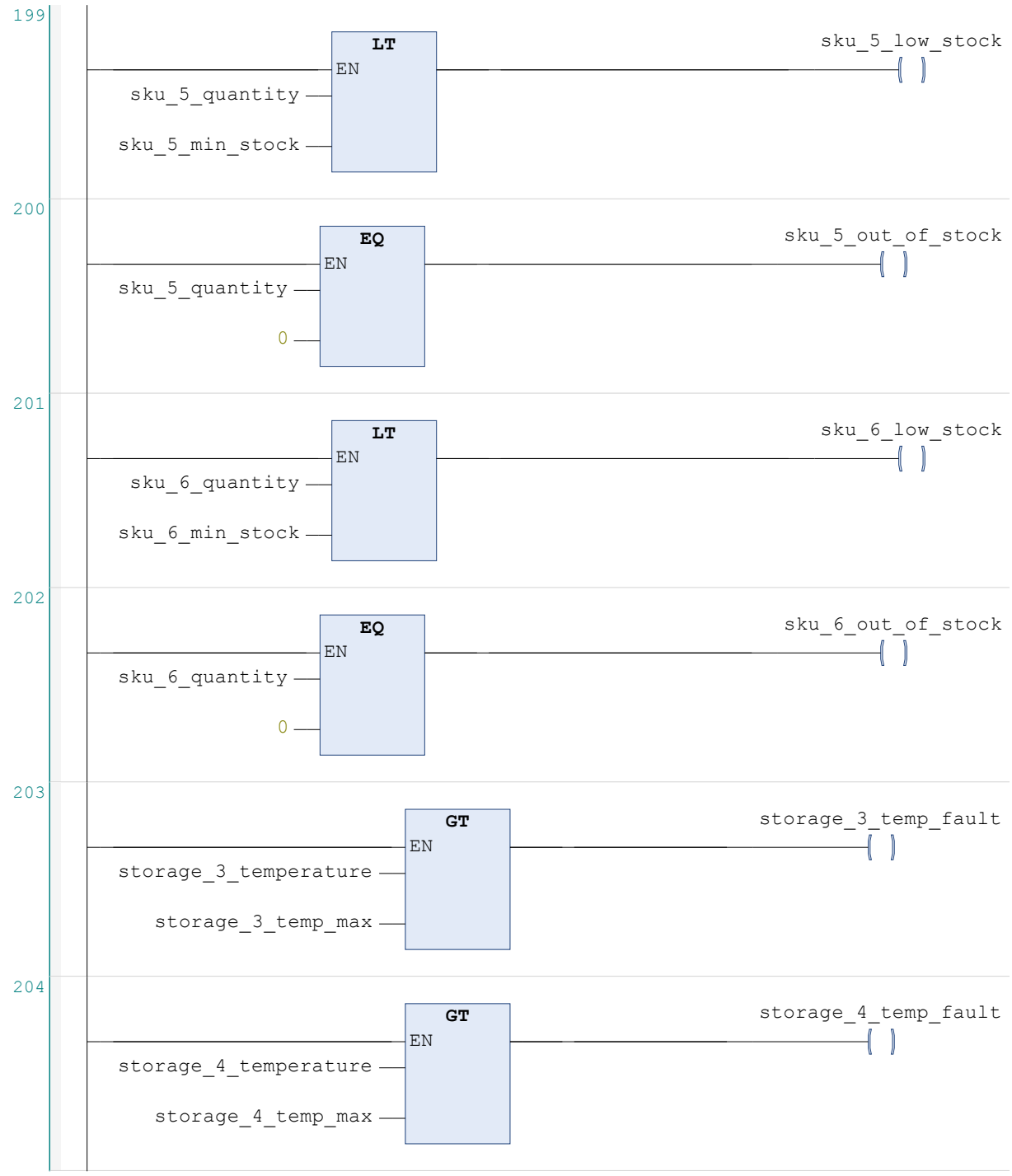

188
inventory_record_added
EQ
EN
=
sorting_sku
4
ADD
EN
ENO
sku_4_quantity
sorting_item_quantity
sku_
189
inventory_record_added
EQ
EN
=
sorting_sku
5
ADD
EN
ENO
sku_5_quantity
sorting_item_quantity
sku_
190
inventory_record_added
EQ
EN
=
sorting_sku
6
ADD
EN
ENO
sku_6_quantity
sorting_item_quantity
sku_
191
LT
EN
sku_1_quantity
sku_1_min_stock
sku_1_low_stock
192
EQ
EN
sku_1_quantity
0
sku_1_out_of_stock

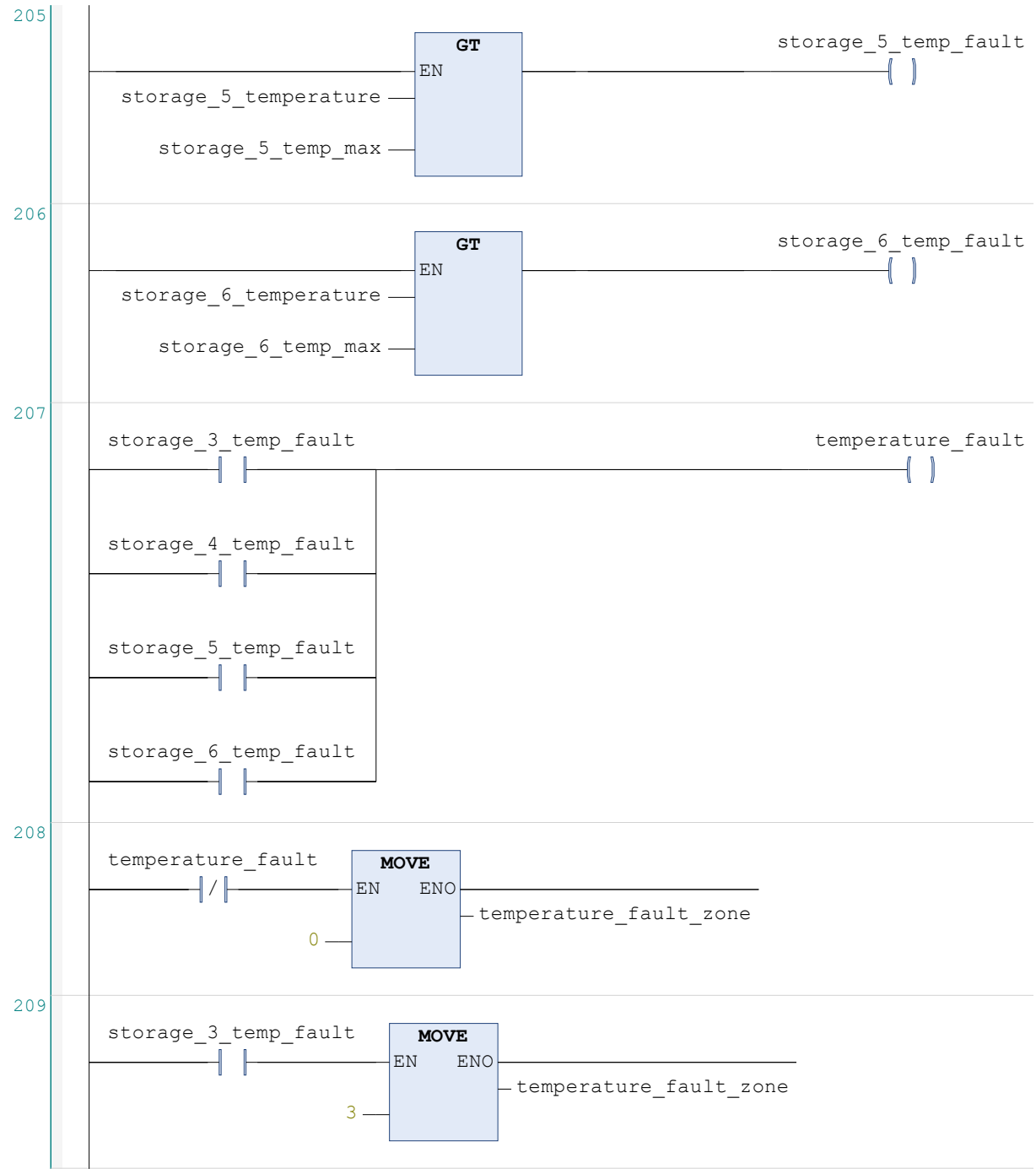
4_quantity

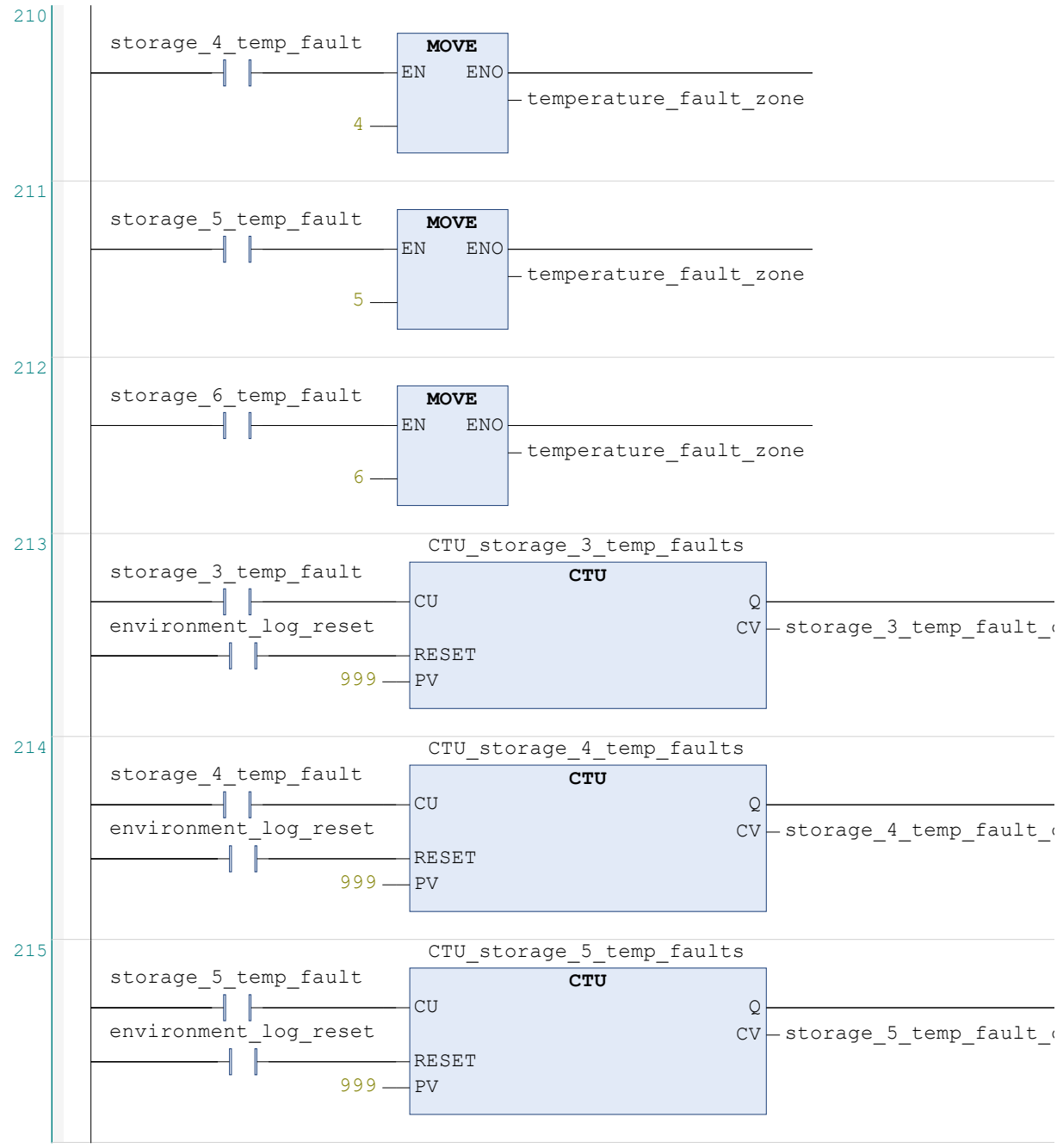
5_quantity

6_quantity


193
LT
EN
sku_2_quantity
sku_2_min_stock
sku_2_low_stock
194
EQ
EN
sku_2_quantity
0
sku_2_out_of_stock
195
LT
EN
sku_3_quantity
sku_3_min_stock
sku_3_low_stock
196
EQ
EN
sku_3_quantity
0
sku_3_out_of_stock
197
LT
EN
sku_4_quantity
sku_4_min_stock
sku_4_low_stock
198
EQ
EN
sku_4_quantity
0
sku_4_out_of_stock


199
LT
EN
sku_5_quantity
sku_5_min_stock
sku_5_low_stock
200
EQ
EN
sku_5_quantity
0
sku_5_out_of_stock
201
LT
EN
sku_6_quantity
sku_6_min_stock
sku_6_low_stock
202
EQ
EN
sku_6_quantity
0
sku_6_out_of_stock
203
GT
EN
storage_3_temperature
storage_3_temp_max
storage_3_temp_fault
204
GT
EN
storage_4_temperature
storage_4_temp_max
storage_4_temp_fault


205
GT
EN
storage_5_temperature
storage_5_temp_max
storage_5_temp_fault
206
GT
EN
storage_6_temperature
storage_6_temp_max
storage_6_temp_fault
207
storage_3_temp_fault
storage_4_temp_fault
storage_5_temp_fault
storage_6_temp_fault
temperature_fault
208
temperature_fault
/
MOVE
EN
ENO
0
temperature_fault_zone
209
storage_3_temp_fault
MOVE
EN
ENO
3
temperature_fault_zone

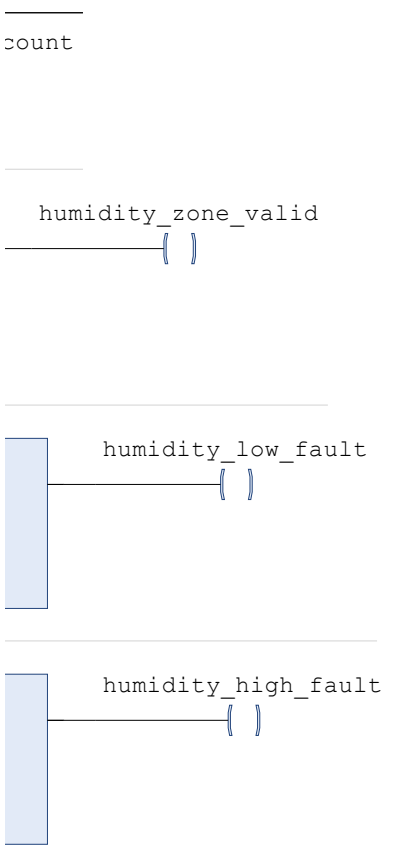

210
storage_4_temp_fault
MOVE
EN
ENO
temperature_fault_zone
4
211
storage_5_temp_fault
MOVE
EN
ENO
temperature_fault_zone
5
212
storage_6_temp_fault
MOVE
EN
ENO
temperature_fault_zone
6
213
CTU_storage_3_temp_faults
storage_3_temp_fault
CTU
CU
Q
CV
storage_3_temp_fault_
environment_log_reset
RESET
999
PV
214
CTU_storage_4_temp_faults
storage_4_temp_fault
CTU
CU
Q
CV
storage_4_temp_fault_
environment_log_reset
RESET
999
PV
215
CTU_storage_5_temp_faults
storage_5_temp_fault
CTU
CU
Q
CV
storage_5_temp_fault_
environment_log_reset
RESET
999
PV

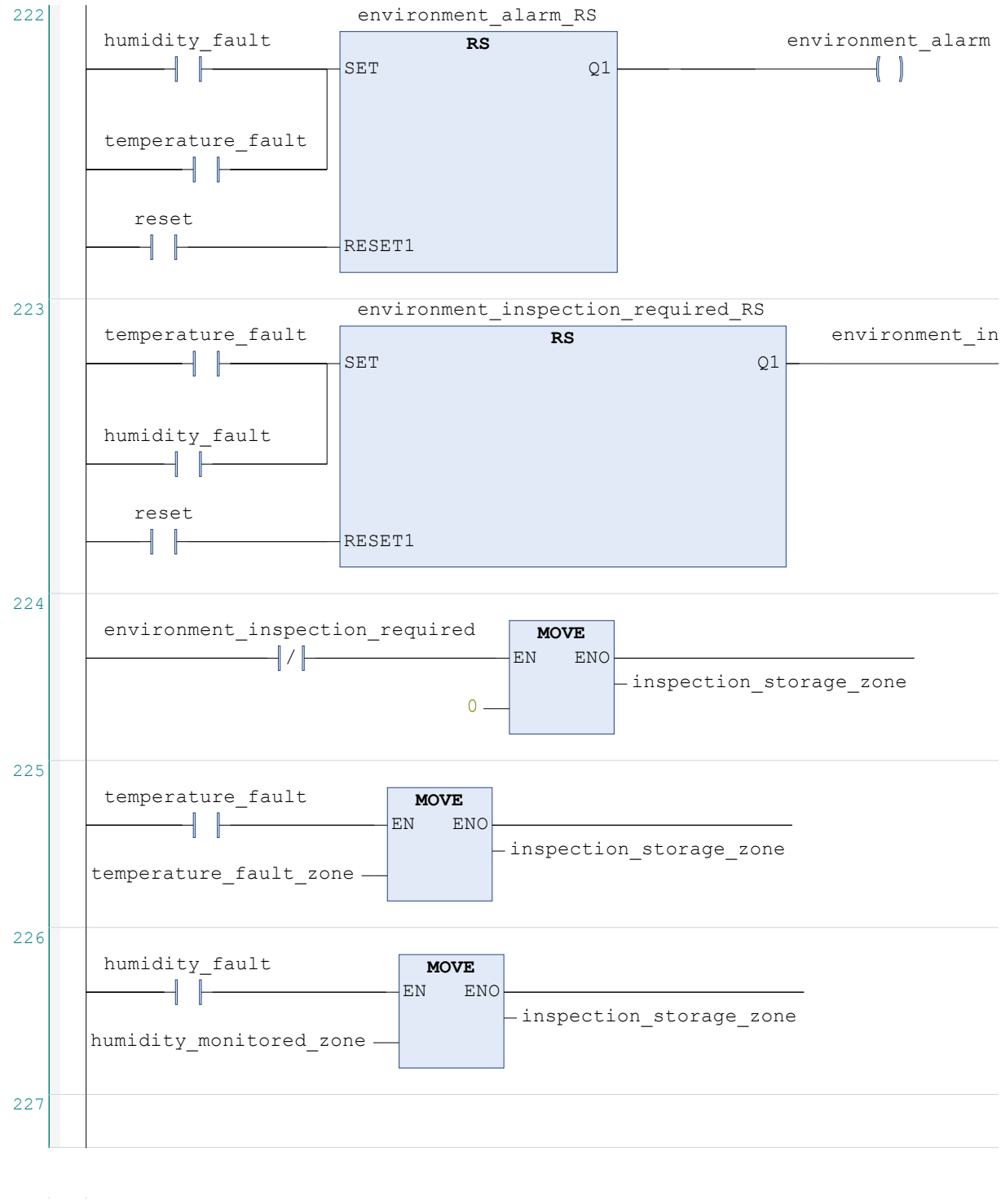
count

count

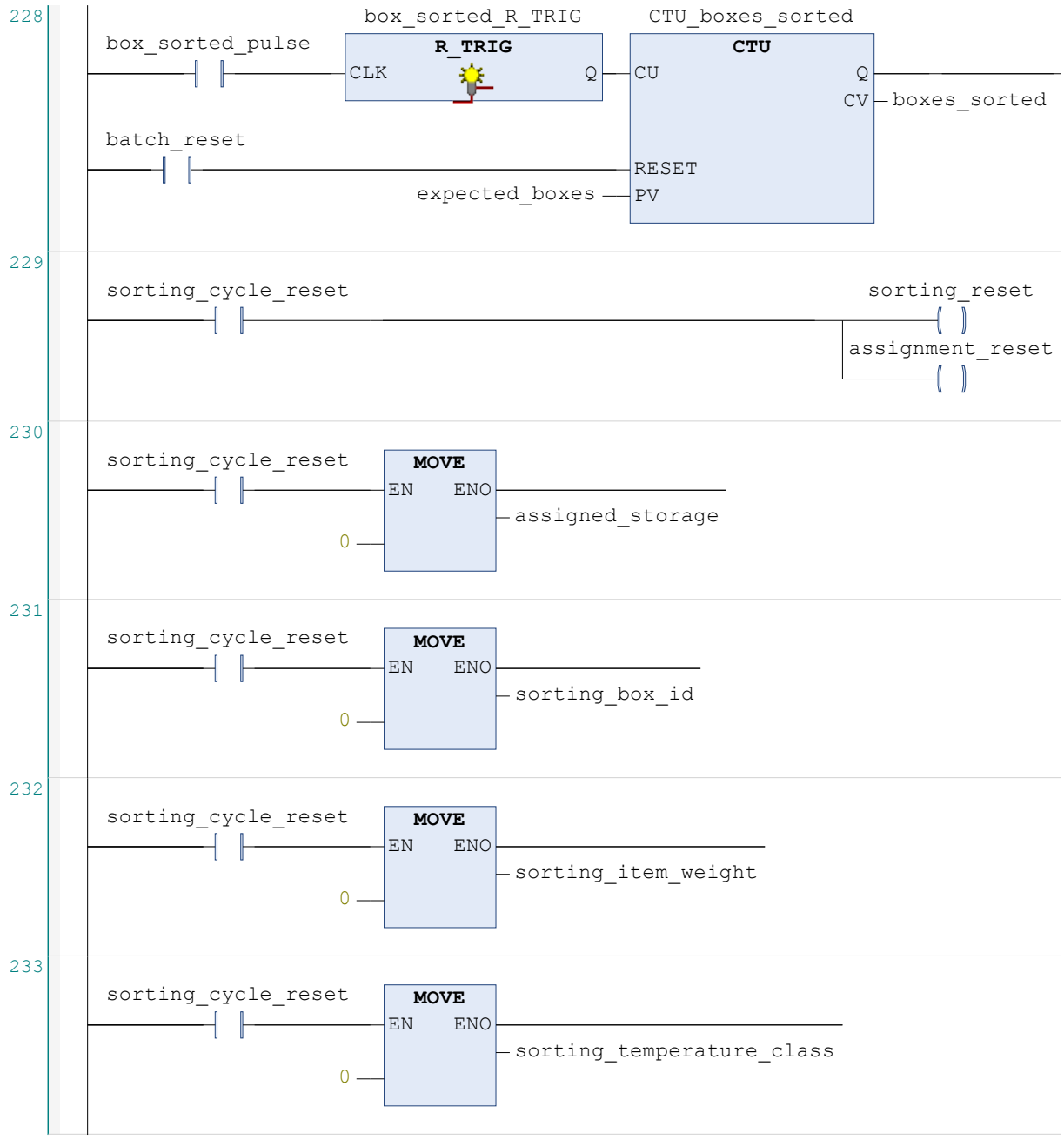
count

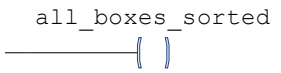

216
CTU_storage_6_temp_faults
storage_6_temp_fault
environment_log_reset
CTU
CU
RESET
PV
999
Q
CV
storage_6_temp_fault_
217
GE
EN
≥
humidity_monitored_zone
1
LE
EN
humidity_monitored_zone
6
218
humidity_monitor_enable
humidity_zone_valid
LT
EN
measured_humidity
humidity_min_limit
219
humidity_monitor_enable
humidity_zone_valid
GT
EN
measured_humidity
humidity_max_limit
220
humidity_low_fault
humidity_high_fault
humidity_fault
221
CTU_humidity_faults
humidity_fault
environment_log_reset
CTU
CU
RESET
PV
999
Q
CV
humidity_fault_count

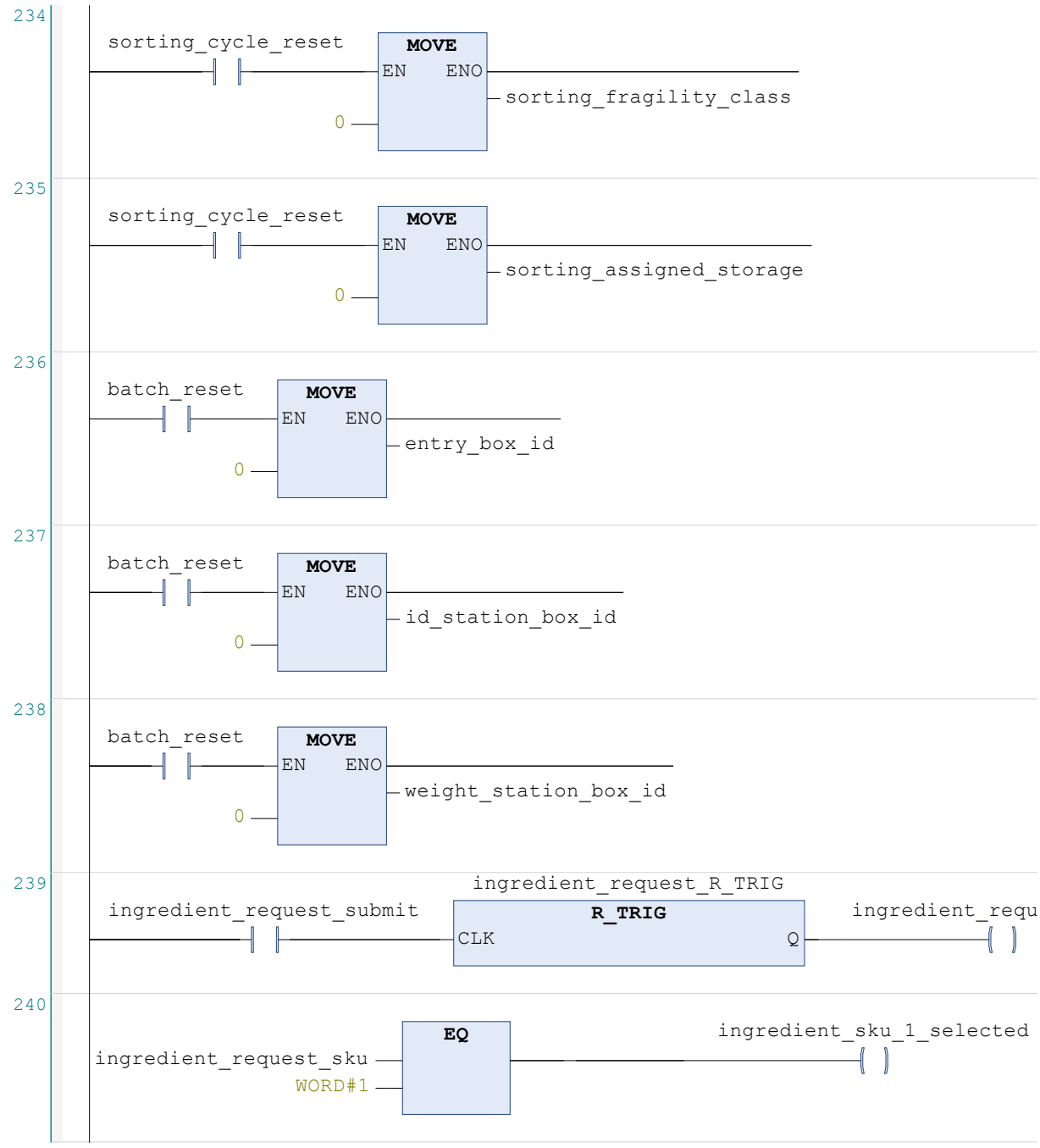

count
humidity_zone_valid
humidity_low_fault
humidity_high_fault


222
environment_alarm_RS
humidity_fault
RS
SET
Q1
environment_alarm
temperature_fault
reset
RESET1
223
environment_inspection_required_RS
temperature_fault
RS
SET
Q1
environment_in
humidity_fault
reset
RESET1
224
environment_inspection_required
/
MOVE
EN
ENO
inspection_storage_zone
0
225
temperature_fault
MOVE
EN
ENO
inspection_storage_zone
temperature_fault_zone
226
humidity_fault
MOVE
EN
ENO
inspection_storage_zone
humidity_monitored_zone
227

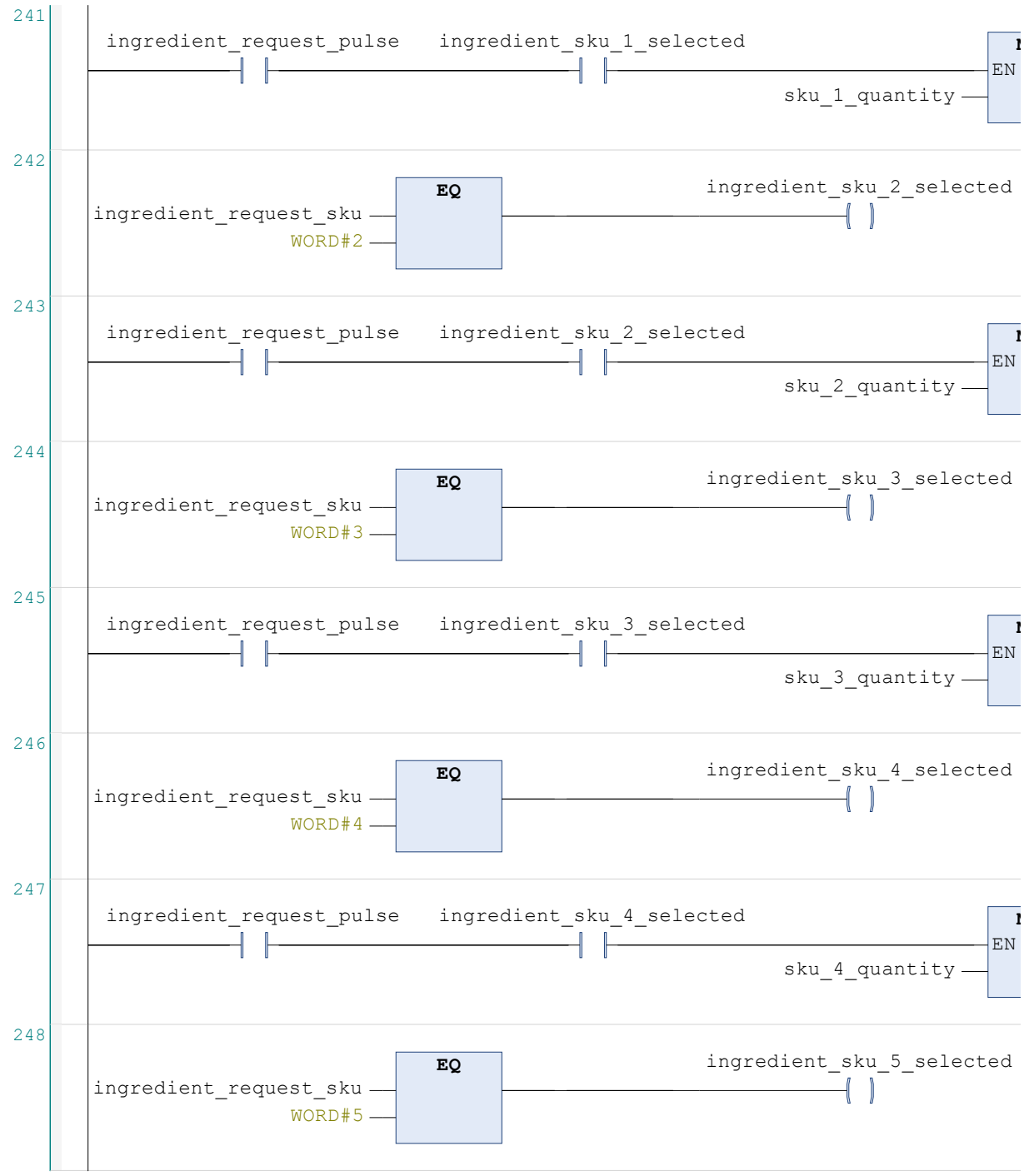
spection_required

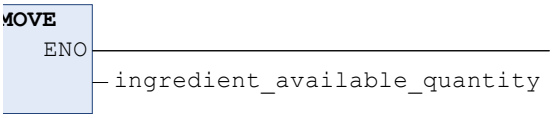

228
box_sorted_R_TRIG
CTU_boxes_sorted
box_sorted_pulse
R_TRIG
CLK
Q
CTU
CU
Q
CV
boxes_sorted
batch_reset
RESET
expected_boxes
PV
229
sorting_cycle_reset
sorting_reset
assignment_reset
230
sorting_cycle_reset
MOVE
EN
ENO
assigned_storage
0
231
sorting_cycle_reset
MOVE
EN
ENO
sorting_box_id
0
232
sorting_cycle_reset
MOVE
EN
ENO
sorting_item_weight
0
233
sorting_cycle_reset
MOVE
EN
ENO
sorting_temperature_class
0

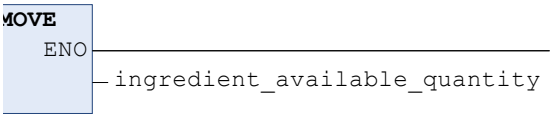

all_boxes_sorted

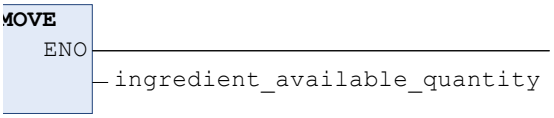

234
sorting_cycle_reset
MOVE
EN
ENO
0
sorting_fragility_class
235
sorting_cycle_reset
MOVE
EN
ENO
0
sorting_assigned_storage
236
batch_reset
MOVE
EN
ENO
0
entry_box_id
237
batch_reset
MOVE
EN
ENO
0
id_station_box_id
238
batch_reset
MOVE
EN
ENO
0
weight_station_box_id
239
ingredient_request_R_TRIG
ingredient_request_submit
R_TRIG
CLK
Q
ingredient_requ
240
EQ
ingredient_request_sku
WORD#1
ingredient_sku_1_selected

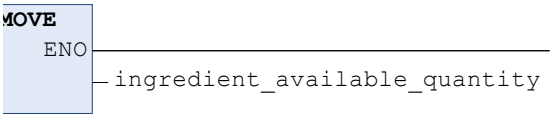
```
est_pulse
```

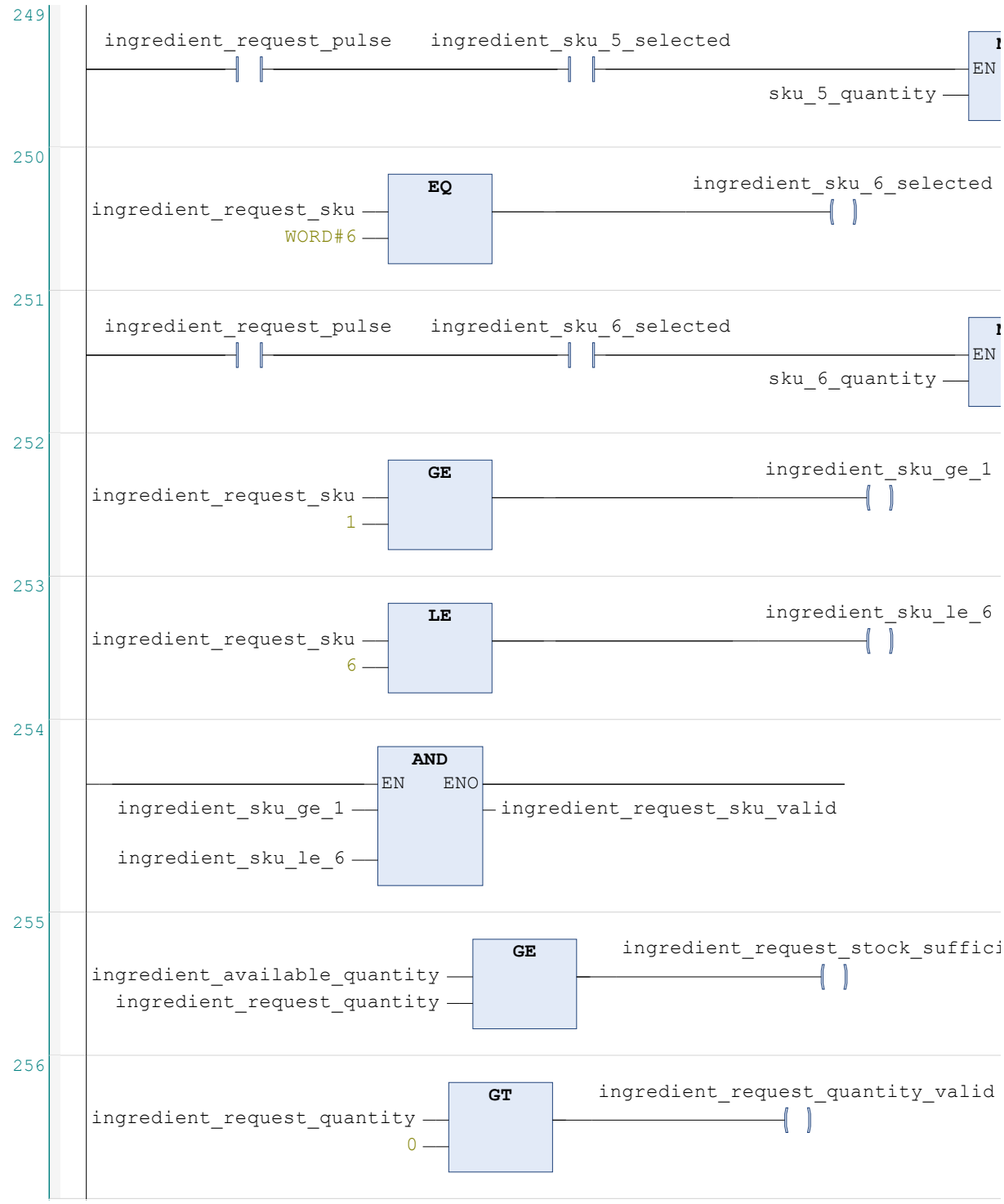

241
ingredient_request_pulse
ingredient_sku_1_selected
EN
sku_1_quantity
242
ingredient_request_sku
WORD#2
EQ
ingredient_sku_2_selected
243
ingredient_request_pulse
ingredient_sku_2_selected
EN
sku_2_quantity
244
ingredient_request_sku
WORD#3
EQ
ingredient_sku_3_selected
245
ingredient_request_pulse
ingredient_sku_3_selected
EN
sku_3_quantity
246
ingredient_request_sku
WORD#4
EQ
ingredient_sku_4_selected
247
ingredient_request_pulse
ingredient_sku_4_selected
EN
sku_4_quantity
248
ingredient_request_sku
WORD#5
EQ
ingredient_sku_5_selected

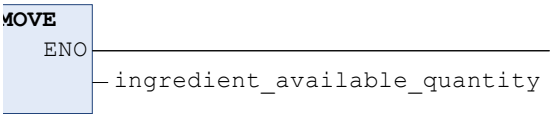

MOVE
ENO
ingredient_available_quantity

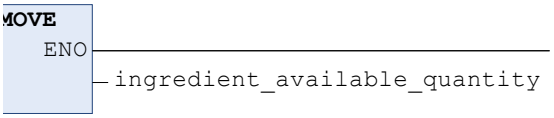

MOVE
ENO
ingredient_available_quantity


MOVE
ENO
ingredient_available_quantity

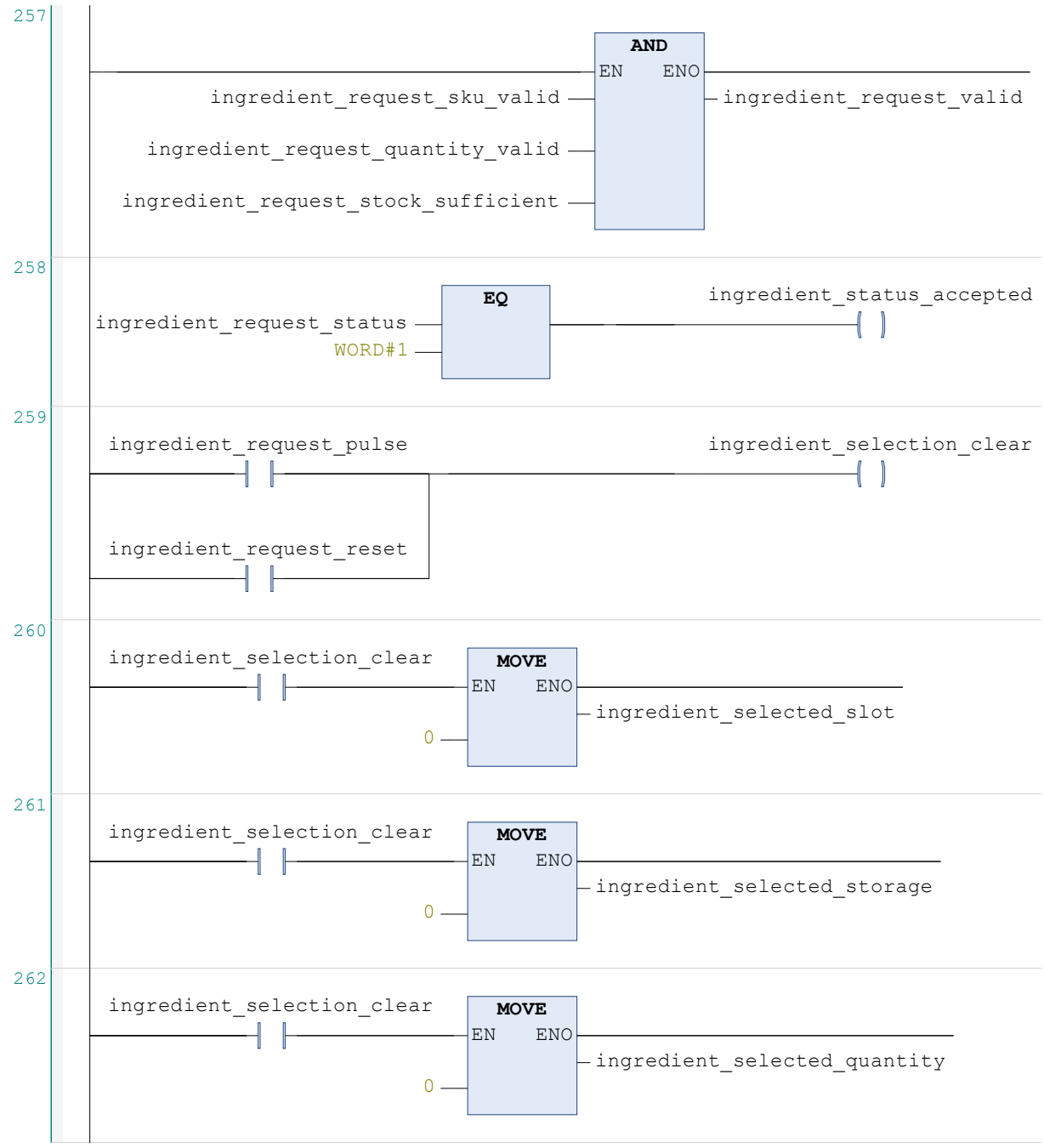

MOVE
ENO
ingredient_available_quantity

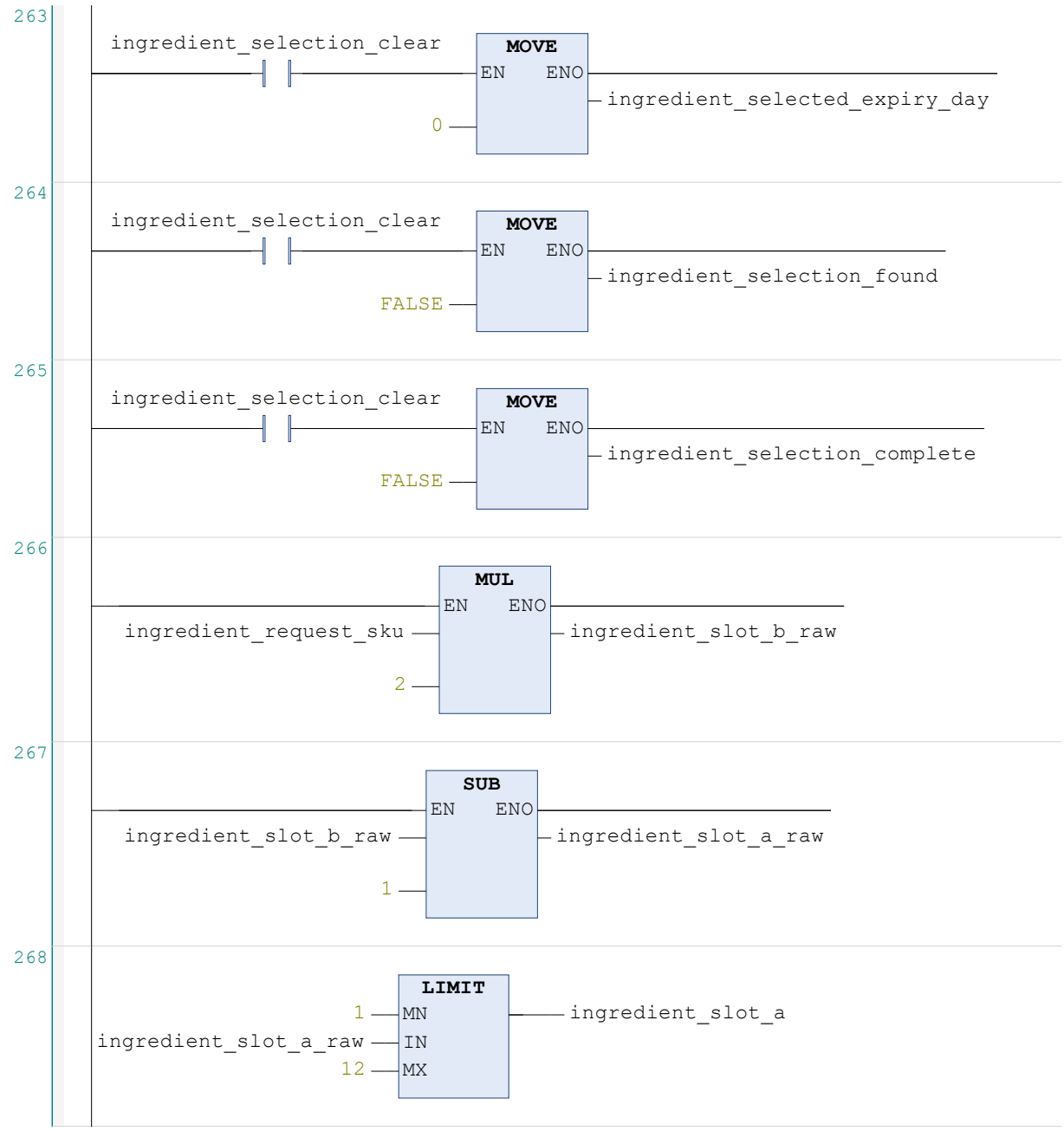

249
ingredient_request_pulse
ingredient_sku_5_selected
EN
sku_5_quantity
250
EQ
ingredient_request_sku
WORD#6
ingredient_sku_6_selected
251
ingredient_request_pulse
ingredient_sku_6_selected
EN
sku_6_quantity
252
GE
ingredient_request_sku
1
ingredient_sku_ge_1
253
LE
ingredient_request_sku
6
ingredient_sku_le_6
254
AND
EN
ENO
ingredient_sku_ge_1
ingredient_sku_le_6
ingredient_request_sku_valid
255
GE
ingredient_available_quantity
ingredient_request_quantity
ingredient_request_stock_suffici
256
GT
ingredient_request_quantity
0
ingredient_request_quantity_valid


MOVE
ENO
ingredient_available_quantity


MOVE
ENO
ingredient_available_quantity

ient


257
AND
EN
ENO
ingredient_request_sku_valid
ingredient_request_quantity_valid
ingredient_request_stock_sufficient
ingredient_request_valid
258
EQ
ingredient_request_status
WORD#1
ingredient_status_accepted
259
ingredient_request_pulse
ingredient_selection_clear
ingredient_request_reset
260
ingredient_selection_clear
MOVE
EN
ENO
0
ingredient_selected_slot
261
ingredient_selection_clear
MOVE
EN
ENO
0
ingredient_selected_storage
262
ingredient_selection_clear
MOVE
EN
ENO
0
ingredient_selected_quantity


263
ingredient_selection_clear
MOVE
EN
ENO
0
ingredient_selected_expiry_day
264
ingredient_selection_clear
MOVE
EN
ENO
FALSE
ingredient_selection_found
265
ingredient_selection_clear
MOVE
EN
ENO
FALSE
ingredient_selection_complete
266
MUL
EN
ENO
ingredient_request_sku
2
ingredient_slot_b_raw
267
SUB
EN
ENO
ingredient_slot_b_raw
1
ingredient_slot_a_raw
268
LIMIT
1
MN
ingredient_slot_a_raw
IN
12
MX
ingredient_slot_a


269
LIMIT
1
MN
ingredient_slot_b_raw
IN
12
MX
ingredient_slot_b
270
EQ
EN
inventory_sku[ingredient_slot_a]
ingredient_request_sku
ingredient_slot_a_sku_matcl
271
EQ
EN
inventory_status[ingredient_slot_a]
1
ingredient_slot_a_statu
272
GE
EN
inventory_quantity[ingredient_slot_a]
ingredient_request_quantity
ingredient_slot_a_quar
273
GT
EN
inventory_expiry_day[ingredient_slot_a]
restaurant_current_day
ingredient_slot_a_e
274
ingredient_request_valid
inventory_occupied[ingredient_slot_a]
ingredi

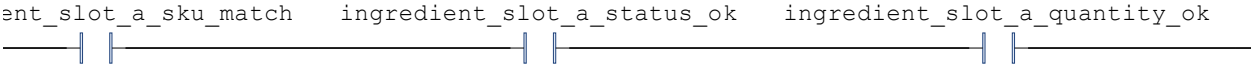
n

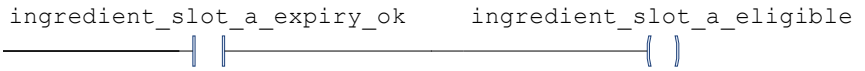
s_ok

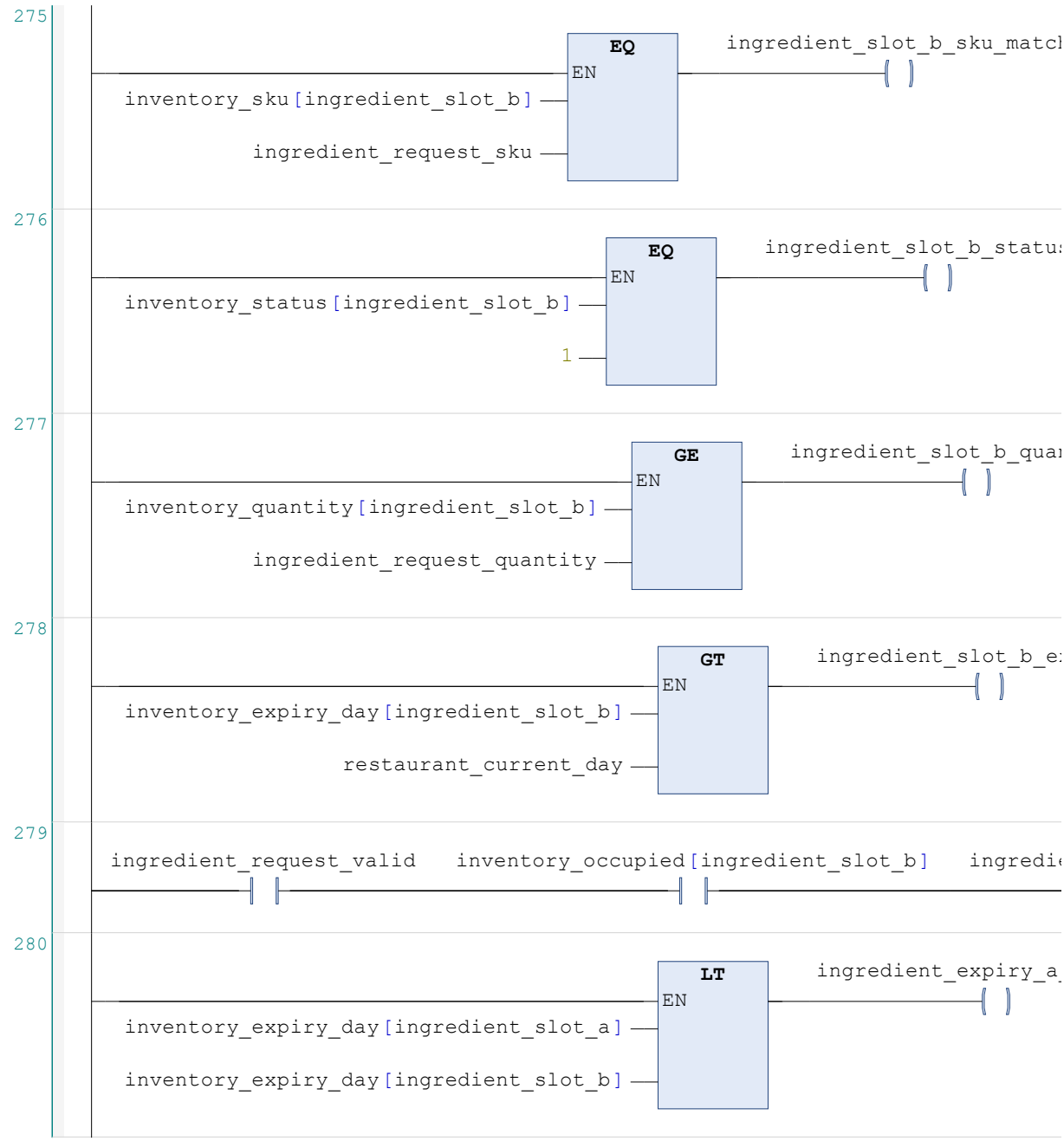
ntity_ok

xpiry_ok

ent_slot_a_sku_match ingredient_slot_a_status_ok ingredient_slot_a_quantity_ok




ingredient_slot_a_expiry_ok
ingredient_slot_a_eligible


275
EQ
EN
inventory_sku[ingredient_slot_b]
ingredient_request_sku
ingredient_slot_b_sku_matcl
276
EQ
EN
inventory_status[ingredient_slot_b]
1
ingredient_slot_b_statu
277
GE
EN
inventory_quantity[ingredient_slot_b]
ingredient_request_quantity
ingredient_slot_b_quar
278
GT
EN
inventory_expiry_day[ingredient_slot_b]
restaurant_current_day
ingredient_slot_b_e
279
ingredient_request_valid
inventory_occupied[ingredient_slot_b]
ingredi
280
LT
EN
inventory_expiry_day[ingredient_slot_a]
inventory_expiry_day[ingredient_slot_b]
ingredient_expiry_a

n

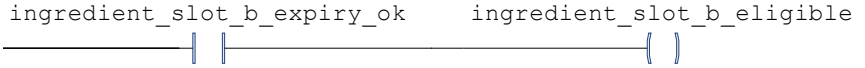
s_ok

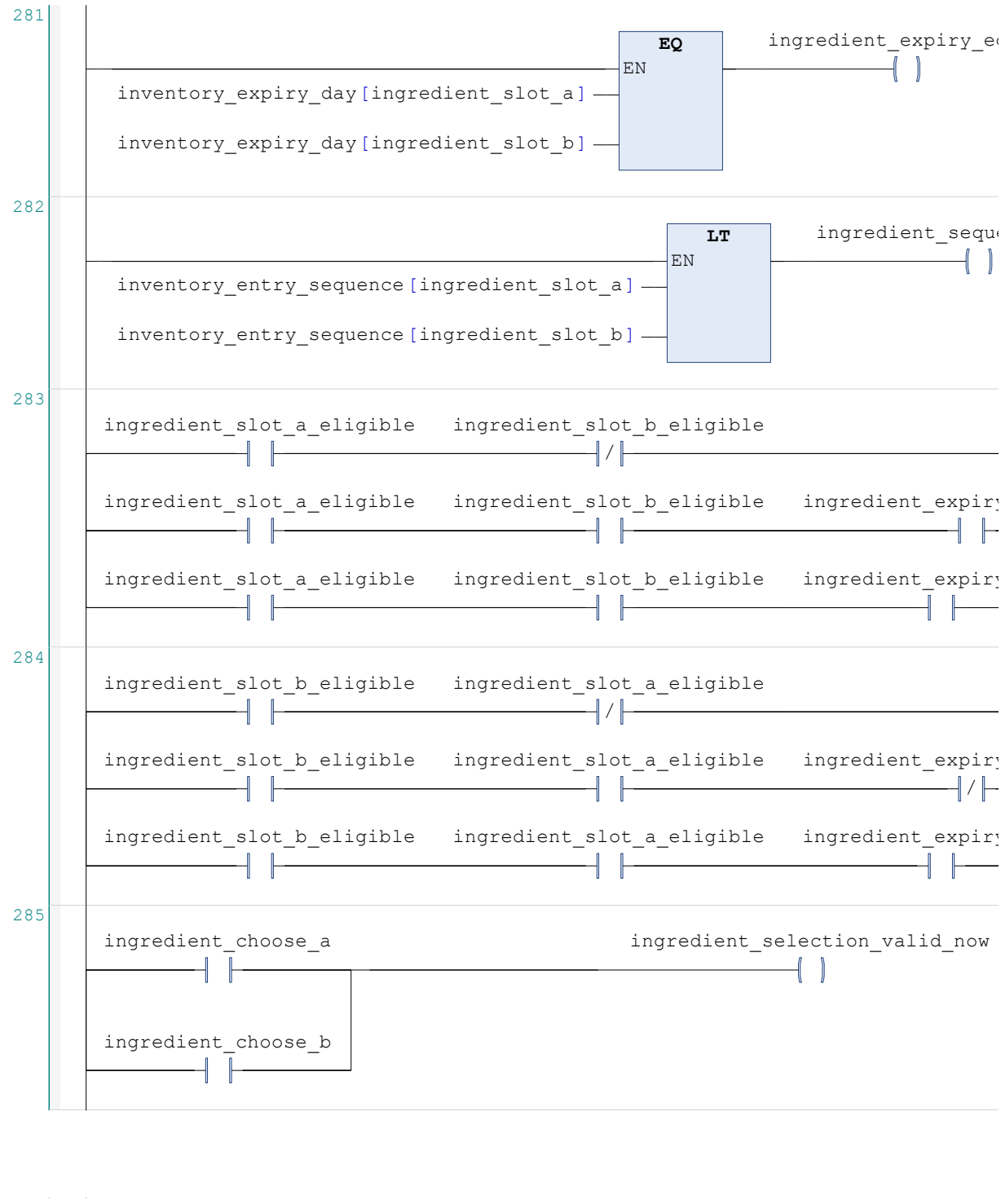
ntity_ok

xpiry_ok

ent_slot_b_sku_match ingredient_slot_b_status_ok ingredient_slot_b_quantity_ok

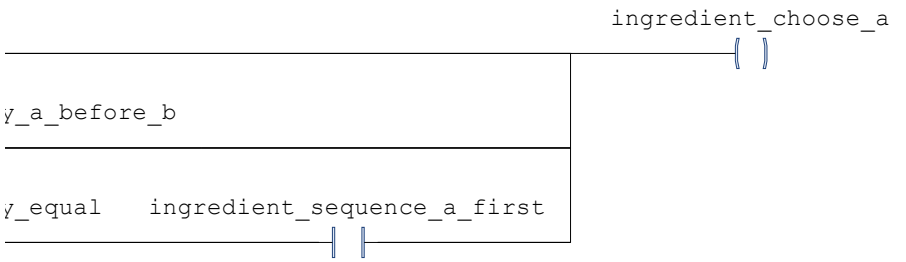
_before_b

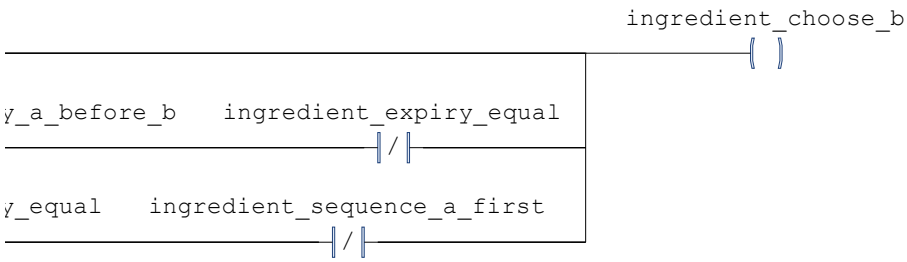

ingredient_slot_b_expiry_ok
ingredient_slot_b_eligible

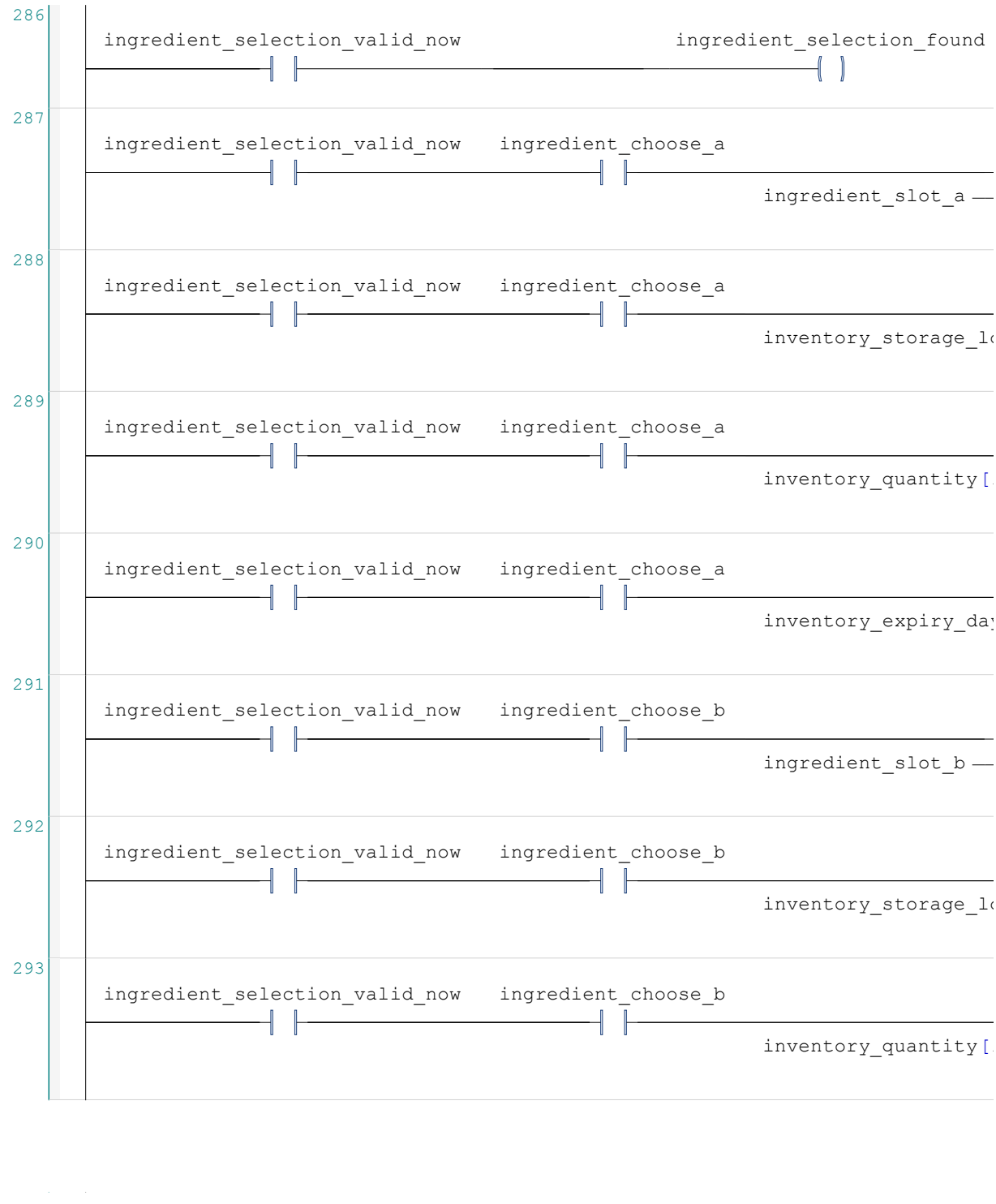

281
EQ
EN
ingredient_expiry_e
inventory_expiry_day[ingredient_slot_a]
inventory_expiry_day[ingredient_slot_b]
282
LT
EN
ingredient_sequ
inventory_entry_sequence[ingredient_slot_a]
inventory_entry_sequence[ingredient_slot_b]
283
ingredient_slot_a_eligible
ingredient_slot_b_eligible
ingredient_slot_a_eligible
ingredient_slot_b_eligible
ingredient_expiry
ingredient_slot_a_eligible
ingredient_slot_b_eligible
ingredient_expiry
284
ingredient_slot_b_eligible
ingredient_slot_a_eligible
ingredient_slot_b_eligible
ingredient_slot_a_eligible
ingredient_expiry
ingredient_slot_b_eligible
ingredient_slot_a_eligible
ingredient_expiry
285
ingredient_choose_a
ingredient_selection_valid_now
ingredient_choose_b

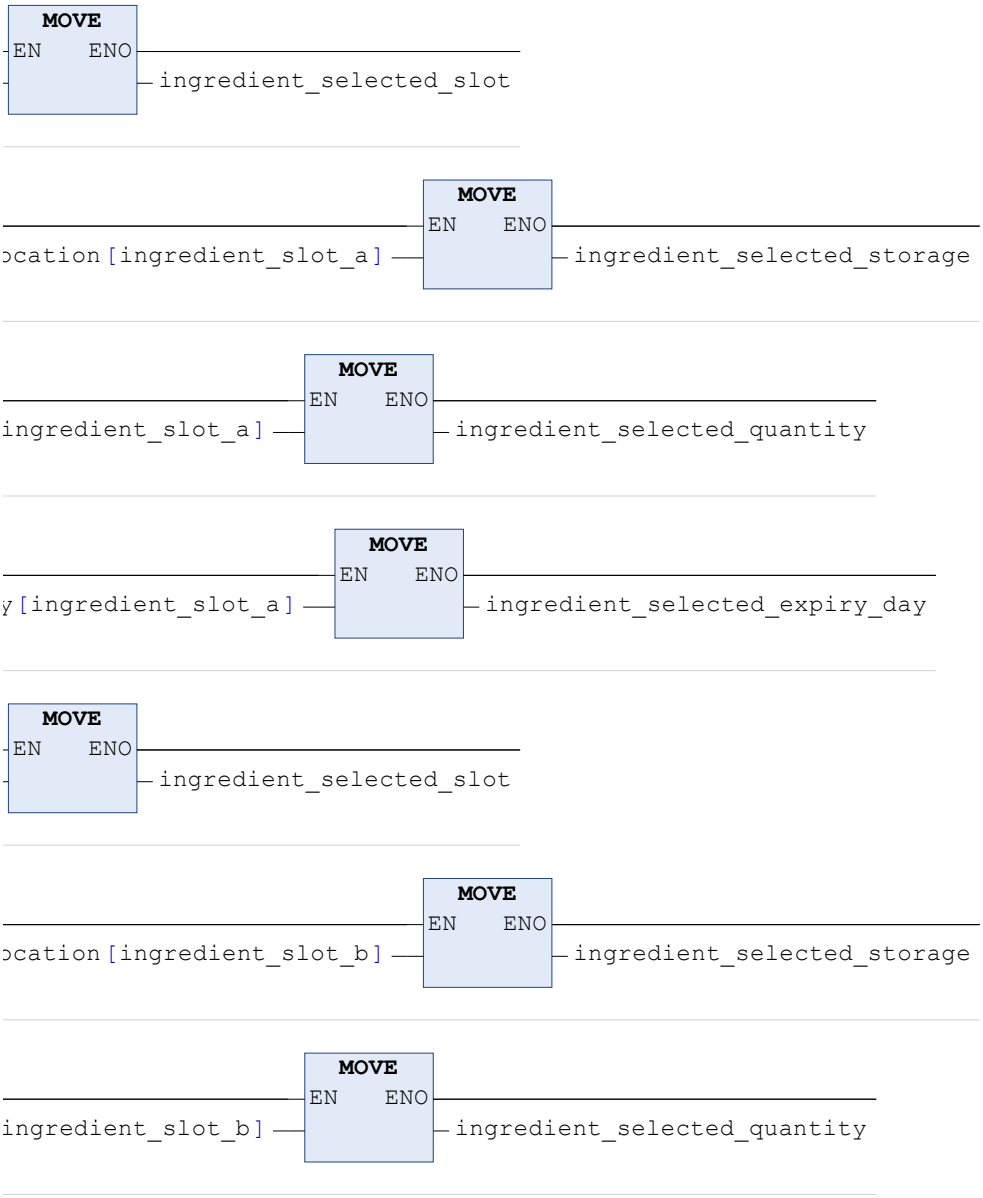
qual

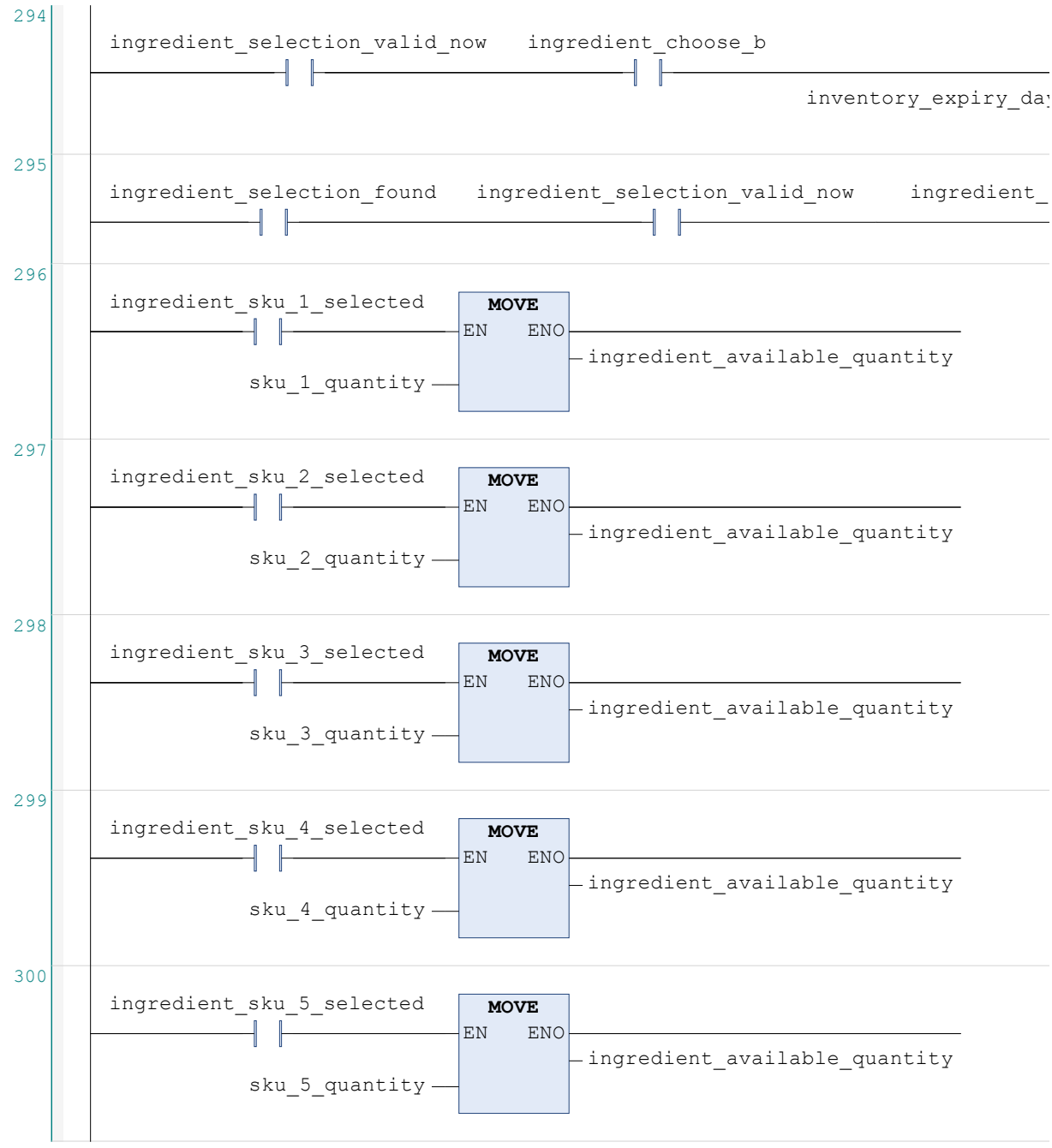
ence_a_first

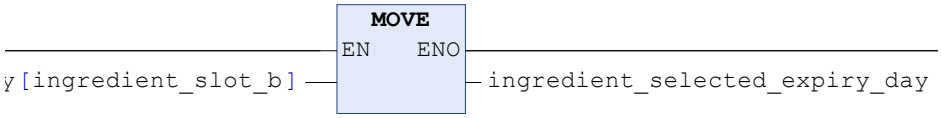


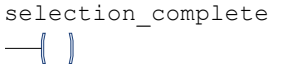


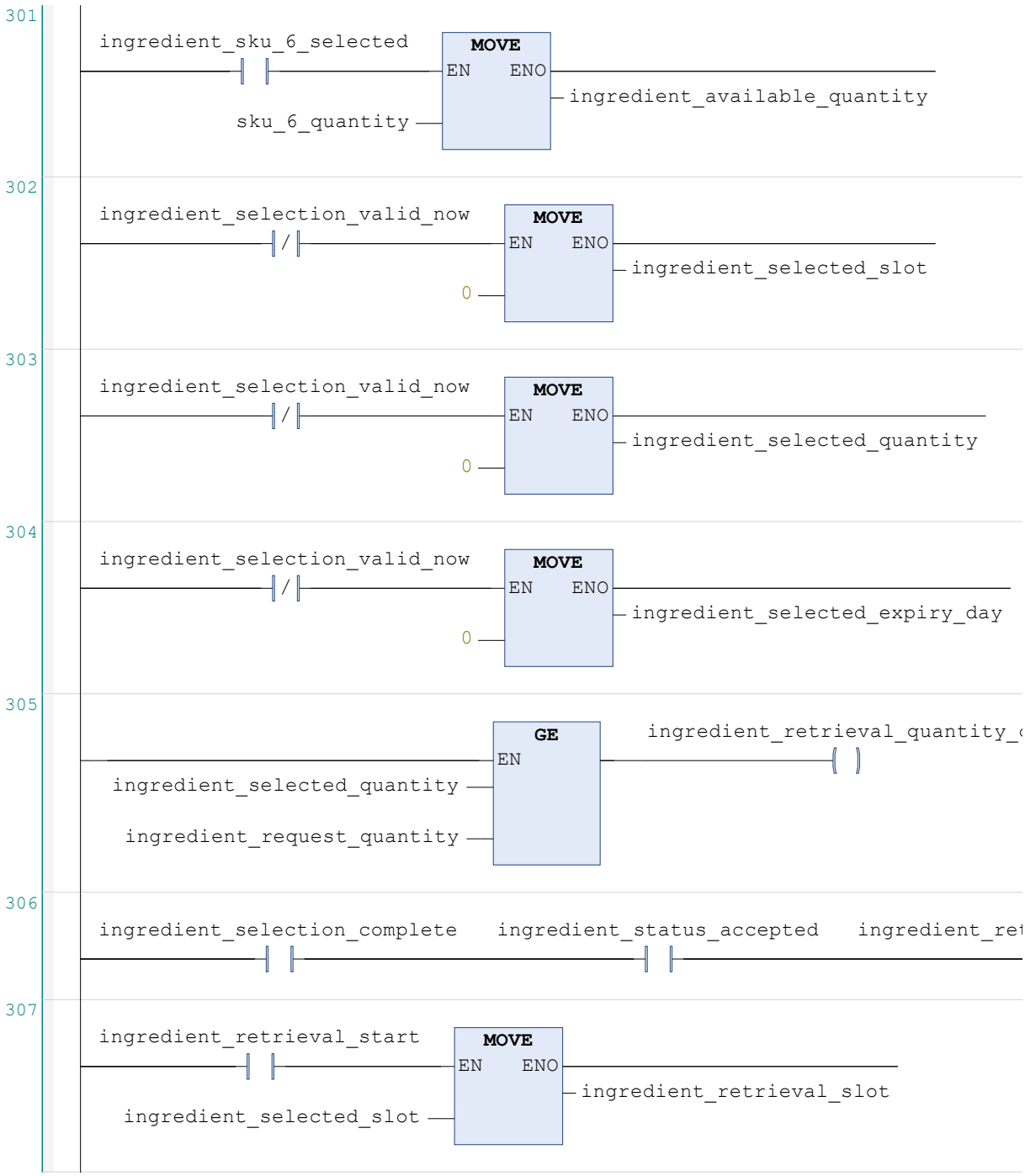

286
ingredient_selection_valid_now
ingredient_selection_found
287
ingredient_selection_valid_now
ingredient_choose_a
ingredient_slot_a
288
ingredient_selection_valid_now
ingredient_choose_a
inventory_storage_l
289
ingredient_selection_valid_now
ingredient_choose_a
inventory_quantity[
290
ingredient_selection_valid_now
ingredient_choose_a
inventory_expiry_da
291
ingredient_selection_valid_now
ingredient_choose_b
ingredient_slot_b
292
ingredient_selection_valid_now
ingredient_choose_b
inventory_storage_l
293
ingredient_selection_valid_now
ingredient_choose_b
inventory_quantity[


MOVE
EN
ENO
ingredient_selected_slot
ocation[ingredient_slot_a]
ingredient_selected_storage
ingredient_slot_a]
ingredient_selected_quantity
y[ingredient_slot_a]
ingredient_selected_expiry_day
MOVE
EN
ENO
ingredient_selected_slot
ocation[ingredient_slot_b]
ingredient_selected_storage
MOVE
EN
ENO
ingredient_slot_b]
ingredient_selected_quantity

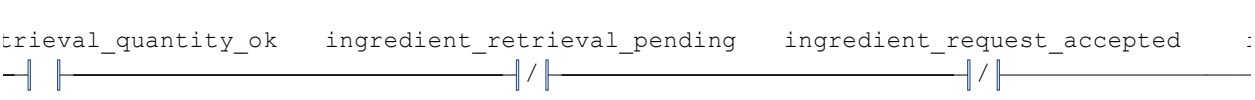

294
ingredient_selection_valid_now
ingredient_choose_b
inventory_expiry_da
295
ingredient_selection_found
ingredient_selection_valid_now
ingredient_
296
ingredient_sku_1_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_1_quantity
297
ingredient_sku_2_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_2_quantity
298
ingredient_sku_3_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_3_quantity
299
ingredient_sku_4_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_4_quantity
300
ingredient_sku_5_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_5_quantity

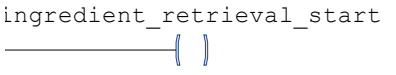

MOVE
EN
ENO
y[ingredient_slot_b]
ingredient_selected_expiry_day

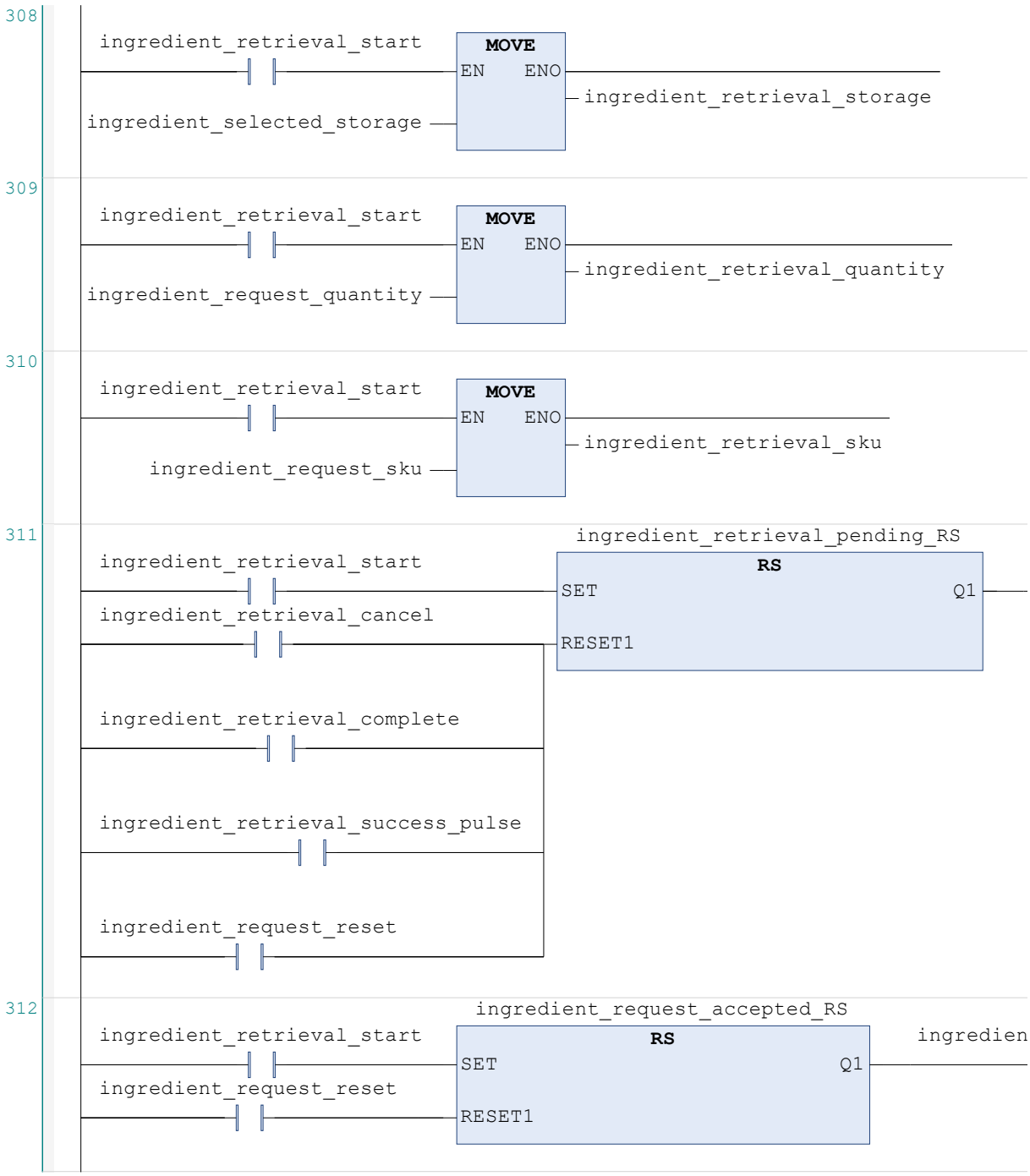

selection_complete

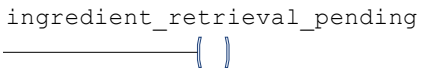

301
ingredient_sku_6_selected
MOVE
EN
ENO
ingredient_available_quantity
sku_6_quantity
302
ingredient_selection_valid_now
MOVE
EN
ENO
ingredient_selected_slot
0
303
ingredient_selection_valid_now
MOVE
EN
ENO
ingredient_selected_quantity
0
304
ingredient_selection_valid_now
MOVE
EN
ENO
ingredient_selected_expiry_day
0
305
GE
EN
ingredient_retrieval_quantity_
ingredient_selected_quantity
ingredient_request_quantity
306
ingredient_selection_complete
ingredient_status_accepted
ingredient_re
307
ingredient_retrieval_start
MOVE
EN
ENO
ingredient_retrieval_slot
ingredient_selected_slot

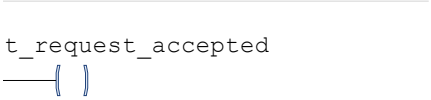
ok

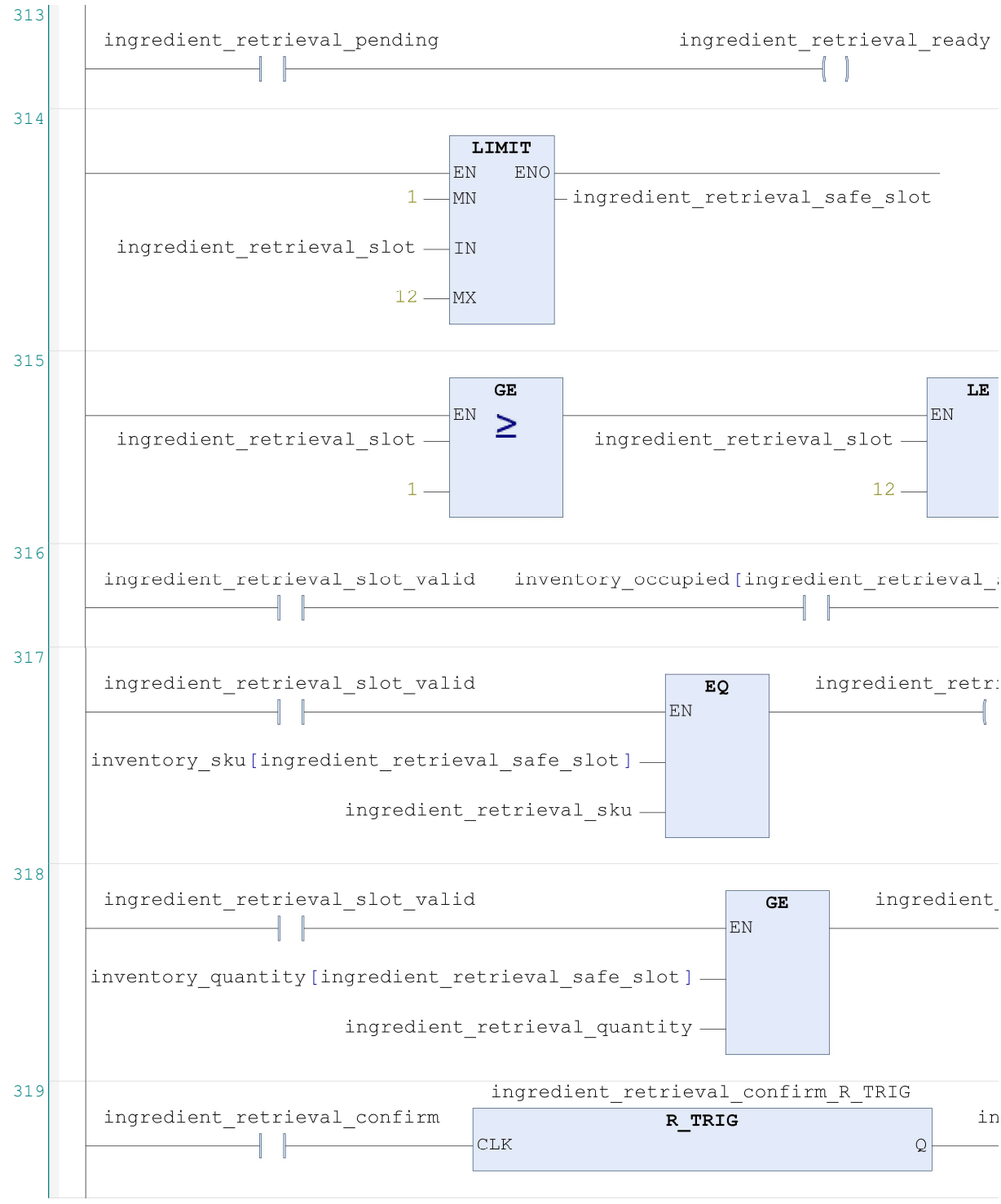

trieval_quantity_ok
ingredient_retrieval_pending
ingredient_request_accepted

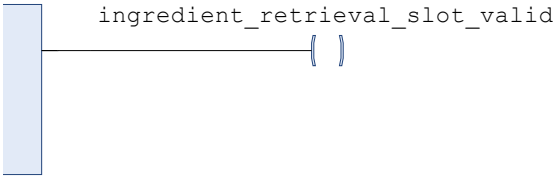

ingredient_retrieval_start

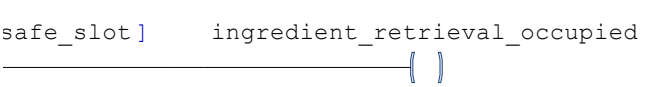

308
ingredient_retrieval_start
MOVE
EN
ENO
ingredient_retrieval_storage
ingredient_selected_storage
309
ingredient_retrieval_start
MOVE
EN
ENO
ingredient_retrieval_quantity
ingredient_request_quantity
310
ingredient_retrieval_start
MOVE
EN
ENO
ingredient_retrieval_sku
ingredient_request_sku
311
ingredient_retrieval_pending_RS
ingredient_retrieval_start
RS
SET
Q1
ingredient_retrieval_cancel
RESET1
ingredient_retrieval_complete
ingredient_retrieval_success_pulse
ingredient_request_reset
312
ingredient_request_accepted_RS
ingredient_retrieval_start
RS
ingredien
SET
Q1
ingredient_request_reset
RESET1

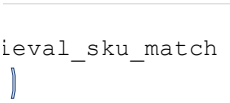

ingredient_retrieval_pending

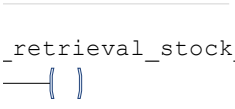

t_request_accepted

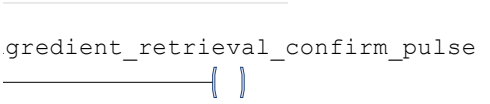

313
ingredient_retrieval_pending
ingredient_retrieval_ready
314
LIMIT
EN
ENO
1
MN
ingredient_retrieval_safe_slot
ingredient_retrieval_slot
IN
12
MX
315
GE
EN
≥
ingredient_retrieval_slot
1
LE
EN
ingredient_retrieval_slot
12
316
ingredient_retrieval_slot_valid
inventory_occupied[ingredient_retrieval_
317
ingredient_retrieval_slot_valid
EQ
EN
ingredient_retr
inventory_sku[ingredient_retrieval_safe_slot]
ingredient_retrieval_sku
318
ingredient_retrieval_slot_valid
GE
EN
ingredient_
inventory_quantity[ingredient_retrieval_safe_slot]
ingredient_retrieval_quantity
319
ingredient_retrieval_confirm_R_TRIG
ingredient_retrieval_confirm
R_TRIG
CLK
Q
in


ingredient_retrieval_slot_valid


safe_slot]
ingredient_retrieval_occupied


ieval_sku_match


_retrieval_stock_ok


gredient_retrieval_confirm_pulse


320
GT
EN
ingredient_retrieval_quantity
0
ingredient_retrieval_locked_qu
321
EQ
EN
=
ingredient_retrieval_sku
1
MOVE
EN
ENO
sku_1_quantity
ingredie
322
EQ
EN
=
ingredient_retrieval_sku
2
MOVE
EN
ENO
sku_2_quantity
ingredie
323
EQ
EN
=
ingredient_retrieval_sku
3
MOVE
EN
ENO
sku_3_quantity
ingredie
324
EQ
EN
=
ingredient_retrieval_sku
4
MOVE
EN
ENO
sku_4_quantity
ingredie
325
EQ
EN
=
ingredient_retrieval_sku
5
MOVE
EN
ENO
sku_5_quantity
ingredie

```
uantity_ok
```

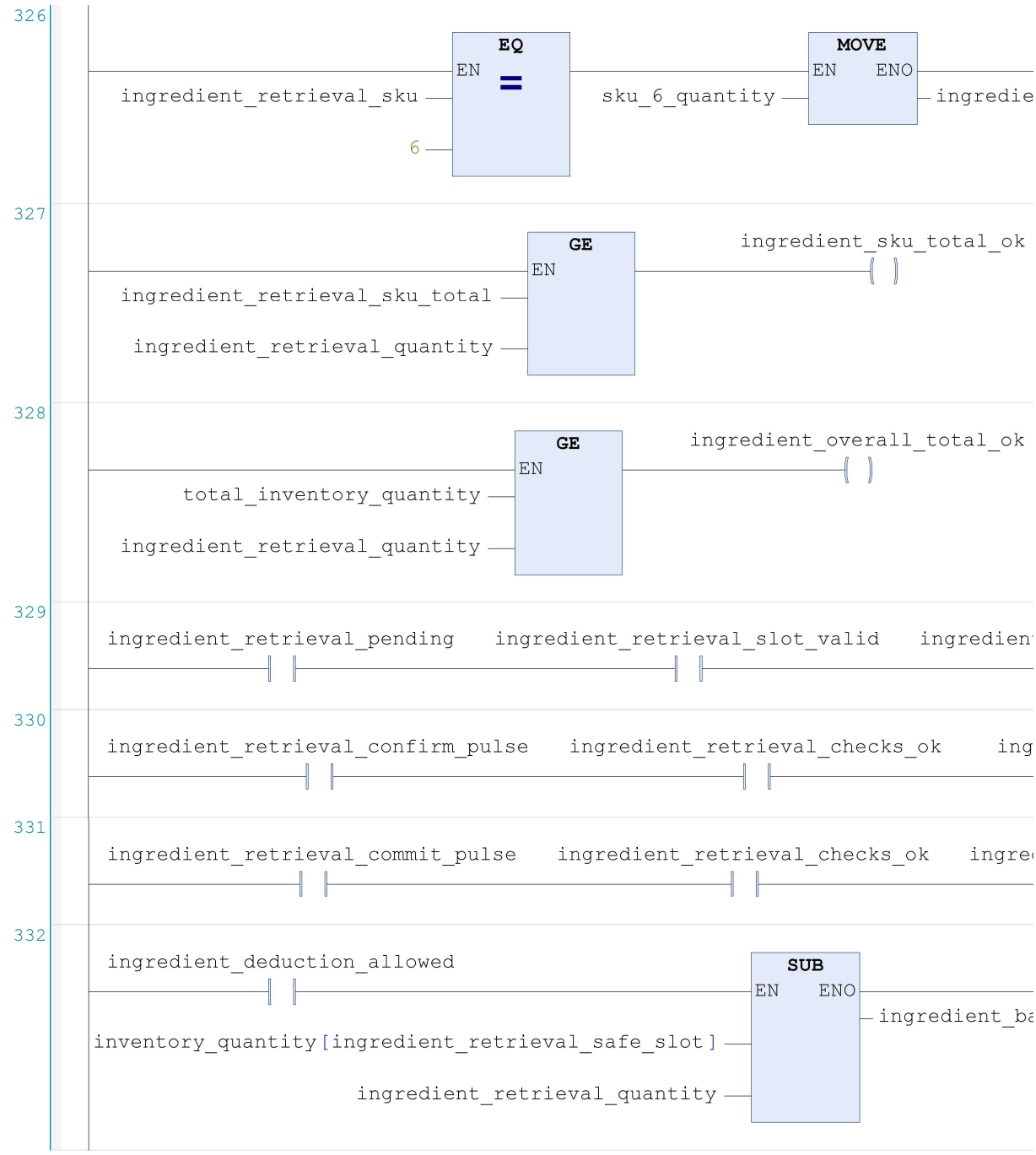
```
nt_retrieval_sku_total
```

```
nt_retrieval_sku_total
```

```
nt_retrieval_sku_total
```

```
nt_retrieval_sku_total
```

```
nt_retrieval_sku_total
```


326
EQ
EN
=
ingredient_retrieval_sku
6
MOVE
EN
ENO
sku_6_quantity
ingredie
327
GE
EN
ingredient_retrieval_sku_total
ingredient_retrieval_quantity
ingredient_sku_total_ok
328
GE
EN
total_inventory_quantity
ingredient_retrieval_quantity
ingredient_overall_total_ok
329
ingredient_retrieval_pending
ingredient_retrieval_slot_valid
ingredien
330
ingredient_retrieval_confirm_pulse
ingredient_retrieval_checks_ok
ing
331
ingredient_retrieval_commit_pulse
ingredient_retrieval_checks_ok
ingre
332
ingredient_deduction_allowed
SUB
EN
ENO
ingredient_ba
inventory_quantity[ingredient_retrieval_safe_slot]
ingredient_retrieval_quantity

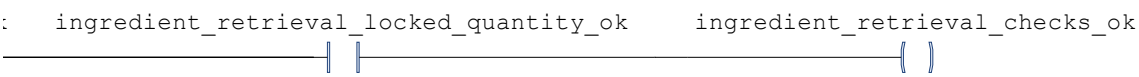
nt_retrieval_sku_total

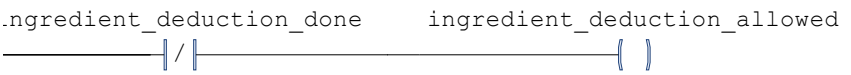
t_retrieval_occupied ingredient_retrieval_sku_match ingredient_retrieval_stock_ok

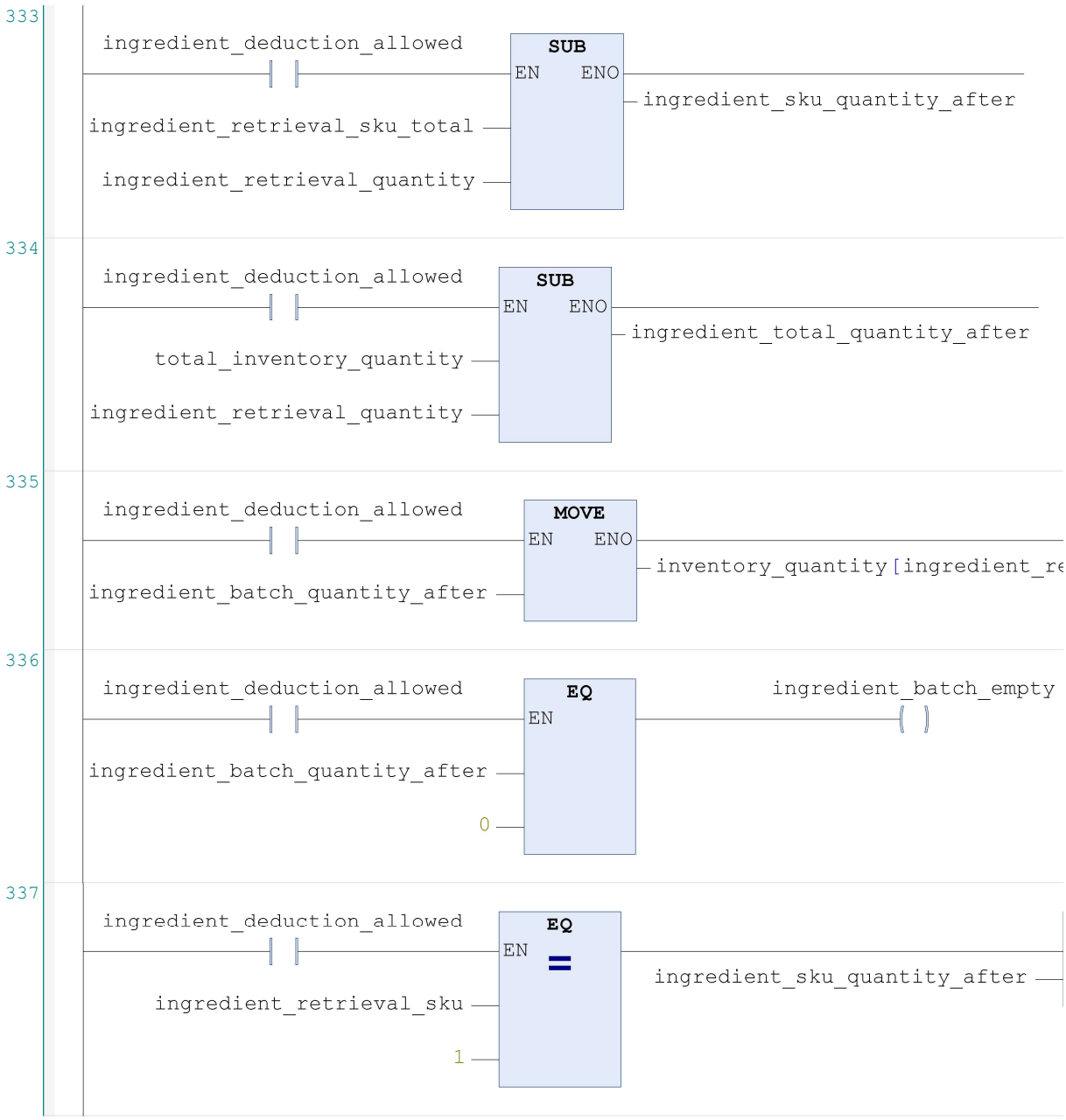
redient_retrieval_commit_pulse

dient_sku_total_ok ingredient_overall_total_ok ingredient_retrieval_sku_valid i

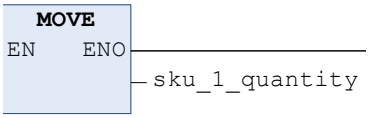
ıtch_quantity_after

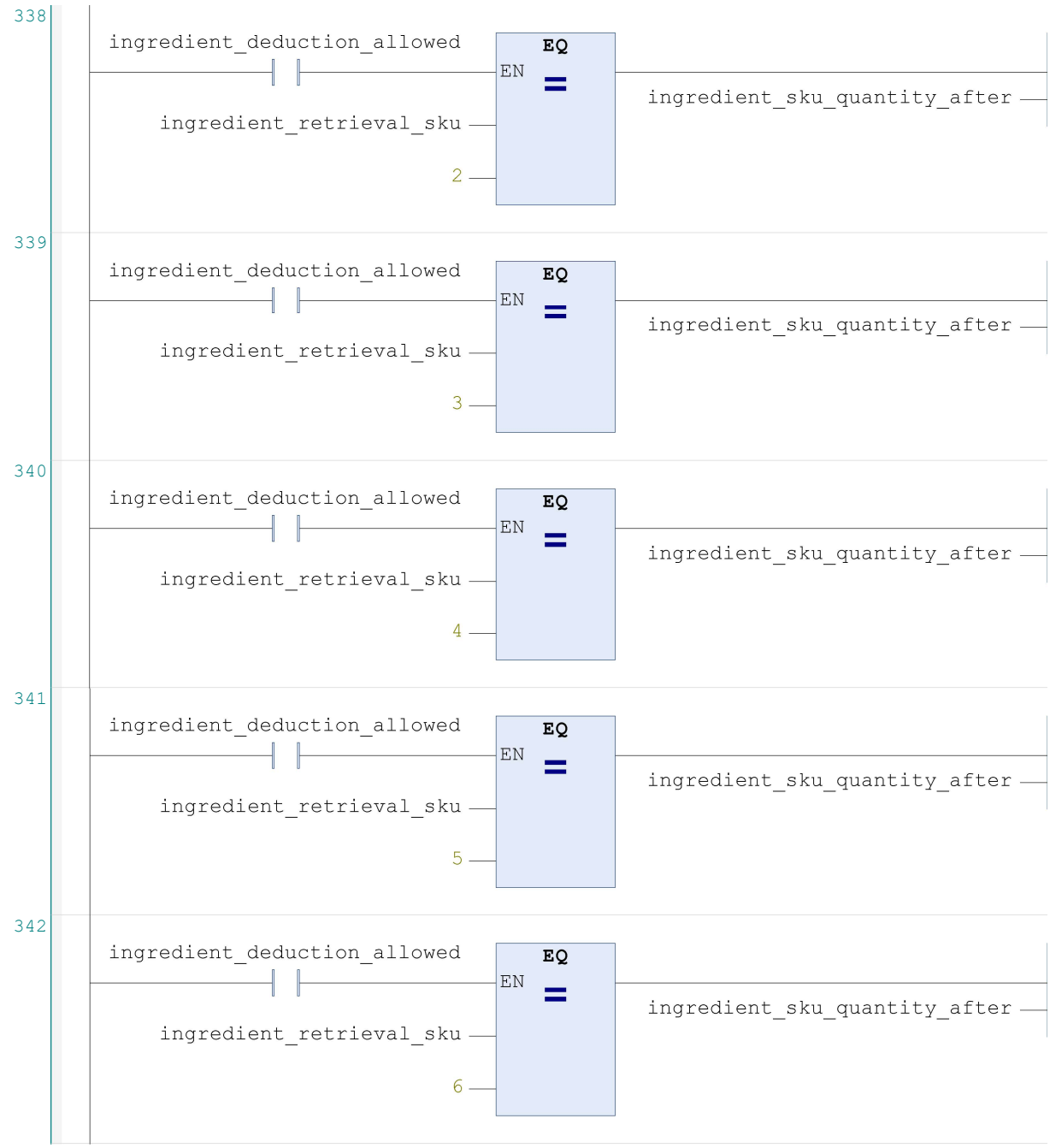

ingredient_retrieval_locked_quantity_ok
ingredient_retrieval_checks_ok

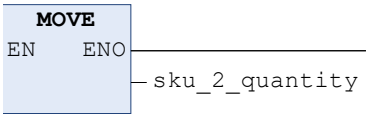

ngredient_deduction_done
ingredient_deduction_allowed

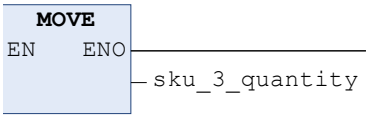

333
ingredient_deduction_allowed
SUB
EN
ENO
ingredient_sku_quantity_after
ingredient_retrieval_sku_total
ingredient_retrieval_quantity
334
ingredient_deduction_allowed
SUB
EN
ENO
ingredient_total_quantity_after
total_inventory_quantity
ingredient_retrieval_quantity
335
ingredient_deduction_allowed
MOVE
EN
ENO
inventory_quantity[ingredient_re
ingredient_batch_quantity_after
336
ingredient_deduction_allowed
EQ
EN
ingredient_batch_empty
ingredient_batch_quantity_after
0
337
ingredient_deduction_allowed
EQ
EN
=
ingredient_sku_quantity_after
ingredient_retrieval_sku
1

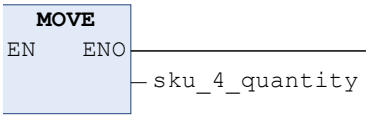
```
etrieval_safe_slot]
```

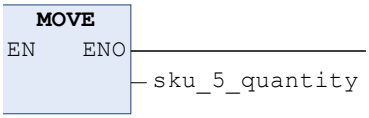


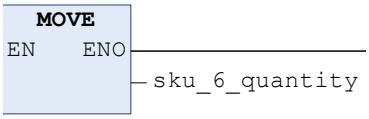

338
ingredient_deduction_allowed
EQ
EN
=
ingredient_retrieval_sku
2
ingredient_sku_quantity_after
339
ingredient_deduction_allowed
EQ
EN
=
ingredient_retrieval_sku
3
ingredient_sku_quantity_after
340
ingredient_deduction_allowed
EQ
EN
=
ingredient_retrieval_sku
4
ingredient_sku_quantity_after
341
ingredient_deduction_allowed
EQ
EN
=
ingredient_retrieval_sku
5
ingredient_sku_quantity_after
342
ingredient_deduction_allowed
EQ
EN
=
ingredient_retrieval_sku
6
ingredient_sku_quantity_after


MOVE
EN
ENO
sku_2_quantity

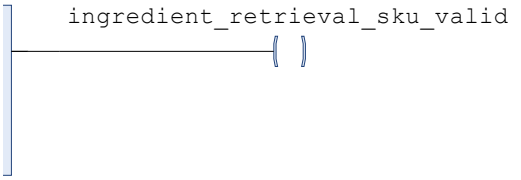

MOVE
EN
ENO
sku_3_quantity

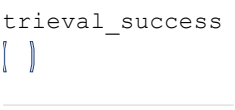

MOVE
EN
ENO
sku_4_quantity

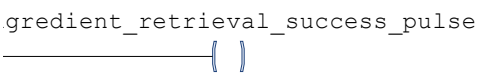

MOVE
EN
ENO
sku_5_quantity

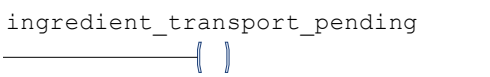

MOVE
EN
ENO
sku_6_quantity

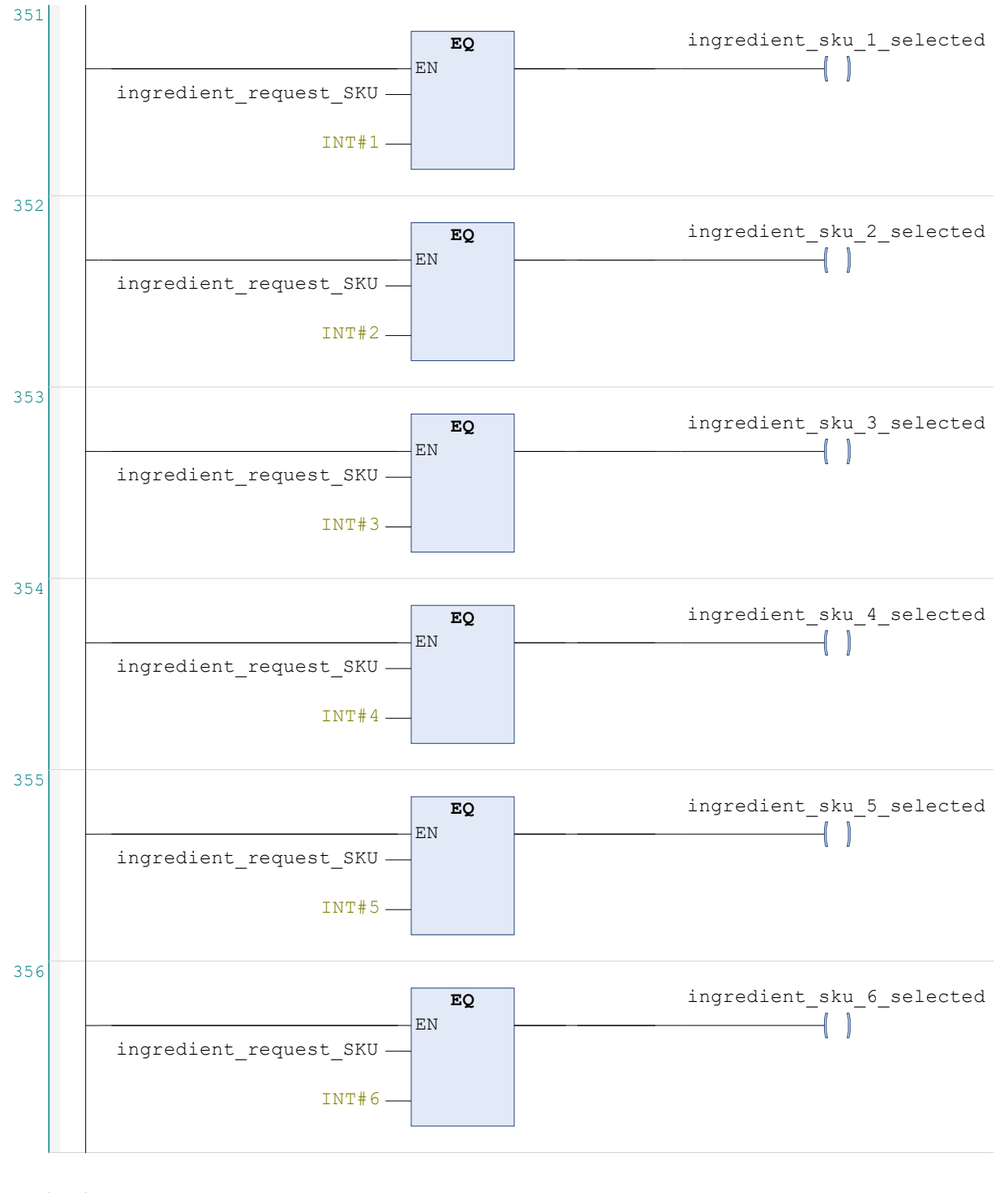

343
ingredient_deduction_allowed
MOVE
EN
ENO
total_inventory_quantity
ingredient_total_quantity_after
344
ingredient_deduction_allowed
ingredient_deduction_done
S
345
GE
EN
≥
ingredient_retrieval_sku
1
LE
EN
ingredient_retrieval_sku
6
346
ingredient_request_reset
ingredient_deduction_done
R
347
ingredient_deduction_done
ingredient_retrieval_complete
348
ingredient_deduction_done
ingredient_retrieval_complete
ingredient_re
349
ingredient_retrieval_success_R_TRIG
ingredient_retrieval_success
R_TRIG
CLK
Q
in
350
ingredient_transport_pending_RS
ingredient_retrieval_success_pulse
RS
SET
Q1
kitchen_transport_request
RESET1

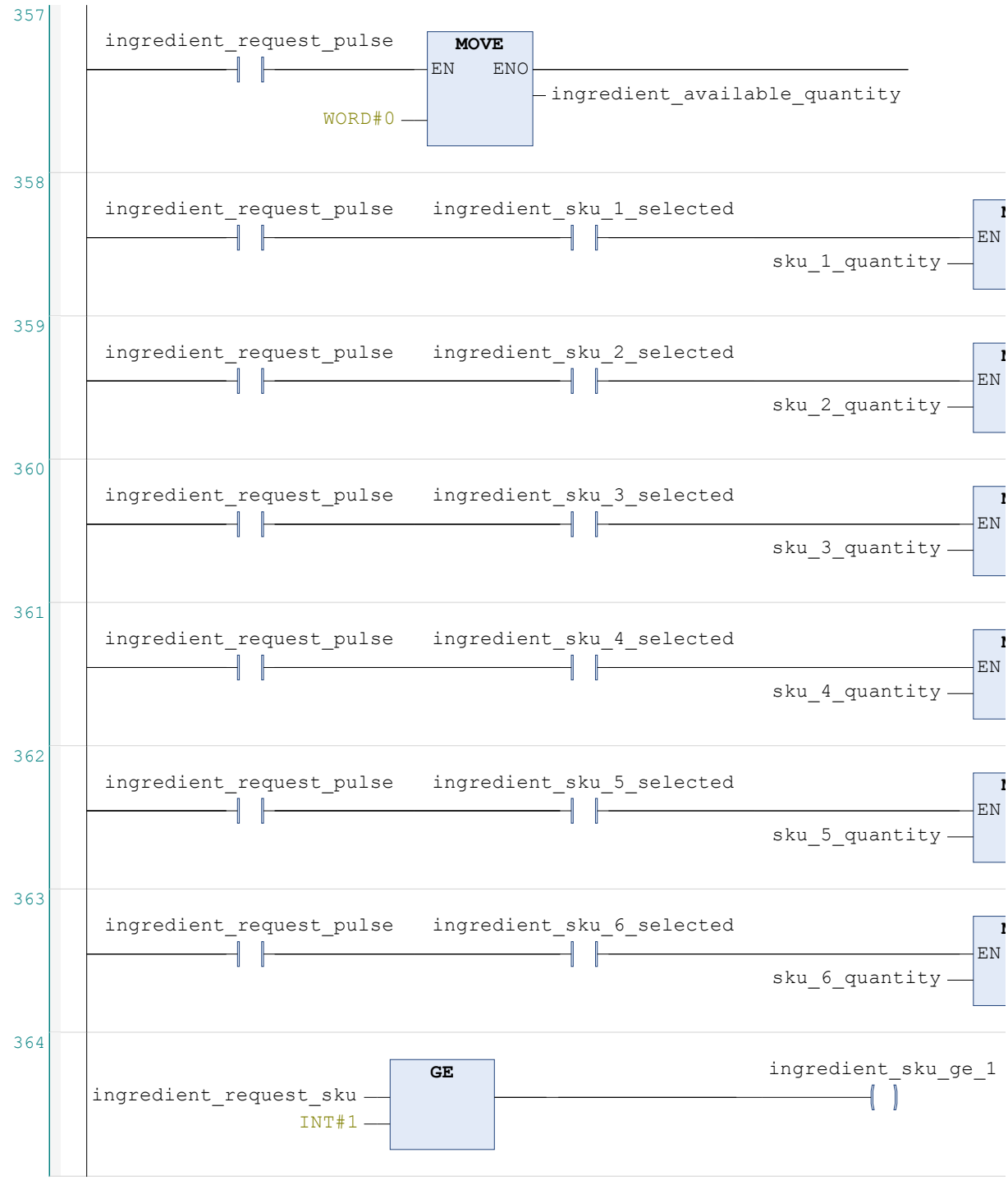

ingredient_retrieval_sku_valid

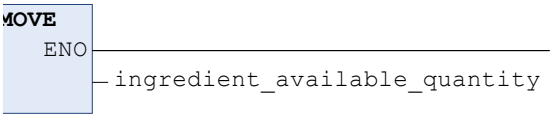

trieval_success

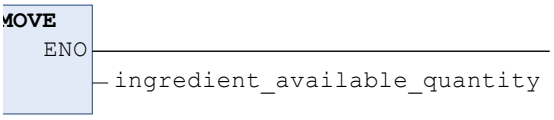

gredient_retrieval_success_pulse

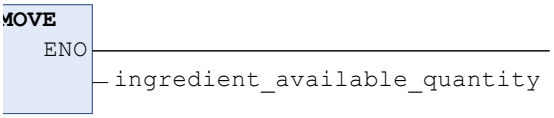

ingredient_transport_pending

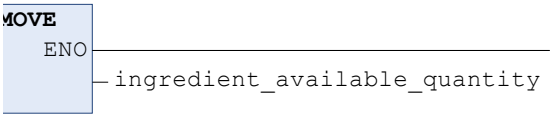

351
EQ
EN
ingredient_request_SKU
INT#1
ingredient_sku_1_selected
352
EQ
EN
ingredient_request_SKU
INT#2
ingredient_sku_2_selected
353
EQ
EN
ingredient_request_SKU
INT#3
ingredient_sku_3_selected
354
EQ
EN
ingredient_request_SKU
INT#4
ingredient_sku_4_selected
355
EQ
EN
ingredient_request_SKU
INT#5
ingredient_sku_5_selected
356
EQ
EN
ingredient_request_SKU
INT#6
ingredient_sku_6_selected

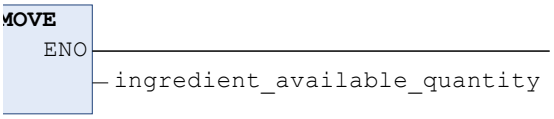

357
ingredient_request_pulse
MOVE
EN
ENO
ingredient_available_quantity
WORD#0
358
ingredient_request_pulse
ingredient_sku_1_selected
EN
sku_1_quantity
359
ingredient_request_pulse
ingredient_sku_2_selected
EN
sku_2_quantity
360
ingredient_request_pulse
ingredient_sku_3_selected
EN
sku_3_quantity
361
ingredient_request_pulse
ingredient_sku_4_selected
EN
sku_4_quantity
362
ingredient_request_pulse
ingredient_sku_5_selected
EN
sku_5_quantity
363
ingredient_request_pulse
ingredient_sku_6_selected
EN
sku_6_quantity
364
GE
ingredient_request_sku
INT#1
ingredient_sku_ge_1

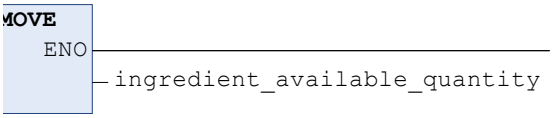

MOVE
ENO
ingredient_available_quantity

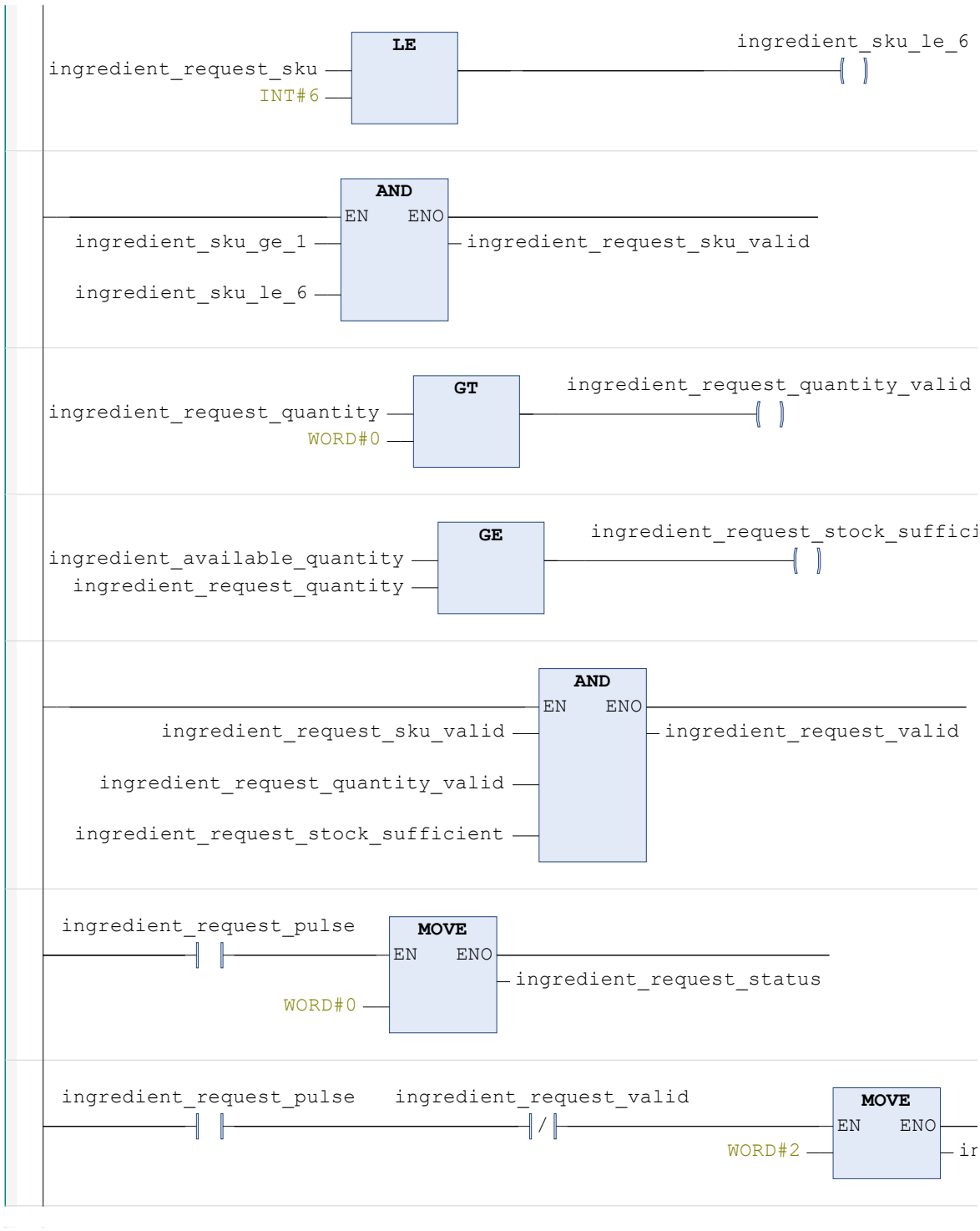

MOVE
ENO
ingredient_available_quantity


MOVE
ENO
ingredient_available_quantity


MOVE
ENO
ingredient_available_quantity

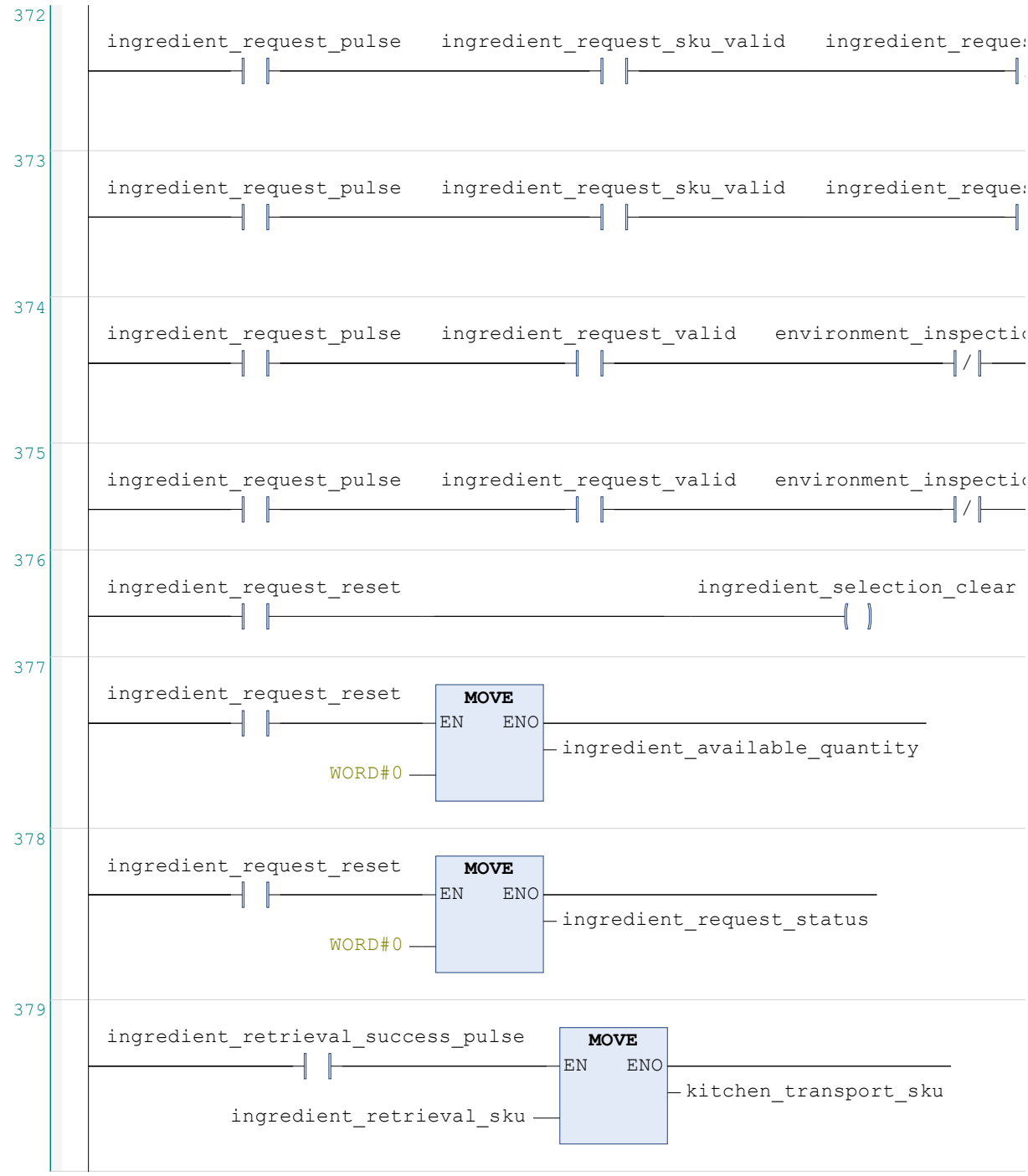

MOVE
ENO
ingredient_available_quantity

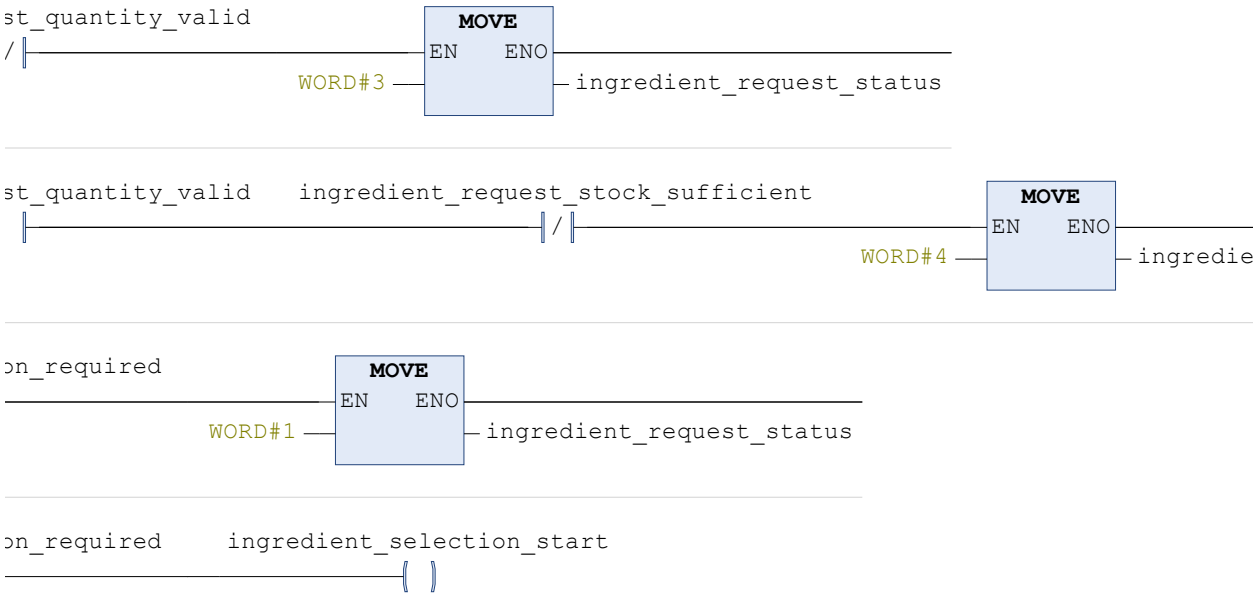

MOVE
ENO
ingredient_available_quantity


365
LE
ingredient_request_sku
INT#6
ingredient_sku_le_6
366
AND
EN
ENO
ingredient_sku_ge_1
ingredient_sku_le_6
ingredient_request_sku_valid
367
GT
ingredient_request_quantity
WORD#0
ingredient_request_quantity_valid
368
GE
ingredient_available_quantity
ingredient_request_quantity
ingredient_request_stock_suffici
369
AND
EN
ENO
ingredient_request_sku_valid
ingredient_request_quantity_valid
ingredient_request_stock_sufficient
ingredient_request_valid
370
ingredient_request_pulse
MOVE
EN
ENO
WORD#0
ingredient_request_status
371
ingredient_request_pulse
ingredient_request_valid
MOVE
EN
ENO
WORD#2

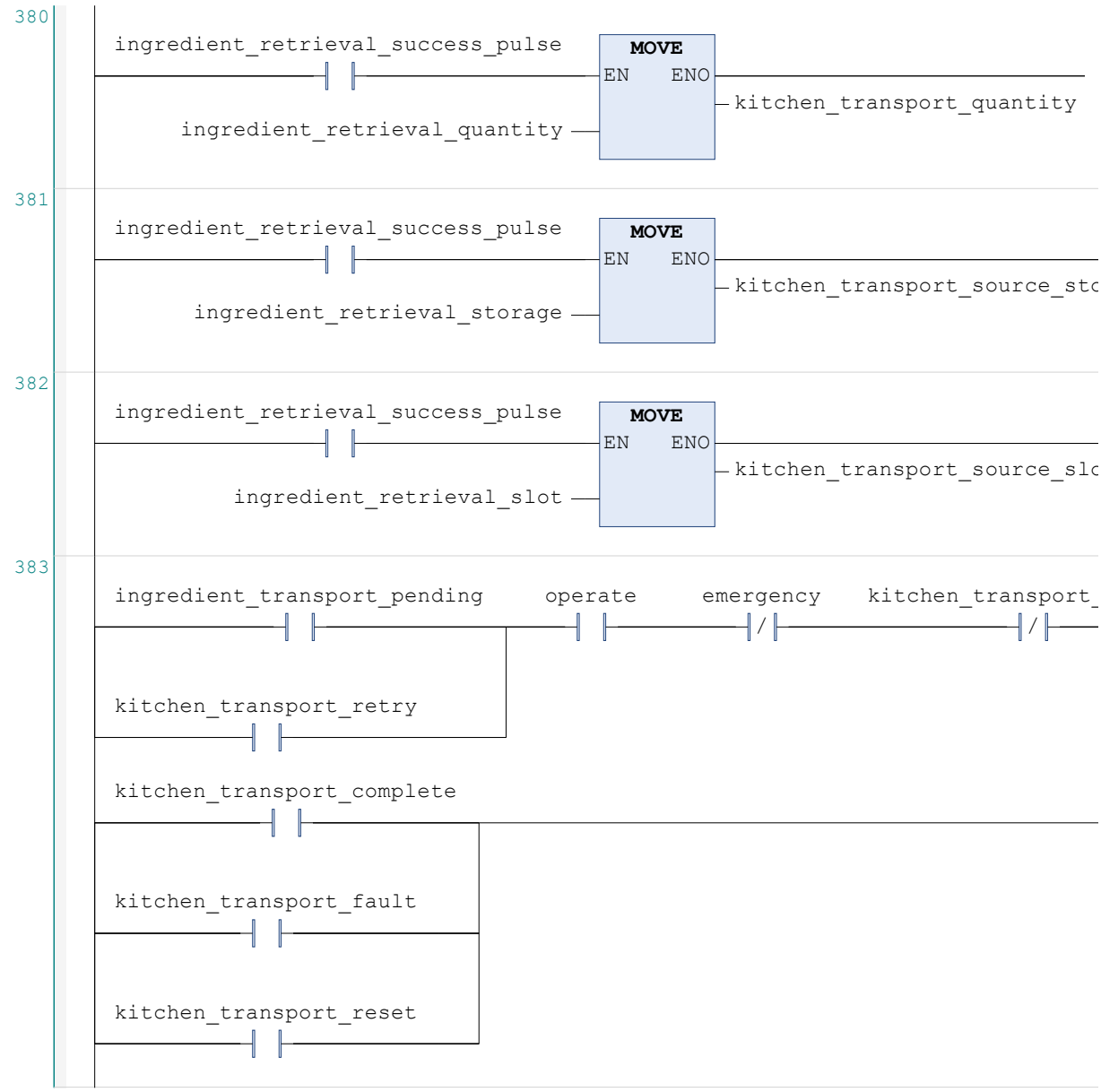
ient

ıgredient_request_status


372
ingredient_request_pulse
ingredient_request_sku_valid
ingredient_reque
373
ingredient_request_pulse
ingredient_request_sku_valid
ingredient_reque
374
ingredient_request_pulse
ingredient_request_valid
environment_inspecti
375
ingredient_request_pulse
ingredient_request_valid
environment_inspecti
376
ingredient_request_reset
ingredient_selection_clear
377
ingredient_request_reset
MOVE
EN
ENO
WORD#0
ingredient_available_quantity
378
ingredient_request_reset
MOVE
EN
ENO
WORD#0
ingredient_request_status
379
ingredient_retrieval_success_pulse
MOVE
EN
ENO
ingredient_retrieval_sku
kitchen_transport_sku

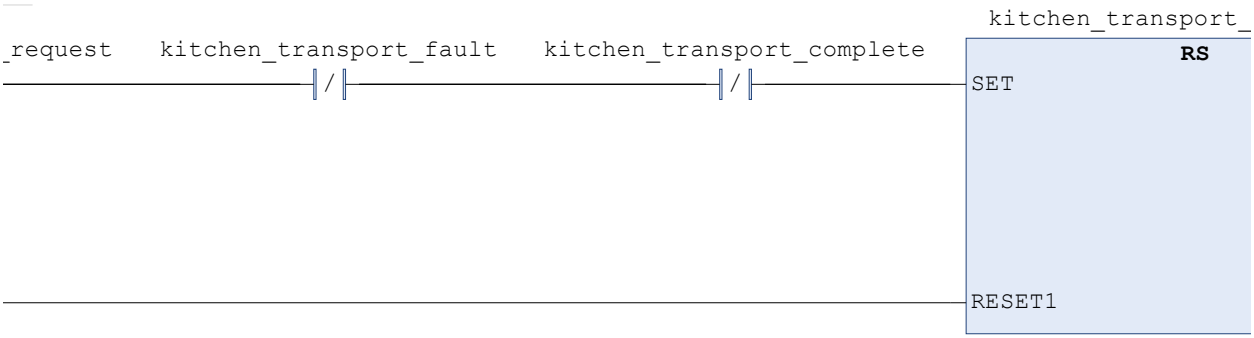

st_quantity_valid
MOVE
EN
ENO
WORD#3
ingredient_request_status
st_quantity_valid
ingredient_request_stock_sufficient
MOVE
EN
ENO
WORD#4
ingredie
on_required
MOVE
EN
ENO
WORD#1
ingredient_request_status
on_required
ingredient_selection_start

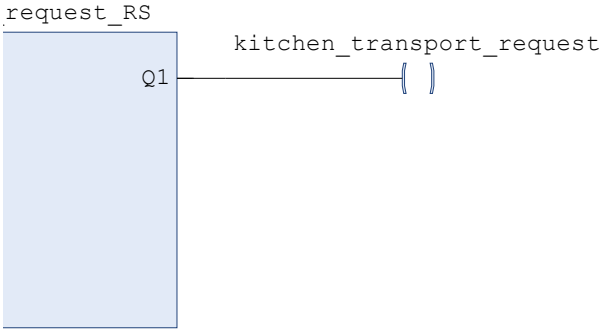
```
nt_request_status
```

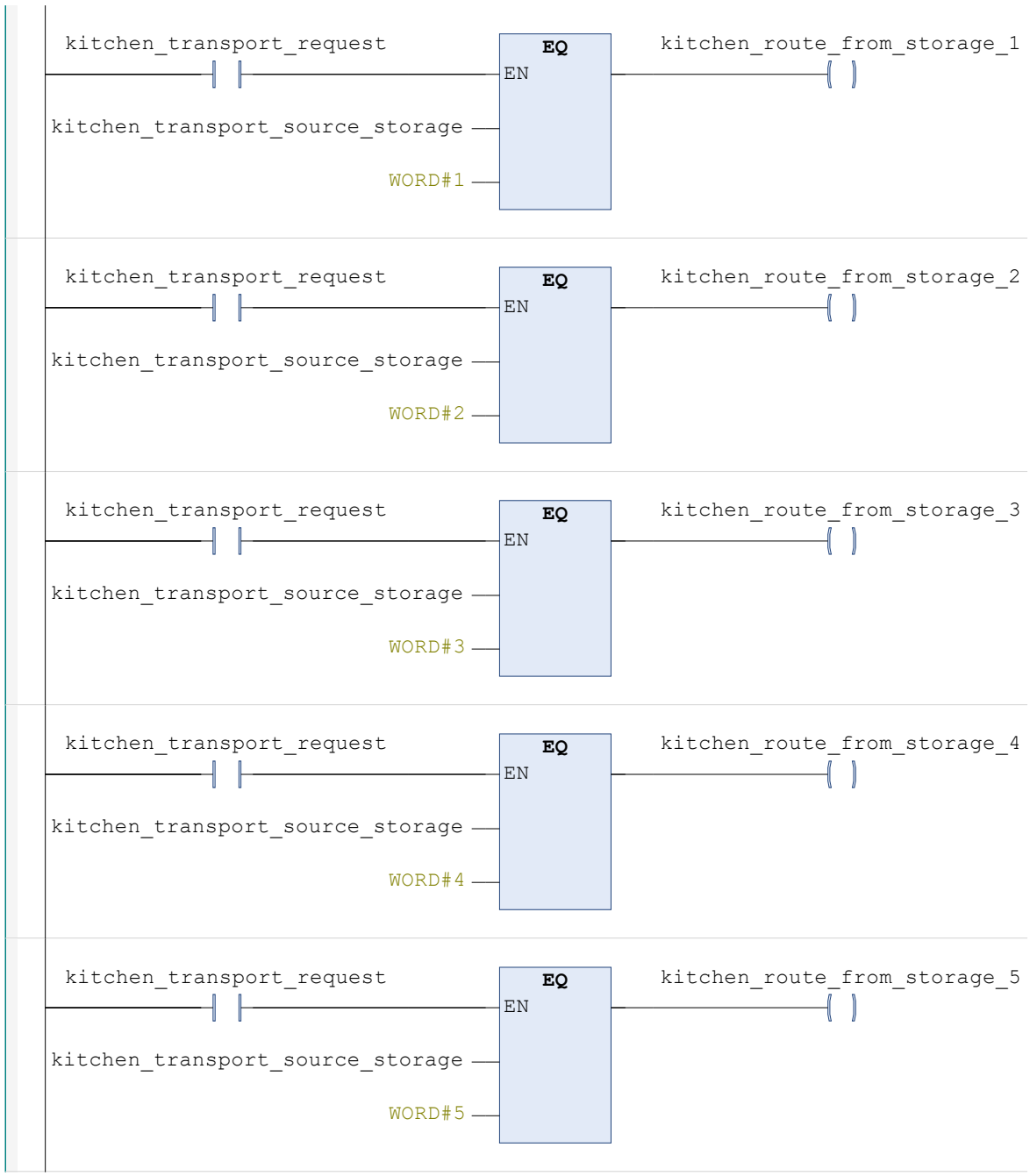

380
ingredient_retrieval_success_pulse
MOVE
EN
ENO
kitchen_transport_quantity
ingredient_retrieval_quantity
381
ingredient_retrieval_success_pulse
MOVE
EN
ENO
kitchen_transport_source_stc
ingredient_retrieval_storage
382
ingredient_retrieval_success_pulse
MOVE
EN
ENO
kitchen_transport_source_slc
ingredient_retrieval_slot
383
ingredient_transport_pending
operate
emergency
kitchen_transport_
kitchen_transport_retry
kitchen_transport_complete
kitchen_transport_fault
kitchen_transport_reset

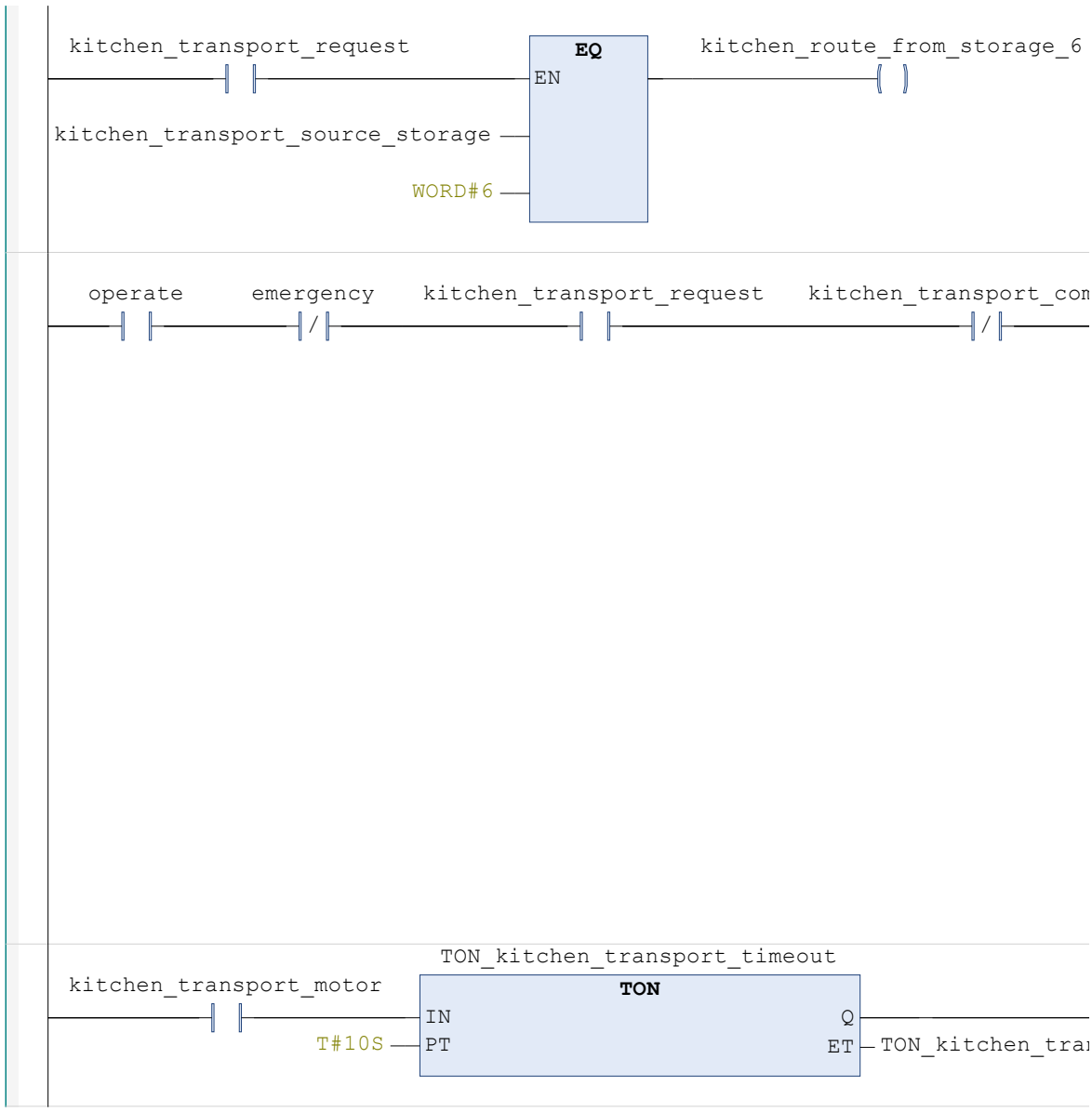
›rage

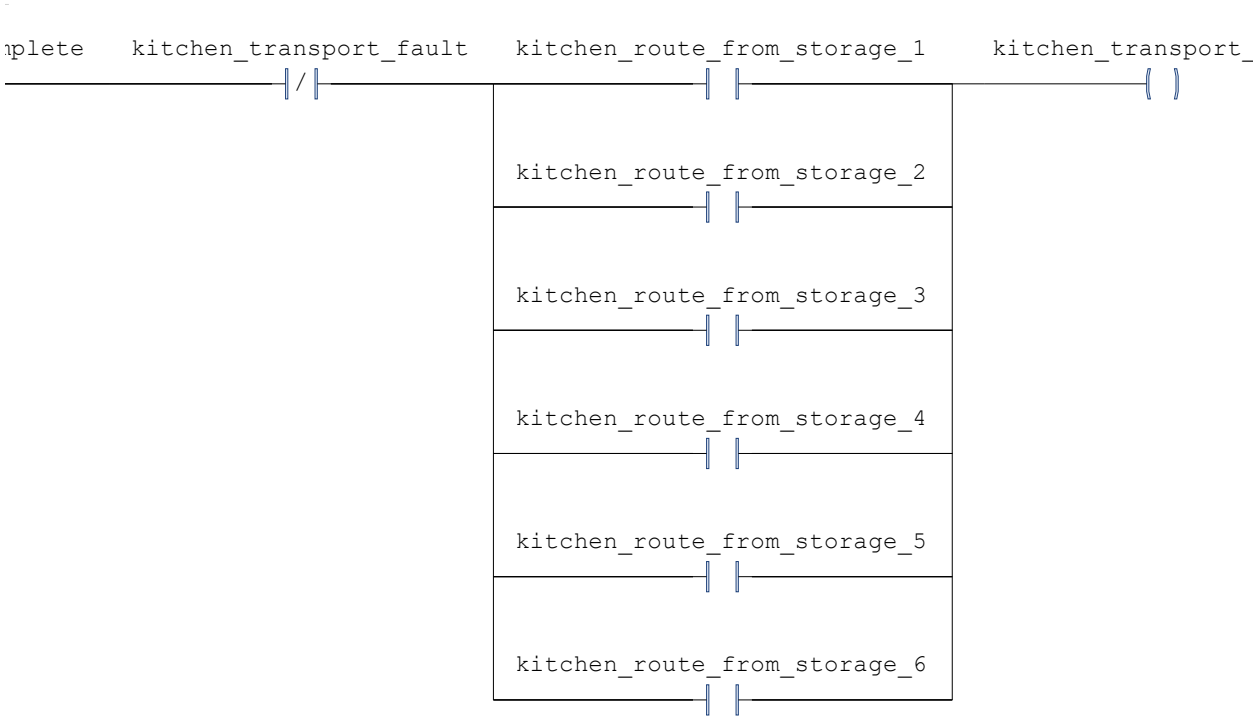
›t

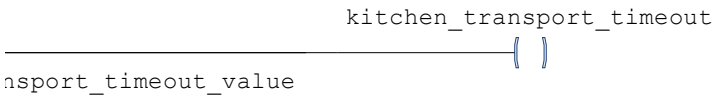

kitchen_transport_
_request
kitchen_transport_fault
kitchen_transport_complete
RS
SET
RESET1


request_RS
Q1
kitchen_transport_request


384
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_1
kitchen_transport_source_storage
WORD#1
385
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_2
kitchen_transport_source_storage
WORD#2
386
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_3
kitchen_transport_source_storage
WORD#3
387
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_4
kitchen_transport_source_storage
WORD#4
388
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_5
kitchen_transport_source_storage
WORD#5

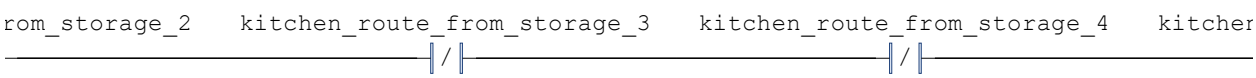

389
kitchen_transport_request
EQ
EN
kitchen_route_from_storage_6
kitchen_transport_source_storage
WORD#6
390
operate
emergency
kitchen_transport_request
kitchen_transport_com
391
TON_kitchen_transport_timeout
kitchen_transport_motor
TON
IN
PT
T#10S
Q
ET
TON_kitchen_tra

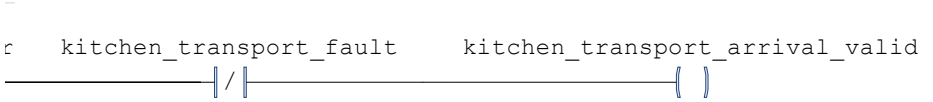

nplete
kitchen_transport_fault
kitchen_route_from_storage_1
kitchen_transport_
kitchen_route_from_storage_2
kitchen_route_from_storage_3
kitchen_route_from_storage_4
kitchen_route_from_storage_5
kitchen_route_from_storage_6

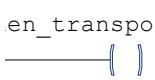

kitchen_transport_timeout
nsport_timeout_value

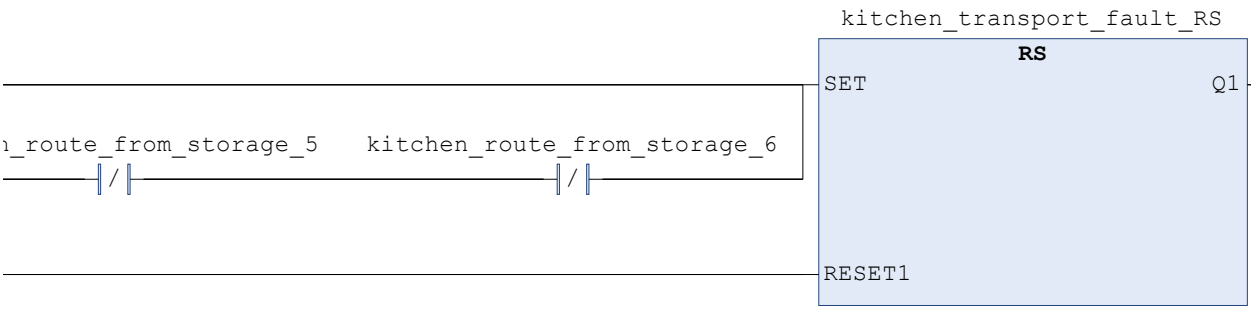
```
motor
```

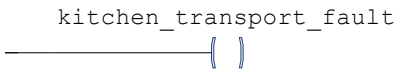

392
kitchen_transport_timeout
kitchen_transport_request
kitchen_route_from_storage_1
kitchen_transport_reset
393
kitchen_arrival_R_TRIG
kitchen_arrival_sensor
R_TRIG
CLK
Q
kitchen_arrival_pulse
394
kitchen_arrival_pulse
kitchen_transport_request
395
kitchen_transport_complete_RS
kitchen_transport_arrival_valid
RS
SET
Q1
kitchen_transport_reset
RESET1
396
order_submit_R_TRIG
order_submit
R_TRIG
CLK
Q
order_submit_pulse
397
GT
EN
order_meal_valid
order_input_meal_id
INT#0
398
GT
EN
order_quantity_valid
order_input_quantity
WORD#0

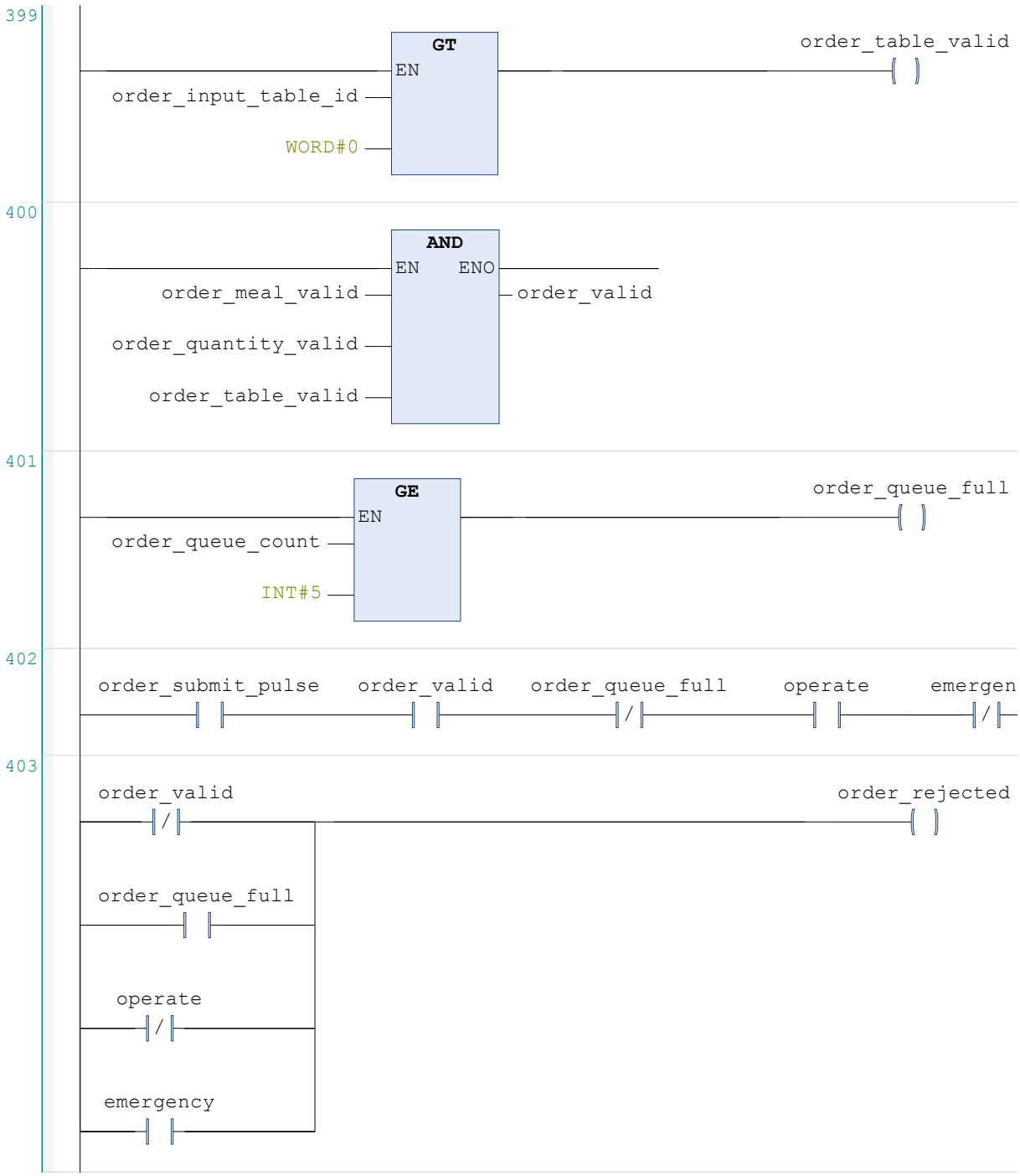

rom_storage_2
kitchen_route_from_storage_3
kitchen_route_from_storage_4
kitcher

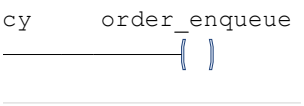

r
kitchen_transport_fault
kitchen_transport_arrival_valid

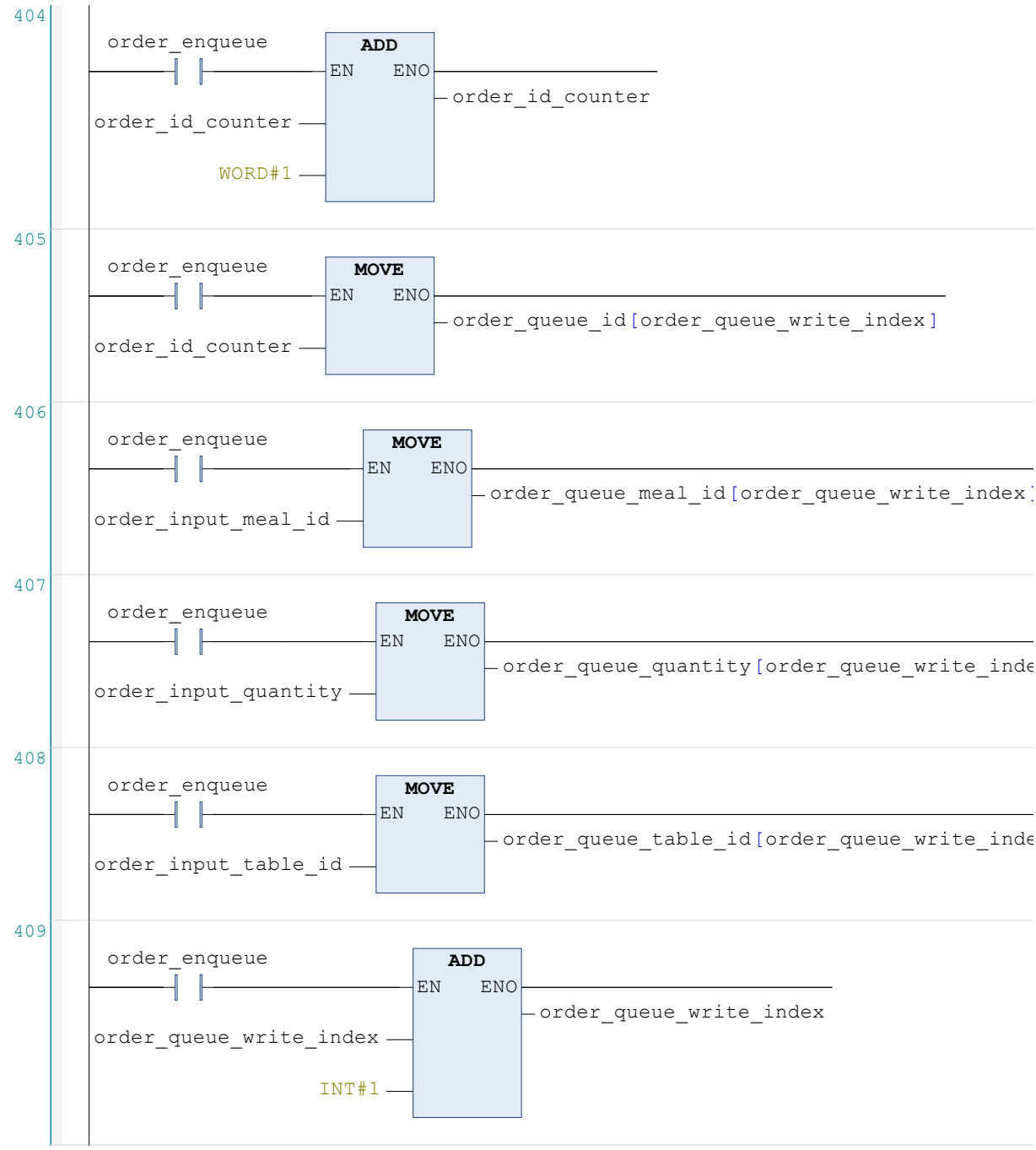

en_transport_complete

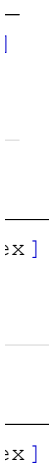

kitchen_transport_fault_RS
RS
SET
Q1
ı_route_from_storage_5
kitchen_route_from_storage_6
/
/
RESET1


kitchen_transport_fault


399
GT
EN
order_input_table_id
WORD#0
order_table_valid
400
AND
EN
ENO
order_meal_valid
order_quantity_valid
order_table_valid
order_valid
401
GE
EN
order_queue_count
INT#5
order_queue_full
402
order_submit_pulse
order_valid
order_queue_full
operate
emergen
403
order_valid
order_rejected
order_queue_full
operate
emergency


cy
order_enqueue

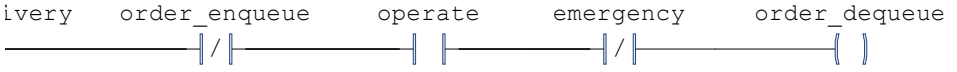

404
order_enqueue
ADD
EN
ENO
order_id_counter
order_id_counter
WORD#1
405
order_enqueue
MOVE
EN
ENO
order_queue_id[order_queue_write_index]
order_id_counter
406
order_enqueue
MOVE
EN
ENO
order_queue_meal_id[order_queue_write_index]
order_input_meal_id
407
order_enqueue
MOVE
EN
ENO
order_queue_quantity[order_queue_write_inde
order_input_quantity
408
order_enqueue
MOVE
EN
ENO
order_queue_table_id[order_queue_write_inde
order_input_table_id
409
order_enqueue
ADD
EN
ENO
order_queue_write_index
order_queue_write_index
INT#1




410
GT
EN
>
order_queue_write_index
INT#5
MOVE
EN
ENO
INT#1
order_queue_write_
411
ADD
EN
ENO
order_queue_count
INT#1
order_queue_count
412
order_enqueue
order_accepted
413
GT
EN
order_queue_count
INT#1
order_queue_has_items
414
order_queue_has_items
kitchen_order_active
/
kitchen_order_ready_for_del
/
415
order_dequeue
MOVE
EN
ENO
order_queue_id[order_queue_read_index]
kitchen_order_current_id
416
order_dequeue
MOVE
EN
ENO
order_queue_meal_id[order_queue_read_index]
kitchen_order_currer

index

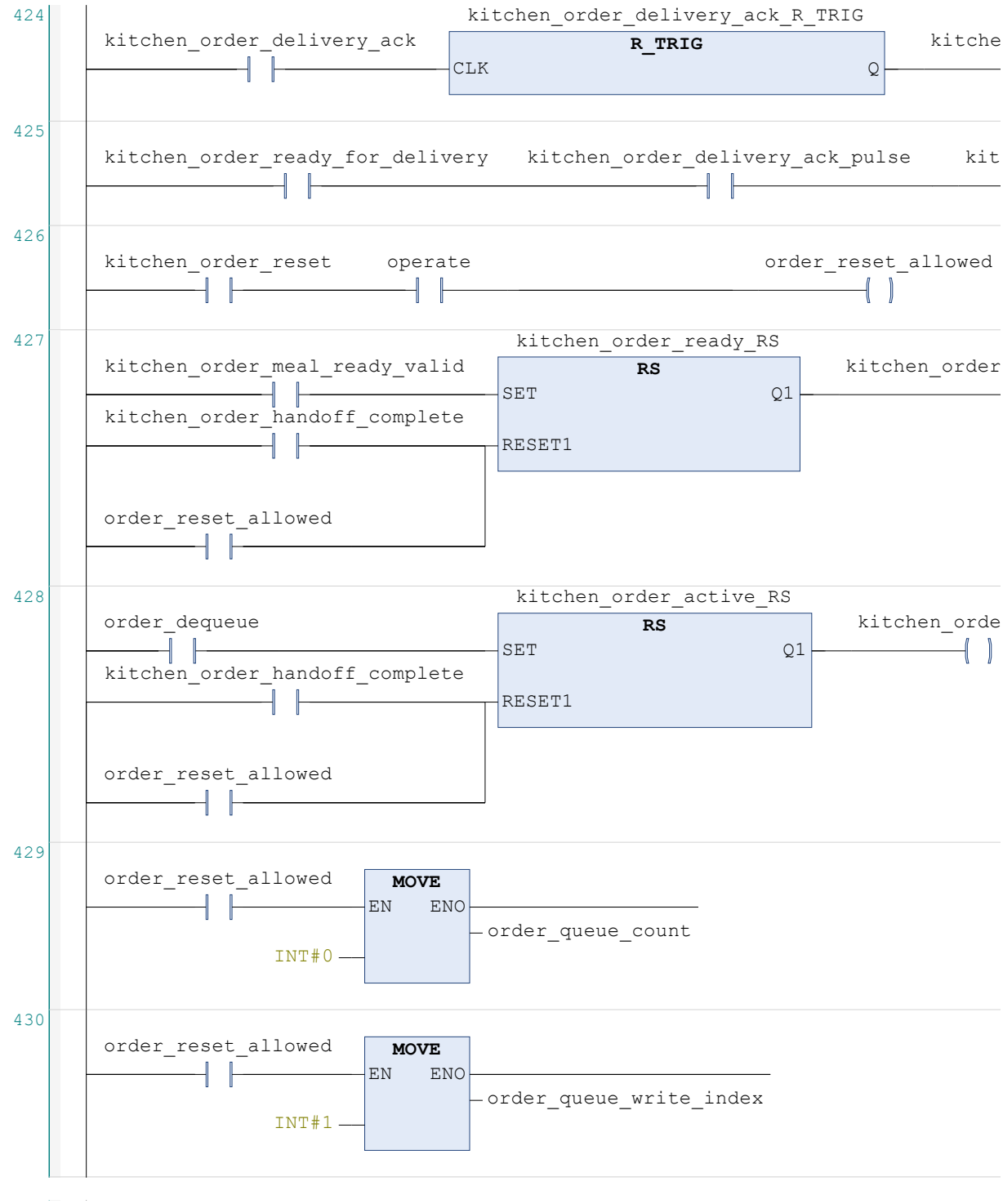
ivery order_enqueue operate emergency order_dequeue

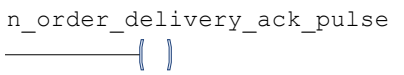


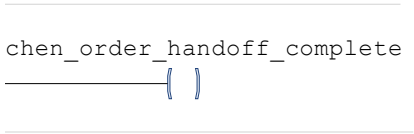
ıt_meal_id

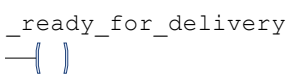

417
order_dequeue
MOVE
EN
ENO
order_queue_quantity[order_queue_read_index]
kitchen_order_curre
418
order_dequeue
MOVE
EN
ENO
order_queue_table_id[order_queue_read_index]
kitchen_order_curre
419
order_dequeue
ADD
EN
ENO
order_queue_read_index
order_queue_read_index
INT#1
420
GT
EN
>
order_queue_read_index
INT#5
MOVE
EN
ENO
INT#1
order_queue_read_in
421
SUB
EN
ENO
order_queue_count
order_queue_count
INT#1
422
kitchen_order_meal_ready_R_TRIG
kitchen_order_meal_ready
R_TRIG
CLK
Q
kitchen_or
423
kitchen_order_meal_ready_pulse
kitchen_order_active
kitchen_order_read
/

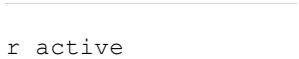

ent_quantity
ent_table_id
dex
der_meal_ready_pulse
y_for_delivery
operate
operate
kitchen_order_meal_ready_valid

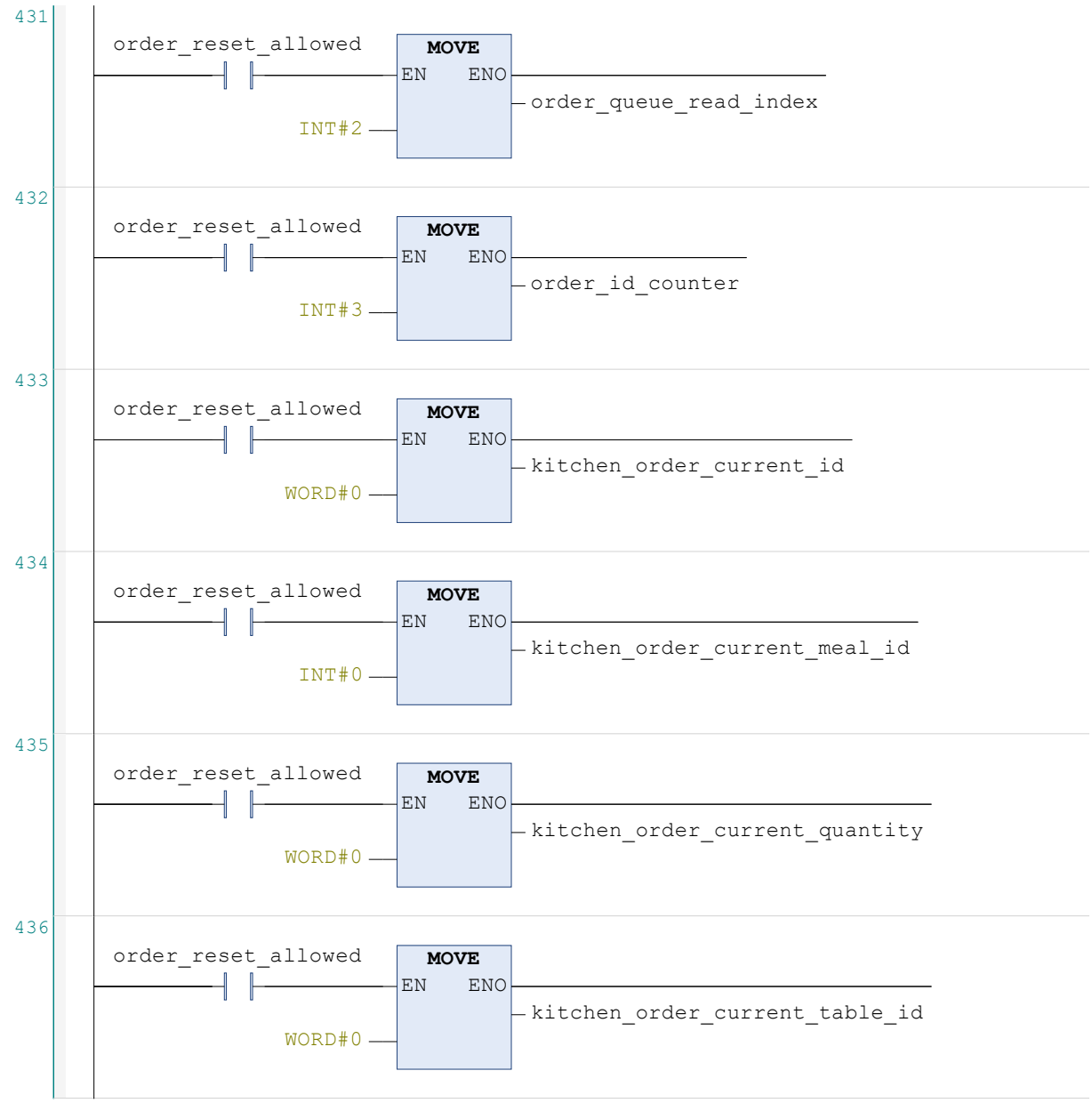

424
kitchen_order_delivery_ack_R_TRIG
kitchen_order_delivery_ack
R_TRIG
CLK
Q
kitche
425
kitchen_order_ready_for_delivery
kitchen_order_delivery_ack_pulse
kit
426
kitchen_order_reset
operate
order_reset_allowed
427
kitchen_order_ready_RS
kitchen_order_meal_ready_valid
RS
SET
Q1
kitchen_order
kitchen_order_handoff_complete
RESET1
order_reset_allowed
428
kitchen_order_active_RS
order_dequeue
RS
kitchen_orde
SET
Q1
kitchen_order_handoff_complete
RESET1
order_reset_allowed
429
order_reset_allowed
MOVE
EN
ENO
order_queue_count
INT#0
430
order_reset_allowed
MOVE
EN
ENO
order_queue_write_index
INT#1

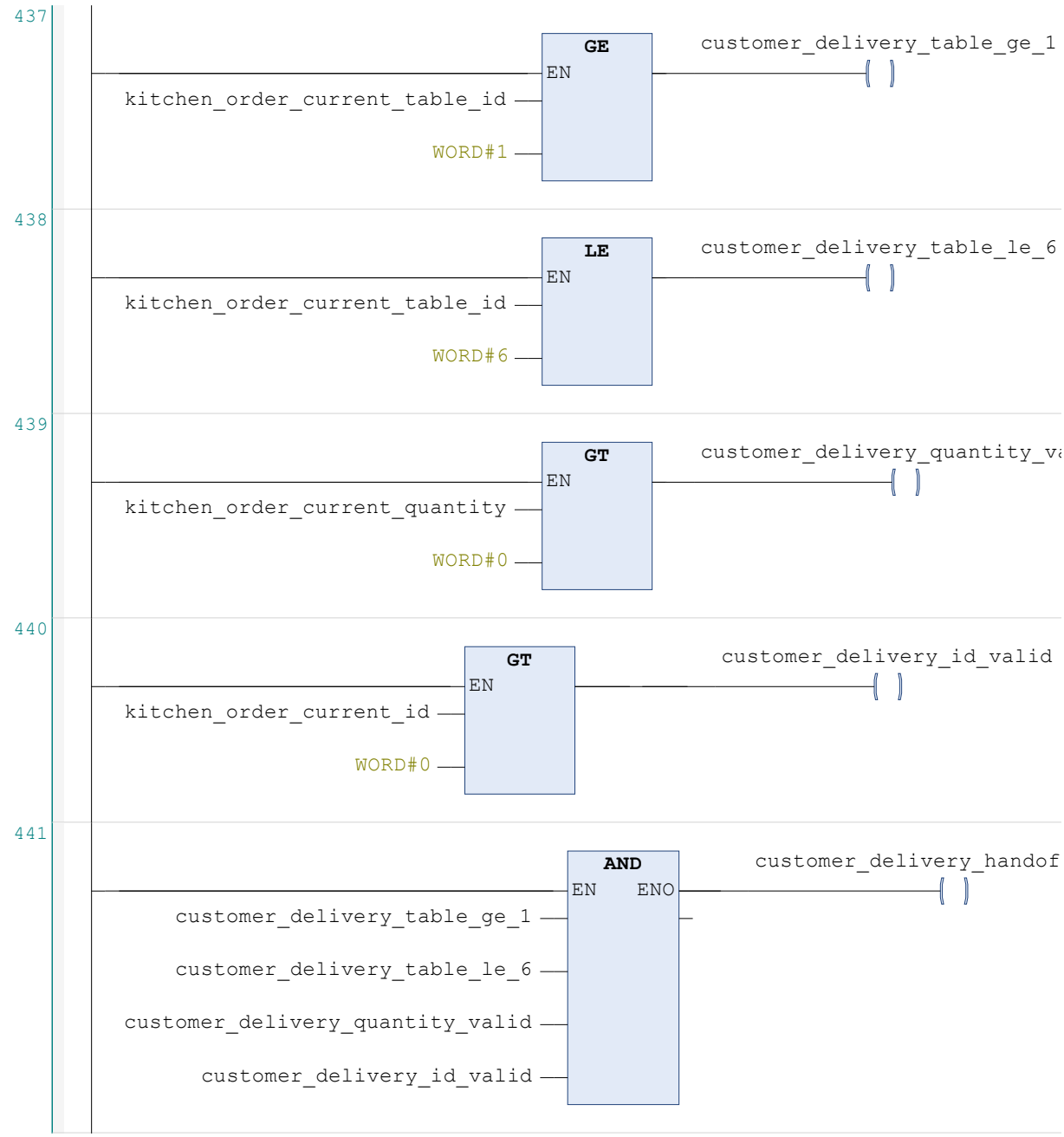

n_order_delivery_ack_pulse


chen_order_handoff_complete


_ready_for_delivery

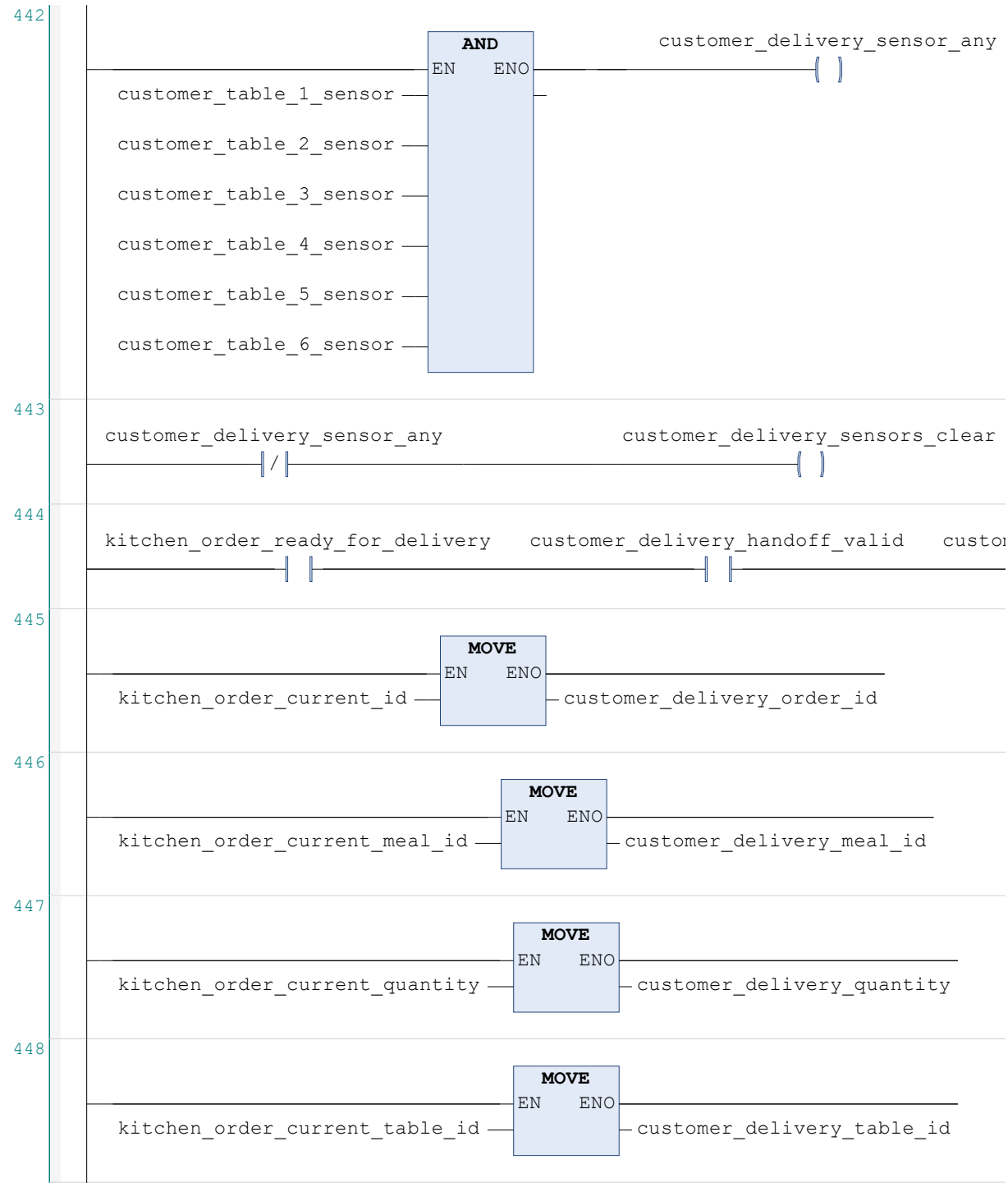

r_active

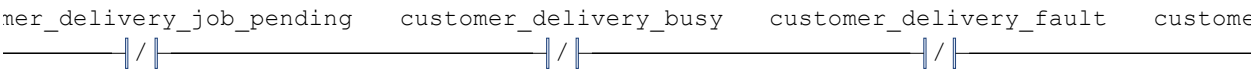

431
order_reset_allowed
MOVE
EN
ENO
INT#2
order_queue_read_index
432
order_reset_allowed
MOVE
EN
ENO
INT#3
order_id_counter
433
order_reset_allowed
MOVE
EN
ENO
WORD#0
kitchen_order_current_id
434
order_reset_allowed
MOVE
EN
ENO
INT#0
kitchen_order_current_meal_id
435
order_reset_allowed
MOVE
EN
ENO
WORD#0
kitchen_order_current_quantity
436
order_reset_allowed
MOVE
EN
ENO
WORD#0
kitchen_order_current_table_id

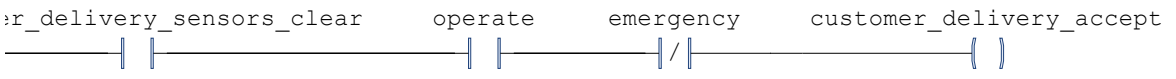

437
GE
EN
kitchen_order_current_table_id
WORD#1
customer_delivery_table_ge_1
438
LE
EN
kitchen_order_current_table_id
WORD#6
customer_delivery_table_le_6
439
GT
EN
kitchen_order_current_quantity
WORD#0
customer_delivery_quantity_va
440
GT
EN
kitchen_order_current_id
WORD#0
customer_delivery_id_valid
441
AND
EN
ENO
customer_delivery_table_ge_1
customer_delivery_table_le_6
customer_delivery_quantity_valid
customer_delivery_id_valid
customer_delivery_handof

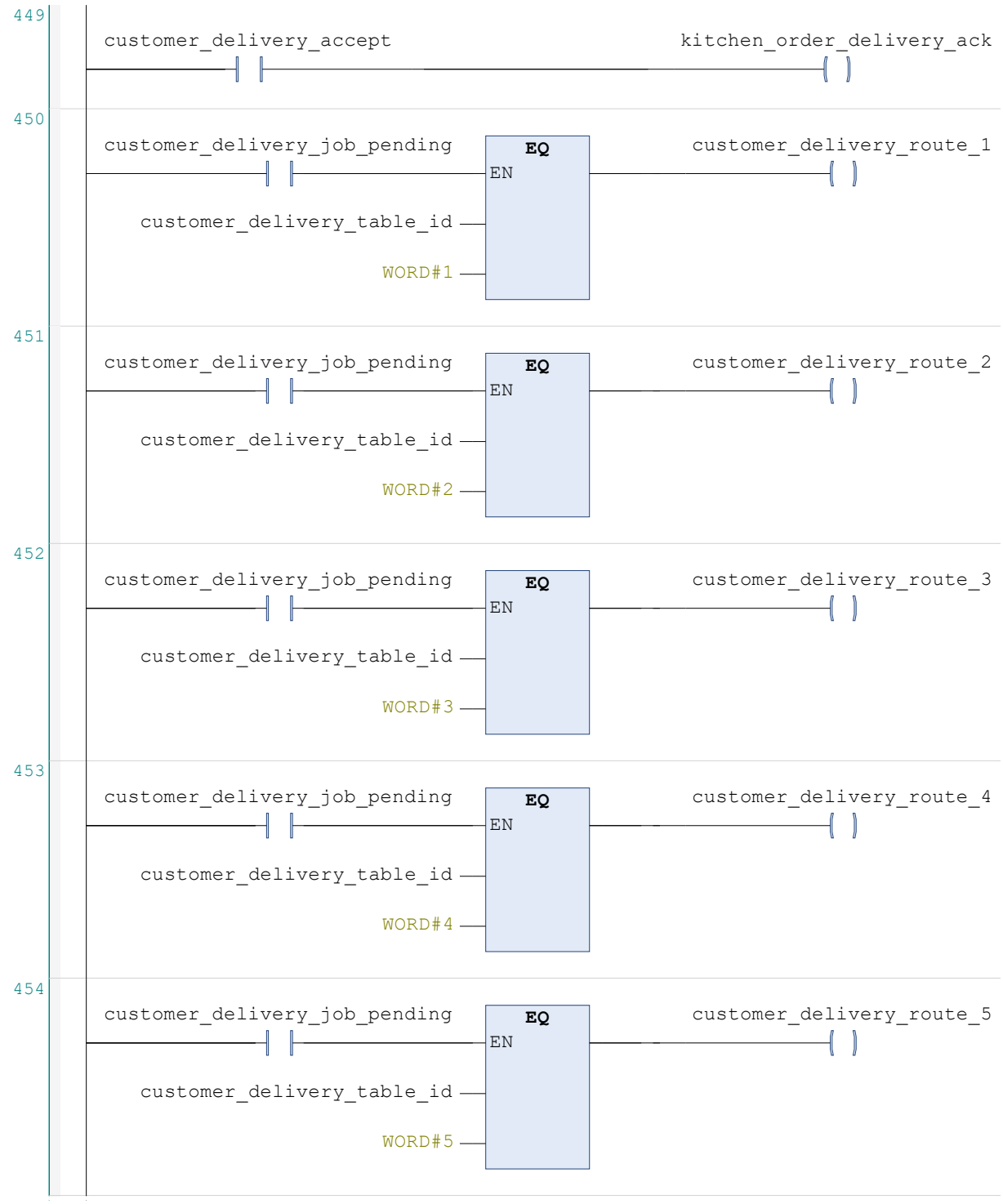
alid

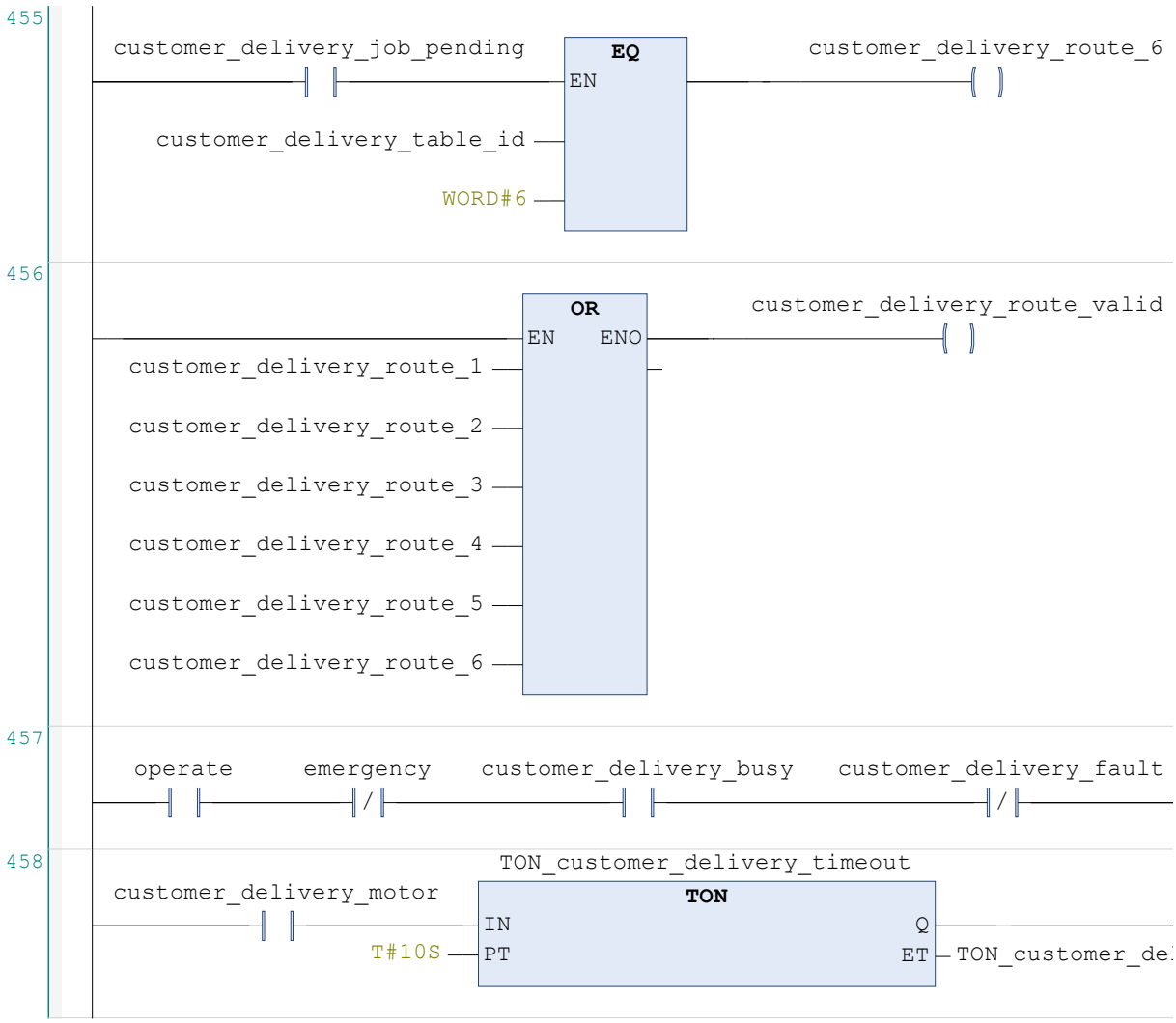
f_valid

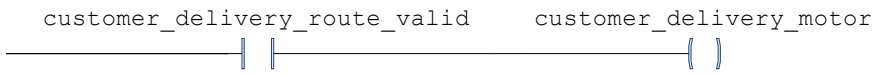

442
AND
EN
ENO
customer_table_1_sensor
customer_table_2_sensor
customer_table_3_sensor
customer_table_4_sensor
customer_table_5_sensor
customer_table_6_sensor
customer_delivery_sensor_any
443
customer_delivery_sensor_any
/
customer_delivery_sensors_clear
444
kitchen_order_ready_for_delivery
customer_delivery_handoff_valid
445
MOVE
EN
ENO
kitchen_order_current_id
customer_delivery_order_id
446
MOVE
EN
ENO
kitchen_order_current_meal_id
customer_delivery_meal_id
447
MOVE
EN
ENO
kitchen_order_current_quantity
customer_delivery_quantity
448
MOVE
EN
ENO
kitchen_order_current_table_id
customer_delivery_table_id

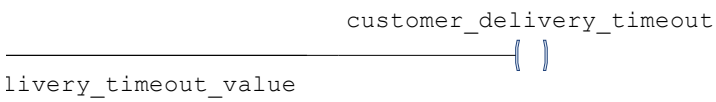

mer_delivery_job_pending
customer_delivery_busy
customer_delivery_fault
custome

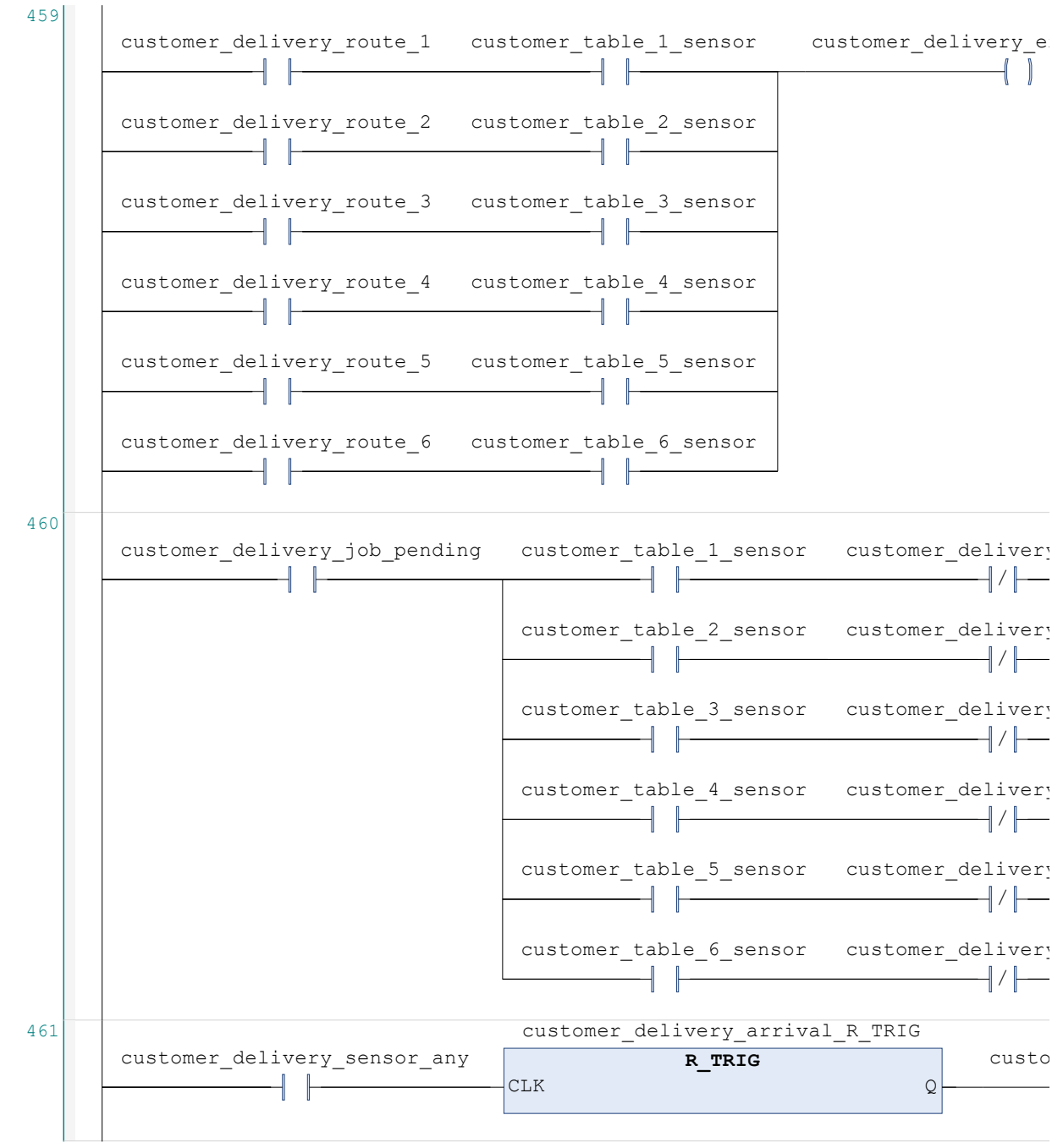

er_delivery_sensors_clear
operate
emergency
customer_delivery_accept


449
customer_delivery_accept
kitchen_order_delivery_ack
450
customer_delivery_job_pending
EQ
EN
customer_delivery_table_id
WORD#1
customer_delivery_route_1
451
customer_delivery_job_pending
EQ
EN
customer_delivery_table_id
WORD#2
customer_delivery_route_2
452
customer_delivery_job_pending
EQ
EN
customer_delivery_table_id
WORD#3
customer_delivery_route_3
453
customer_delivery_job_pending
EQ
EN
customer_delivery_table_id
WORD#4
customer_delivery_route_4
454
customer_delivery_job_pending
EQ
EN
customer_delivery_table_id
WORD#5
customer_delivery_route_5

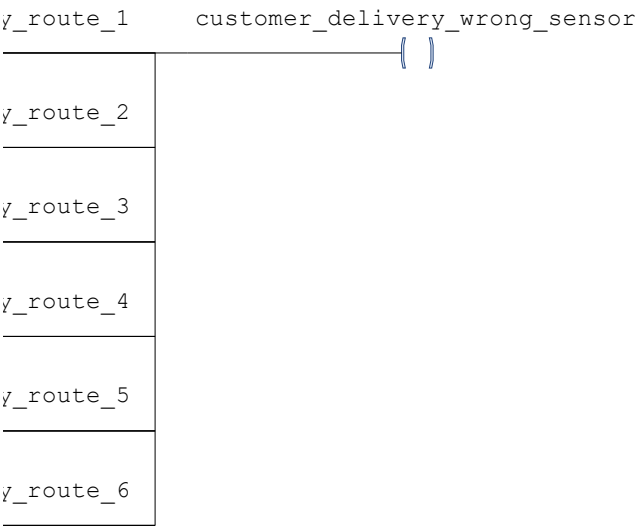

455
customer_delivery_job_pending
EQ
EN
customer_delivery_route_6
customer_delivery_table_id
WORD#6
456
OR
EN
ENO
customer_delivery_route_valid
customer_delivery_route_1
customer_delivery_route_2
customer_delivery_route_3
customer_delivery_route_4
customer_delivery_route_5
customer_delivery_route_6
457
operate
emergency
customer_delivery_busy
customer_delivery_fault
458
TON_customer_delivery_timeout
customer_delivery_motor
TON
IN
PT
Q
ET
T#10S
TON_customer_de


customer_delivery_route_valid
customer_delivery_motor

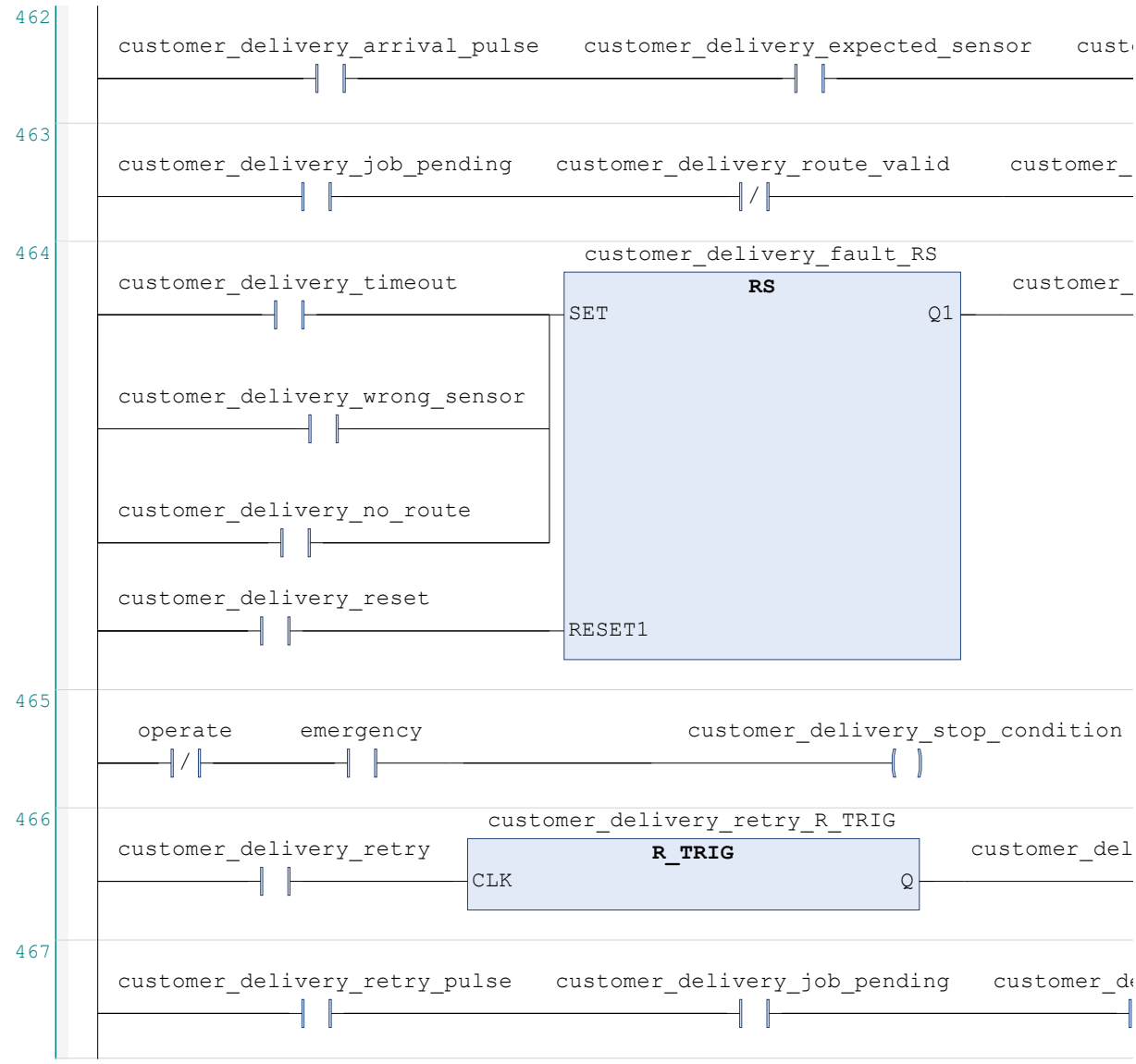

customer_delivery_timeout
livery_timeout_value

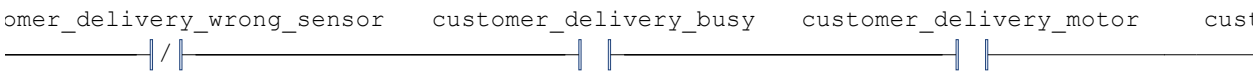

459
customer_delivery_route_1
customer_table_1_sensor
customer_delivery_e
customer_delivery_route_2
customer_table_2_sensor
customer_delivery_route_3
customer_table_3_sensor
customer_delivery_route_4
customer_table_4_sensor
customer_delivery_route_5
customer_table_5_sensor
customer_delivery_route_6
customer_table_6_sensor
460
customer_delivery_job_pending
customer_table_1_sensor
customer_delivery
customer_table_2_sensor
customer_delivery
customer_table_3_sensor
customer_delivery
customer_table_4_sensor
customer_delivery
customer_table_5_sensor
customer_delivery
customer_table_6_sensor
customer_delivery
461
customer_delivery_arrival_R_TRIG
customer_delivery_sensor_any
R_TRIG
CLK
Q
custo

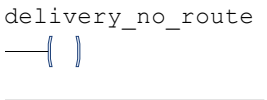
xpected_sensor

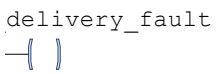


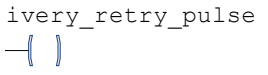
mer_delivery_arrival_pulse

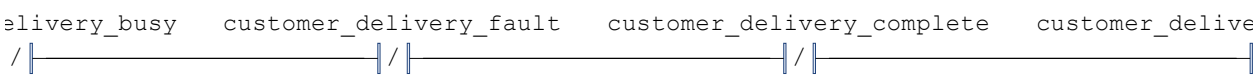

462
customer_delivery_arrival_pulse
customer_delivery_expected_sensor
cust
463
customer_delivery_job_pending
customer_delivery_route_valid
customer_
464
customer_delivery_fault_RS
customer_delivery_timeout
RS
SET
Q1
customer_
customer_delivery_wrong_sensor
customer_delivery_no_route
customer_delivery_reset
RESET1
465
operate
emergency
customer_delivery_stop_condition
466
customer_delivery_retry_R_TRIG
customer_delivery_retry
R_TRIG
CLK
Q
customer_del
467
customer_delivery_retry_pulse
customer_delivery_job_pending
customer_d

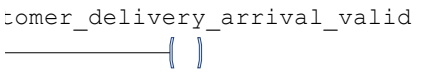

omer_delivery_wrong_sensor
customer_delivery_busy
customer_delivery_motor
cust

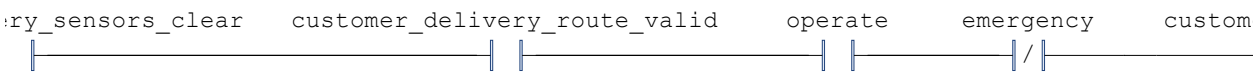

delivery_no_route


delivery_fault

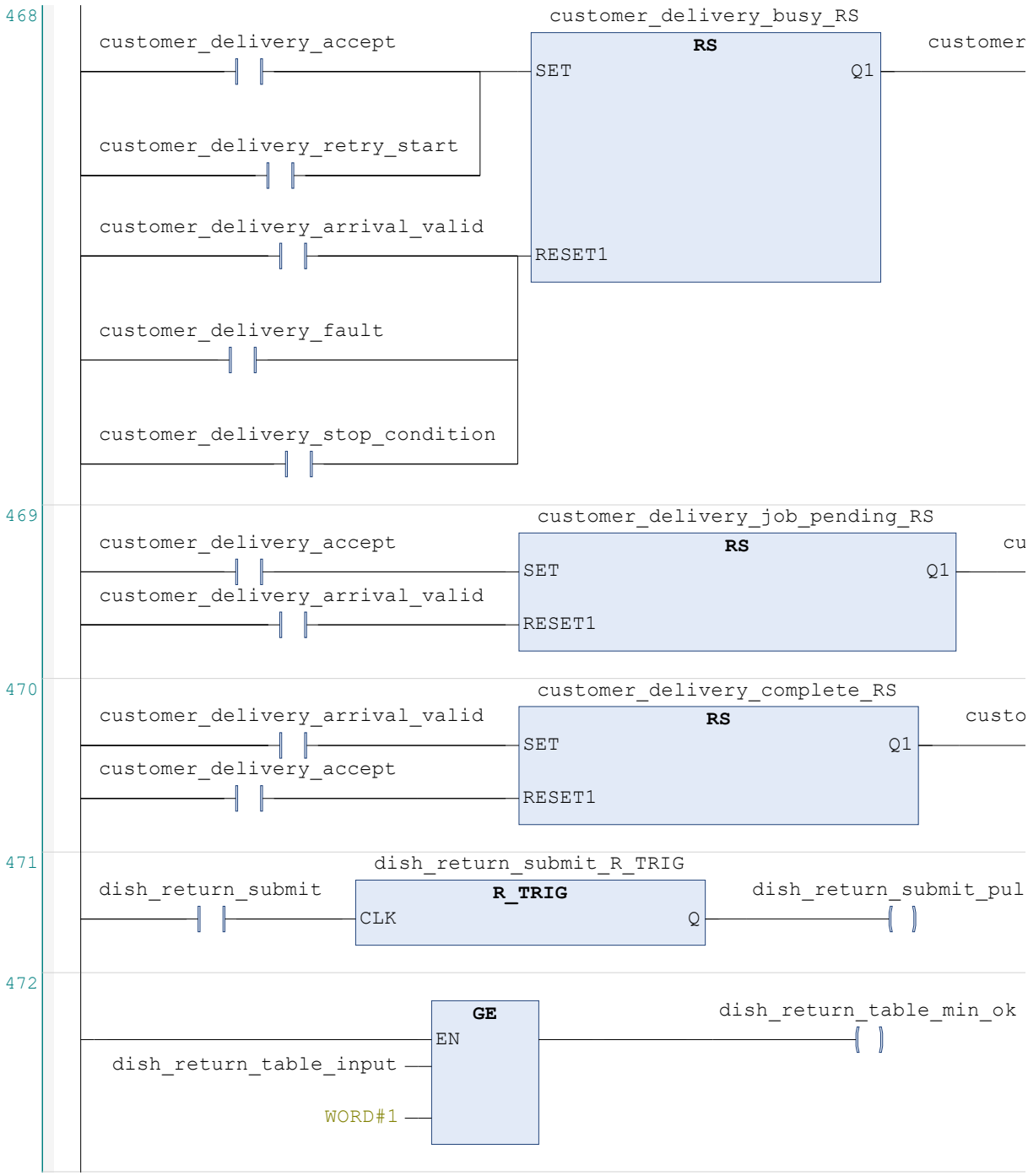

ivery_retry_pulse

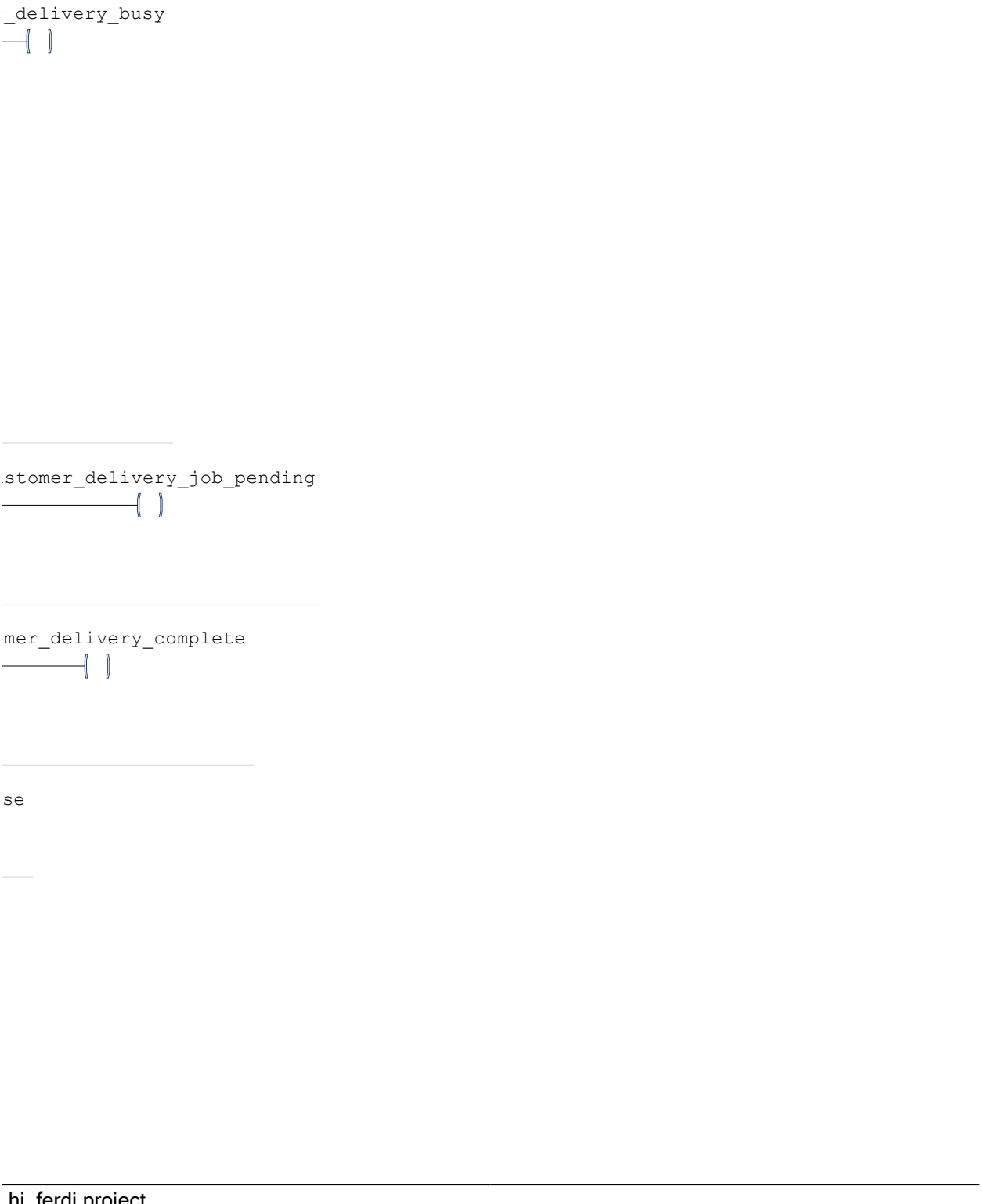

elivery_busy
customer_delivery_fault
customer_delivery_complete
customer_delive

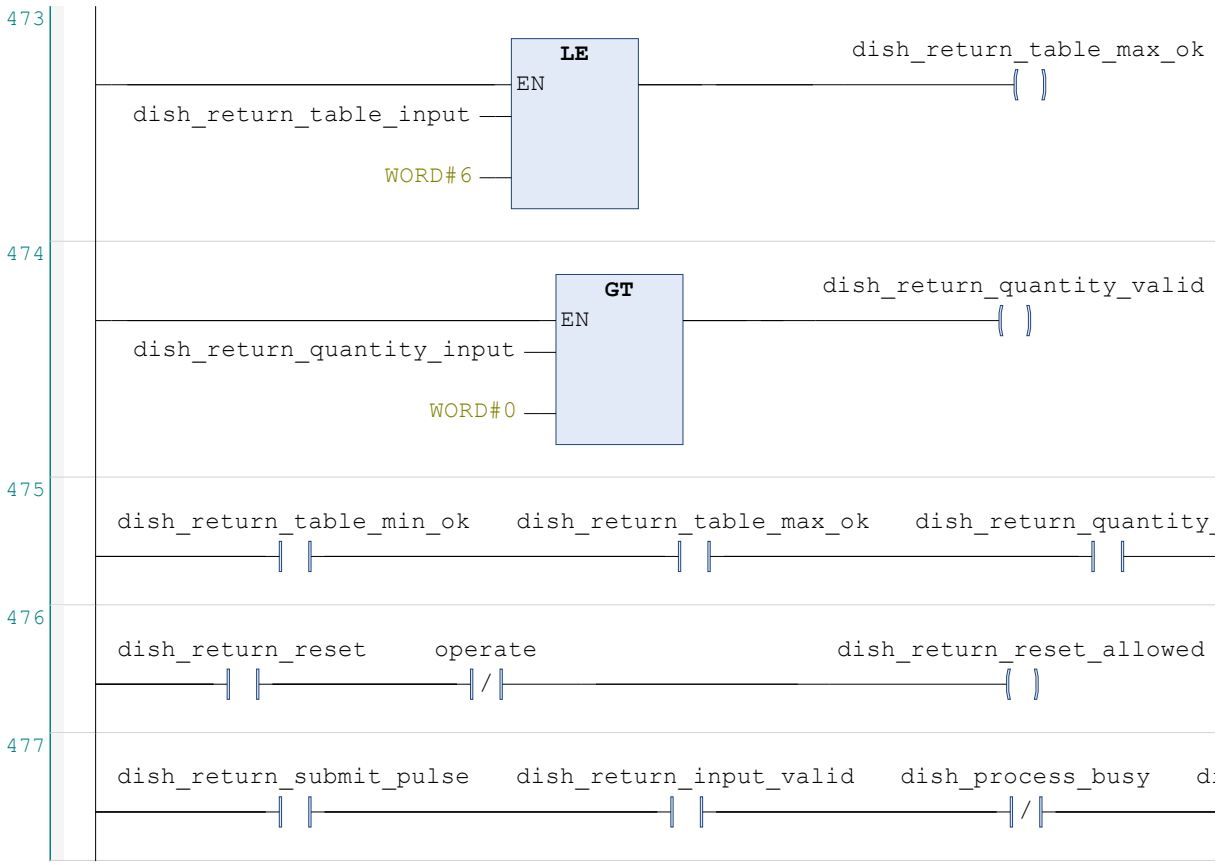

tomer_delivery_arrival_valid

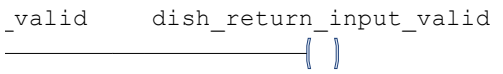

ry_sensors_clear
customer_delivery_route_valid
operate
emergency
/
custom

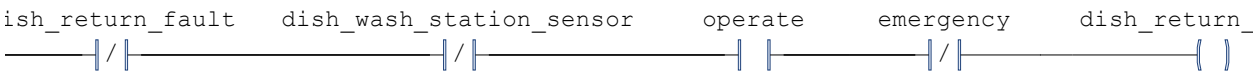
er_delivery_retry_start
( )


468
customer_delivery_accept
customer_delivery_retry_start
customer_delivery_arrival_valid
customer_delivery_fault
customer_delivery_stop_condition
customer_delivery_busy_RS
RS
SET
Q1
RESET1
customer
469
customer_delivery_accept
customer_delivery_arrival_valid
customer_delivery_job_pending_RS
RS
SET
Q1
RESET1
cu
470
customer_delivery_arrival_valid
customer_delivery_accept
customer_delivery_complete_RS
RS
SET
Q1
RESET1
custo
471
dish_return_submit
dish_return_submit_R_TRIG
R_TRIG
CLK
Q
dish_return_submit_pul
472
dish_return_table_input
WORD#1
GE
EN
dish_return_table_min_ok

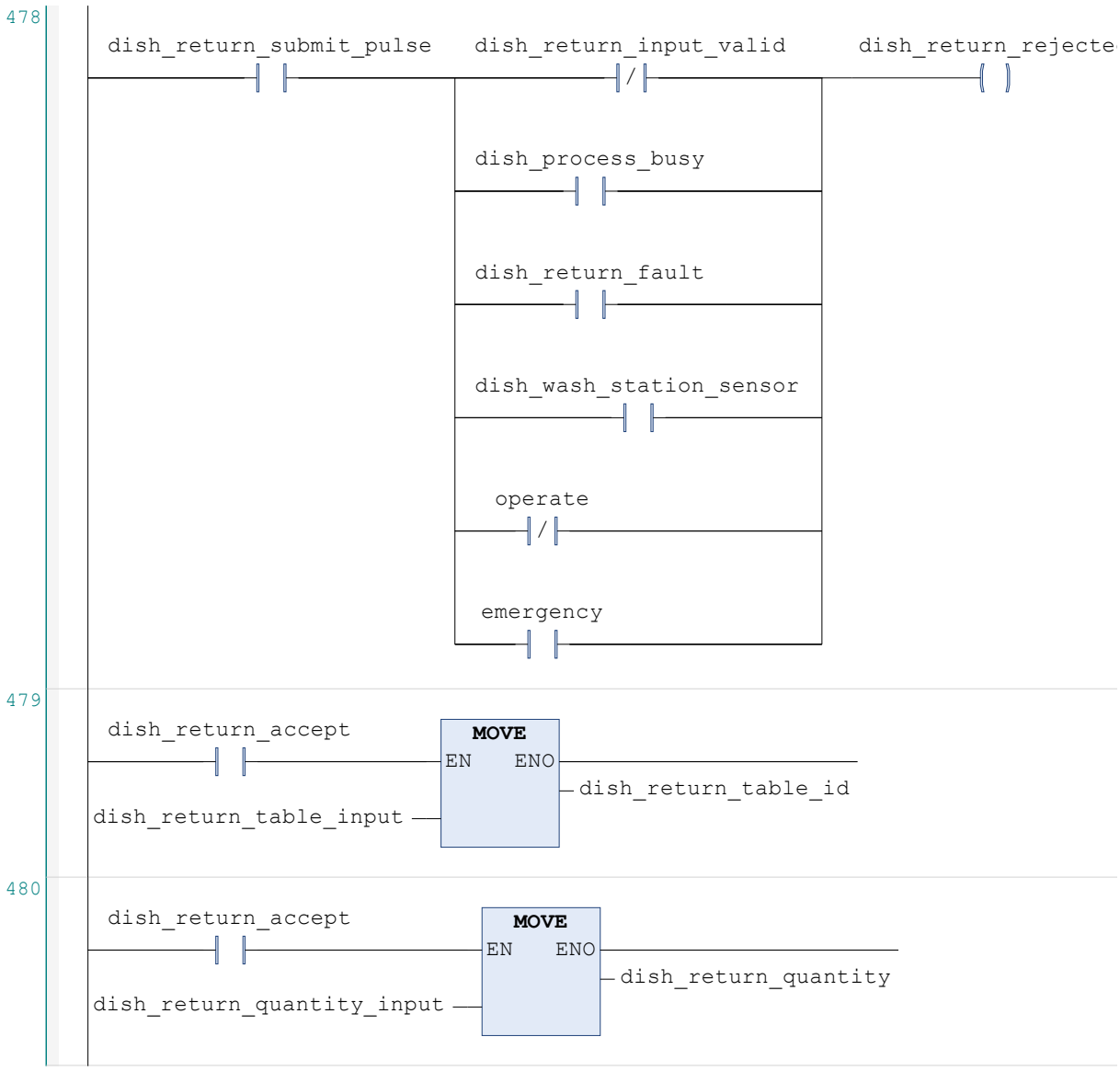

_delivery_busy
stomer_delivery_job_pending
mer_delivery_complete
se


473
LE
EN
dish_return_table_input
WORD#6
dish_return_table_max_ok
474
GT
EN
dish_return_quantity_input
WORD#0
dish_return_quantity_valid
475
dish_return_table_min_ok
dish_return_table_max_ok
dish_return_quantity_
476
dish_return_reset
operate
dish_return_reset_allowed
477
dish_return_submit_pulse
dish_return_input_valid
dish_process_busy

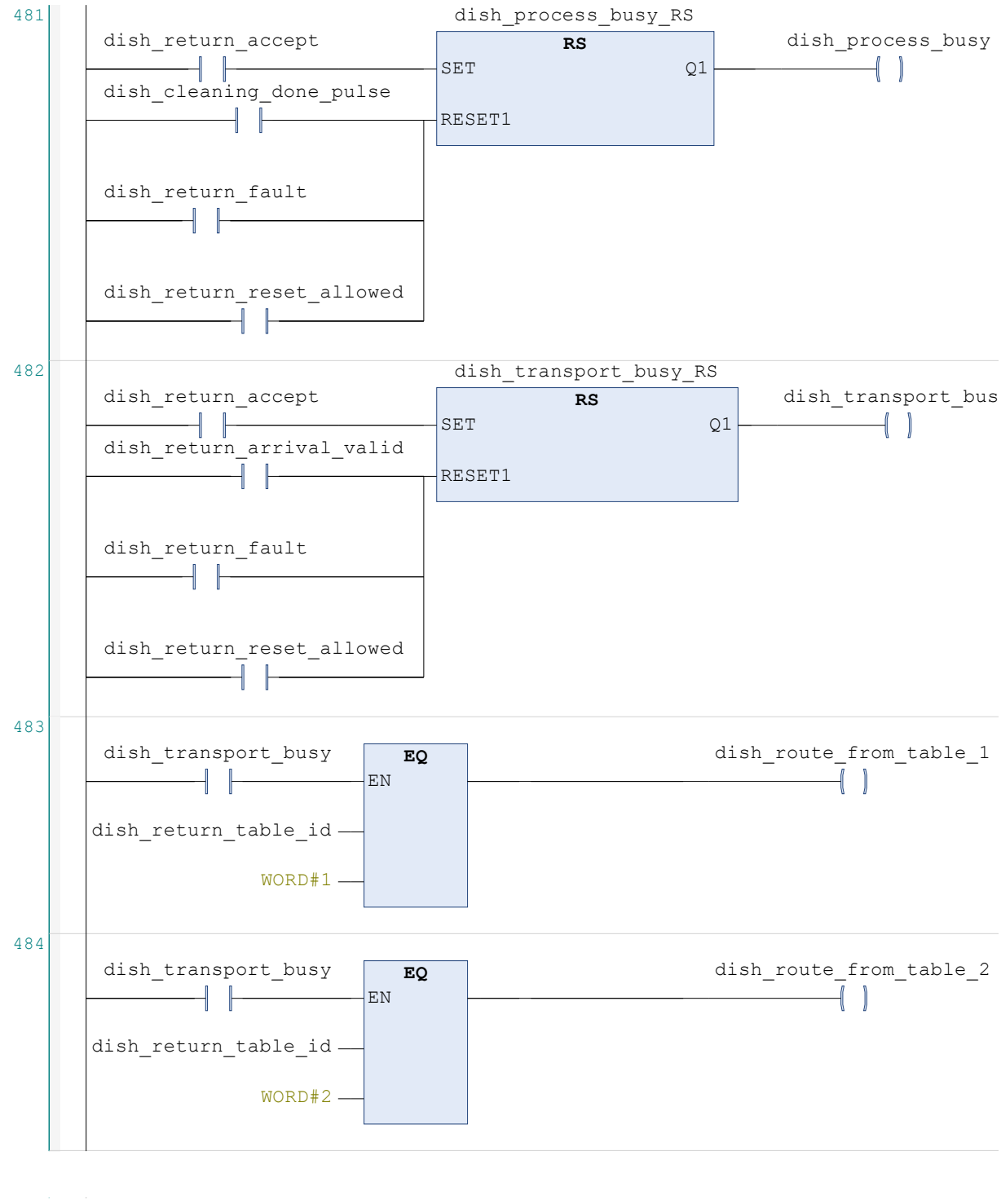

_valid
dish_return_input_valid


ish_return_fault
dish_wash_station_sensor
operate
emergency
dish_return_

```
accept
```

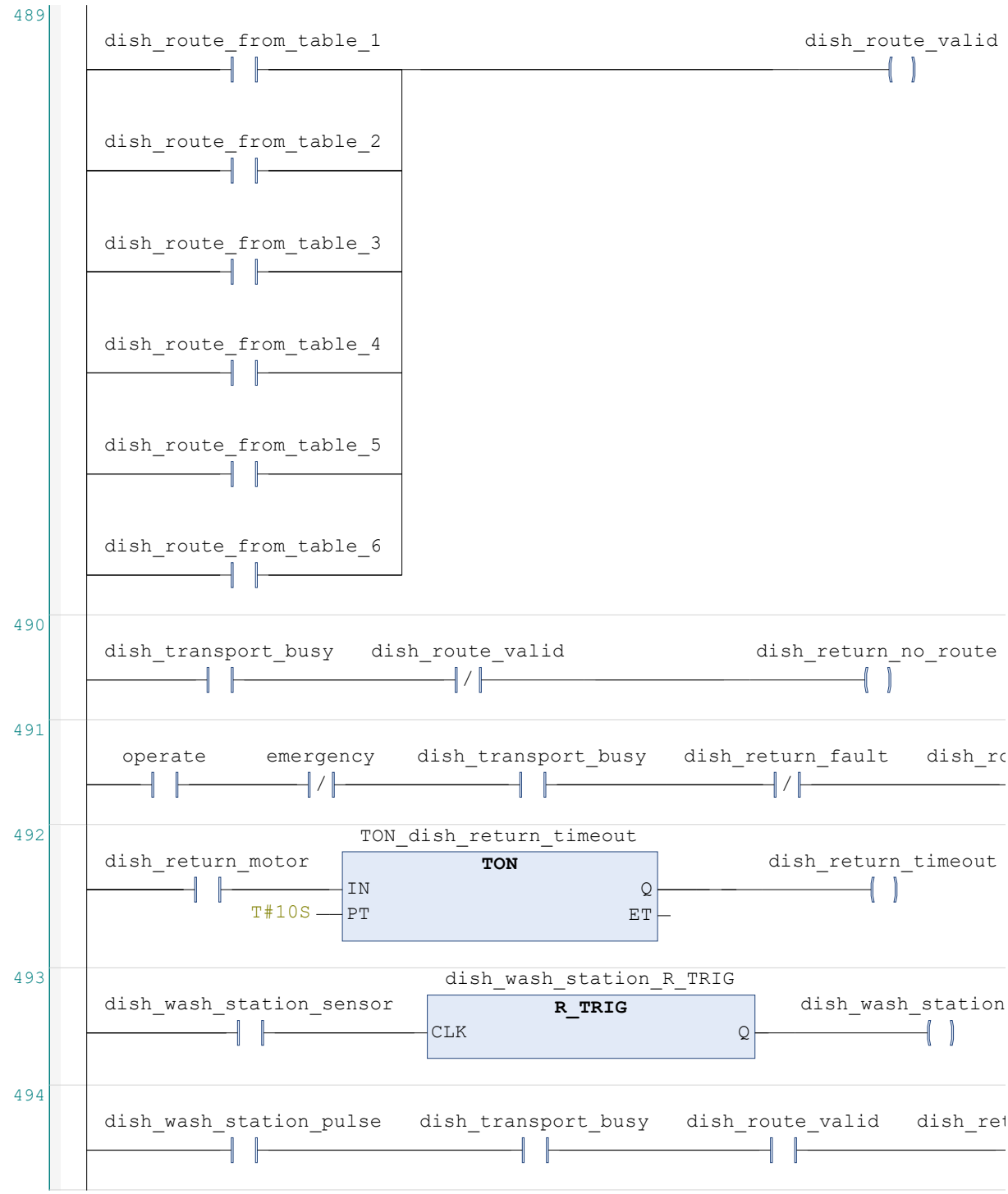

478
dish_return_submit_pulse
dish_return_input_valid
dish_return_rejecte
dish_process_busy
dish_return_fault
dish_wash_station_sensor
operate
emergency
479
dish_return_accept
MOVE
EN
ENO
dish_return_table_id
dish_return_table_input
480
dish_return_accept
MOVE
EN
ENO
dish_return_quantity
dish_return_quantity_input

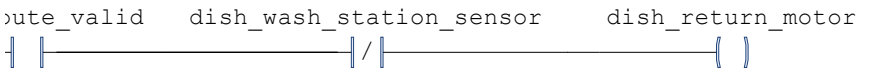
```
d
```


481
dish_process_busy_RS
RS
dish_return_accept
SET
Q1
dish_process_busy
dish_cleaning_done_pulse
RESET1
dish_return_fault
dish_return_reset_allowed
482
dish_transport_busy_RS
RS
dish_return_accept
SET
Q1
dish_transport_bus
dish_return_arrival_valid
RESET1
dish_return_fault
dish_return_reset_allowed
483
dish_transport_busy
EQ
EN
dish_route_from_table_1
dish_return_table_id
WORD#1
484
dish_transport_busy
EQ
EN
dish_route_from_table_2
dish_return_table_id
WORD#2

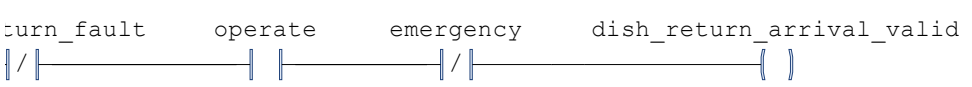
y

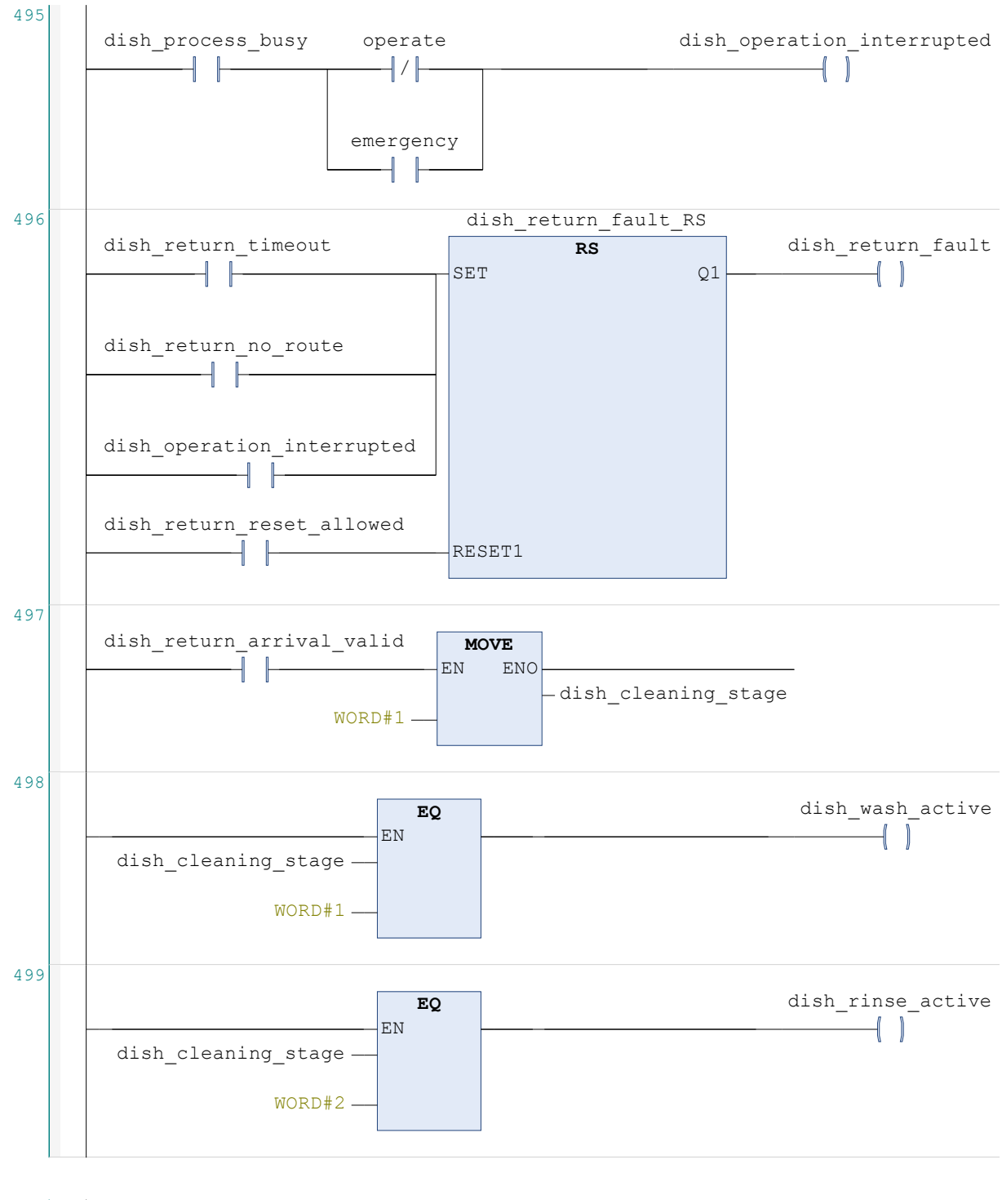

485
dish_transport_busy
EQ
EN
dish_return_table_id
WORD#3
dish_route_from_table_3
486
dish_transport_busy
EQ
EN
dish_return_table_id
WORD#4
dish_route_from_table_4
487
dish_transport_busy
EQ
EN
dish_return_table_id
WORD#5
dish_route_from_table_5
488
dish_transport_busy
EQ
EN
dish_return_table_id
WORD#6
dish_route_from_table_6

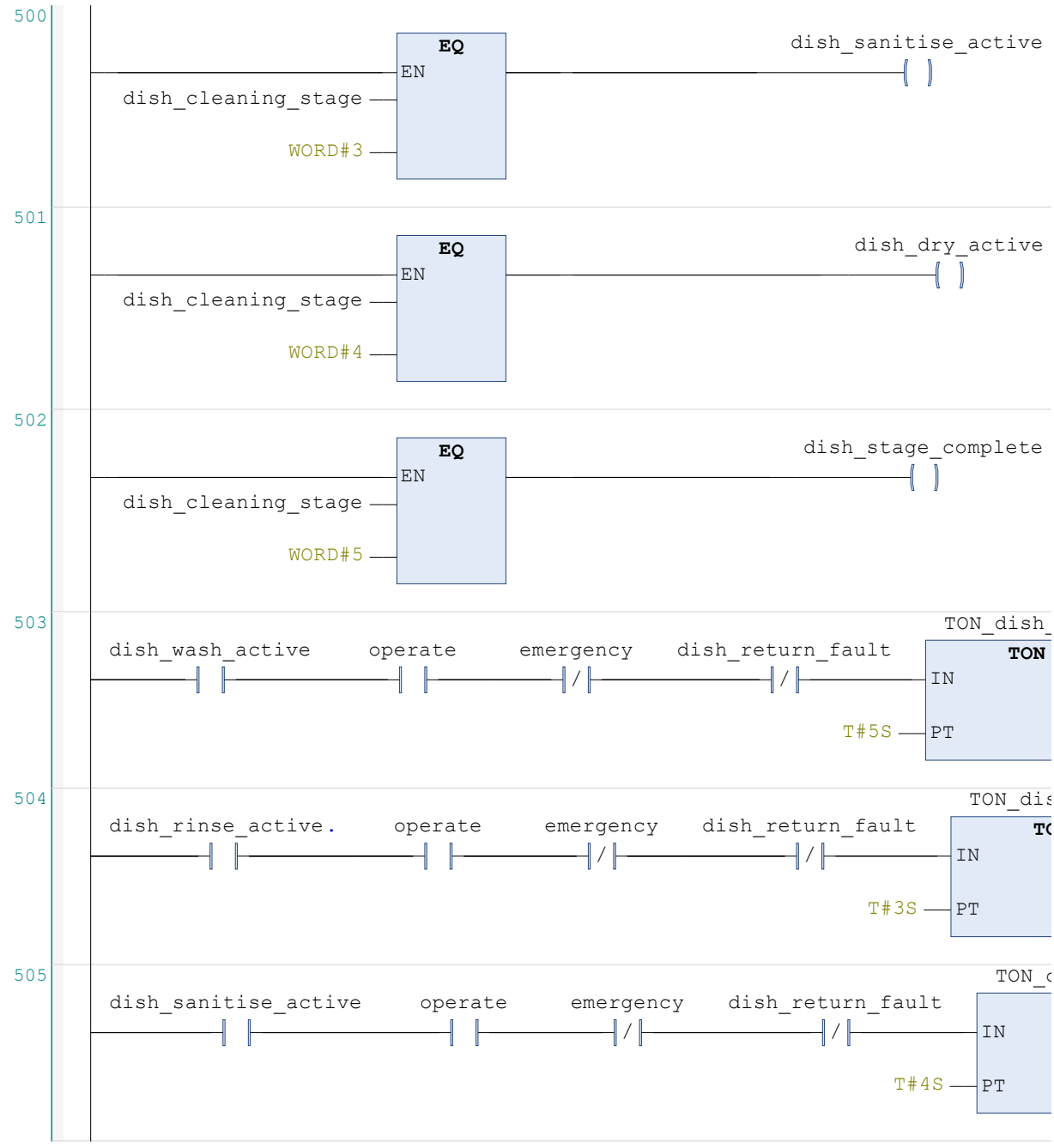

489
dish_route_from_table_1
dish_route_valid
dish_route_from_table_2
dish_route_from_table_3
dish_route_from_table_4
dish_route_from_table_5
dish_route_from_table_6
490
dish_transport_busy
dish_route_valid
dish_return_no_route
491
operate
emergency
dish_transport_busy
dish_return_fault
dish_rc
492
TON_dish_return_timeout
dish_return_motor
TON
IN
PT
T#10S
Q
ET
dish_return_timeout
493
dish_wash_station_R_TRIG
dish_wash_station_sensor
R_TRIG
CLK
Q
dish_wash_station
494
dish_wash_station_pulse
dish_transport_busy
dish_route_valid
dish_ret

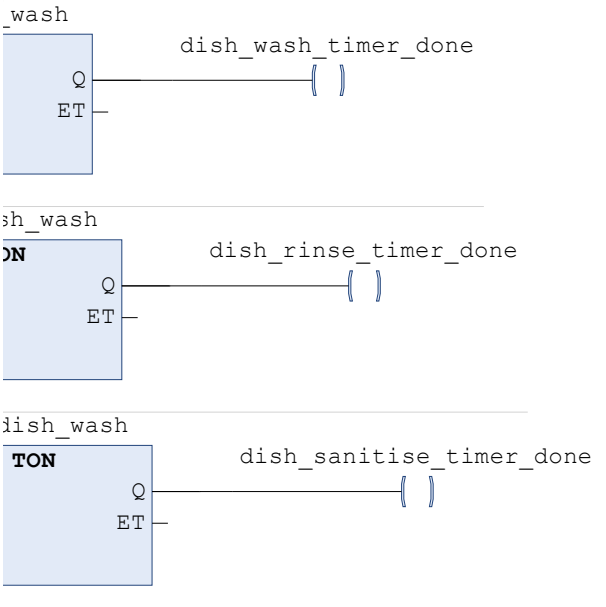

ute_valid
dish_wash_station_sensor
dish_return_motor

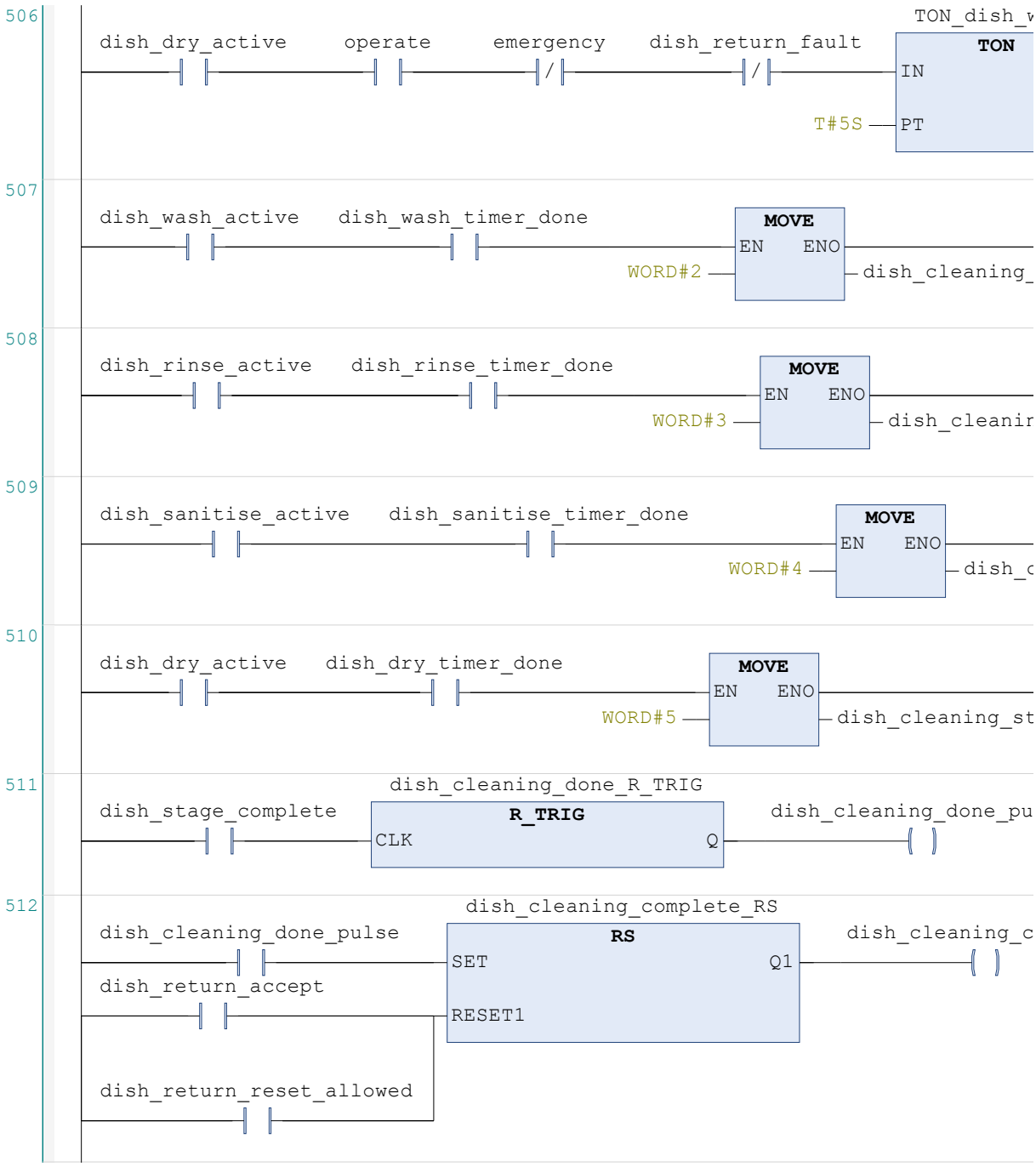
_pulse


turn_fault
operate
emergency
dish_return_arrival_valid

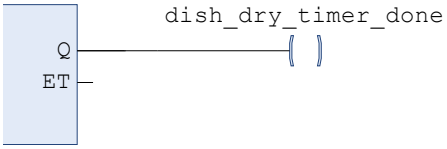

495
dish_process_busy
operate
/
dish_operation_interrupted
emergency
496
dish_return_fault_RS
RS
dish_return_timeout
SET
Q1
dish_return_fault
dish_return_no_route
dish_operation_interrupted
dish_return_reset_allowed
RESET1
497
dish_return_arrival_valid
MOVE
EN
ENO
dish_cleaning_stage
WORD#1
498
EQ
EN
dish_wash_active
dish_cleaning_stage
WORD#1
499
EQ
EN
dish_rinse_active
dish_cleaning_stage
WORD#2


500
EQ
EN
dish_cleaning_stage
WORD#3
dish_sanitise_active
501
EQ
EN
dish_cleaning_stage
WORD#4
dish_dry_active
502
EQ
EN
dish_cleaning_stage
WORD#5
dish_stage_complete
503
TON_dish_
dish_wash_active
operate
emergency
dish_return_fault
TON
IN
T#5S
PT
504
TON_dis
dish_rinse_active.
operate
emergency
dish_return_fault
IN
T#3S
PT
505
TON_
dish_sanitise_active
operate
emergency
dish_return_fault
IN
T#4S
PT


_wash
dish_wash_timer_done
Q
ET
sh_wash
ON
dish_rinse_timer_done
Q
ET
dish_wash
TON
dish_sanitise_timer_done
Q
ET


506
TON_dish_w
dish_dry_active
operate
emergency
dish_return_fault
TON
IN
T#5S
PT
507
dish_wash_active
dish_wash_timer_done
MOVE
EN
ENO
WORD#2
dish_cleaning_
508
dish_rinse_active
dish_rinse_timer_done
MOVE
EN
ENO
WORD#3
dish_cleanir
509
dish_sanitise_active
dish_sanitise_timer_done
MOVE
EN
ENO
WORD#4
dish_c
510
dish_dry_active
dish_dry_timer_done
MOVE
EN
ENO
WORD#5
dish_cleaning_st
511
dish_cleaning_done_R_TRIG
dish_stage_complete
R_TRIG
CLK
Q
dish_cleaning_done_pu
512
dish_cleaning_complete_RS
dish_cleaning_done_pulse
RS
SET
Q1
dish_cleaning_c
dish_return_accept
RESET1
dish_return_reset_allowed

vash



_stage

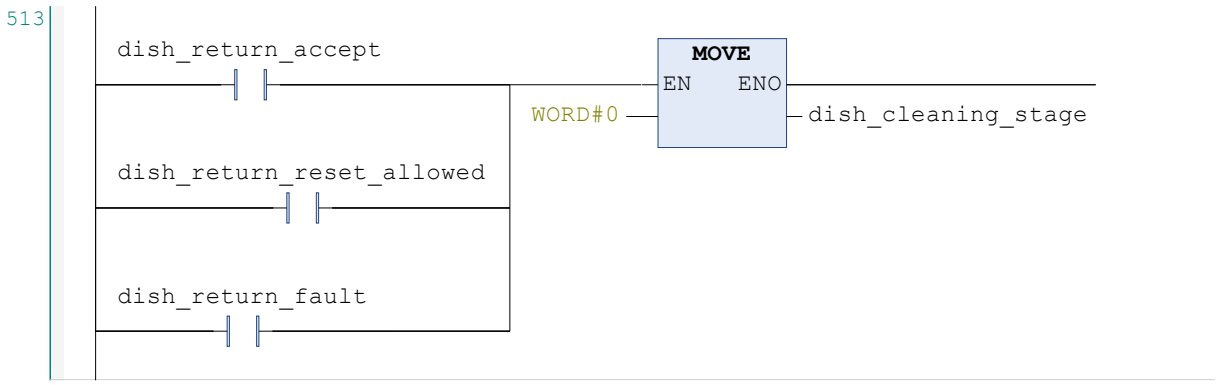
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dish_return_reset_allowed
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dish_cleaning_stage